



NCR Regional Development Plan 2017-2022



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NCR Regional Development Plan 2017-2022

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REGIONAL DEVELOPMENT COUNCIL NATIONAL CAPITAL REGION

RDC RESOLUTION NO. 3 Series of 2018

APPROVING AND ADOPTING THE REGIONAL DEVELOPMENT PLAN OF THE NATIONAL CAPITAL REGION (RDP-NCR) 2017-2022

WHEREAS, pursuant to Executive Order No. 5 (S. 2016) adopting the long term vision or the *AmBisyon Natin 2040*, the President issued Memorandum Circular No. 12 (S.2016) which directs the formulation of the Philippine Development Plan and the Public Investment Program for the planning period 2017-2022;

WHEREAS, in adherence to the President's directive, MMDA formed sectoral planning committees to prepare the various chapters of the plan to ensure that all sectors are represented and actively involved in the plan formulation process, taking into consideration special areas of concern including but not limited to: Climate Change Adaptation and Mitigation, Disaster Risk Reduction, Gender Mainstreaming and Development, among others;

WHEREAS, the RDC-NCR has conducted series of bilateral consultations to solicit inputs with regional line agencies and private sectors;

WHEREAS, a validation workshop was conducted on 29 January 2018 wherein the latest draft of the Plan was presented and validated by the stakeholders;

WHEREAS, consistent with the President's instructions, the herein attached Regional Development Plan of the National Capital Region (RDP-NCR) 2017-2022 (*see Annex A*) is anchored on the *AmBisyon Natin 2040*, the PDP 2017-2022, the MMDA Resolution No. 12-28, s. 2012 otherwise known as the "*Metro Manila Greenprint Vision 2030*" and includes with it the assessment and challenges, strategies, and programs, activities and projects that are necessary in shaping the development path and priorities of the region in the next five (5) years;

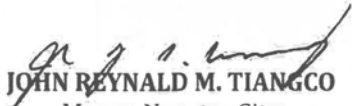
WHEREAS, the Regional Line Agencies issued their concurrence to the latest draft of the RDP-NCR 2017-2011, consistent to their respective mandates, thrust and priorities, and now presents the same for the Council's approval;

NOW, THEREFORE, BE IT RESOLVED, the Council in session duly assembled, recognizing the merit and importance of the Philippine Government's overall long-term goals to achieve "a prosperous, predominantly middle-class society where no one is poor; our peoples shall live long and healthy lives, be smart and innovative, and shall live in a high-trust society, the Regional Development Council pursuant to Section 6 of RA 7924 and E.O. No. 113, s. of 2002, hereby **APPROVES AND ADOPTS** the Regional Development Plan (RDP) of the National Capital Region (NCR) 2017-2022.

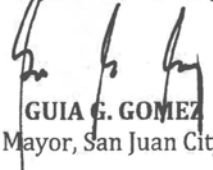
This Resolution shall take effect June 21, 2018.

APPROVED this 21st day of June 2018 in Makati City Metro Manila, Philippines.

DANILO DELAPUZ LIM
Chairman, MMDA



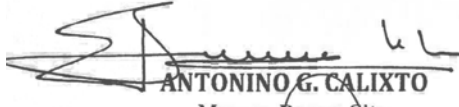
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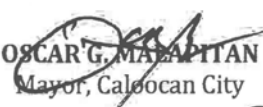
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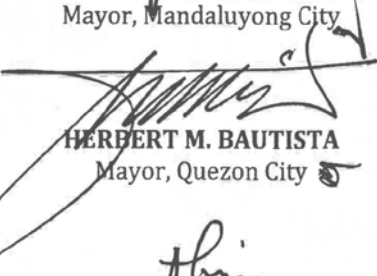
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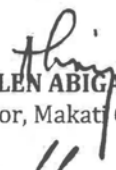
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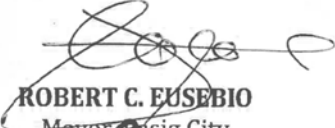
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EDWIN L. OLIVAREZ
Mayor, Parañaque City



LUIS I. BUSTAMANTE
President, Metro Manila Vice Mayors' League

Message



Earlier this year, we launched the Philippine Development Plan (PDP) 2017-2022 through the initiative of the National Economic and Development Authority (NEDA). The PDP serves as our medium-term blueprint towards attaining a more secure life for our people in the next 25 years.

To complement the PDP, we have also formulated Regional Development Plans 2017-2022. This will provide direction in policy formulation for the next six years as we steer public and private investments to the regions.

We intend to place regional development at the center of our socio-economic development strategy. By creating more jobs improving social services, encouraging innovation and connecting the countryside to growth centers, we will reduce poverty and accelerate development in rural areas.

The RDPs will also prioritize accelerating infrastructure development, protecting our natural resources, addressing criminality and illegal drugs, and ensuring peace and security in the regions.

I commend the Regional Development Councils for aligning their respective regional plans to our PDP through fruitful collaboration with local government units and the private sector. I am confident that through the RDPs, we can realize our goal of laying a solid foundation for a stronger and more resilient nation for future generations.


RODRIGO ROA DUTERTE
President
Republic of the Philippines

Message



With regional and local development being one of the main thrusts of President Rodrigo R. Duterte's socioeconomic development agenda, the Philippine Development Plan (PDP) 2017-2022, the first medium-term plan anchored on a long-term vision (*AmBisyon Natin 2040*), was specifically designed to cultivate growth and reduce inequality between the regions. This can be achieved by directing development to key areas throughout the country and connecting these growth centers to rural areas

As such, the Regional Development Plans (RDPs), as accompanying documents to the PDP, will be an important tool in guiding both public and private investments that will catalyze growth in the regions. It will also serve as our blueprint in laying down the three main pillars of *Malasakit, Pagbabago, at Patuloy na Pag-unlad* that will help us build a secure, comfortable, and strong rooted life for all Filipinos by 2040.

I would like to express my gratitude to the Regional Development Councils (RDCs) for their leaderships in the crafting of the RDPs, and the coordinating various development efforts in the regions. Finally, we seek the support of our local government units, regional institutions, and private institutions to support the realizations of our plans as we venture towards the creation of prosperous, peaceful and resilient communities



ERNESTO M. PERNIA
Secretary of Socioeconomic Planning

Preface



The Regional Development Plan for the National Capital Region (RDP-NCR) 2017-2022 is dedicated to all our partners in development and to the residents of Metro Manila who are the workforce behind our region's sustained growth and development. The main objective of this Plan is to provide the general development framework that will guide the region's sectoral growth priorities and strategies in support of the overall long-term vision: "Metro Manila for all; green, connected, resilient; offering talent and opportunity; processing knowledge and delivering services at home and abroad," geared towards transforming Metro Manila into a highly competitive Asian metropolis in the future.

This Plan is fully committed in contributing concretely for the fulfillment of President Rodrigo Roa Duterte's 0+10 Point Socio-Economic Agenda, PDP 2017-2022 targets for the realization of the 2030 Sustainable Development Agenda (SDA) and *AmBisyon Natin 2040*.

The development path of Metro Manila is challenged by the impact of rapid urbanization on land use, increasing demand for health and social services, inadequate and modern infrastructure to support the demands of a growing population, and to mitigate the effects of climate change.

However, with the unified efforts of all stakeholders in development and by strengthening public-private partnership initiatives, substantiated by the provision of modern infrastructure, quality education and more income opportunities, the government's goal of providing the public with better social services and adequate growth opportunities across all sectors will not be far behind.

The RDP-NCR 2017-2022 is not an MMDA plan, but a Plan for Metro Manila owned by all our cooperating partners and completed by various stakeholders, who, without reservation, invested their time, effort and talent in a formulation process characterized by in-depth analysis of sectoral challenges and opportunities as well as thorough discussions of development strategies and prioritization of development interventions.

As RDC-NCR Chairman, I dedicate this Plan to the people of Metro Manila, appealing to everyone to strive harder and cooperate more for a better Metropolitan Manila towards a brighter future and a more prosperous nation for our people.

"MABUHAY PO TAYONG LAHAT"


DANILO DELAPUZ LIM
Chairman
RDC -NCR and MMDA

PART I

INTRODUCTION



The background of the slide is a gradient of blue and red wavy lines, creating a sense of motion and depth. The blue is a deep, rich color, while the red is a vibrant, fiery red. The waves flow from the bottom left towards the top right, creating a dynamic and modern aesthetic.

01

Introduction

Introduction

On October 24, 2016, the Office of the President issued Memorandum Circular No. 12, s. 2016, mandating the National Economic and Development Authority (NEDA) to take the lead in formulating the Philippine Development Plan (PDP) and its accompanying document, the Public Investment Program (PIP), covering the period 2017-2022.

In the absence of a NEDA Regional Office in the National Capital Region (NCR), the Metro Manila Development Authority (MMDA), serving as the Regional Development Council (RDC) technical secretariat for NCR, led the Regional Development Plan (RDP) formulation process in early 2017.

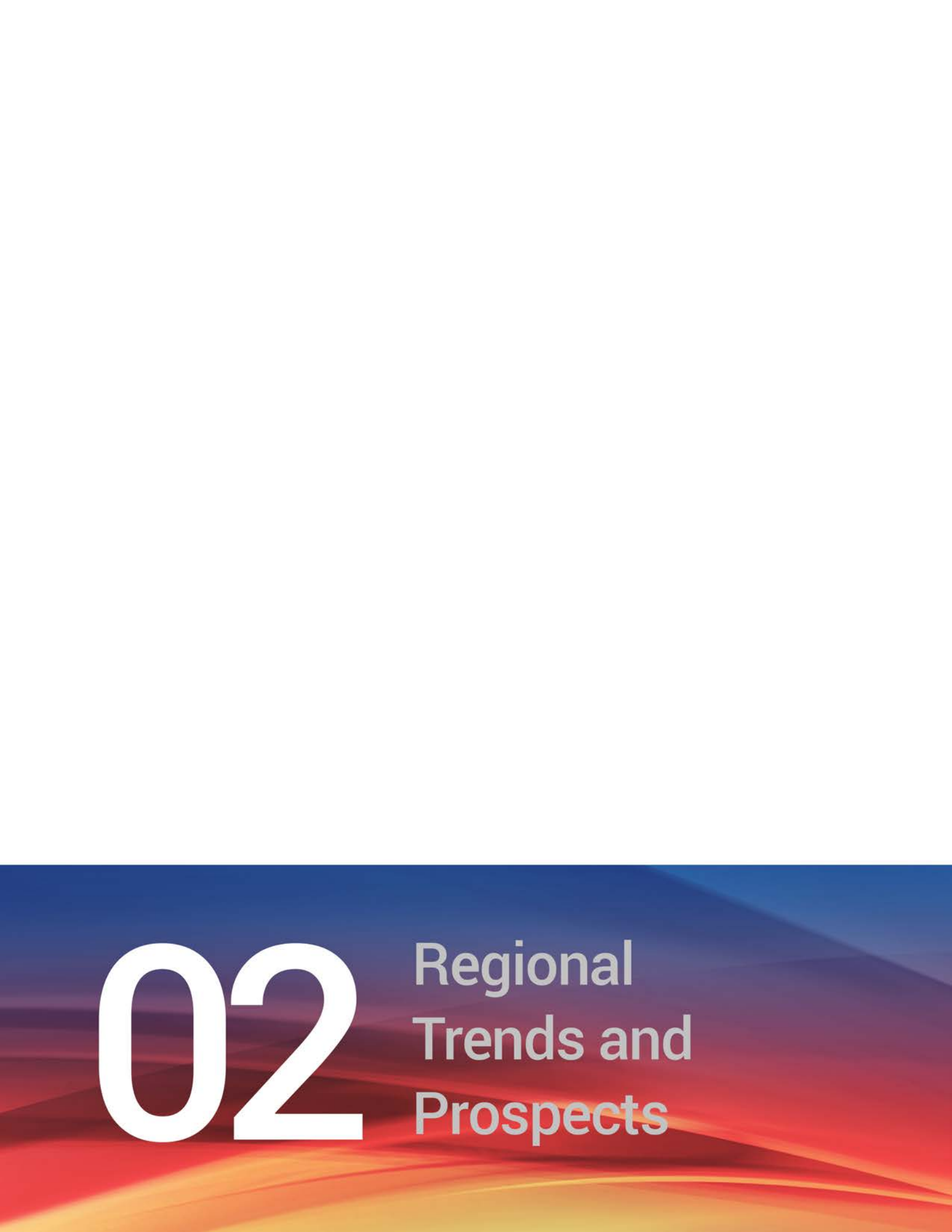
The formulation of the RDP-NCR started during the time of then OIC-Chairman Thomas M. Orbos in 2016. The draft chapters of the RDP-NCR 2017-2022, which focused on the sectoral assessment and review of opportunities and challenges in the region, served as the working draft for the consultative meetings and writeshops with regional line agencies, the private sector, and the local government units (LGUs).

Series of consultative meetings and workshops with cooperating partners were conducted to further improve and refine the Plan Chapters. A validation workshop was also conducted to ensure that cross-cutting concerns among thematic areas are adequately covered.

The RDP-NCR contains sectoral assessment and outlines the opportunities and challenges that face each thematic areas. The Metro Manila Greenprint 2030 which identified the long-term vision for Metro Manila also guided the crafting of the development direction and priority strategies for the region.

The RDP, as a development instrument, provides a strong basis for the determination of a strategic framework which will effectively guide the identification of priority programs, projects, and policy interventions to operationalize the RDP-NCR. This in turn will help ensure the attainment of development targets, in support of President Rodrigo Roa Duterte's 0-10 Point Socioeconomic Agenda, *AmBisyon Natin 2040*, PDP 2017-2022, and 2030 Agenda for Social Development.

This plan is intended to be used by stakeholders and partners as a tool to build on the promise of the region to transform Metro Manila into a highly competitive Asian metropolis in the future.



02

Regional
Trends and
Prospects

Regional Trends and Prospects

The National Capital Region (NCR), more popularly known as Metro Manila is the 18th largest megacity in the world.¹

It is the seat of the Philippine Government and is considered as the country's premier region being the center of political, cultural, and economic activities, as well as being the key driver of urban development in the Philippines.

The region, as one of the 12 defined metropolitan areas in the country,² exerts the most significant impact on commerce, finance, media, art, fashion, research, technology, education, and entertainment, both locally and abroad.

Similar to other premier urban centers in the world, the region attracts migrants from all over the country in search of better economic opportunities, jobs, and education. This urbanization trend of migration has significantly influence not only the land-use pattern within the metropolis, but also the quality and delivery of basic services to its people such as public transportation, education, health, and housing.

This rapid urban growth in the NCR applies tremendous pressure on the existing social and economic system of the metropolis, as well as aggravates existing environmental concerns such as air and water pollution, waste management, health and sanitation, and inadequacy of land and water resources in the urban areas.

These issues are further complicated by the effects of climate change which bring forth bigger problems for the metropolis, such as extreme weather conditions and vulnerabilities to natural disasters (e.g., flooding and landslides).

NCR is the country's political, financial, and economical capital, accounting for 36.4 percent of the Gross Domestic Product in 2017. Metro Manila ranks 15th among the largest urban agglomeration in the world. In addition, the rapid pace of development of Metro Manila's peri-urban areas further underlines the dominance of the region.

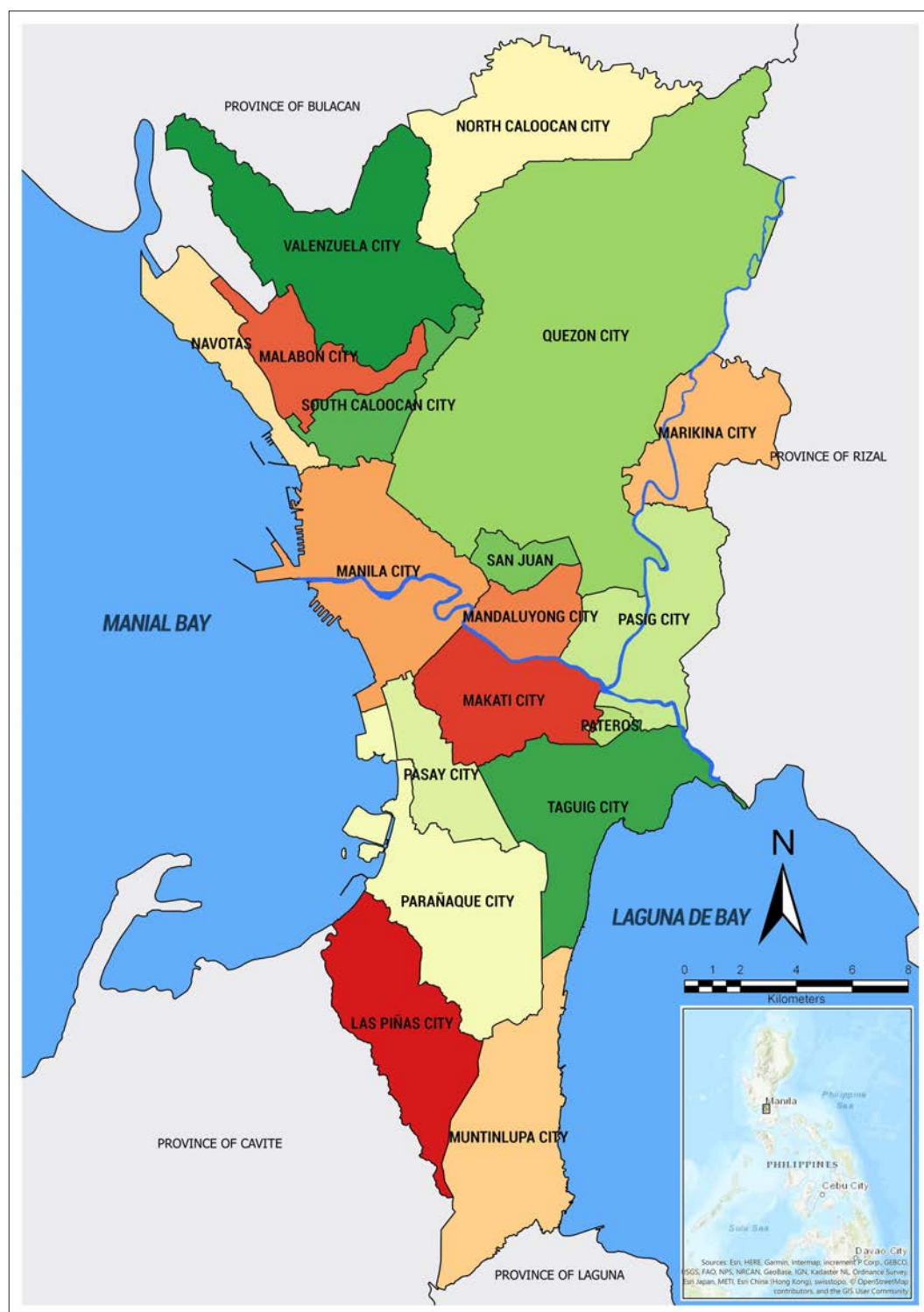
While Metro Manila leads in economic development among urban areas in the country, it lags with other Asian cities. Lack of planning contributes to the region's failure to account for its rapid population growth, environmental degradation, lack of infrastructure, and slow service provision, to name a few, costing Metro Manila the competitiveness it enjoyed five decades ago. The efficiency of the region is debilitated by traffic congestion, urban blight, periodic flood and many other factors. However, NCR remains the dominant urban center in the country. It has a pivotal development role in the country's competitiveness and growth. The development trajectory of the country will continue to depend largely on the performance of the region.

Given the demographic and socioeconomic trends, the condition in the region is expected to improve if the tourism, infrastructure, industry, and services sectors will be harnessed.

¹United Nations. (2016). The World's Cities in 2016 Data Book.

²"Building Globally Competitive Metro Areas in the Philippines." National Economic Development Authority. August 30, 2007.

Figure 2.1. Location Map of the National Capital Region



Tourism and Infrastructure

The NCR is rich in cultural resources as evident by its various tangible and intangible heritage. It is home to many cultural heritage sites and top museums housing numerous artifacts and artworks which are representatives of the diverse cultures in the capital.

NCR, being the main gateway to many destinations in the Philippines, registers the highest tourist arrivals in the country as it serves as the drop-off point due to the presence of international and domestic airports and seaports. However, it fails to attract tourists as a first choice destination based on the average length of tourist stay which is only a maximum of two days.

To unleash NCR's tourism potential and attract neighbour countries, the Metro Manila Greenprint 2030 aims to make the region the first choice destination for newly emerging affluent citizens of East Asia for shopping, cultural tourism, visits to heritage sites, conventions, meetings, and entertainment.

To achieve this goal, the following are the identified principles indicated in the Metro Manila Greenprint 2030:

- Provide visitors with a variety of activities to participate in and destinations to explore
- Assure visitors of their personal safety
- Ensure visitors have access to high-quality infrastructure and high levels of convenience

There is a need to identify and promote key activities and destinations such as cultural heritage sites, major shopping destinations, emerging integrated resorts in Manila Bay waterfront, world-class Meetings, Incentives, Conferences, and Events facilities, and a convenient central staging location for beach tourism.

To encourage tourist arrivals, the region needs to have modernized infrastructures that will make travelling seamless such as airports, seaports, and roads, which are the backbone of tourism industry. Good working infrastructures will make travelling seamless, especially in using the Tourism Highway for the continuous influx of tourists in Metro Manila.

The improved transport infrastructure will be complemented by tourism-related auxiliary facilities, such as accommodations offering world-class facilities and services, information centers, and the like.

Industry and Services

The following strategies will be carried out to enhance the industry and services sectors in the region:

Maintain and enhance the existing industry in Metro Manila.

The industry sector may not be the prime economic driver in the region, but it would trigger investments in support of certain types of activities that are beneficial to the economy.

Improve the development of the peri-urban areas in Metropolitan Manila.

With regard to peri-urban livelihood opportunities, there is a need to focus public investment and promote public-private partnership initiatives to improve

infrastructure and systems that will support globally competitive activities such as in the areas of electronics, manufacturing, and automotive assembly.

Create industry-specific incubators and platforms to catalyze growth.

Industry-specific cluster development should be supported through setting up of small firm incubators and/or platforms that allow cross-firm knowledge-sharing, ease of access to resources, and “seed” funding from either public and private sources.

These platforms will provide small businesses with access to funding and resources that they may not have the capability to obtain independently.

03

Overlay of Economic
Growth, Demographic
Trends, and Physical
Characteristics

Overlay of Economic Growth, Demographic Trends, and Physical Characteristics

The Regional Development Plan - National Capital Region (RDP-NCR) 2017-2022, consistent with the development goals of the Philippine Development Plan (PDP) 2017-2022, aims for the realization of inclusive development which is anchored on *AmBisyon Natin 2040*. The plan defines the desired spatial distribution of the human activities in the region considering economic, social, political, and environmental conditions. Below are the discussions on the region's economic and demographic trends and physical conditions which are important considerations in the Plan's implementation.

Location

The NCR is strategically located in the middle of the Luzon Island, on the eastern coast of Manila Bay, and at the mouth of Pasig River. It lies along the rugged lands of the Marikina Valley and the Sierra Madre mountains in the east, bounded by the Manila Bay in the west, Central Luzon in the north, and Laguna de Bay in the south.

Land Area and Composition

The NCR is the smallest region in the country with a land area of only 636 square kilometers occupying less than one percent (0.19%) of the total land area of the country of 343,282.00 square kilometers.

The NCR is composed of 17 local government units (LGUs) namely Manila City, Pasay City, Caloocan City, Quezon City, Makati City, Mandaluyong City, Malabon City, San Juan City, Navotas City, Muntinlupa City, Valenzuela City, Las Piñas City, Parañaque City, Taguig City, Marikina City, Pasig City, and the Municipality of

Pateros.

In terms of land area, Quezon City is the biggest LGU with 166.20 square kilometers or more than one forth of the total land area of NCR.

Land Use Patterns

In NCR, land use has largely been in response to the socioeconomic demands of the growing population and increasing density of informal settlements in city centers.

Having a polycentric urban structure as evident in the presence of several distinct central business districts Metro Manila is relatively unique in East Asia. Metro Manila, however, is very much shaped by the private sector, with most LGUs accepting a passive land use development role. It is only through the zoning ordinances that LGUs control land use development with major interventions such as Catalytic Infrastructure/Development being left to external or higher actors.¹

¹The Metro Manila Greeprint 2030: Building a Vision

Table 3.1. Metro Manila LGUs Primary Function and Regional Roles

LGU	PRIMARY FUNCTION (PER CLUP*)	SELF-DESCRIBED REGIONAL ROLE	METROPOLITAN IMPACT
Caloocan	Suburban Residential Area	None	Secondary
Las Piñas	Suburban Residential Area	None	Secondary
Makati	Hub for Business Nationwide	National/Global Business Hub	Primary
Mandaluyong	N/A	N/A	N/A
Manila	Historic Regional Center	Hub for Business Nationwide	Primary
Marikina	Suburban Mixed Use Area	Bedroom Community	Secondary
Muntinlupa	Suburban Mixed Used Area	Gateway to CALABARZON	Secondary
Navotas	Suburban Residential Area	Fishing Industry Hub of MM	Tertiary
Paranaque	Emerging Mixed Use Hub	None	Primary
Pasay	Established Mixed Use Hub	Cross Roads of MM	Primary
Pasig	Emerging Regional Center	Emerging Regional Center	Primary
Pateros	Urban Residential Area	Residential Enclave	Tertiary
Quezon City	Suburban Res + Emerging Commercial Hub	Green, Knowledge, and Health Hub	Primary
San Juan	Urban Residential Area	None	Tertiary
Taguig	Mixed Use	Emerging International Business Hub	Primary
Valenzuela	Residential + Industrial Area	Northern Gateway to MM	Secondary

**Comprehensive Land Use Plan*

Moreover, LGUs largely adopt planning roles independent from other LGUs with Primary / Metropolitan Role LGUs such as Makati City, Manila City, Pasay City, Quezon City, Taguig City and Secondary Role LGUs such as Las Pinas City, Caloocan City, Muntinlupa City, and Marikina City (*see Table 3.1*).

Looking at the current and proposed Comprehensive Land Use Plans (CLUPs) of the 17 LGUs of NCR, it is evident that the metropolis is primarily residential in nature. The metropolis also has a large industrial as well as parks and open space portions within its borders (*see Table 3.2*).

Population

In the latest 2015 population census conducted by the Philippine Statistics Authority (PSA), NCR registered as the 2nd most populous region among 18 administrative regions in the country having 12,877,253 or 12.75 percent of the total 100,981,437 population. The

region is also considerably crowded with a population density of 20,785 persons per square kilometer compared to the national figure of 337 persons per square kilometer.

The 12.8 million population who resides in Metro Manila balloons to an estimated 15 million during daytime when residents from nearby provinces such as Bulacan, Rizal, Laguna, and Cavite come to the metropolis for purposes of schooling, working, transacting business, or simply for pleasure or recreation.

Quezon City, Caloocan City and Manila City have the highest population, while Manila City, Caloocan City and Pateros top the list in terms of population density (*see Table 3.3*).

Despite the increasing trend in population, the population in NCR grew slower than the surrounding provinces particularly CALABARZON. This can be attributed to the development of new housing sites in surrounding provinces adjacent to Metro Manila.

Table 3.2. Metro Manila Land Use Classification per City/Municipality (in hectares)

Land Use	Makati City CLUP/ZO 2013-2023	Pasig City CLUP/ZO 2014-2022	Quezon City CLUP/ZO 2015-2025	Pasay City CLUP/ZO 2014-2028	Muntinlupa City CLUP/ZO 2017-2023	Navotas City CLUP/ZO 2003-2013	Marikina City CLUP/ZO 2000-2010	Municipality of Pateros CLUP/ZO 2000-	Malabon City CLUP/ZO 1995-2015	Parañaque City CLUP/ZO 1997-2000	San Juan City CLUP/ZO 2001-2014	Manila City CLUP/ZO 2005-2020	Caloocan City CLUP 2015-2025	Las Piñas City CLUP 2015-2025	Taguig City CLUP/ZO 2000-2021	Valenzuela City CLUP/ZO 2009-2019	Mandaluyong City CLUP/ZO 1995-2020	TOTAL
Residential	844.00	1,955.91	6,697.45	380.60	2,113.81	426.73	813.06	114.88	589.14	2,308.64	353.96	867.29	2,747.42	2,053.69	2,158.32	2,600.06	389.25	27,414.21
Socialized Housing	-	-	-	-	69.55	-	-	-	-	-	-	-	-	-	253.00	40.30	-	362.85
Commercial	390.00	157.85	1,310.23	130.57	658.73	52.79	132.82	33.52	289.87	549.96	98.87	1,719.19	220.88	342.20	-	384.74	241.50	6,555.86
Industrial	-	336.92	891.68	23.85	62.48	230.10	280.57	-	547.27	532.77	5.09	96.72	395.10	43.17	280.00	1,054.90	38.19	4,818.81
Institutional	57.00	97.81	1,112.05	73.99	81.02	27.72	72.60	10.31	96.33	60.20	41.82	780.74	150.81	62.82	436.00	16.95	90.66	3,268.83
Special Urban Development	-	-	649.49	959.90	-	-	-	-	-	-	-	-	-	-	-	-	-	1,609.39
Utility	10.00	18.93	227.58	453.30	31.10	-	373.80	-	-	104.82	-	-	68.53	5.44	-	8.11	-	1,301.61
Parks and Open Space	24.00	23.87	2,618.30	1.40	135.86	21.75	61.91	5.05	14.62	114.93	-	258.85	102.77	101.68	243.00	-	116.60	3,844.59
Cemetery	87.00	14.03	85.13	3.72	38.98	-	-	-	14.17	124.62	-	-	74.75	20.56	-	11.82	7.87	482.65
Road	655.00	384.61	2,237.77	308.78	-	-	-	21.23	-	-	94.53	-	639.15	413.90	-	-	181.77	5,036.74
Waterways	-	135.68	182.85	26.21	-	-	-	-	-	34.08	-	-	-	36.04	421.88	-	-	836.74
Buffer Zone	-	-	-	3.03	-	-	-	-	-	-	-	-	-	-	-	-	-	3.03
Fishponds	317.91	-	-	20.00	-	317.91	-	-	20.00	-	-	-	-	-	-	334.33	-	1,010.15
Cultural	-	-	-	22.31	-	-	-	-	-	-	-	-	-	-	-	3.60	-	25.91
Villamor Golf Courses	-	-	-	59.62	-	-	-	-	-	-	-	-	-	-	-	-	-	59.62
Heritage Preservation Zone	3.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3.00
Riverside Development Zone	25.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	25.00
Planned Unit Development	4.00	-	-	-	369.65	-	-	-	-	827.00	-	-	-	-	-	-	-	1,200.65
Special Mixed-Use Zone	417.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	417.00
Special Precincts for Urban Redevelopment	85.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	85.00
Disaster Resiliency Initiatives for Vulnerable Enclaves	23.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	23.00
Government Center Zone	7.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	7.00
Recreational	104.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	104.00
Vacant	-	-	-	-	-	-	-	-	-	-	-	-	933.99	-	-	-	34.54	968.53
Others	-	-	-	-	-	-	-	-	-	-	-	469.47	-	-	-	-	-	469.47
Mixed Use	-	40.26	-	-	-	-	-	-	-	-	-	-	-	209.05	-	-	-	249.31
Tourism Site	-	6.96	-	-	-	-	-	-	-	-	-	-	-	10.06	1.00	-	-	18.02
Urban Core Zone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	344.00	-	-	344.00
Entertainment Zone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	301.00	-	-	301.00
Military Zone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	100.00	-	-	100.00
Waste Disposal	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4.61	-	4.61
Idle land	-	3,04.98	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3,014.98
Roads, Rivers and Creeks	-	-	-	-	-	413.81	-	-	-	-	-	-	-	-	-	-	-	413.81
TOTAL	3,052.91	6,029.96	16,112.53	2,467.28	3,974.99	1,077.00	1,734.76	184.99	1,571.40	4,657.00	594.27	4,192.26	5,333.40	3,298.61	4,538.20	4,459.42	1,100.38	64,379.36

Table 3.3. Population of the Philippines, by Region, 2000, 2010, and 2015

REGION	TOTAL POPULATION		
	2000	2010	2015
PHILIPPINES	76,506,928 ^a	92,337,852 ^b	100,981,437 ^c
NATIONAL CAPITAL REGION (NCR)	9,932,560	11,855,975	12,877,253
CORDILLERA ADMINISTRATIVE REGION (CAR)	1,365,412	1,616,867	1,722,006
REGION I (ILOCOS REGION)	4,200,478	4,748,372	5,026,128
REGION II (CAGAYAN VALLEY)	2,813,159	3,229,163	3,451,410
REGION III (CENTRAL LUZON)	8,204,742	10,137,737	11,218,177
REGION IV-A (CALABARZON)	9,320,629	12,609,803	14,414,774
REGION IV-B (MIMAROPA)	2,299,229	2,744,671	2,963,360
REGION V (BICOL REGION)	4,686,669	5,420,411	5,796,989
REGION VI (WESTERN VISAYAS)	3,645,315	4,194,579	4,477,247
REGION VII (CENTRAL VISAYAS)	4,576,865	5,513,514	6,041,903
NEGROS ISLAND REGION (NIR)*	3,695,811	4,194,525	4,414,131
REGION VIII (EASTERN VISAYAS)	3,610,355	4,101,322	4,440,150
REGION IX (ZAMBOANGA PENINSULA)	2,831,412	3,407,353	3,629,783
REGION X (NORTHERN MINDANAO)	3,505,708	4,297,323	4,689,302
REGION XI (DAVAO REGION)	3,676,163	4,468,563	4,893,318
REGION XII (SOCCSKSARGEN)	3,222,169	4,109,571	4,545,276
AUTONOMOUS REGION IN MUSLIM MINDANAO (ARMM)	2,803,045	3,256,140	3,781,387
REGION XIII (Caraga)	2,095,367	2,429,224	2,596,709

Notes:

a. Population counts for the regions do not add up to the national total. It includes 18,989 persons residing in the areas disputed by the City of Pasig (NCR) and the province of Rizal (CALABARZON); and 2,851 Filipinos in Philippine Embassies, Consulates, and Mission Abroad.

b. Population counts for the regions do not add up to the national total. It includes 2,739 Filipinos in Philippine Embassies, Consulates, and Mission Abroad.

c. Population counts for the regions do not add up to the national total. It includes 2,134 Filipinos in Philippine Embassies, Consulates, and Mission Abroad.

*Created into a region under Executive Order (EO) No. 183, s. 2015 and abolished by EO 38, s. 2017

Economic Development

For the past years, Metro Manila has consistently dominated the country's economic activities with a percentage share of 36.2 percent, 36.5 percent, and 36.6 percent from 2014 - 2016, respectively. Its economic power makes the region the country's premier center for finance and commerce, and its performance is critical to the overall development and growth of the country.

In 2016, the NCR continues to have the largest share of the country's gross domestic product (GDP) at 36.6 percent. It was followed by CALABARZON with a 16.8 percent share and Central Luzon with a 9.5 percent share.

With regards to economic growth, all 17 regions posted positive growth from 2015 to 2016, with Eastern Visayas' economy growing the fastest with 12.4 percent, followed by Central Luzon at 9.5 percent. The NCR recorded only 7.5 percent growth.

In terms of contribution to the GDP growth rate of 6.9 percent in the same year, the NCR contributed the most with 2.7 percentage points, followed by Central Luzon (0.9 percentage point), and CALABARZON (0.8 percentage point).

Poverty

According to the PSA, poverty incidence among Filipinos registered at 26.3 percent in the first semester of 2015. This means that 26.3 percent of Filipinos don't have the income needed to meet their basic food and non-food needs.

Regionally, poverty incidence from 2006 to 2015 went up in five regions namely: NCR, CALABARZON, Eastern Visayas, SOCCSKSARGEN, and ARMM.

On the other hand, subsistence incidence among Filipinos, or the proportion of Filipinos whose income fall below the food threshold, declined to 12.1 percent for the first semester of 2015 from 13.4 percent for the same period in 2012. According to PSA data, this equates to: a family of five in Metro Manila needs ₱10,431.00 a month to meet basic and non-food needs. On the other hand, MIMAROPA registered the lowest per capita poverty threshold in the first semester of 2015 with poverty threshold of ₱8,414 per month.

04

Regional
Development
Framework

Regional Development Framework (Metro Manila Greenprint 2030)

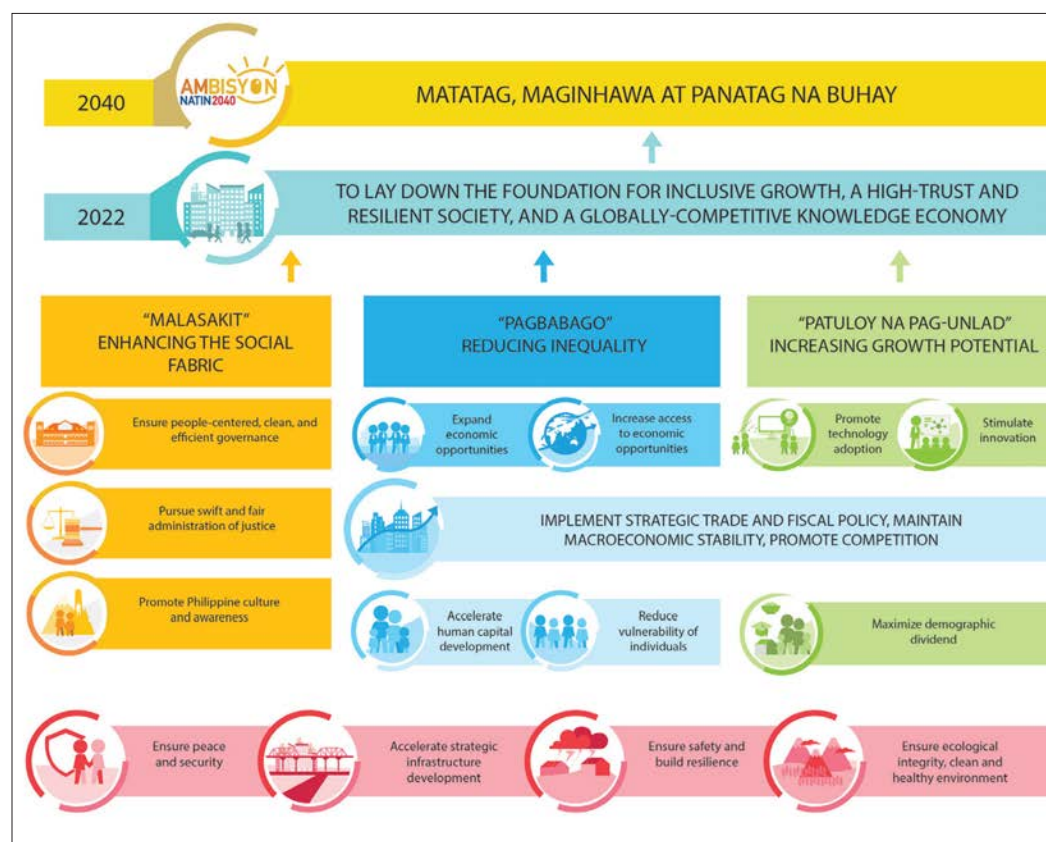
The Regional Development Plan-National Capital Region (RDP-NCR) 2017-2022 was formulated in consonance with the Philippine Development Plan (PDP) 2017-2022 which is anchored on the President's 0-10 Point Socioeconomic Agenda and is geared towards the attainment of the country's long term vision – *AmBisyon Natin 2040*.

The RDP-NCR follows the development agenda of the PDP 2017-2022, highlighting the role of the regions in achieving national development goals and objectives.

In pursuit of a *matatag, maginhawa at panatag na buhay* as depicted in *AmBisyon Natin 2040*, the Region's development framework is anchored on the goal of the PDP 2017-2022 “to lay down the foundation for inclusive growth, a high trust and resilient society, and globally competitive knowledge economy.”

The RDP-NCR 2017-2022 shall adopt the broad strategies of the PDP 2017-2022 (refer to Figure 4.1).

Figure 4.1. PDP 2017-2022 Overall Strategic Framework



Toward enhancing the Region's moral and social fabric *Malasakit*, efforts at promoting cultural diversity and its value shall be pursued (*Chapter 5*).

Reducing inequality *Pagbabago* by pursuing a two-pronged approach of expanding economic opportunities in industry and services (*Chapter 6*) by accelerating human capital development (*Chapter 7*) and reducing vulnerabilities of the poor (*Chapter 8*); and by promoting disaster risk- and climate-proof housing and resettlement facilities to promote the safety and security of the communities (*Chapter 9*).

Furthermore on increasing growth potential *Patuloy na Pag-unlad*, the likelihood of

accelerating the Region's growth potential shall be further increased by maximizing the gains from the demographic dividend (*Chapter 10*).

Growth of the regional economy shall be further accelerated and sustained by moving into a knowledge economy, where technology adoption shall be promoted and innovation encouraged (*Chapter 11*).

As the Region's foundations for sustainable development, all of the foregoing strategies shall be supported by ensuring security, public order, and safety (*Chapter 12*); accelerating infrastructure development (*Chapter 13*); and ensuring ecological integrity, and a clean and healthy environment (*Chapter 14*).

Metro Manila Greenprint 2030

The regional development framework is anchored on the long-term development vision stated in the Metro Manila Greenprint 2030:

“Metro Manila for all; green, connected, resilient; offering talent and opportunity; processing knowledge and delivering services at home and abroad.”

The development framework is designed to transform Metro Manila into a highly competitive East Asian metropolis sustainably delivering adequate services and promoting higher standards of living.

Metro Manila is a city not only for the rich, the poor, and the middle class, but a city for all. By “green”, this means a livable and orderly city that takes care of the environment for the generations to come. By “connected” we refer to seamless access of people within and among 17 cities as well as neighboring emerging urbanized provinces; and “connected” 24/7 to the world (KPO/BPO, overseas Filipinos and their host communities/tourists). And by “resilient”, we want Metro Manila to be responsive and adaptive, while pro-actively managing the effects of climate change, disasters, and natural calamities.

The vision will be achieved through a two-pronged approach which divides goals into strategic areas.

The first strategic area identifies major economic opportunities that could improve livelihoods in Metro and Mega Manila.

The second strategic area focuses on physical interventions that will attract and sustain competitive industries and talent (including returning OFs who have acquired the skills and expertise while living and/or working abroad, as well as doctors,

nurses, dentist, engineers, architects, accountants, surveyors, and tour guides from ASEAN member states) by making Metro Manila an inclusive, connected, and resilient metropolis. These strategic areas complement and reinforce each other.

The goals in each area are supported by principles established through best global practices, as well as opportunities where the principles can be applied in Metro and Mega Manila.

Leveraging opportunities for information technology-business process outsourcing (IT-BPO). Metro Manila having a goal of becoming a global center for knowledge processing, through continuous wide scale upgrading of human resources and the provision of global leading urban environments and infrastructure for information technology and business processing outsourcing (IT-BPO) activities, the combination of which will position Metro Manila to provide world-class services at home and abroad.

For Metro Manila to become a global center for knowledge processing, the following strategies will be implemented:

- Emphasize the importance of moving up the value chain and increasing global market share.
- Enhance IT-BPO human resource development through public-private partnerships.
- Create world-class urban environments to support IT-BPO industry development.

Unleashing tourism potential and attracting newly rich neighbours. To make Metro Manila the first-choice destination of emerging affluent citizens of East Asia for shopping, cultural tourism, visits to

heritage sites, conventions, meetings, and entertainment, the following principles should be implemented:

- Provide visitors with a variety of activities to participate in and destinations to explore.
- Assure visitors of their personal safety.
- Ensure visitors have access to high-quality infrastructure and high levels of convenience.

Reclaiming high-value services and promoting high-potential economic clusters. Reclaim Metro Manila's past role as a leading high-value services center in Asia in areas such as higher education, medical services, tourism, and cultural economy; and promote economic clusters with high potential to create jobs for lower-income populations.

The following strategies will be implemented in the region to reclaim high-value services and promote high-potential economic clusters:

- Ensure visitors have access to high-quality infrastructure and high levels of convenience.
- Ensure the availability of well-trained, specialized workers to meet growing demand in key industries.
- Create industry-specific incubators and platforms to catalyze growth in key areas.
- Promote economic clusters, particularly those with high potential for job creation.

Connecting Metro Manila. Focus major development initiatives around urban and peri-urban nodes.

To achieve the said synergy, one of the identified strategies is to focus on future large-scale investments in mass rapid transit and real estate around existing urban nodes.

The government is keen on investing in mass rapid transit projects that will provide its residents with safer and convenient transfer from one node to another.

Through the years, the government, with the help of the private sector had invested in massive infrastructure projects specifically on mass rapid transit, such as the Manila Light Rail Transit System Line 1 and 2 and the Manila Metro Rail Transit System Line 3.

The capacity expansion and extension of these rail systems are necessary considering that these systems are operating beyond their capacity. The following are the identified investments that are expected to augment existing facilities and respond to the current pressing demand:

1. North-South Commuter Rail Project (heavy rail from Bulacan to Laguna)
2. MRT 7 (North Avenue to San Jose Del Monte)
3. Manila–Makati–Pasay–Parañaque–Taguig Mass Transit System (MMPPT MTS)
4. Manila Corridor Improvement Project (Quezon Avenue Bus Rapid Transit System)
5. C-5 Corridor Improvement Project (high-quality bus system along C-5)

Connect urban and peri-urban nodes via an efficient, safe, and reliable multi-modal transport system

Design transportation system as the primary driver of spatial development in the Mega Manila area.

Improve levels of service on existing mass rapid transit, including integrating payment systems and coordinating schedules.

Accelerate investment and expansion of mass rapid transit system.

Reorganize bus routes and services to complement rail-based trunk routes and to connect nodes where rail connections are unavailable, while promoting the use of cleaner fuels and vehicle technologies.

Reorganize jeepney and action utility vehicle services to feed, rather than compete with), major trunk routes and key nodes, while providing short-distance intra-nodal travel using cleaner fuels and vehicle technologies.

Protect existing and proposed transport rights-of-way.

Making Metro Manila Inclusive. To create inclusive communities through mixed-use, mixed-income, and variable density nodal development capitalizing on opportunities to engage stakeholders and utilize land re-adjustment practices across Metro Manila's existing socioeconomic spatial structure.

Leverage opportunities to unlock huge land values through in-situ redevelopment and catalytic infrastructure development.

Areas in Metro Manila occupied by informal settlements are generally perceived to have low land value relative to its immediate surroundings. The presence of informal settlers decreases the value of land and discourages investors to develop the encroached areas. However, once this land value is unlocked, the revenues from the development and market value of the land will exponentially increase in relation to the size and the location of the informal settlement that has been cleared.

The following strategies will be implemented to unlock land values in Metro Manila:

Fast-track the development of underutilized lands that can be developed for commercial, residential or mixed-uses. LGUs will identify idle lands in their Comprehensive Land Use Plans (CLUPs), enumerate strategies and identify projects to develop

these lands. City governments will put in tremendous amount of infrastructure and social investments in the said areas which will result in marked improvements in accessibility and mobility, health, sanitation, and environmental hygiene. Improvements in the mass transportation system as well as provision of social, educational, and protective institutions in areas will boost the market value of areas and the surrounding regions.

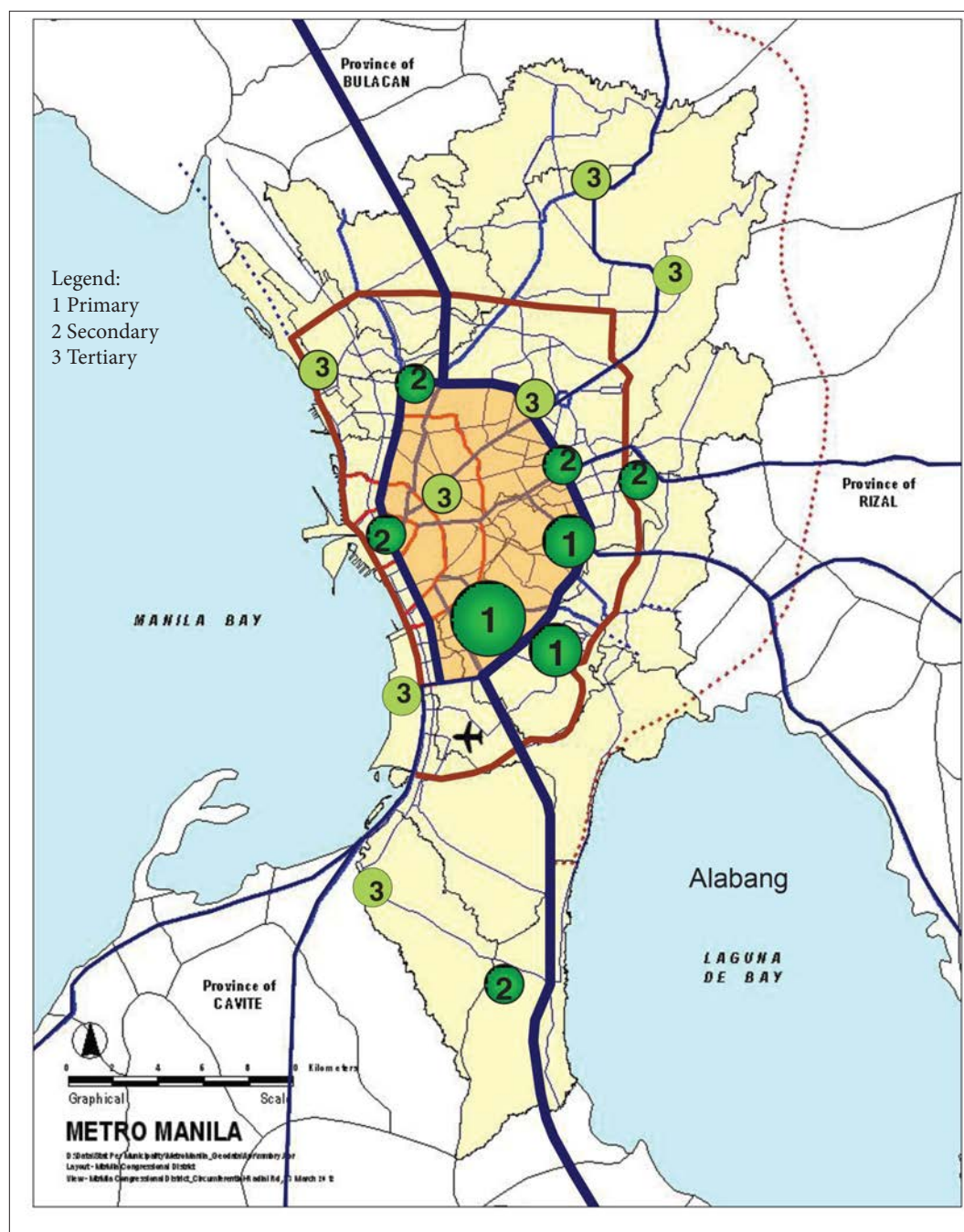
Develop socialized housing sites identified by the local government units in the CLUPs.

Strict monitoring of the compliance of LGUs in the provision of Socialized Housing Zone as per Executive Order 841, s. 2009 will be done. This will address the current housing backlog of low-income groups by channeling underutilized lands into production of housing units. LGUs must also implement the 0.5 percent Socialized Housing Tax and the use of proceeds from the sale of sequestered properties for socialized housing programs, as stated in the Urban Development and Housing Act (Republic Act 7279).

Prioritize development of potential areas/growth centers that have informal settlements. Informal settler families (ISFs) living in the identified growth centers will be relocated to attract local and foreign investors who can further develop the areas. Furthermore, the local government units will provide special incentives for investors who will express their interest to take part in the development of the identified growth centers. City governments will also identify these in the CLUPs/ZOs as areas for priority development to ensure fast growth and expansion in these centers (*see Figure 4.2*).

Allocate more funds for the development of infrastructure and basic facilities in the local government. City governments will prioritize transfer after the word infrastructure development sourcing of more funds that will be allocated for

Figure 4.2. Metropolitan Centers and Sub-Centers



Source: Metro Manila Greenprint 2030

infrastructure development. Areas that require immediate services, particularly those that do not have schools and health centers will be given utmost priority. Transit-oriented development initiatives, with particular focus on more efficient mass transportation system and infrastructure, will be developed in order to guide economic growth of the areas.

City-wide Informal Settlement Upgrading

Upgrading of informal settlement in a large-scale creates more tangible result and wider impact. Development of housing of the ISFs and the low-income group will create better social and economic opportunities for all regardless of income.

To promote informal settlement upgrading, the following strategies will be implemented:

Expedite processing and approval of socialized housing loans and finance borrowing schemes. To encourage low-income groups to secure their own housing units, the government will streamline approval of loans of capable and interested groups. Reasonable discounts and promos will be offered to borrowers and practical payment schemes will be proposed. Partnership with the private sector will be strengthened for lower interest on housing loans. A “one-stop-shop” will be also put-up in each government agency for the processing of all necessary documents and requirements.

Leverage opportunities to support peoples’ proposals for housing projects. Social Housing Finance Corporation, Presidential Commission for the Urban Poor, and other government agencies will facilitate the fast approval of peoples’ plans. These proposals are in-city resettlement so the city government must also give utmost priority in the allocation of lands or in their identified socialized housing zone, as

prescribed in the CLUP for the said peoples’ plans.

Provide incentives for socialized housing developers and land owners. Lower tax will be imposed among developers and project contractors of socialized housing projects to attract them to invest in the construction of housing for low-income groups. Faster official transaction with the city government that is related to development of socialized housing projects must be given utmost attention.

Implement legislative measures that promote the development of informal settlements. The passage of legislative measures to ensure appropriate allocation of funds needed to implement the government’s socialized housing programs is an important initiative that can be pushed in the legislative branch of government. The amendment to RA 7835 or the Comprehensive and Integrated Shelter Financing Act of 1994 is an important law that needs to be revisited.

Equally important are the amendments to specific provisions of the Urban Development and Housing Act of 1992 (RA 7279) that cover the beneficiary selection criteria, particularly giving top priority to renters as beneficiaries to resettlement and shelter assistance programs and stiffer sanctions to informal settler syndicates and LGU/barangay officials that through poor governance, contribute to the continued growth of informal settlements within their respective jurisdiction.

Explore the possible application of the New Town Development or the “Township” for more sustainable resettlement sites. City governments will identify available large tract of land required for the creation of an entirely new town with balanced housing development, complete with livelihood opportunities from industrial and commercial establishments, infrastructure

facilities, access to utilities, transportation, and other basic social services and amenities such as hospitals, schools, public market, recreation, and parks.

Developing a Resilient Metropolis

The Philippines experiences various types of hazards due to its physical environment and geographical location which is along the Pacific Ring of Fire, making it vulnerable to earthquakes, tsunamis, and volcanic eruptions, and flooding.

Among these hazards, Metro Manila was reported to be highly vulnerable to typhoons and floods. These occurrences damage properties, cause epidemics and widespread diseases, and worse, result to loss of lives. Moreover, any type of disaster and regardless of its magnitude, without question disrupt economic activities thereby affecting the overall growth and development in the region.

Based on the study conducted by the World Wide Fund for Nature entitled “Climate Change Vulnerability Mapping for Southeast Asia”, Metro Manila was ranked as one of the most vulnerable cities in Asia in terms of exposures to natural hazard. According to the study which assessed the vulnerability of 530 urban centers in Southeast Asian countries, namely Indonesia, Thailand Cambodia, Laos, Malaysia, Vietnam, and the Philippines, Metro Manila (National

Capital Region) is ranked 7th among the most vulnerable urban areas to climate change. Indonesian cities dominated the list, with 7 of them included in the top 9.

Given these circumstances, the Greenprint 2030 aims to adopt a comprehensive approach to disaster resilience that includes decisive action and investment in risk reduction infrastructures to improve the living conditions and livelihoods of those in danger areas.

The principles distinguished were:

- Promote better understanding of disaster risks and their impacts on sustainability and competitiveness
- Encourage investment in disaster risk reduction programs
- Engage community level participation in adaptation and mitigation efforts
- Implement risk-sensitive land use guidelines and encourage private sector cooperation to prevent future encroachment
- Build redundancy in lifeline systems

Migration and development. This RDP, recognizing the role of overseas Filipinos and their families in development, will protect their rights, improve their wellbeing, strengthen their engagement in local governance and development, and facilitate their eventual reintegration to their communities of origin or choice.

PART II

ENHANCING SOCIAL FABRIC ("MALASAKIT")



05

Promoting
Awareness and
Valuing Cultural
Diversity

Promoting Awareness and Valuing Cultural Diversity

Culture is a manifestation of the freedom of belief and expression and is a human right to be accorded with due respect and allowed to flourish.¹ Anchored on the constitutional provisions that Filipino culture is based on the principle of unity in diversity,² subsequent statutes and cultural policies and programs recognize that each locality has a distinct cultural identity.³ Culture reflects and shapes values, beliefs, and aspirations,⁴ thereby promoting a high-trust society and enhancing the social fabric or *malasakit* while defining the identity of the nation⁵ and of the locality.⁶

Metro Manila or the National Capital Region (NCR), with a population of 12.8 million,⁷ is the most densely populated city in the Philippines. Being the country's political, economic, and educational center, it has attracted rural migrants, making it home to a lot of ethnolinguistic groups from all over the country. This has consequently made Manila a melting pot of cultures.

This Plan recognizes the challenges and appreciates the advantages that is brought about by the diversity of cultures in the region and its role in development by (1) promoting awareness and appreciation of local culture and values, (2) promoting the development of Filipino creativity as an instrument for social cohesion and advancement of the creative economy, and (3) strengthening culture-sensitive governance and development.

Assessment and Challenges

The NCR is the center of culture, economy, education, and government of the Philippines. The prospect of better economic opportunities and access to public facilities and services encourage a lot of rural dwellers to move to the capital. This rural to urban migration since the

1970's has made Metro Manila a melting pot of different Filipino cultures, with most if not all of the ethnolinguistic groups represented, which makes NCR culture rich and vibrant. However, this diversity also makes it difficult to define.

¹As stated in Section 2 of Republic Act 7356 or the Law Creating the National Commission for Culture and the Arts

²Section 14, Article XIV of the 1987 Philippine Constitution: "The State shall foster the preservation, enrichment, and dynamic evolution of a Filipino national culture based on the principle of unity in diversity in a climate of free artistic and intellectual expression."

³As stated in Section 5 of RA 7356

⁴As stated in Section 3 of RA 7356

⁵*Ibid*

⁶As stated in Section 5 of RA 7356

⁷Philippine Statistics Authority. (2015). *Census of Population*

The NCR is also rich in other cultural resources which can be seen through its various tangible and intangible heritage. It is home to many iconic Philippine heritage structures such as the Malacañang Palace, the Bahay Nakpil-Bautista in Quiapo, the Quezon Memorial Circle, the Cultural Center of the Philippines (CCP), to name a few (see Figure 5.1). Most of the country's top museums such as the National Museum, Metropolitan Museum of Manila, Ayala Museum, which are found in the NCR, also hold numerous artifacts and artworks which are representatives of the diverse cultures in the capital. Communities in Metro Manila also hold many traditional practices, which are part of their intangible heritage such as the traslacion of the Black Nazarene in Quiapo, the Santacruzán, balut-making in Pateros, etc.

The NCR is home to cultural and art spaces and a hub for the creative industries. Major performance venues such as the CCP, the Meralco Theater, RCBC Theater, Solaire Theater, PETA Center, can be found in Manila. Many designers, artists, cultural experts are also based in the capital.

To ensure the safeguarding of heritage and promotion of culture and arts in the NCR while taking into account its role in achieving inclusive, sustainable, and people-centered development, the following challenges need to be addressed in the medium term:

Difficulty in identifying an NCR culture.

The demographic situation of the NCR makes it difficult to identify a sole cultural identity specific to the region. Unlike other regions which are composed of relatively homogenous group of indigenous people, the NCR is home to a myriad of ethnolinguistic groups brought by rural-urban migration. As the country's economic and political

center, many people from other parts of the country flock to the region to look for greener pastures and greater opportunities. As such, it can be observed that there are a handful of ethnolinguistic groups which predominate the region's population—mainly the Tagalogs, Cebuanos, Warays, and Ilokanos.

Additionally, foreign nationals are part of the diverse demographic make-up of the region. Unsurprisingly, this was caused by the rich historical ties forged by trade relations with the Chinese, Arabs, and Malays, along with centuries of colonial rule by the Spanish and the Americans. Nowadays, the NCR is also home to a lot of foreign migrants from Korea, Japan, India, Iran, etc.

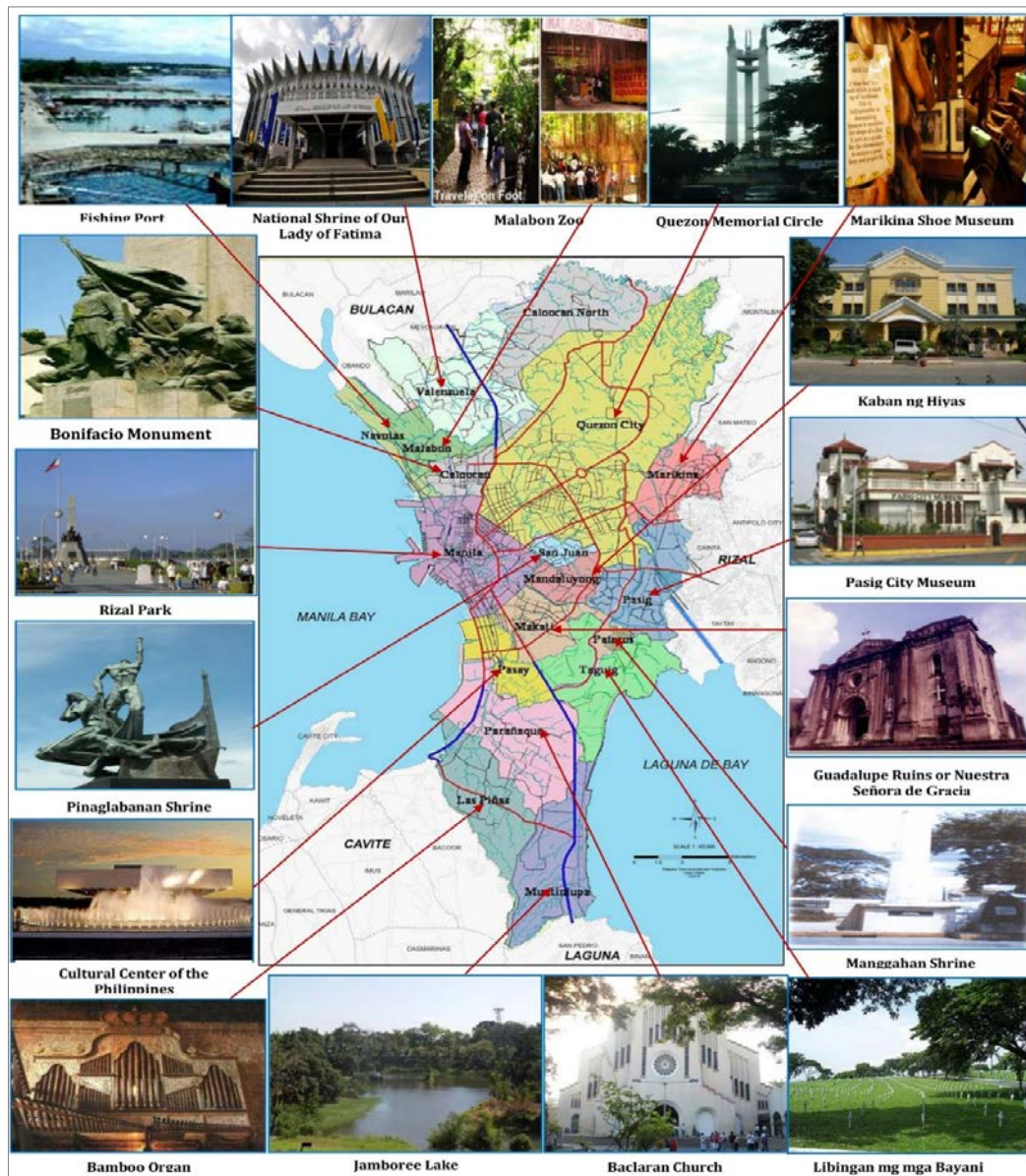
This unique history and experience of the region led to a lack of sense of belongingness following a multicultural environment, causing for many to become less appreciative of cultural heritage, less participative in creative activities, and less culturally sensitive. Furthermore, the insufficiency of administrative direction, legislative priorities, and financial assistance for the cultural sector exacerbate the situation.

Inadequacy and inaccessibility of documentations on culture and arts studies.

To safeguard and promote the rich culture and heritage of the NCR, there is a need to have a complete assessment of the region's cultural resources, covering both the tangible and intangible. This can be done by conducting cultural mapping⁸, a research technique and tool used to record distinct peoples' tangible and intangible cultural assets within local communities. The primary goal of cultural mapping is to help communities recognize, celebrate, and support cultural diversity for economic,

⁸Cultural mapping is a valuable tool for identifying a community's strengths and its resources. This process can help as communities move into the planning and implementation phase by identifying early the resources, efficiencies, and links between arts and cultural groups, as well as their common aspirations and values. Retrieved from https://www.creativecity.ca/database/files/library/cultural_mapping_toolkit.pdf

Figure 5.1. Major Cultural Heritage Sites in Metro Manila



social, and regional development.⁹ Some local government units (LGUs) such as Pasay, Makati, Mandaluyong, and Quezon City have their inventories of cultural resources, while others are in the process of conducting their cultural mapping. There are also numerous publications, documentaries, and researches made about the NCR, however there is a need to have a database of these resources to make it more accessible to the public.

Lack of understanding on the value of cultural heritage. The low awareness on culture and embedded values is a challenge not only seen in the region but also in the entire country. The NCR, as mentioned in the previous sections, has a rich culture and heritage. However, channels of learning about our culture and values have not been effective. Popular celebrations and festivities including shows and movies also failed to promote positive values of the region. It is important for community members to learn about their heritage as this will lead to deeper understanding, appreciation, and sense of ownership to it. This will help establish the identity of the people in the region, strengthen their sense of belongingness, and motivate them to advocate for the safeguarding and promotion of the cultural heritage of the NCR. *“A culturally sensitive and gendered approach can empower marginalized individuals and communities to participate in cultural and political life. New imaginations of the urban can transform citizens’ sense of place and sense of self.”*¹⁰

Low cultural sensitivity towards a multicultural environment. Multiculturalism can be likened to a double-edged sword. It can be beneficial or detrimental depending on how it is appreciated and utilized. In the case of the NCR, there were instances when multiculturalism had bred divisiveness

among members of different ethnolinguistic groups and nationalities. It creates a scenario of prejudice, segregation and, at times, conflict.

These kinds of attitude are usually enacted by some members of the dominant groups towards members of marginalized sectors (i.e. women, persons with disabilities, etc.). These are further reinforced through mass media notably in television shows, movies, and commercial advertisements.

Low participation in creative activities. Despite the existence of spaces, both public and privately-controlled, where civic and creative (or cultural) activities are conducted, it has been observed that there is a low turn-out rate in activities with cultural significance such as artistic performances, concerts, art workshops, book fairs, and literary festivals. However, a considerable number of people participate in activities such as local film festivals, specifically Metro Manila Film Festival.

Lack of appreciation for the creative industries. There is a general lack of appreciation of the value of culture as a driver of development. In the Philippines, its value is often equated with entertainment, but it is much more than that. It includes audiovisual products, design, new media, performing arts, publishing, and visual arts. In the NCR, the creative industries are driven by the private sector. The country is not lacking talent, as evidenced by renowned Filipino designers and animators. Government support is important in pushing forward the cultural and creative sectors so that it could compete in one of the most rapidly growing sectors of the world economy. The creative industry is a highly transformative sector in terms of income generation, job creation, and export earnings. Between 2002 and 2011, developing-countries

⁹“A culturally sensitive and gendered approach can empower marginalized individuals and communities to participate in cultural and political life. New imaginations of the urban can transform citizens’ sense of place and sense of self.” Retrieved from <http://www.unescobkk.org/culture/tools-and-resources/tools-for-safeguarding-culture/culturalmapping/>

¹⁰ United Cities and Local Governments. (2016). *Why must culture be at the heart of sustainable urban development?*. Retrieved from http://www.agenda21culture.net/sites/default/files/files/documents/en/culture_sd_cities_web.pdf

averaged 12.1 percent annual growth in exports of creative goods.¹¹

Lack of public spaces for creative activities and cultural promotion. The Creative Hubs' Potential of the Philippines study of the British Council conducted by the Ateneo Art Gallery, reported that although Filipino creatives are socially connected, they are spatially fragmented.¹² There is not one physical space that can be called the Philippines' Creative District. Unlike England's Soho, Amsterdam's Jordaan, and Belgrade's Savamala, the Philippine's creative district at most can only be situated in virtual space or through the network of connections of the various art groups and design firms. After being exposed to Yokohama's Creative City, 98B Collaboratory's Creative Director Mark Salvatus, reckons that the problem lies in lack of proper urban planning and zoning – there is no deliberate sectioning of a locale dedicated to the creative industries (British Council and Ateneo Art Gallery, 2017). Hence, the need to identify and develop creative districts which can help bring together all creative firms in the metropolis to share a common space where ideas, innovations, art, and culture can flow vibrantly.

NCR needs to capitalize on its creative talent and resources by providing creative hubs or spaces where designers and innovators can freely exchange ideas and collaborate on projects that will benefit their immediate communities. These spaces, which can be

scattered in key areas in the various cities, may be connected via a network of creative hubs which will also include museums, art galleries, studios, ateliers, and public and private parks. These places can be platforms of both permanent or temporary and indoor or outdoor exhibitions. These hubs will also host workshops for the community where they are located so the locals can engage both as students and as collaborators, with the creatives who use the spaces.

It is important for the local government units to consider their local tourism brand in identifying and developing public spaces for creative activities and cultural promotion (*Table 5.1*).

Insufficient administrative and/or legislative measures, and financial support to safeguard and promote local cultural heritage. While the mandate of cultural agencies over cultural properties under their respective jurisdiction are prescribed in several laws, the role of LGUs in promotion of local cultural heritage are not spelled out in specific terms and remains to be an optional power under the Local Government Code of 1991.

Despite efforts by the national and local governments to implement the heritage laws, different factors have been challenging the region. The weak implementation and lack of understanding of the laws and regulations threaten the safeguarding of the cultural resources of the region.

¹¹ UNESCO. (14 November 2013). *Creative industries boost economies and development, shows UN Report*. Retrieved from <http://www.unesco.org/new/en/media-services/in-focus-articles/creative-industries-boost-economies-and-development-shows-un-report/>

¹² British Council and Ateneo Art Gallery. (2017). *Fostering Communities: The Creative Hubs' Potential in the Philippines*.

Table 5.1 Local Tourism Brand of Metro Manila LGUs

LGU	Tourism Brand
Caloocan City	Tao ang Una
Las Piñas City	Bamboo Organ
Makati City	Financial Capital of the Philippines
Malabon City	Home of the Original Tricycle Tours
Mandaluyong City	Tiger City
Manila City	Sulong Maynila
Marikina City	Shoe Capital of the Philippines
Muntinlupa City	Yan ang Tama Yan ang Muntinlupa (Emerald City of the South)
Navotas City	Fishing Capital of the Philippines
Parañaque City	A Mega City by the Bay
Pasay City	Travel City
Pasig City	The Green City
Municipality of Pateros	Isang Pateros Disiplinadong Pamayanan Maunlad na Bayan
Quezon City	The Philippine City of Asia
San Juan City	Historic City of Excellence
Taguig City	Probinsiyudad (Think Big, Dream Big, Taguig)
Valenzuela City	Tayo na Valenzuela

Strategic Framework

The region will promote local culture and values by increasing the level of cultural awareness, inculcating values for the common good, promoting the development of Filipino creativity as an instrument for social cohesion, and advancement of the creative economy, and intensify national pride among Filipinos, including those

living and/or working overseas. These will be included in the development plans of all local government units in Metro Manila which will be supported by adequate resources, capacity building for government agencies on culture-sensitivity, and increased engagement with stakeholders.

Targets

Core outcome and output indicators that will be monitored to ensure progress toward valuing diverse cultures, cultivating creativity, inculcating values of the common good,

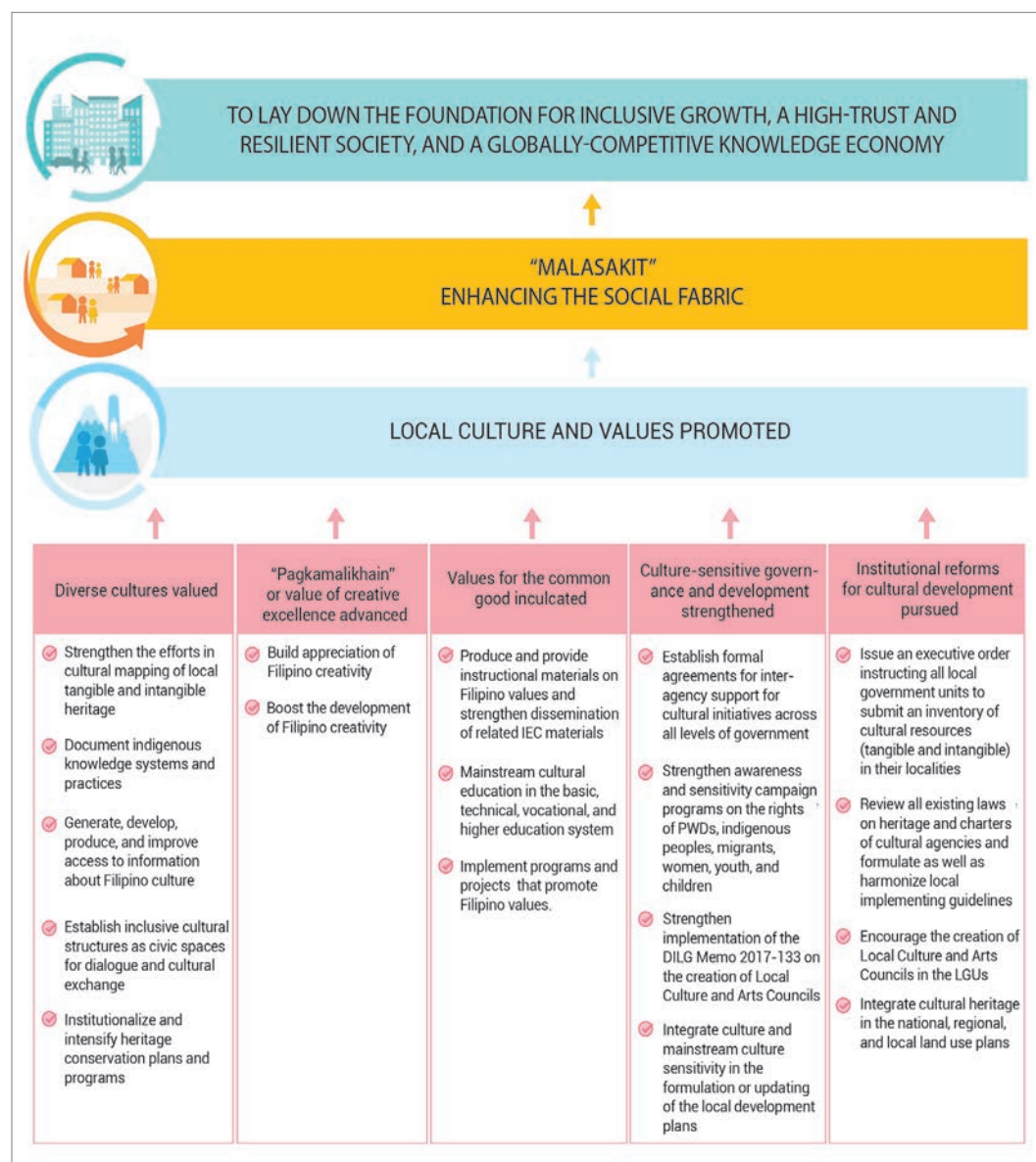
strengthening culture-sensitive governance and development, and pursuing institutional reforms for cultural development are presented in *Table 5.2*.

Table 5.2. Plan Targets to Promote Awareness and Value Cultural Diversity

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Number of cultural mapping implemented	2017	0	7	NCCA, LGUs (Cities of Las Piñas, Malabon, Marikina, Muntinlupa, and San Juan)
Number of curriculum-integration projects to mainstream cultural education in the basic, technical, and higher education system	2017	0	8	DepED, LGUs (Cities of Las Piñas, Marikina, Pasay, and San Juan)
Number of instructional materials on intangible heritage developed	2017	0	1	LGU (City of Parañaque)
Number of activities promoting local culture and history conducted	2017	0	12	NCCA, NHCP DOT, LGUs (Cities of Las Piñas, Malabon, Muntinlupa, and San Juan)
Number of cultural events and festivals conducted	2017	350	1,710	LGUs (Cities of Malabon, Manila, Marikina, and Quezon)
Number of inclusive cultural structures established	2017	0	6	LGUs (Cities of Parañaque, and Pasay)
Number of established community-based cultural groups showcasing culture, arts, crafts, and music	2017	0	2	LGUs (Cities of San Juan and Malabon)
Number of important cultural heritage sites and structures preserved and/or restored conducted	2017	0	13	NHCP, LGUs (Cities of Manila, Marikina, Muntinlupa, Parañaque, and Pasay)
Number of activities conducted to increase level of awareness of Filipino values, cultural diversity, creativity, and culture-sensitivity	2017	520	3,809	LGUs (Cities of Manila, Marikina, Parañaque, Muntinlupa, and Quezon)
Number of projects that develop Filipino creativity as tool for social cohesion and impetus for creative industries conducted	2017	120	451	LGUs (Cities of Quezon and Malabon)
Number of local artists appreciated and documented	2017	0	1	LGU (City of San Juan)
Number of IEC materials produced and disseminated (magazines, AVPs, posters, among others)	2017	0	460,000	LGU (Muntinlupa City)
Number of local ordinance or Executive Order declaring cultural structures as civic spaces issued	2017	0	1	LGU (Muntinlupa City)
Number of comprehensive tourism and heritage development plan created	2017	0	1	LGU (Muntinlupa City)
Number of LGUs with institutionalized and intensified heritage conservation plans and programs	2017	0	5	LGUs (Cities of Parañaque and Pasig)

Strategies

Figure 5.2. Strategic Framework to Promote Awareness and Value Cultural Diversity, 2017-2022



To value diverse cultures

Strengthen the efforts in cultural mapping of local tangible and intangible heritage.

To promote and preserve the culture and identity of the NCR, there is a need to define its parameters first. This can be done by identifying the values, traditions, and other cultural assets of the region that contribute to its culture and identity. There is a need to map-out the tangible and intangible assets of the NCR. Data gathered from the mapping shall be used for the planning activities of the region.

Document indigenous knowledge systems and practices.

As a melting pot of cultures from the different parts of the country, the NCR is home to migrant indigenous communities. Related to the mapping of the intangible heritage of the region, it is important to document the indigenous knowledge systems and practices of the native Manileños, as well as those of the migrant indigenous communities, as these practices form part of the identity of the region.

Generate, develop, produce, and improve access to information about Filipino culture.

Information materials on Filipino culture will be generated and produced, and access to these will be improved. This may be done through the enhancement of the Filipiniana collection of public libraries, the creation of databases of available resource materials in public libraries, and the creation of a portal that will enable the uploading of these databases and sharing it among the LGUs in the region. Various modes of communication will be used to disseminate and popularize Filipino culture in the NCR. This strategy will enhance people's knowledge of Filipino culture and create a sense of ownership and national identity among the peoples in NCR.

Establish inclusive cultural structures as civic spaces for dialogue and cultural exchange.

LGUs will be encouraged to establish cultural infrastructure and facilities which shall be used as venues for fostering cultural understanding and social cohesion. These include museums, parks, libraries, cultural, art galleries, and other cultural spaces.

Institutionalize and intensify heritage conservation plans and programs.

There is a need to develop a regional culture awareness program that will be cascaded at the local levels, as well as the framework for Culture and Heritage Conservation and Development Plans. To ensure that these will be developed and sustained, a Cultural Conservation and Development Fund must be legislated to allocate a percentage of agency and LGU budget for culture-related programs and activities.

To advance the value of creative excellence or "Pagkamalikhain"

Build appreciation of Filipino creativity.

There is a need to develop the sense of aesthetics, creativity, and appreciation of Philippine culture. Encouraging and supporting local artists and their creative initiatives will advance creativity and artistic excellence in the NCR. Moreover, exposing the younger generation to these creative activities will encourage them to promote culture and arts of the country.

Boost the development of Filipino creativity.

This could be achieved by encouraging the production of creative works, investing in human development of its stakeholders, supporting their product and developing the industry in order to realize the potentials of the creative process. Support for the development of the creative industries in the NCR, as well as those from the regions, can be given with the provision

of a venue that will showcase the creative outputs, which can contribute to economic growth.

To inculcate values for the common good

Produce and provide instructional materials on Filipino values and strengthen dissemination of related IEC materials. Increase government efforts to promote values for the common good. To reach the population, instructional materials will be produced and disseminated to advance the citizens' awareness of our culture and values. Dissemination of these materials through various media channels (mass and social media) shall also be strengthened.

Mainstream cultural education in the basic, technical, vocational, and higher education system. In mainstreaming culture in the education system, there shall be close coordination and support from the Department of Education, and the Commission on Higher Education. In line with its thrust of promoting culture-based Education, the DepEd shall strengthen the contextualization of its K to 12 basic education curriculum. This will ensure that education programs are truly inclusive, culture-sensitive, and relevant to the aspirations of the various communities in the region.

Implement programs and projects that promote Filipino values. To reach the population, projects, activities, and programs that promote Filipino values will be developed and implemented across all levels of government.

To strengthen culture-sensitive governance and development

Establish formal agreements for inter-agency support for cultural initiatives across all levels of government. Inter-agency cooperation shall be formalized to support and ensure the sustainability of culture and arts activities across all levels of government in the region.

Strengthen the awareness and sensitivity campaign programs on the rights of persons with disabilities (PWDs), indigenous peoples (IPs), migrants, women, youth, and children. The government needs to be open to draw in engaging marginalized groups (i.e., IPs) to the development discourse to enrich and broaden development perspectives. There is a need to strengthen awareness of the rights of the PWDs, IPs, migrants, women, youth, and children to promote the protection of their rights.

Strengthen implementation of the DILG Memo 2017-133 on the creation of Local Culture and Arts Councils. In line with the implementation of the National Cultural Heritage Act of 2009 (RA 10066) Local Culture and Arts Councils shall be created and/or strengthened in local governments units to plan, develop, and implement programs for the promotion of local cultural heritage and arts, that encourage the development of artistic talent, creativity, and innovation for cultural growth and the advancement of the creative industries.

Integrate culture and mainstream culture sensitivity in the formulation or updating of the local development plans. The integration of culture in local development plans will ensure that it is mainstreamed in all sectors of society. In relation to this, LGUs will undertake and allocate the necessary budget for cultural awareness and conservation programs.

To pursue institutional reforms for cultural development

Issue an executive order instructing all local government units to submit an inventory of cultural resources (tangible and intangible) in their localities. In accordance with the Heritage Law (RA 10666) and 2003 Convention for the Safeguarding of the Intangible Cultural Heritage, all LGUs shall submit an inventory of cultural resources in their localities.

Review all existing laws on heritage i.e., National Cultural Heritage Law (RA 10066) and charters of cultural agencies and formulate as well as harmonize local implementing guidelines. There are a lot of existing cultural laws at different levels of government (national and local laws and ordinances, etc.) It would be good to review

these and see if these are consistent and not contradicting each other. The review will also reveal whether any of these laws need to be dropped or amended.

Encourage the creation of Local Culture and Arts Councils in the LGUs. Establish offices for culture and arts in the LGUs with a mandate to protect local cultural heritage and art.

Integrate cultural heritage in the national, regional, and local land use plans. Integration of culture and heritage in the regional and local land use plans is important to ensure the safeguarding, protection, and conservation of cultural resources of the localities. There is also a need to ensure the alignment of the structure and facilities to be built and programs to be implemented to these plans.

PART III

INEQUALITY-REDUCING TRANSFORMATION ("PAGBABAGO")



06

Expanding Economic
Opportunities in
Industry and
Services through
Trabaho at Negosyo

Expanding Economic Opportunities in Industry and Services through *Trabaho at Negosyo*

There is an urgent need to sustain the efforts to expand economic opportunities in industry and services as they play a vital role in the National Capital Region's (NCR) economic growth and development.

Moreover, the region plays the pivotal role in national governance being the seat of the national government, and where majority of non-governmental organizations (NGOs) and sectoral/industry organizations/associations of national coverage are also located. It is also home to top notch academic institutions which serve as the training ground for most individuals and aspiring entrepreneurs both local and international, major medical institutions, specialized hospitals, specialty clinics and other wellness and fitness facilities, and the center for media networks and telecommunications with the headquarters of the major television networks (2014-2016 NCR MSME Development Plan).

Assessment and Challenges

Micro, Small and Medium Enterprises (MSMEs)

NCR is the bastion of premier central business districts, modern infrastructure system, the gateway for economic and entertainment activities as well as an international gateway for various economic activities which provide a pool of opportunities for the MSMEs in the region.

In terms of business establishments (*Table 6.1*), NCR accounts for 191,955 or 21 percent of the total 900,914 Philippine business establishments, contributing 37 percent of the Philippine's GDP and 37 percent to Philippine employment (2015, PSA).

Table 6.1. Contribution of NCR Establishments

INDICATORS	PHILIPPINES	NCR	PERCENT (%) SHARE OF NCR
Total Business Establishments	900,914	191,955	21
Contribution to Employment	7,766,689	2,911,554	37
Contribution to Value Added	8,126,403,433	2,977,477,320	37

Table 6.2. NCR MSMEs versus Philippines MSMEs

	PHILIPPINES		NCR		PERCENT (%) SHARE OF NCR
	NUMBER	PERCENT (%)	NUMBER	PERCENT (%)	
MSME	896,839	99.5	190,166	99.1	21.3
Micro	806,609	89.5	157,677	82.1	
Small	86,367	9.6	30,850	16.1	
Medium	3,863	0.4	1,639	0.9	
Large	4,075	0.5	1,789	0.9	43.9
Total	900,914	100	191,955	100%	21.2

Table 6.3. Breakdown of MSMEs in NCR According to Sector

REGION/SECTION	NO. OF ESTABLISHMENTS		PERCENT (%) SHARE OF NCR
	PHILIPPINES	NCR	
AGRICULTURE, HUNTING, FORESTRY & FISHING			
Agriculture, Forestry, and Fishing	8,378	240	2.9
INDUSTRY SECTOR			
Mining and Quarrying	932	104	11.2
Manufacturing	115,810	20,429	17.6
Construction	3,337	1,414	42.4
Electricity, Gas, Steam, and Air Conditioning Supply	1,117	119	10.6
SERVICE SECTOR			
Transport and Storage	5,134	2,187	42.6
Information and Communication	35,295	8,269	23.4
Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles	417,514	80,683	19.3
Financial and Insurance Activities	34,549	7,968	23.1
Real Estate Activities	9,688	4,374	45.1
OTHER SERVICES			
Water Supply; Sewerage, Waste Management and Remediation Activities	1,312	168	12.8
Accommodation and Food Service Activities	119,112	24,540	20.6
Professionals, Scientific and Technical Activities	17,471	5,832	33.3
Administrative and Support Service Activities	15,924	5,883	36.9
Education	17,435	4,410	25.2
Human Health and Social Work Activities	27,107	7,419	27.4
Arts, Entertainment, and Recreation	14,625	2,349	16.1
Other Service Activities	56,916	15,567	27.4

The potential of MSME in promoting business investments and job opportunities should be adequately supported, especially so that NCR has a total of 190,166 MSMEs which is 21 percent of the 900,914 total MSMEs in the country, according to PSA 2015 report (*Table 6.2*). The total percentage of micro enterprises in NCR is slightly lower than that of the Philippines at 82 percent as against 90 percent but it is worth noting that the percentage of small enterprises in NCR is almost double (16% versus 10%) while medium and large are more than double that of the Philippines.

In terms of contribution to job generation, the 2015 PSA report discloses that the NCR MSMEs generated a total of 1,471,465 jobs, accounting to about 31 percent of the 4,784,870 jobs generated by the services sector in the country for CY 2015.

Table 6.3 also reflects the breakdown of NCR MSMEs according to sector. The wholesale and retail trade, repair of motor vehicles and motorcycles is the number one (1) scope/coverage followed by accommodation and food service activities.

High cost of doing business. The cost of doing business in the region remained to be high not to mention the tedious procedural requirements for securing business permit which discourage investors from putting up or expanding businesses. Overhead costs such as labor, transportation, communications, electricity and water are higher in Metro Manila than in other regions. Also, establishments in other regions, particularly those in the economic zones enjoy tax cuts, tax holidays and other benefits.

Vulnerability of the regional economy to external shocks. Metro Manila, being the center of commerce and industry in the country, is seen as the showcase for the world market. Therefore, the multi-sectoral events and issues concerning the region can definitely affect international trade and business environment in the region.

Disruptions from man-made and natural calamities, increasing traffic congestion, environmental degradation, and crime may affect the economic dynamism of the region. Reduction in traffic congestion will facilitate easy and fast access to areas in the metropolis. This may be translated to increased productivity as less time will be wasted idling in traffic. Traffic in Metro Manila is equivalent to ₱277 million daily (\$2.4 billion a year) lost man-hours.

Insufficient institutional support to business.

Lack of Business Continuity Plans. Contingency planning still remains a new concept among business sectors, despite the fact that the region is exposed to different kinds of natural and man-made disasters such as earthquakes and flooding.

Aside from natural hazards and phenomena, there are also human-induced disasters,

such as fire, bombing, stampede/mobs, and the like, that pose great threat to a highly populated Metro Manila.

Given these hazards, there are potentially immense dangers in the region especially on damages of properties and infrastructure, loss of labor hours and income, cause of epidemics and widespread diseases that results in substantial loss of life. In addition, disasters disrupt economic activities which affect the overall growth and development in the region.

Disasters compromise capital, logistics, product market and labor that greatly impact business continuity and recovery. It can also leave behind physical damage and interruptions in supply and labor which can cause temporary business closure. For instance, according to MMEIRS¹ in case a 7.2 magnitude earthquake occurs, it was estimated that approximately ₱2.3 trillion worth of economic losses and disrupted national socio-economic activities will be realized.

Tourism

Tourism is recognized as an important growth engine being one of the major contributors to the country's total foreign exchange receipts. The UN World Tourism Organization (WTO) stated that "tourism is a primary vehicle for job creation and economic regeneration."²

Tourism steers socio-economic activities by contributing directly in the creation of opportunities for more local employment, in harnessing entrepreneurial capabilities and promoting business opportunities for new tourism products and services, especially in and around local tourist sites and destinations.

¹ Metro Manila Earthquake Impact Reduction Study conducted by Japan International Cooperation Agency (JICA, MMDA and PHIVOLCS

² <http://www.eturbonews.co/8526/tourism-and-g20-summit>

Figure 6.1. Visitors Arrivals to the Philippines per Country of Origin January-August 2016



In 2016, the country welcomed 502,739 tourists for the month of August, 4.59 percent higher than the arrivals of 480,689 in August last year. It can be noted that this is the first time that this month surpassed the 500,000 visitor volume.

By regional grouping, East Asia is the biggest source of arrivals with 302,812 arrivals with a market share of 60.23 percent. This figure rose by 4.97 percent relative to August 2015 arrivals of 288,483. Korea, Japan and China which comprised the top 3 markets for August accounted for 54 percent of the total volume.

North America provided the second biggest influx of tourists with 65,402 arrivals, forming 13.01 percent of the total tourist traffic. This volume showed 10.42 percent growth vis-à-vis its arrivals of 59,230 for the same period last year. The ASEAN region accounted for 35,223 arrivals, providing 7.01 percent of the total. The Australasia/Pacific region supplied 20,789 arrivals with a share of 4.14 percent to the total.

The largest chunk of arrivals for August came from Asia, covering 68.86 percent of

the total or 346,204 visitors. Arrivals from the Americas contributed 13.11 percent³ of total inbound volume equivalent to 65,904. On the other hand, 42,339 arrivals (8.42%) was recorded from Europe while some 20,789 arrivals (4.14%) were from Australasia/Pacific (Figure 6.1).

In NCR, the average hotel occupancy rate was decreased by 0.39 percentage point in 2015 from 67.21 percent in 2014 to 66.95 percent in 2015 (Table 6.4). On the other hand, the average length of stay during night time becomes longer from an average of 2.46 nights in 2014 to 2.85 nights in 2015.⁴

Metro Manila is not a priority destination for holidays/vacation tourists. In 2015, the Philippine travel and tourism industry contributed a total of ₱1.09 trillion to the local economy, equivalent to about 8.2 percent of the country's GDP⁵ based on the Philippine Tourism Satellite Account, PSA.

The NCR recorded a total of 1,417,075 visitors in 2015. 848,152 or equivalent to 59.85 percent were foreign travellers while the remaining 40.15 percent or 568,923 are composed of overseas Filipinos and

³ Statistics Economic Analysis and Info Management, OTDPRIM, TD-DOT

⁴ 2016 Regional Social and Economic Trends-NCR, PSA Regional Statistical Services Office

⁵ <https://business.inquirer.net/208861/tourism-contributes-p1-4t-to-gdp>

Table 6.4. Metro Manila Hotel Status, 2014-2015

REGION/HOTEL CLASSIFICATION	NUMBER OF HOTELS		AVAILABLE ROOMS			AVERAGE OCCUPANCY (%)		AVERAGE LENGTH OF STAY (IN NIGHTS)	
	2014	2015	2014	2015	% Inc./Dec.	2014	2015	2014	2015
National Capital Region	79	87	15,374	16,800	9.27	67.21	66.95	2.46	2.85
Deluxe	21	24	8,426	9,168	8.81	71.11	71.60	2.74	2.71
First Class	8	8	1,620	1,770	9.26	58.83	59.09	2.14	2.19
Standard*	43	46	5,042	5,436	7.81	63.96	62.51	2.51	2.42
Economy*	7	9	286	496	73.41	59.73	60.14	1.65	1.57

Note: Includes DOT accredited and non-accredited hotels

Source: Metro Manila Hotels

domestic travellers.

The pocket open skies policy allows the airlines to use other airports outside Metro Manila. Since Metro Manila is known as a shopping destination and not a beach destination, visitors who intend to go on a vacation holiday directly fly to their holiday/beach destination and skip Metro Manila. This is the reason why the average length of tourist stay in Manila is only two (2) days.

Inadequate tourism infrastructure support facilities. Good working infrastructures will make travelling seamless. The international and domestic airports and seaports in Metro Manila serve as the main gateway to many destinations in the Philippines. However, at present, the airport has limited capacity and the seaport has a limited docking facility. A world class airport and seaport will provide travel convenience to the passengers.

Insufficient security measures for visitors/tourists. A good image of the country would enhance the country's attraction as a tourism destination and could contribute to the increase of socio-economic benefits which can be derived from tourism. There should be a collaboration among government agencies and local government units to achieve our goal to improve the overall peace and order and to create a favorable image that the Philippines is a safe destination.

Inadequate mechanism to generate tourism statistics. Data and statistics are crucial to planning. At present, there is lack of tourism statistical data despite the provision in the Tourism Act (RA 9593) Section 38 on "Shared Responsibilities of National and Local Governments".

Port Operation and Management

To support the ever-growing logistics sector, the existing condition of transportation and traffic situation must be analysed against the perceived problems of congestion of the Port of Manila, particularly, the two main international ports, South Harbor (SH) and Manila International Container Terminals (MICT) where at least 82 percent of the country's foreign containerized cargoes in 2016 were handled. North Harbor (NH), on the other hand, was able to handle almost 50 percent of the country's domestic containerized cargoes on the same year.

A port is said to be congested, when ships are queuing and waiting to obtain a suitable berth to load/discharge cargoes. In the case of SH and MICT, "congestion" appears in other form. The queueing exists at the landside or at their gates. Container trucks that come in and out of these terminals are estimated to be least 9,000 per day. They tend to be hemmed in at the port gates, line along port roads and spill over to Metro Manila streets. These artificial bottlenecks

Table 6.5. Performance /Congestion Indicators, 2016 and 2017

	BERTH OCCUPANCY RATE (%)			CRANE PRODUCTIVITY (MOVES/HOUR)			IMPORT DWELL TIME (DAYS)			YARD UTILIZATION LEVEL (%)		
	2015	2016	2017	2015	2016	2017	2015	2016	2017	2015	2016	2017
South Harbor	56	58	62	24	26	26	9	7	8	66	68	71
North Harbor	54	62	43	25	26	23	3	3	2	45	47	43
MICT	74	69	60	22	27	29	9	6	6	67	62	61

were created in 2014 due to a limitation of cargo velocity along the truck routes imposed through a daytime truck ban vis-à-vis the 24-hour operations of SH and MICT. Most of the containers handled at SH and MICT are carried to/from Metro Manila cities and neighboring provinces of Bulacan, Cavite, Laguna and Pampanga. A few and seasonal cargoes are transhipped to Batangas, Cebu, Iloilo and Davao. A diversion of container vessel traffic to Subic and Batangas Ports supposedly to decongest SH and MICT did not appear to be the most economical option. Trucking still appears to be the most viable mode of transporting containers from/to SH and MICT. The origin of exports at the Manila Ports, are found to be mostly from NCR/Metro Manila (37.75%), Cavite (24.30%) and North of NCR (14.94%)⁶. Only a small portion of exports originates from Laguna, Rizal, Batangas and other areas of Southern Tagalog. On the other hand, imports, which comprise about 80 percent of total port traffic, are destined to NCR/Metro Manila (55.03%), Cavite (9.64%) and North of NCR (14.74%)⁶. With these distributions, a successful diversion of containers to Subic and/or Batangas will not necessarily ease traffic in Metro Manila as the bulk of the same containers will eventually find their way back in Metro Manila routes with a higher hauling cost to boot. Also, these origins/destinations of containers do not support hauling via a railway system, which is most viable, in terms of economies of scale, for long haul transport.

In 2014, the Philippine Ports Authority (PPA) directed the terminal operators of

SH and MICT to adopt a vehicle booking system in accepting and delivering import and export containers. The terminal operators thus implemented the Terminal Appointment Booking System (TABS) in collaboration with the government, MMDA and supply-chain stakeholders. The on-line implementation of TABS practically transferred physical queueing of trucks from port gates to the internet. TABS reduced the number of cargo trucks plying Metro Manila city streets during peak hours, and the off-peak hours were utilized. TABS practically distributed container trucks in a 24/7 basis. Acceptance and withdrawal of container cargoes by/from Port of Manila done through the TABS have dramatically contributed to logistical efficiency.

Terminal operations has also improved. While there is a parallel increase in vessel calls and cargo throughput at the same capacity, the productivity of both SH and MICT were sustained within desirable levels. The average yard utilization level at SH is 68 percent and 71 percent in 2016 and 2017, respectively, and MICT has recorded 61 percent and 62 percent for the same period (*Table 6.5*).

The Philippine container throughput has not dipped in 2017 despite the Mindanao crisis. The monthly average of combined container traffic of MICT and SH for the period January-August has actually increased significantly from 175,521 Twenty-Foot Equivalent Unit (TEUs) in 2016 to 249,172 TEUs in 2017 or a soar of 41.96 percent. Yet, truck route traffic

⁶Feasibility Study and Strategic Plan for the Optimum Utilization of Batangas Port-July 2015

is manageable due to the TABS. On the other hand, the domestic cargo throughput being handled by NH has been increasing significantly from 15,233,359.71 metric tons in 2014 to 26,562,144 metric tons in 2017 or an increase of 74.36 percent.

South Harbor and MICT terminals are the traditional top earners of the PPA sharing 56 percent of its 5-year revenue from 2010-2014, or ₱27 billion of the total PPA revenue of ₱49 billion. In the following years, PMO-NCR South, under which the administration of South Harbor and MICT fall, kept its position as top earners generating a revenue of ₱7.382 billion and ₱7.810 billion in 2015 and 2016, respectively. South Harbor is operated by the Asian Terminals Inc (ATI), and MICT by the International Container Terminals Inc. (ICTSI) while North Harbour is being managed and operated by Manila North Harbour Port, Inc. (MNHPI). The remittances of ATI and ICTSI of government share made up the huge share to government coffers.

The administration and the infrastructure development by PPA of other 140 major

ports nationwide are subsidized by revenues coming from South Harbor and MICT. These other PPA ports are vital to the life of regional economy. In line with this, the economic benefits from SH and MICT are critical to the national port system.

Contribution to Job Generation

Aside from the billions annually contributed to the national coffers, port of Manila provides direct employment to a total of 3,287 executives, professionals, technical, skilled and unskilled workers. Salaried workers are definite patrons of various trades (*Table 6.6*).

Table 6.6. Number of Direct Employment Generated

	PPA	ICTSI	ATI	MNHPI
South Harbor	67			
North Harbor	56			
MICT	9			
Total	134	1,531	1,622	1,902

Source: PPA Personnel Inventory as of 30 September 2017; ICTSI 2016 Annual Report; ATI-HR Record

Table 6.7. Type of Indirect Employment Generated

1	PPA Outsourced Personnel*		
	1.1 Technical & Administration (214)	1.2 Janitorial (75)	1.3 Security Guards
2	Portworkers, skilled		
	2.1 Stevedores	2.4 Porters	2.7 Drivers, Rolling Cargo
	2.2 Arrastre	2.5 Crane Operator	2.8 Painters, vessel
	2.3 Lineman	2.6 Forklift Operator	
3	Other workers, unskilled		
	3.1 Garbage collectors	3.3 Parking attendants	3.5 Food vendors/attendants
	3.2 Street sweepers	3.4 Washers	
4	Port Ancillary Service Providers, employees and workers		
	4.1 Custom Brokerage	4.10 Import/Export	4.19 Shop
	4.2 Bunkering	4.11 Lay-up Agents	4.20 Tank Cleaning
	4.3 Cargo Checking	4.12 Lighterage	4.21 Tele Communications
	4.4 Cargo Forwarding	4.13 Maritime Hauling	4.22 Tours & Travel
	4.5 Container Repair	4.14 Marine Survey	4.23 Transportation
	4.6 Crewing/Manning	4.15 Ship Chandling	4.24 Tug Assistance
	4.7 CY/CFS	4.16 Ship Repair	4.25 Water Supply
	4.8 Food Service	4.17 Shipping Agent	4.26 Water Taxi
	4.9 Fumigation	4.18 Shipping & Manning Agent	
5	Third Party Contractors, employees and workers		
	5.1 Engineering Contractors	5.3 Suppliers	
	5.2 Service Contractors	5.4 Parking Operators	

*Combined work body at PMOs NCR South, NCR North and Batangas as of 30 Sept 2017

The outsourcing of personnel and services in the administration and operation ports have also offered job opportunities as well as other businesses that supports and complements port operation and management (*Table 6.7*).

The number of indirect employment is difficult to quantify. A simple basis is PMO-NCR's record of pedestrian and vehicle passes issued to legitimate port users. An average of 2,000 annual pedestrian passes were issued from 2014 to 2016, while a total of 1,641 have already been issued during the first semester of 2017. Occasional and peak season operations generated jobs to more than 20,000 workers yearly as indicated by the temporary passes secured from PMO NCR South.

In PMO NCR North, the issuance of permit to operate has increased from 393 in 2016 to 688 on the following year or a soar of 75 percent indicative that trade and business inside the port are thriving.

The economic dynamics at port of Manila, indeed invites road traffic, especially when certain transactions and activities cannot presently be done outside the ports.

Capacity and Potential of Existing Port Facilities

South Harbor spreads to an area of more

than 85 hectares, excluding the wharf zone. Directly facing the Manila Bay, it is protected by a 880-meter breakwater at the South and another 2.3 kilometers at the West. Its proximity to the four (4) anchorage basins and 200-meter wide fairway channel steps up its advantage to foreign container feeder vessels, cruise vessels and foreign vessels calling on diplomatic purposes. Five (5) finger piers has a total of 30 berths with length overall ranging from 82.00 meters to 600 meters, and draft from 10.00 meters to 13.00 meters (*Table 6.8*).

On the other hand, MICT has a total area of 105.17 hectares, with 94.08 hectares developed for its operation. It has a 1,300 meter straight wharf divided into five berths and another new berth spanning 450 meters, with draft ranging from 12.10 meters to 13.60 meters. It has three (3) anchorage berths that can accommodate bulk carriers. MICT inner basin and boundaries are protected by breakwater with an average height of 2.0 meters.

There are 18 regular container liners calling at South Harbor and 62 at MICT.

In terms of container volume, port of Manila has a combined capacity of 4.0 million TEUs; 1.20 million TEUs for South Harbor and 2.80 million TEUs at MICT. Its average monthly capacity is 84,865 TEUs in 2016. An increase of 6.54 percent as of September 2017

Table 6.8. Berth Capacity of Existing Ports

	NUMBER OF BERTHS	BERTH LENGTH (LOA, IN METERS)	DRAFT (IN METERS)
South Harbor	30		
Pier 3	5	360	12.00
Pier 5	8	600	12.00
Pier 9	5	362 / 101	10.00
Pier 13*	7	381 / 82.70	10.00
Pier 15**	5	376 / 103	13.00
MICT	6		
Berth 1	1	250	12.10
Berth 2	1	250	12.30
Berth 3	1	250	13.30
Berth 4	1	250	13.60
Berth 5	1	300	13.00
Berth 6	1	500	13.30

*Non-Containerized, PCG, PN **Non-Containerized, Cargos

results from minor improvements inside South Harbor which primarily involved the reconfiguration of yard layout and access roads to improve mobility and safety.

Nine (9) quay cranes and several heavy lift/yard transfer equipment with varying capacity and capability can presently sustain the cargo-handling requirement of South Harbor. The port is also provided with reefer stations and dangerous cargo areas for handling specialized cargoes, and complemented by a truck holding area with 250 slots. All five (5) port gates exits into Bonifacio Drive.

At MICT, the vast area offers 33.75 hectares of stacking area along its Berths 1 to 6 in addition to its reefer area capable of accommodating 96 TEUs. Truck holding facility has a total area of 6.67 hectares with 400 slots for 40' truck and trailer. Access to the port is via Road 10 and IPB Road through North Harbor domestic port.

Actual Utilization of Port Facilities

In 2016, the combined container vessel calls of SH and MICT was 3,439. As of end of September 2017, recorded calls is 2,411 marking a decline of 7 percent compared to the same period last year. The rise of larger container vessels calling at Port of Manila explains the decline in vessel traffic statistics. Monthly average in 2017 is 107 at SH, and 160 at MICT.

The 15 operational berths at SH and 6 at MICT has accommodated these container vessels, while still maintaining an acceptable Berth Occupancy Rate of 62 percent at SH and 64 percent at MICT. The rate is lower in 2016 for the same period in SH at 57 percent but higher at MICT 68 percent.

While container vessel calls are declining, container volume is steadily rising which is indicative of constant demand of cargoes of Metro Manila industries. As of September

2017, container traffic has reached 2.52 million which is a 6.0 percent higher than the volume for the same period in 2016. This registry has exceeded the 268,165 TEUs monthly average in 2016, yet the peak season for cargo throughput is expected in the last quarter of the year.

The volume and velocity of cargo movement from the vessels to the yard has to be analysed against the velocity of cargo deliveries and withdrawal at the port. Average yard utilization as of September 2017 of 64 percent is almost the same as the utilization level for the entire 2016. The yearly peak season has just started in October 2017.

The dreaded recurrence of the 2014 port congestion is indicated by the above figures. PPA, on its part has instituted measures within its mandate to prevent it. While port development and other engineering solutions could be an option, container terminals and other mega infrastructures cannot be built as quickly as imposing road traffic rules and regulations and easing transactions to facilitate cargo movement inside and outside the port.

In the Port Operation and Management the following are the challenges that needs to be addressed:

- Container road velocity is slowed down by local traffic enforcement and other factors such as DPWH road weighbridges.
- Inadequate container depot of shipping lines.
- LGU's traffic woes versus its support to Terminal Appointment Booking System (TABS).
- Existence of informal settler families (ISFs) within the Ports and its peripheral areas.
- Inadequate road connectors.
- Stripping and stuffing inside the port which causes congestion thereby affecting day-to-day operations.

Labor and Employment

Population Profile

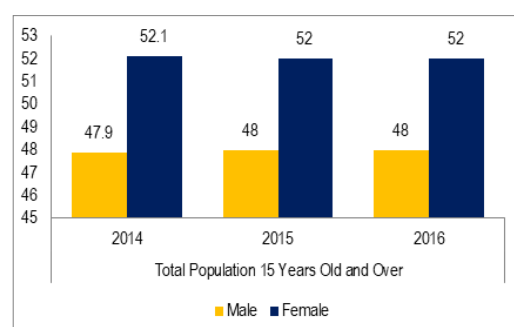
As shown in *Table 6.9* the total population of 15 years old and over in the country was estimated at 68.7 million in 2016. From that national figure, NCR was the second among the regions with the highest number of persons 15 years and older at 8.9 million.

In terms of distribution by gender, majority of the total population as in *Figure 6.2* were consistently females at 52 percent since 2014. The remaining 48 percent were males with just a slight increase from 47.9 percent in 2014 to 48.0 percent both in 2015 and 2016.

Table 6.9. Total Population 15 Years Old and Over (in thousands): Philippines and NCR (October 2014-2016)⁷

	2014	2015	2016
Philippines	64,257	66,622	68,743
NCR	8,221	8,277	8,961

Figure 6.2. Total Population 15 Years and Over by gender in NCR: (October 2014-2016)⁸



Labor Force Participation Rate

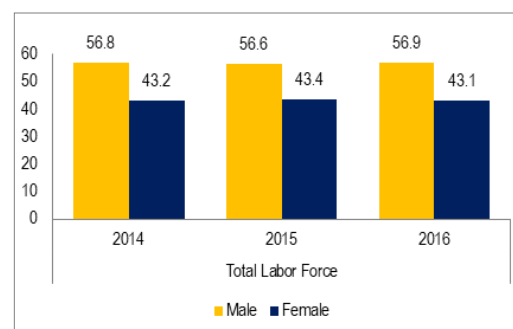
As shown in *Table 6.10*, NCR's Labor Force Participation Rate had increased to 63.0 percent in 2016. This is higher than the 62.4 percent in October 2015. As observed, the increase could be attributed to the upsurge of the population by approximately 684,000.

From the regional figure in *Table 6.10*, 56.9 percent were mostly males (*Figure 6.3*). This showed an increase in the male distribution by 0.3 percent in 2016. Females, however, comprised the remaining 43.1 percent in 2016, which showed a decline in percentage by 0.3 percent in 2016.

Table 6.10. Labor Force Participation Rate: Philippines and NCR (October 2014-2016)⁹

	2014	2015	2016
Philippines	64.3	63.3	63.6
NCR	64.4	62.4	63

Figure 6.3. Total Labor Force Participation in NCR by Gender, October 2014-2016¹⁰



⁷ Philippine Statistics Authority, October 2014-2016 (Final Results)

⁸ Ibid

⁹ Ibid

¹⁰ Ibid

Employment Rate

Despite an increase in 2016 by 1.2 percent, the region's employment rate still shows an upward trend since 2014 (Figure 6.4). Majority of the employed individuals in 2016 were males at 56.5 percent—a 0.4 percent improvement compared in the previous year (Figure 6.5). There were only 43.5 percent employed females in 2016, which only showed a 0.4 percent decline.

Figure 6.4. Employment Rate in NCR, October 2014-2016¹¹

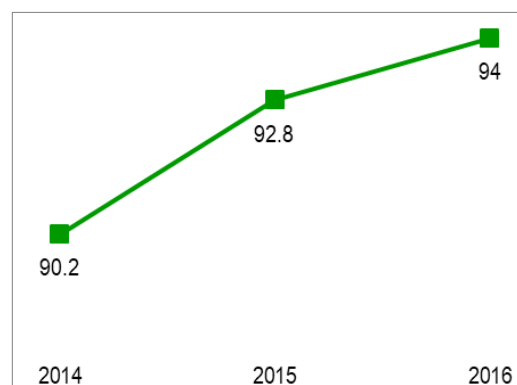
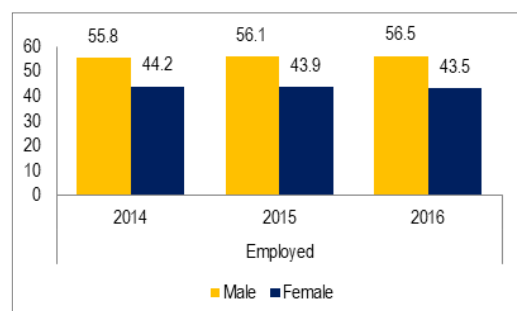


Figure 6.5. Employment rate in NCR by Gender, October 2014-2016¹²



Unemployment Rate

Unemployment rate in the region showed a downward trend since 2014 (Figure 6.6). This translated an improvement in the region, with its lowest rate by 6.0 percent in 2016. In the same manner, the region also bettered the national figure in 2016, which was only 4.7 percent.

Despite improvement, majority of the unemployed in the region were mostly males, which comprised 63 percent of the population in 2016—a 0.1 percent higher than the previous year. Unemployed females in NCR represented the 37 percent and had remained unchanged as it was in 2015 (Figure 6.7).

Figure 6.6. Unemployment Rate in NCR, October 2014-2016¹³

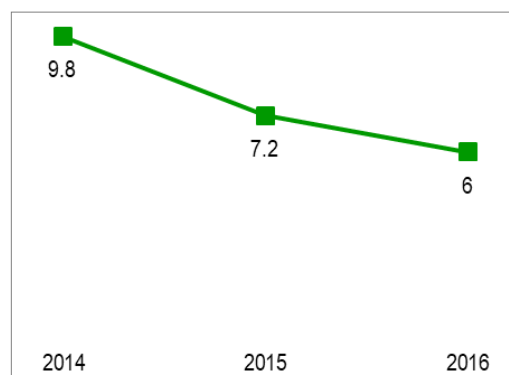
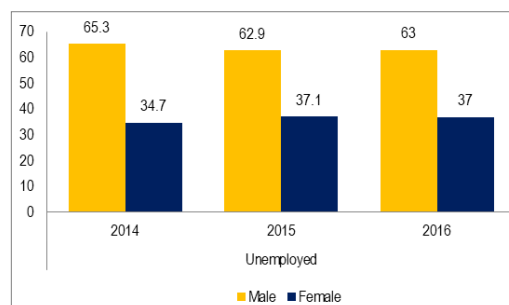


Figure 6.7. Unemployment Rate in NCR by Gender, October 2014-2016¹⁴



¹¹ Philippine Statistics Authority, October 2014-2016 (Final Results)

¹² Ibid

¹³ Ibid

¹⁴ Ibid

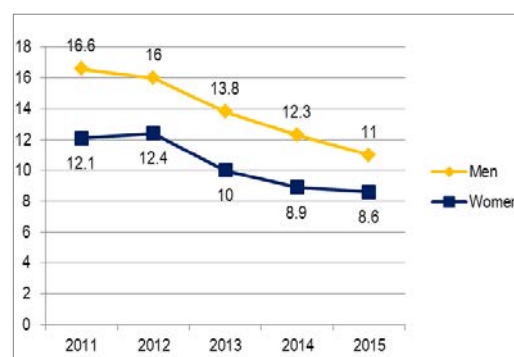
Underemployment Rate

NCR's previous rate in 2015 significantly declined by 5.4 percent in 2016—recorded the lowest among other regions.¹⁵ In latest Gender Statistics data released by PSA on 29 November 2016, 11.0 percent of males and 8.6 percent of females in NCR were reportedly underemployed. Further, the report also revealed the significant downward trend of underemployment in the region since 2011.

Table 6.11. Underemployment Rate: Philippines and NCR (October 2014–2016)¹⁶

	2014	2015	2016
Philippines	18.7	17.6	18
NCR	11.4	12.9	7.5

Figure 6.8. Underemployment Rate in NCR by Gender, October 2014–2016¹⁷



Youth Employment

Youth refers to age group between 15 and 24 as defined by the United Nations (UN); however, the working definition of youth may be viewed differently from other nations, subject to several factors such

as culture, politics, among others. In the Philippines, the youth sector is between 15–30 years old as defined in Republic Act (RA) No. 8044.¹⁸

The youth sector of ages 15–24 years old was estimated at 19.5 million whereas 15–30 years of age was at 28.9 million. In NCR, 15–24 years were estimated at 672 million whereas 15–30 years old at 1.5 million (*Table 6.12*).

Table 6.12. Youth Household Population 2015: Philippines and NCR (in thousands)¹⁹

YOUTH	PHILIPPINES	NCR
15–24 years	19,522	2,067
15–30 years	28,919	3,365

Key Employment Generators in NCR

Aviation. The aviation sector will continue to rise in 2016, with expansion plans in progress to keep up with the demand for air travel both in domestic and international operations. The industry is also expected to stay robust through engagement in various agreements with neighboring countries to improve aircraft facilities and resources, boost passenger traffic numbers, among others.²⁰

Out of the 71 airports in the country, the most popular one is situated in the NCR—The Ninoy Aquino International Airport, which is located in Pasay City. Said airport consisted of four terminals and can accommodate both international and domestic flights on a daily basis, making it the busiest airport in the country.

¹⁵ Labor Force Participation, Employment, Unemployment and Underemployment Rates, by Region: October 2016

¹⁶ Philippine Statistics Office, Oct 2014–2016 (Final Results)

¹⁷ 2016 Gender Statistics on Labor and Employment, Philippine Statistics Authority

¹⁸ Bureau of Labor and Employment Statistics, "LabStat Updates," Vol. 7, No. 19, December 2003

¹⁹ Youth Household Population (15–24 and 15–30 y/old) by Region, Philippines: 2011–2015, PSA (November 2016)

²⁰ <http://business.inquirer.net/204935/airlines-seen-further-expanding-in-2016>

The industry is seen as a good training ground for aspiring pilots²¹ due to increasing demand for air travel in the region.

Construction. The construction industry is anticipated to sustain its growth in the region. With the construction of various residential projects all over NCR, residential sales will become stronger among the high-end market, as they are more prone to leasing and renting properties in the region.²² Further, the Duterte administration focuses on infrastructure projects (e.g. road improvement, water and sanitation, transportation, among others)—therefore, construction firms in the region will increase demands for skilled workers.²³

Information Technology and Business Process Management (IT-BPM). The information technology-business process management industry is the largest employer in the country with 70 percent of the BPO firms located in NCR. Although the industry is expected to rise in the years to come, the Contact Center Association of Philippines (CCAP) had projected growth in other regions. In a study conducted by IT & Business Process Association et al., the IT-BPM companies are increasing their workforce throughout the Philippines, with approximately a quarter of a million jobs created outside NCR by end of 2015.²⁴

Hotel, Restaurant and Tourism. The Hotel, Restaurant, and Tourism industry shows a positive outlook in the country and NCR. Tourist arrivals in the country had climbed to 1.2 million visitors or 10.9 percent on the

first two months of 2017 versus 2016. These tourists are coming from South Korea, USA, China, Japan and Australia.

To date, over 575 accommodation establishments with more than 53,045 rooms are operating all over Metro Manila. A total of 17 accommodation establishments are under construction²⁵. This therefore translates a demand for individuals who would like to have a career opportunity in the said industry.

Health and Wellness. The health and wellness industry is bound to progress in the region given the increasing demand for health care, mostly by aging populations due to changing lifestyles and upsurge of various diseases. Major firms with extensive medical coverage are in quest for quality, affordable, and accessible healthcare for their employees as one of the retention strategies. Further, this also entails tapping the IT experts to provide innovative solutions to digitize some of the traditional hospital services, such as allowing health and wellness professionals to immediately access patient records and efficiently deliver other medical services.

Wholesale Retail and Trade. The wholesale and retail industry has become one of the country's largest employers at approximately 19 percent of the country's total employment.²⁶ Among other regions, NCR, was the top employer in the wholesale retail and trade industry according to the Annual Survey of Philippine Business and Industry (ASPBI) report that was released in September 2017.²⁷

²¹ <http://www.wccaviation.com/overview-airline-industry-philippines/>

²² <http://www.wccaviation.com/overview-airline-industry-philippines/>

²³ <http://primer.com.ph/blog/2017/04/29/4-growing-industries-in-the-philippines/> and <http://www.ibtimes.ph/dole-high-demand-skilled-workers-construction-industry-5551>

²⁴ The Philippine IT-BPM Sector Roadmap 2022: http://www.boi.gov.ph/files/Roadmaps/TID/Executive%20Summary%20Accelerate%20PH%20Future-Ready%20Roadmap%202022_with%20corrections.pdf

²⁵ Based from initial reports on DOT accommodation capacity Survey 2017

²⁶ <http://www.dlsu.edu.ph/research/centers/aki/participant/trainings/workingPapers/2015-21.pdf>

²⁷ <https://psa.gov.ph/content/2015-annual-survey-philippine-business-and-industry-aspbi-wholesale-and-retail-trade-repair>

Table 6.13. Total Deployed Overseas Filipino Workers by Type of Hiring, 2015

TYPE	2016	2015	PERCENT (%) CHANGE
Landbased	1,669,511	1,437,875	16.11
New Hires	582,216	515,217	13.12
Rehires	1,086,695	922,658	17.78
Seabased	442,820	406,531	8.93
Total	2,112,331	1,844,406	14.53

The labor sector faces the following two major challenges:

- Need for adequate access for employment opportunities to minimize unemployment and underemployment through labor market information.
- Jobs and skills mismatch needing strong coordination and linkage in the academe sector and industries in key employment generators.

Overseas Filipinos

Based on latest estimates of the Commission on Filipinos Overseas (CFO), there are a total of 10.2 Million Overseas Filipinos (OFs) worldwide. Of this number, temporary migrants which include OFWs and irregular or undocumented Filipinos accounted for more than half of the said number.

In 2016, records of the Philippine Overseas Employment Administration (POEA) showed

that a total of 2,112,331 Overseas Filipino Workers were deployed in different parts of the globe which recorded a 14.53 percent increase from 2015 deployment figures. Of this number, 79 percent are land based and 21 percent are seafarers (*Table 6.13*).

Overall, the Middle East and Asian region host the bulk of land based OFWs or 95.59 percent of the total number deployed in 2015 such as the Kingdom of Saudi Arabia, the United Arab Emirates, Singapore Qatar and Hongkong (*Table 6.14*).

For the same period the top destination of new hire OFWs from NCR, are Saudi Arabia, followed by Qatar, Hongkong, Taiwan and the United Arab Emirates.

In NCR, a total of 212,827 or 11.53 percent of the total OFWs were deployed in 2015. 124,657 OFWs or 58.6 percent are land based and 88,170 or 41.4 percent are seabased workers (*Table 6.15*).

Table 6.14. Deployed New Hires Landbased Overseas Filipino Workers by Sex and World Group, 2015

MAJOR WORLD GROUP	MALE	FEMALE	TOTAL
Africa	6,033	569	6,602
Americas	3,941	1,842	6,602
Asia	38,339	94,553	132,892
Europe	1,279	2,959	4,229
Middle East	148,830	210,817	359,647
Ocenia	4,035	599	4,364
Trust Territories	1,259	422	1,681
Not elsewhere classified	55	54	109
Total	203,762	311,455	515,217

Table 6.15. Deployed OFWs by Type of Hiring, NCR 2015

TYPE OF HIRING	PHILIPPINES	NCR	NCR % TO TOTAL
Landbased	1,437,875	124,657	8.66%
New Hires	515,217	62,615	
Rehires	922,658	62,042	
Seabased	406,531	88,170	21.68%
TOTAL	1,844,406	212,827	11.53%

Table 6.16. Types of Migrants

TYPES OF MIGRANT	2011-2015			2012-2016		
	MALE	FEMALE	TOTAL	MALE	FEMALE	TOTAL
Emigrants	40,313	62,437	101,875			
Filipino Marriage Migrants	2,300	26,066	28,366			
AU Pairs				47	1,483	1,530
Exchange Visitors Program (EVP)				1,509	2,906	4,415

Top skill categories of new hire OFWs from this region are in the elementary occupations which include domestic helpers and cleaners / helpers in offices and hotels and other establishments. Notable is the inclusion of highly skilled workers such as manufacturing and civil engineering laborers, building and related electricians, welders and flame cutters. A substantial number of nursing professionals were likewise sourced from the region (*Table 6.16*).

Available administrative data on other types of migrants reveal that for the period 2011-2015 a total of 101,875 emigrants (62,437 Female and 40,313 Male) and 28,366 Filipino marriage migrants (26,066 Female and 2,300 Male) from the NCR registered with the CFO. Over a 5 year period covering the year 2012-2016, 1,530 AU pairs (1,483 Female and 47 Male) and 4,415 Exchange

Visitors Program (EVP) participants (2,906 Female and 1,509 Male) from the region also registered with the CFO (*Table 6.17*).

In NCR, there are a total of 130,241 migrants for the period 2011-2015 and over a year period covering the year 2012-2016 a total of 5,945 were registered with the CFO.

The Overseas Filipinos are exposed to the following challenges:

- Need for continuous skills upgrading and capacity-building for OFWs and their families.
- Lack of mechanism to promote the savings and investment mind-set of the OFWs and their families and to utilize savings in more productive economic opportunities.
- Proliferation of illegal recruitment and/or trafficking in persons.

Strategic Framework

Targets

The following table presents the medium-term targets for industry and services. Gross value-added in services and industry will be improved. Employment generation in

these sectors will be increased. The targets correspond to the subsector outcomes outlined below.

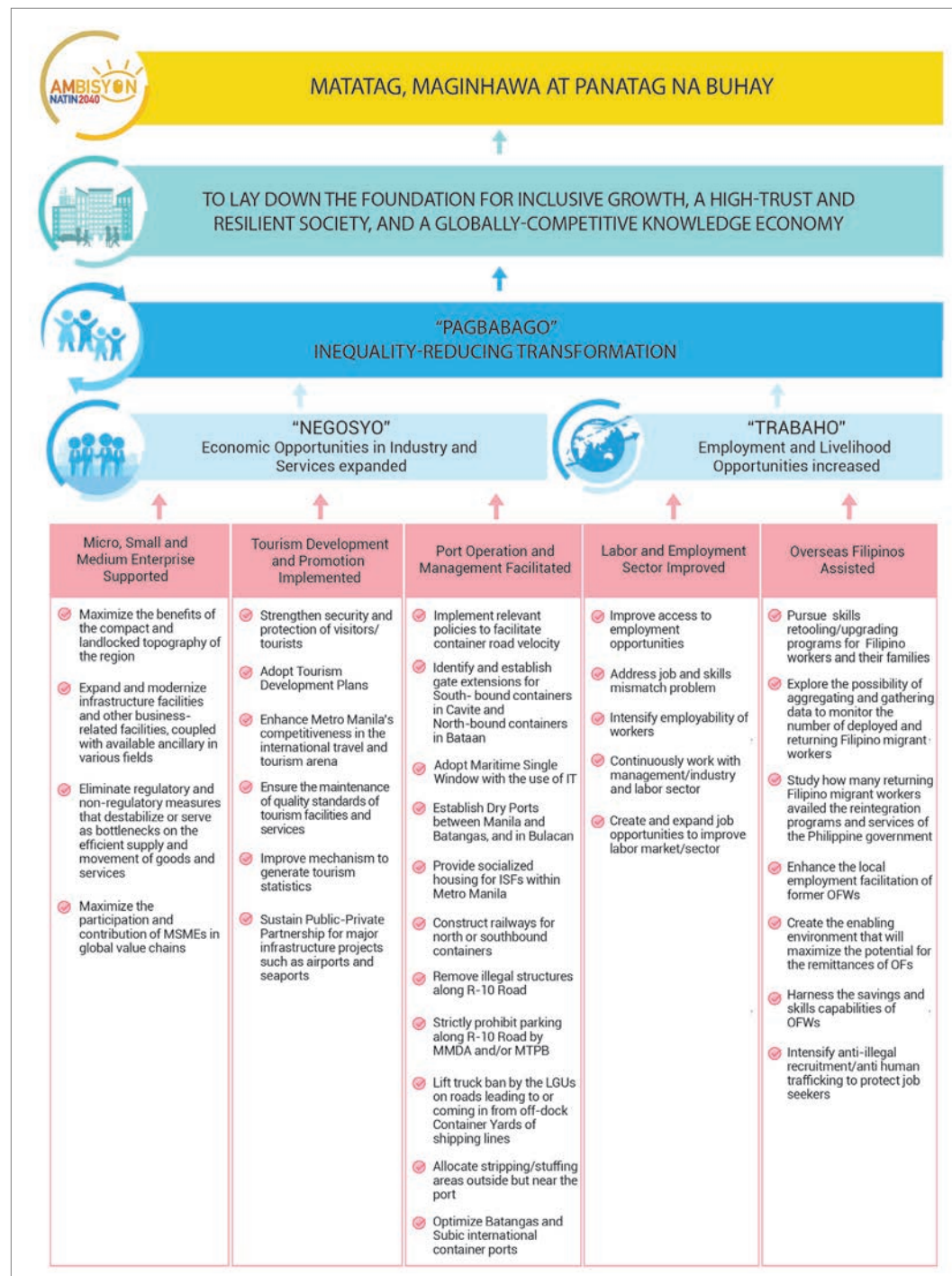
Table 6.17. Plan Targets to Expand Economic Opportunities in Industry and Services 2017-2022

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	Year	VALUE		
Medium and Small/Micro Enterprises				
Total business establishments	2015	191,955	Increased	PSA
Contribution to employment	2015	2,911,554	Increased	PSA
Contribution to value added	2015	2,977,477,320	Increased	PSA
Jobs generated by MSMEs ('000 persons)	2015	1,471,465	Increased	PSA
Number MSMEs assisted				DTI
Proportion of small-scale industries (enterprises) in total industry value added increased				
Tourism Development and Promotion				
Number of tourist arrivals in Metro Manila	2015	1,417,075	Increased	DOT
Number of Available Rooms in Metro Manila	2015	16,800	Increased	DOT
Port Operation and Management				
Number of direct employment generated	2016-2017	3,287		PPA
Number of indirect employment generated	2014-2016	2,000		
Labor and Employment				
Labor force participation rate	2015	62.4	Increased	PSA
Employment rate	2015	92.8%	Increased	PSA
Unemployment rate	2015	7.2	Decreased	PSA
Underemployment rate	2015	12.9%	Decreased	PSA
Overseas Filipinos and their Families				
Number of OFWs deployed				POEA
Land-based	2015	124,657		
Sea-based	2015	88,170		
Total	2015	212,827		

Strategies

As part of reducing inequality, the RDP 2017-2022 aims to increased economic opportunities in the industry and services sectors through trabaho at negosyo.

Figure 6.9. Strategic Framework to Expand Economic Opportunities in Industry and Services, 2017-2022



To support micro, small and medium enterprise

Maximize the benefits of the compact and landlocked topography of the region, which can make the movement of people, goods, and services easier.

Expand and modernize infrastructure facilities and other business-related facilities, coupled with available ancillary in various fields making it an ideal place to do business. The modernization of infrastructure facilities, the availability of a critical mass of service providers (BDS) and the strong presence of social/consumer networks are opportunities that businesses in the region can take advantage of. The growing outsourcing sector and the developing healthcare sectors are among those that provide job opportunities in the region.

Eliminate regulatory and non-regulatory measures that destabilize or serve as bottlenecks on the efficient supply and movement of goods and services. Provide conducive environment in creating competitive services sector.

Maximize the participation and contribution of MSMEs in global value chains

- Forging strong partnership and strategic alliances with the private sector, academe and other government agencies that would promote inclusive growth through social entrepreneurship and extend the reach of government to facilitate faster and convenient access by MSMEs to its services.
- Intensifying cooperation with Head Office Bureaus to maximize resources in the implementation of programs and services for MSMEs development in NCR.
- Adopting value chain approach to MSME Development focused on identified priority sectors.

To implement tourism development and promotion

Strengthen security and protection of visitors/tourists. The Tourism Act of 2009 provides for the institutionalization of a Tourism Security Force within Philippine National Police (PNP) to assist in maintaining peace and order within the areas of high tourism traffic. The strengthening of DOT's TOPCOP (Tourist-Oriented Police for Community Order and Protection) Project in partnership with the Philippine National Police, called for the creation of special police units which aims to enhance the security of tourists in various travel destinations throughout the country. The TOPCOP personnel are given special trainings on proper handling of tourist related security and safety issues.

Adopt Tourism Development Plans aligned with the National Tourism Development Plan. Both the national agencies and local government units should adopt their respective tourism development plans in order to develop first-rate products and cope up with impeccable services to draw more tourists.

Enhance Metro Manila's competitiveness in the international travel and tourism arena.

Metro Manila is the major gateway to the country's 7,107 alluring islands. Metro Manila must be promoted as a vibrant tourist destination. By developing more tourism products and activities through collaborative efforts of the national government agencies and local government units that will make Metro Manila competitive with our neighboring Asian countries.

Ensure the maintenance of quality standards of tourism facilities and services. Enforce a national standard for the operation and maintenance of primary and secondary tourism enterprises. Likewise, prescribe minimum levels of quality and efficiency in order to ensure that facilities and services are

maintained in accordance with acceptable local and international standards.

Improve mechanism to generate tourism statistics. There must be a coordination between national and local governments to generate Metro Manila's tourism statistical/data specifically on the number of hotel guests and museum visitors in a certain LGU.

Sustain Public-Private Partnership for major infrastructure projects such as airports and seaports to support tourism growth in air services and provide greater connectivity to key source markets. Construction of a world class airports and seaports must be a priority of the national government agencies to boost the tourism industry.

To facilitate port operation and management

Implement relevant policies to facilitate container road velocity (such as the TABs booking system) and improved port management such as clearing of port roads, establishment of new truck routes and relocation of DPWH weighbridges

Identify and establish gate extensions for South-bound containers in Cavite and North-bound containers in Bataan whereby containers destined to South or North of NCR Manila would be transported by barge from SH and MICT terminals and gated out from said gate extensions. This will dramatically reduce container traffic along the Metro Manila truck routes. The resultant economic benefits to this scheme, such as additional labor opportunities, would be vast and wide. These would also encourage the establishment of distribution centers outside of congested Manila.

Adopt Maritime Single Window with the use of IT.

Establish Dry Ports between Manila and Batangas, and in Bulacan where empty containers need not return to the ports of Manila but repositioned instead from the ports of Batangas and/or Subic.

Provide socialized housing for ISFs within Metro Manila. If relocation is outside Metro Manila, to provide livelihood programs or job opportunities for the ISFs.

Construct railways for north or southbound containers. Construction of alternate roads and viaducts for laden trucks/vehicles moving in/out of the port to/from their destinations.

Remove illegal structures along R-10 Road, i.e.: junk shops, vulcanizing shops, repair shops, etc.

Strictly prohibit parking along R-10 Road by MMDA and/or Manila Traffic and Parking Bureau (MTPB)

Lift truck ban by the LGUs on roads leading to or coming in from off-dock Container Yards of shipping lines

Allocate stripping/stuffing areas outside but near the port

Optimize Batangas and Subic international container ports to handle thru cargo volume of South Luzon-based and Central/North Luzon-based economic locations

To improve labor and employment

Improve access to employment opportunities through intensified advocacies and appropriate labor market information in partnership with the academe and key industry players that are attuned to labor market trends.

Address job and skills mismatch problem by promoting linkage and coordination between employers and industry stakeholders, academe, national and local governments.

Intensify employability of workers thru career guidance, employment coaching, retooling and retraining, entrepreneurship programs, and productivity improvement programs.

Continuously work with management/industry and labor sector to strengthen tripartite cooperation on enforcement and compliance to general labor standards, occupational safety and health, and security of tenure, among others, and gainfully promote industrial peace.

Create and expand job opportunities to improve labor market/sector.

To give assistance to the Overseas Filipinos

Pursue skills retooling/upgrading programs for Filipino workers and their families to equip them with appropriate skills which will enable them access to more demandable jobs with better work conditions and higher pay.

Explore the possibility of aggregating and gathering data to monitor the number of deployed and returning Filipino migrant workers for policy review and program development.

Study on how many returning Filipino migrant workers availed the reintegration programs and services of the Philippine government.

Enhance the local employment facilitation of former OFWs by requiring private companies to enter into formal agreement with the government to seamlessly hire and recruit qualified OFWs.

Create the enabling environment that will maximize the potential for the remittances of OF's to contribute to local economic and social development through improved access to finance and enhanced financial literacy towards the development of MSMEs. Local governments in the region need to recognize OFs as development and investment partners in line with its priority development plans and investment programs.

Harness the savings and skills capabilities of OFWs towards enabling the local economy to benefit from them.

Intensify anti-illegal recruitment/anti human trafficking to protect job seekers.

07

Accelerating Human
Capital Development

Accelerating Human Capital Development

One of the major resource assets in Metro Manila are its people, being a highly populous region, ranking 2nd highest among all the regions.

The population in Metro Manila is predominantly young. Among the dependent population, approximately 27.5 percent are between 0-14 years old while five percent are 65 years old and above. The economically productive age of 15-64 years old accounts to a high 67.5 percent¹ who technically represent the work force who contributes significantly in producing economic goods and services needed for growth and development.

Sex ratio at birth is high at 104² but decreasing as the population gets older meaning male die younger than their female counterpart resulting to an over-all sex ratio of 96. The dependency ratio is 48.³

The major challenge is to provide adequate opportunities to develop this human capital by strengthening the foundation for a healthy and well-educated citizenship which are important factors for human capital formation. Education lays down the fundamentals of knowledge and mental development while health and nutrition promotion is essential for achieving physical and mental wellness.

Assessment and Challenges

Health

Though the metropolis seems to offer better opportunities to people not available in the countryside, public health issues arise with the increasing prevalence and incidence of communicable and non-communicable diseases as a result of overcrowding or congestion, poor sanitation, pollution, poor housing condition, and unhealthy lifestyle, among others. Moreover, diseases primarily caused by urbanization and globalization such as mental disorder, injuries and accidents, drug and substance abuse are on the rise.

¹DOH-NCR

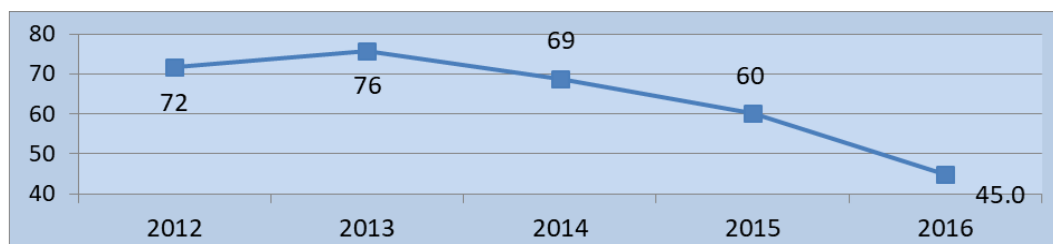
²Male to female ratio is 104:100

³Ibid

Maternal and Child Health

Maternal Health. Maternal mortality is the death of a woman while pregnant or within 42 days after termination of pregnancy. It is a strong indicator for health care status because this arises from the risks attributable to pregnancy and childbirth as well as poor quality of health care services. Carefully examining the trend as shown in *Figure 7.1*, there was a significant decline in Maternal Mortality Ratio (MMR) from 72 maternal deaths per 100,000 livebirths to 45 maternal deaths per 100,000 livebirths.

Figure 7.1. Maternal Mortality Ratio, 2012-2016
Rate per 100,000 Livebirths, National Capital Region



Source: DOH-NCR

Analysis of the direct causes of maternal deaths shows pre-eclampsia, eclampsia and hemorrhage as the top three leading causes. Almost all causes of maternal deaths are preventable. Health related practices that could have contributed to the decline in maternal deaths are the high proportion of births delivered by skilled health professionals (96.5%) and births delivered in a health facility (94.4%), availability and utilization of quality prenatal and postnatal services (Figure 7.2).

The unplanned and uncontrolled pregnancies have contributed to poor maternal and child health. Contraceptive prevalence defined as the percentage of women who are currently using, or whose sexual partner is currently

using, at least one method of contraception, regardless of the method used is usually reported for married or in-union women aged 15 to 49.

The National Capital Region failed to meet the target of 65 percent for Contraceptive Prevalence although the trend increased from 40.7 percent in 2012 to 48.6 percent in 2016 (Figure 7.3). Noteworthy to mention that 4 cities in NCR achieved the target namely Quezon City (89.7%), Navotas (77.3%), Valenzuela (69.9%) and Muntinlupa (67.1%).

Child Health. Child mortality indicators are the most sensitive markers of the general health status of a country. The achievements in terms of attaining the Infant Mortality

Figure 7.2. Skilled Birth Attendants and Facility-Based Deliveries
National Capital Region, 2012-2016 NOH 2016 Target: 90% (FBD & SBA)

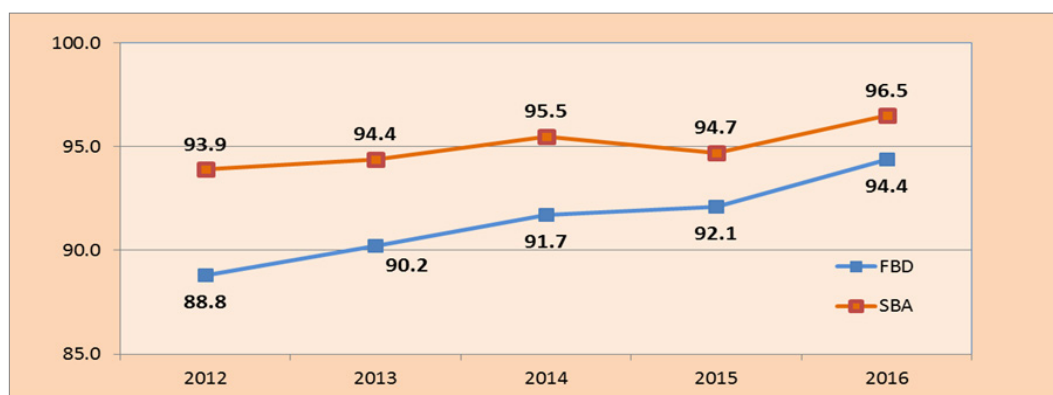
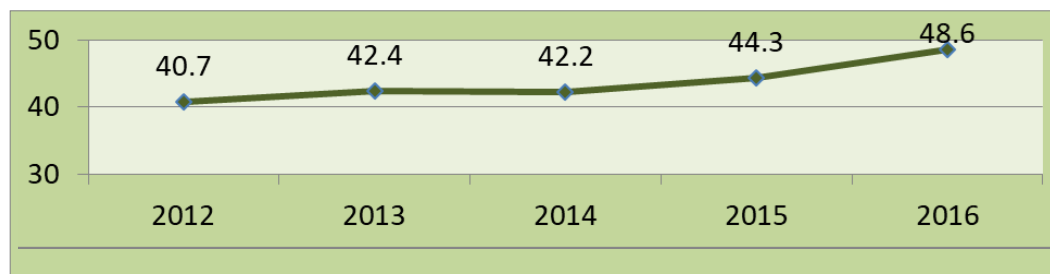


Figure 7.3. Contraceptive Prevalence Rate, National Capital Region, 2012-2016
NOH 2016 Target: 65%



Source: DOH-NCR

Rate (IMR) and Under-five Mortality Rate (UFMR) are substantial. The targets were attained and sustained declining at 14 per 1000 livebirths and 19 per 1,000 livebirths respectively in 2016 (Figures 7.4 and 7.5).

However, sustaining the declining child

mortality becomes a formidable challenge if the Neonatal Mortality Rate (NMR) continues to rise. The neonatal period is the first 28 days of life, the most vulnerable time for child survival. Currently, the NMR is increasing although still on track at <10 per 1,000 livebirths.

Figure 7.4. Infant Mortality Rate, 2012-2016
Rate/1,000 Livebirths, National Capital Region

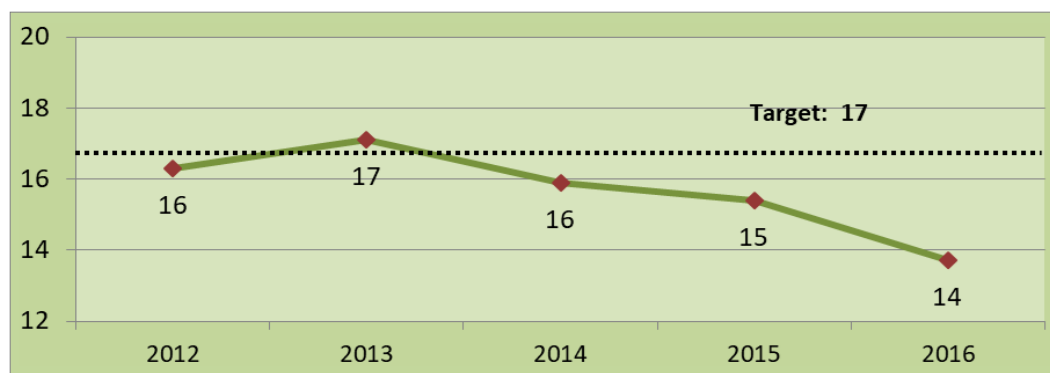
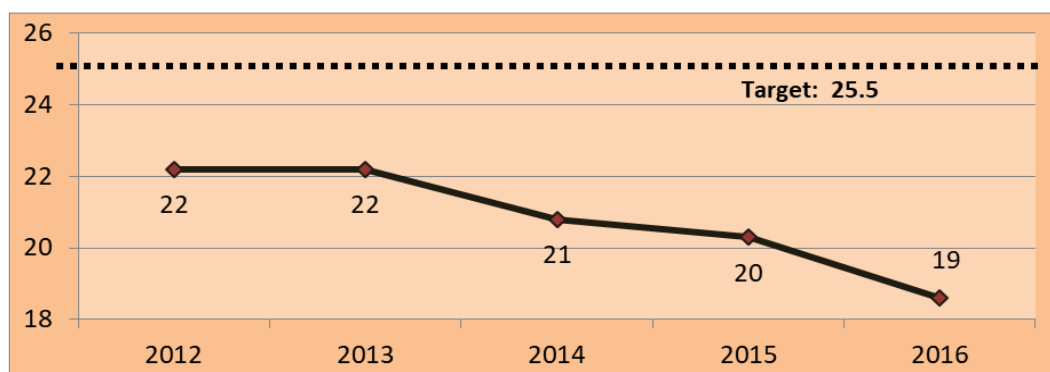


Figure 7.5. Under 5 Mortality Rate, 2012-2016
Rate/1,000 Livebirths, National Capital Region



Source: DOH-NCR

Adolescent Health. Adolescents face many threats to their health and well-being including suicidal tendencies, substance and drug abuse, unexpected/unwanted pregnancies and sexually transmitted infections.

Teen pregnancy is considered an important health issue because both mother and the child are more susceptible to pregnancy related health risks and complications than the women of older age group. Adolescents who are 15-19 years of age are twice as likely to die during pregnancy or childbirth compared to women over 20 years of age; adolescents under 15 years of age are five times more likely to die during pregnancy or childbirth. Moreover, babies born to teen mothers are more likely to be born prematurely and have low birth weight.

The Philippine National Demographic Health Survey of 2013 revealed that 24 percent of young women aged 15-24 in the NCR have begun childbearing; 22 percent had a live birth while 2 percent are pregnant with their first child.

Their sexual curiosity, coupled with poor sexual and reproductive health education and services pose serious problems such as HIV/AIDS.

In May 2017, there were 1,098 new HIV individuals reported to the HIV/AIDS & ART Registry of the Department of Health (DOH); 325 (30%) of whom were among youth aged 15-24 years old and 95 percent of the cases were male. All were infected through sexual contact.

Cumulative report from January 1988 to May 2017 showed 27 percent (12,065) of the 44,010 reported cases belong to the same age group with 84 percent (10,148) reported from January 2012 to May 2017 (*Table 7.1*).

As to geographical distribution, NCR has the most number of reported cases, followed by Regions 4A, 7, 3 and 11 (*Table 7.2*).

This age group, because of their adventurous behavior are also prone to accidents and injuries that could lead to death.

Table 7.1. Quick Facts

DEMOGRAPHIC DATA	MAY 2017	JAN - MAY	JAN 2012 - MAY 2017	CUMULATIVE JAN 1988 -MAY 2017
Total Reported Cases	1,098	4,388	35,646	44,010
Asymptomatic Cases	958	3,874	32,447	39,829
AIDS Cases	140	516	3,199	4,181
Male	1,041	4,172	34,083	40,973 ^a
Female	57	216	1,563	3,026 ^a
Age Range (median)	2-67(28)	1-79 (27)	1-82 (28)	1-82 (28)
Less than 15 y/o	3	13	66	124 ^b
15-24 y/o	325	1,345	10,148	12,065 ^b
25-34 y/o	554	2,207	18,785	22,593 ^b
35-49	185	714	5,834	1,954 ^b
50 y/o & above	31	109	813	1,200 ^b
Pregnant WLHIV	8	29		178
Newly Started on ART	683			
Total PLHIV				20,420
Reported Deaths	15	187		2,156

Source: DOH-NCR

Table 7.2. Percentage of HIV Cases per Region

REGION	MAY 2017 (N=1,098)	JAN-MAY 2017 (N=35,646) ^A	JAN 2012-MAY 2017 (N=35,646) ^A	CUMULATIVE JAN1984 – MAY 2017 (N=44,010) ^B
NCR	404 (37%)	1,564 (36%)	14,754 (41%)	18,463 (42%)
4A	155 (14%)	679 (15%)	5,350 (15%)	6,234 (14%)
7	98 (9%)	396 (9%)	3,407 (10%)	3,981 (9%)
3	108 (10%)	464 (11%)	3,104 (9%)	3,792 (9%)
11	60 (5%)	238 (5%)	2,201 (6%)	2,683 (6%)
ROTC	273 (25%)	1,057 (24%)	6,817 (19%)	7,811 (17%)

Source: DOH-NCR

Health of Older Persons and Persons with Disability (PWD). The proportion of older persons is expected to rise and the increase in the demand for health services by the elderly is therefore inevitable. Health services and programs for the elderly must be put in place and be prioritized.

Chronic diseases are expected to rise and this will require health promotion and disease prevention interventions at the community level. As an intervention, influenza (4,000 vials) and pneumococcal vaccines (61,267 vials) had been distributed for the immunization of the elderly especially the indigents with a 92 percent regional utilization rate.

An eight (8) percent increase in the number of persons with disability (PWD) availing of health services at the local health centers were recorded from 12,398 in 2015 to 13,417 in 2016. In 2016, 825 (43%) health workers were trained on Disability Inclusive Health and Wellness Program.

Coming up with a health registry of PWDs and capability building of our health human resource on providing services for this particular segment of the population is one of the main concerns at the moment.

Presence of triple burden of disease. The presence of the triple burden of disease is evident in the morbidity and mortality picture (All Ages) of the NCR. Both the

prevalence of communicable and non - communicable diseases remains to be high while conditions related to urbanization like injuries, vehicular accidents, mental disorders and substance abuse are on the rise.

Infectious/communicable diseases specifically Respiratory Tract Infection, Bronchitis/ Bronchiolitis, Pneumonia and Tuberculosis continue to be among the 10 leading causes of illness and death in Metro Manila. Lifestyle related Non communicable diseases namely Hypertension, Diabetes, Cardiovascular diseases, Cancer, Hypertension and Cerebrovascular Diseases are also in the list. Injury and Wound (All forms) are among the diseases associated with urbanization.

The triple burden of disease is considered a serious public health concern. The presence of these diseases in our spectrum may be explained by the environmental sanitation conditions in the metropolis, unhealthy lifestyle of people living in urban settings like the lack of physical inactivity, diet, tobacco smoke, pollution, etc. and urbanization and industrialization.

Moreover, as previously discussed, the NCR is the biggest contributor to the diagnosed HIV cases (see *Table 7.2* above). The cities of Quezon, Manila, Makati, Pasig and Caloocan have the most number of HIV cases.

Health inequities or disparity in health outcomes exists between the localities of Metro Manila. The absence of adequate data

and information is a deterrent in determining the difference in health outcomes between the rich and poor or among the socio-economic groups/quintiles.

Access to Quality Health Services. Access to quality health services has improved through initiatives like Health Facility Enhancement Program, Human Resources for Health Deployment Program and the establishment of Service Delivery Networks among others.

A total of 294 health facilities have been completed (renovation, rehabilitation, repaired, newly constructed or expansion) as of 2016. Moreover, about 155 health facilities were given equipment to ensure the provision of quality health services.

Health human resource deployed to the local government units totaled 417 categorized into nurses, medical technologists, midwives, dentists and public health assistants.

To ensure the provision of quality health services and to make health system more responsive to the needs of the Filipinos especially the poor, the Department of Health advocated for the establishment of Service Delivery Networks (SDN).

SDN refers to the network of health facilities and providers within the city-wide health system offering core packages of health care services in an integrated and coordinated manner to priority population.

In NCR, 12 of the 17 local government units in Metro Manila, have established SDN for Maternal, Neonatal and Child Health and Nutrition (MNCHN) and for Human Immuno-deficiency Virus.

The SDN puts in place the health services that certain identified population groups such as the poorest of the poor and those with geographic barriers have difficulty accessing.

Another measure to ensure that quality

services are being delivered is through the accreditation of health facilities by the Philippine Health Insurance Corporation (PhilHealth). As of July 2017, the status of PhilHealth accreditation of the 478 primary care health facilities in NCR are as follows: Primary Care Benefit (PCB)- 466; TB DOTS – 216; Maternity Care Package (MCP) -29; and Animal Bite Treatment Center (ABTC) -7 out of 69 ABTC.

As of CY 2016, enrollment coverage is pegged at 92 percent with 100 percent of National Household Targeting System (NHTS) poor are covered. The average availment rate in the (4) DOH–NCRO retained hospitals is 85.75.

Nutrition

Double Burden of Malnutrition

Metro Manila inspite of its urbanized state experiences the effects of malnutrition. In fact, the region has a double burden of malnutrition in the form of undernutrition and overnutrition affecting vulnerable population groups such as children and women.

Stunting and wasting among under - five children are significant public health problems while overweight and obesity rates among children and adults are higher than the national levels.

Based on the 2015 Updating of the Nutritional Status of Filipino Children conducted by the Food and Nutrition Institute of the Department of Science and Technology (FNRI-DOST), among children less than 5 years old, one (1) out of 10 are underweight (15.1%). This is lower by 6.4 percentage points than the national rate of 21.5 percent. Meanwhile, one (1) in every four (4) children 0-5 years old or 24.9 percent are stunted in the region. This is lower by 8.5 percentage points than the national prevalence of 33.4 percent.

About six (6) out of 100 or 6.4 percent

children 0-5 years old are wasted or thin. Although this rate is lower than the national prevalence rate of 7.1 percent, it is considered to be a public health problem (Figure 7.6).

On the other hand, six (6) out of 100 under-five children or six percent are overweight for their height. This rate is higher by 2.1 percentage points than the national prevalence of 3.9 percent.

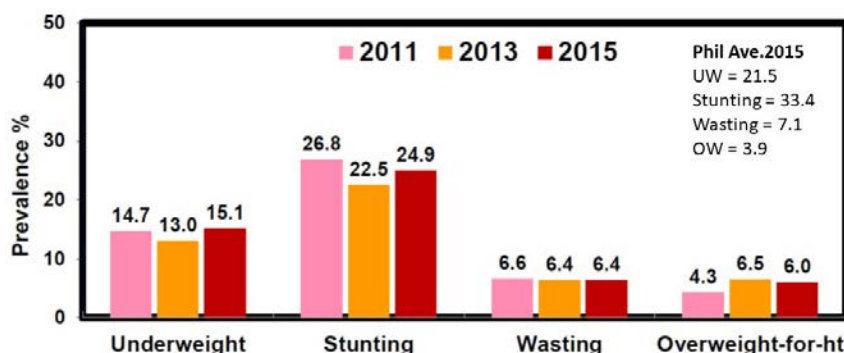
Based on the same survey, children living in rural areas and those from the poorest quintile are more -at-risk to underweight and stunting, while those in urban areas such as Metro Manila and from the richest quintile

are more at-risk to overnutrition (Figure 7.7)

For children 5-10 years old, incidence of underweight children is one (1) in every five (5) children (20.9%) which is lower by 10.3 percentage points than the national rate of 31.2 percent. About one (1) out of five (5) or 20.5 percent children are stunted, lower than the national prevalence of 31.1 percent.

In 2015, seven (7) out of 100 or 7.5 percent of children were found to be wasted or thin. Though the prevalence rate of wasting has continuously decreased from 2011 to 2015, the prevalence of 7.5 percent is still considered to be a public health problem.

Figure 7.6. Prevalence of Malnourished Among Children Less Than 5 Years Old: NCR, 2011, 2013 and 2015



Source: FNRI - DOST

Figure 7.7. Prevalence of Malnourished Among Children Less Than 5 Years Old by Place of Residence and Wealth Quintile: Philippines, 2015



Source: FNRI - DOST

However, the percentage of overweight/obese children 5-10 years old has been increasing from 2011-2015. About one (1) of 10 or 17.9 percent are overweight for their height. This is higher by 9.3 percentage points than the national prevalence of 8.6 percent (*Figure 7.8*).

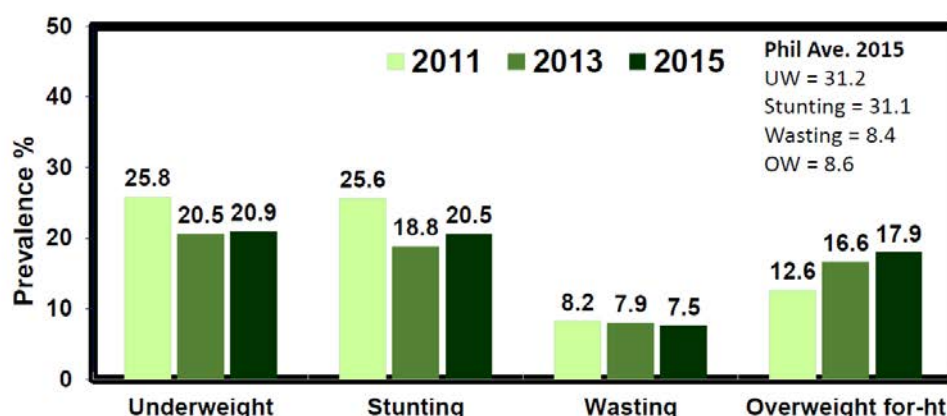
Nutritional status of adolescents ages 10-19 revealed that two (2) out of 10 or 24.3 percent are stunted. This is lower than the national prevalence of 31.9 percent. One (1) out of 10 or 13.5 percent of adolescents

are wasted and thin. This is higher than the national prevalence of 12.5 percent.

Meanwhile, one (1) out of 10 or 16.1 percent are overweight and obese. This is higher by 6.9 percentage points than the national prevalence of 9.2 percent (*Figure 7.9*).

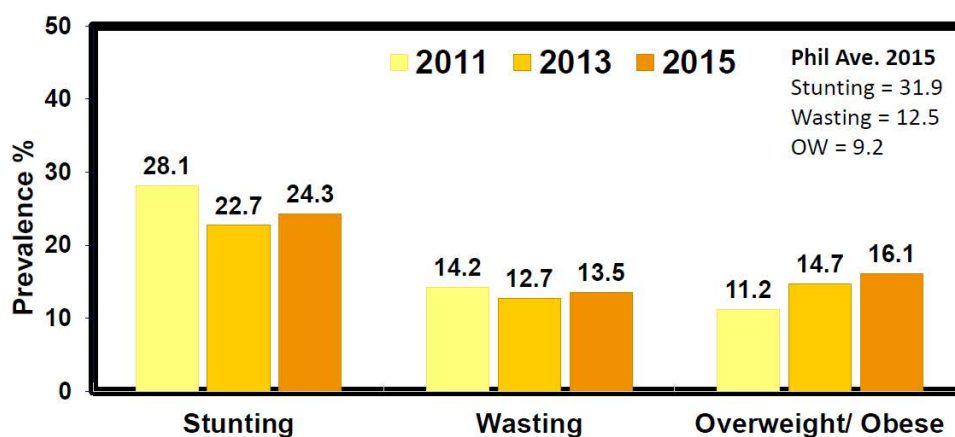
Among adults 20 years old and over, nine (9) out of 100 or 9.3 percent are chronic energy deficient, four (4) out of 10 or 37.7 percent are overweight and obese. The prevalence rates of underweight and overweight among

Figure 7.8. Prevalence of Malnourished among children 5-10 years old: NCR, 2011, 2013, 2015



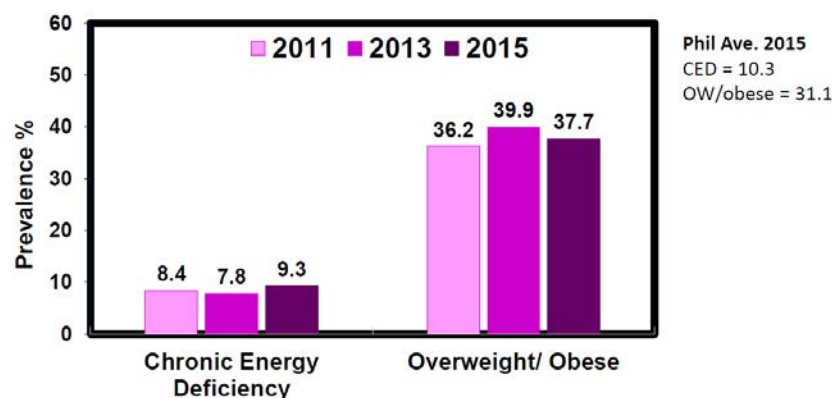
Source: FNRI - DOST

Figure 7.9. Prevalence of Malnourished among children 10-19 years old: NCR, 2011, 2013, 2015



Source: FNRI - DOST

**Figure 7.10. Prevalence of Malnourished among Adults ≥ 20 years old:
NCR, 2011,2013,2015**



adults have a “see-saw” trend from 2011 to 2015. The prevalence of underweight adults increased from 7.8 percent in 2013 to 9.3 percent in 2015. On the other hand, the prevalence of overweight/obesity among adults decreased from 39.9 (2013) to 37.7 percent (2015) (Figure 7.10).

Poor Maternal Nutrition

The poor nutritional status of pregnant women makes them at risk of delivering low birth weight babies and having other negative pregnancy outcomes like stillbirths and miscarriages.

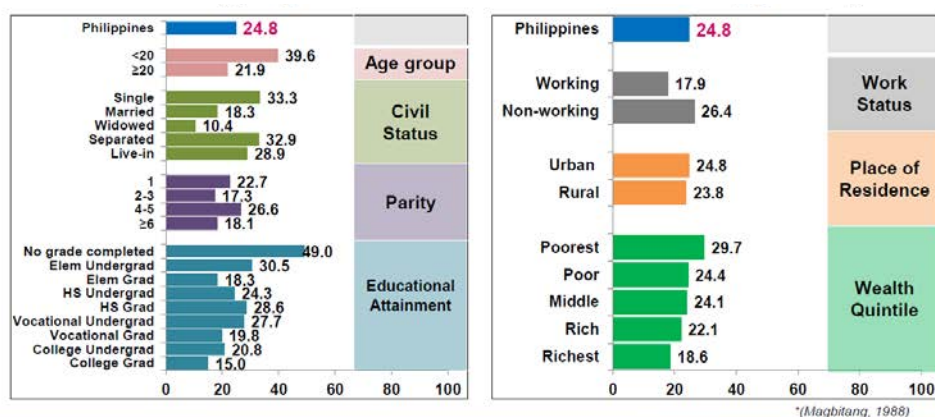
Based on the 2015 FNRI-DOST survey, teenage pregnant mothers (39.6%) in the country were

more nutritionally at-risk than older mothers.

Mothers from the poorest and poor wealth quintiles, not working and more children have higher proportion of nutritionally at-risk and chronic energy deficient pregnant women (Figure 7.11).

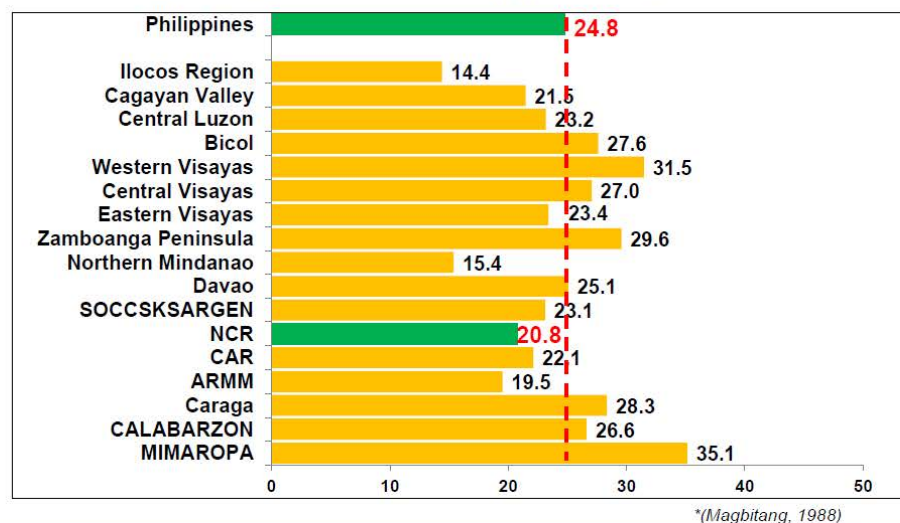
In 2015, 20.8 percent pregnant mothers in Metro Manila were nutritionally at-risk (Figure 7.12) while 15.8 percent and 30.7 percent lactating women were chronic energy deficient (CED) and overweight/obese, respectively. The prevalence of CED is higher by 2.2 percentage points than the national prevalence of 13.6 percent. Prevalence of overweight/obesity among lactating women is much higher

Figure 7.11. Prevalence of Nutritionally-at-risk* pregnant women by socio-demographic characteristic: Philippines, 2015



Source: FNRI - DOST

Figure 7.12. Prevalence of Nutritionally-at-Risk* Pregnant Women by Region: Philippines, 2015



by 8.3 percentage points than the national prevalence of 22.4 percent (Figure 7.13).

Micronutrient Deficiencies

Iron Deficiency Anemia. In 2013, the prevalence of anemia among children 6 months to less than 5 years old is 10.0 percent, lower than the national prevalence of 13.8 percent. Among 6 to 12 years and

old and adolescents 13 to 19 years old in Metro Manila were 7.1 percent and 6.3 percent, respectively. Both prevalence rates are lower by 4.2 and 1.7 percentage points, respectively than of the national prevalence. The prevalence of IDA in all age groups is considered to be “mild” public health significance (Table 7.3).

Meanwhile, the national prevalence rates of anemia among pregnant and lactating

Figure 7.13. Prevalence of Chronic Energy Deficiency and Overweight/Obesity Among Lactating Mothers by Region: Philippines, 2015

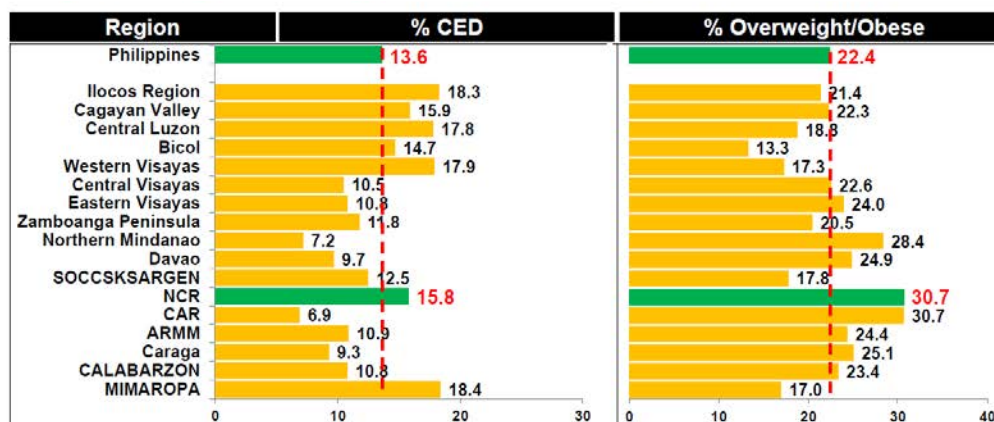


Table 7.3. Iron Deficiencies Anemia, by Age Group, Philippines and NCR, 2013

AGE GROUPS	NCR (%)	PHILIPPINES (%)
6 mos to - 5 years old	10.0	13.8
6-12 years old	7.1	11.1
13-19 years old	6.3	7.7
20-59 years old	8.8	9.3
≥ 60 years old	13.6	20.8

Source: 2013 National Nutrition Survey FNRI –DOST

women continue to decrease from 2003 to 2013. Significant decrease in the prevalence rates of anemia among pregnant and lactating women were noted from 42.5 percent (2008) to 24.6 percent (2013) and 31.6 percent (2008) to 16.7 percent (2013), respectively. Unfortunately, the prevalence rate of anemia per region is not available (Figure 7.14).

In 2013, teenage pregnant mothers were more anemic than older mothers while pregnant women from rich quintile and those residing in urban communities are more at-risk to anemia (Figure 7.15).

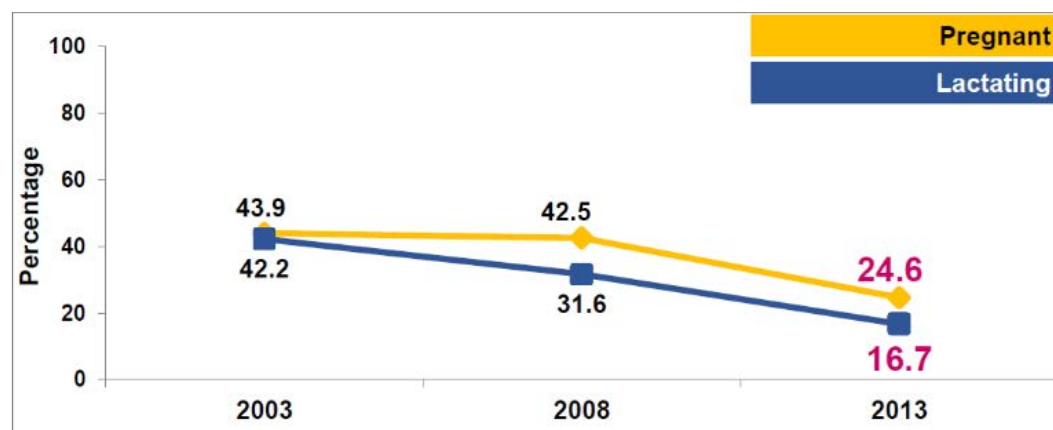
Vitamin A Deficiency. The prevalence of VAD among 5 months to less than 5 years old is 27.3 percent and considered

a “severe” public health problem in NCR. The prevalence of 13.9 percent among 6-12 years old indicates “moderate” public health problem (Table 7.4).

In 2015, only 69.2 percent (estimated 2.2 million) of households in Metro Manila did not meet their caloric requirements. This level is slightly higher than the national percentage of 69.0 percent. While this level is lower than that recorded in 2013 (72.0%) it is higher than the level reported in 2008 (67.7%).

Iodine Deficiency Disorder. In 2013, the prevalence of UIE <50 ug/L among all age groups are considered severe to moderate that corresponds to insufficient iodine intake (Table 7.5).

Figure 7.14. Trends in the Prevalence of Anemia among Pregnant Women and Lactating Mothers with Children 0-36 Months: Philippines, 2003-2013



Source: FNRI - DOST

Figure 7.15. Prevalence of Anemia among Pregnant Women by Age Group, Place of Residence and Wealth Quintile: Philippines, 2013

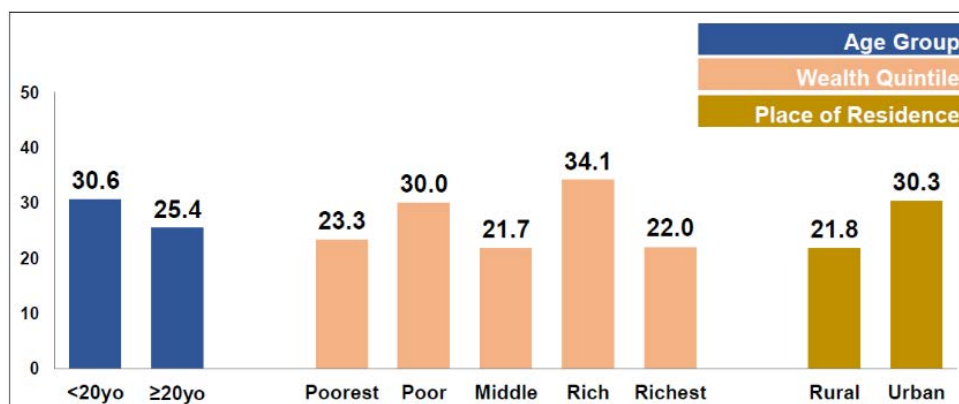


Table 7.4. Vitamin A Deficiency by Age Group, Philippines and NCR, 2013

AGE GROUPS	PHILIPPINES (%)	NCR (%)	PUBLIC HEALTH SIGNIFICANCE
5 months to < 5year old	20.4	27.3	Severe
6-12 years old	10.7	13.9	Moderate
13-19 years old	4.0	9.2	Mild
20-59 years old	2.1	2.4	Mild
< 60 years old	3.0	1.9	No deficiency

Source: 2013 National Nutrition Survey, FNRI-DOST

Table 7.5. Iodine Deficiency Disorder by Age Group, Philippines and NCR, 2013

AGE GROUPS	PHILIPPINES (%)	NCR (%)
6-12 years old	16.4	8.9
13-19 years old	19.8	15.4
20-59 years old	22.4	20.2
< 60 years old	33.7	36.9
Lactating	34.3	27.4
Pregnant	27.0	15.4

Source: 2013 National Nutrition Survey, FNRI-DOST

Hunger and food insecurity remain to be a concern in Metro Manila. Hunger incidence remains to be double - digit in Metro Manila. More families experienced involuntary hunger based on the latest Social Weather Stations Survey on Hunger. The 2017 4th quarter survey found that an estimated 457,000 families in Metro Manila claimed that they had nothing to eat at least once in the past three months. This was three (3) points above the 11.7 percent or around 364,000 recorded in September 2017.

Moderate Hunger refers to those who experienced hunger “Only Once” or “A Few Times” in the last three months, while Severe Hunger refers to those who experienced it “Often” or “Always” in the last three months. The few who did not state their frequency of hunger were classified under Moderate Hunger. Both Moderate Hunger and Severe Hunger increased between September and December 2017. Moderate Hunger rose by 0.7 points, from 9.3 percent in September to 10.0 percent in December. Severe Hunger in

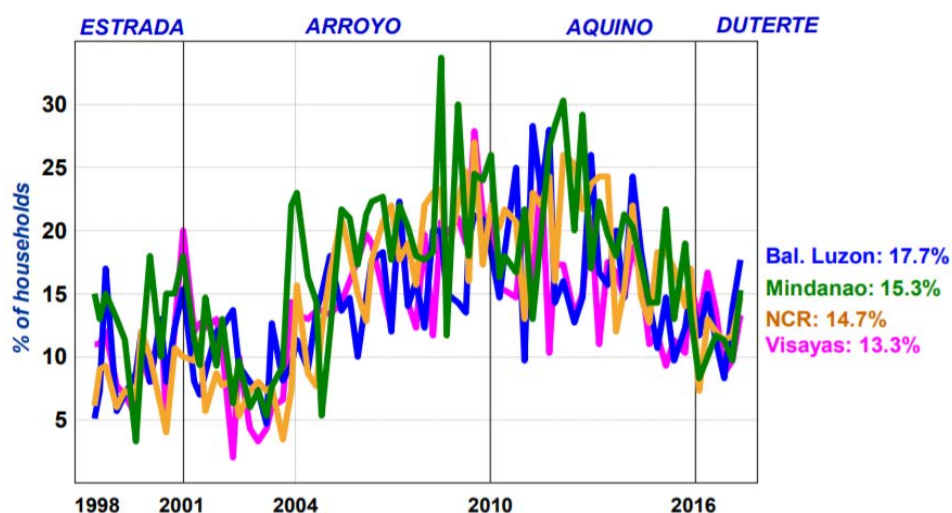
Metro Manila also rose by 2.4 points, from 2.3 percent in September to 4.7 percent in December.

Total hunger or a combination of moderate and severe hunger showed a seesaw trend from 17.0 percent in 2015, 13.0 percent in

2016 and 14.7 percent in 2017, respectively.

An Average Hunger rate of 12.4 percent for 2017 is 0.4 points below the 12.8 percent in 2016. This is the lowest since the 8.8 percent in 2004 (Figure 7.16, Figure 7.17 and Table 7.6).

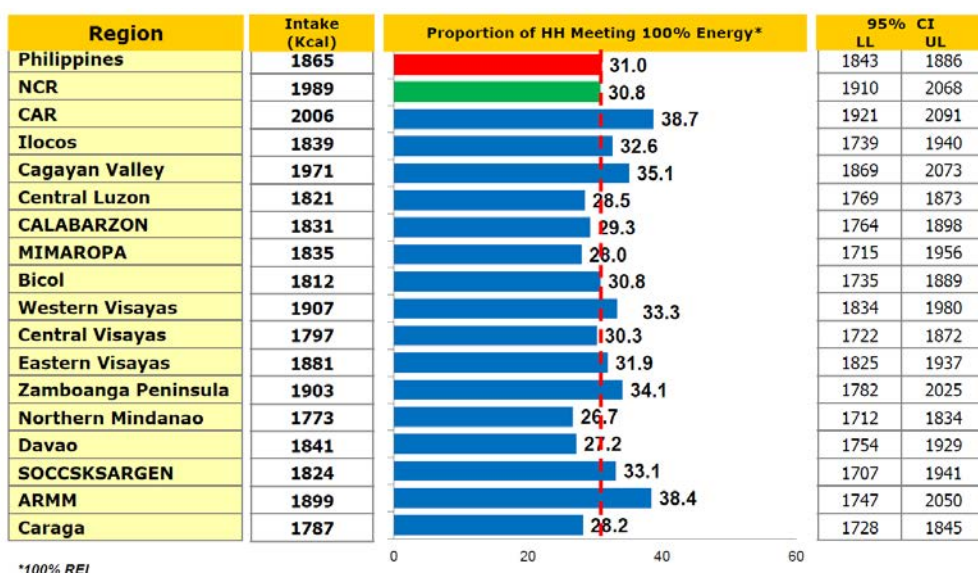
Figure 7.16. Total Hunger, By Area, Jul 1998 - Dec 2017



Note: Don't Know and Refused responses are not shown. Q: Nitong nakaraang 3 buwan, nangyari po ba kahit minsan na ang inyong pamilya ay nakaranas ng gutom at wala kayong makain? Kung OO: Nangyari po ba 'yan ng MINSAN LAMANG, MGA ILANG BESES, MADALAS O PALAGI?

Source: Fourth Quarter 2017 Social Weather Report Dec. 2-16, 2017, National Survey

Figure 7.17. Mean One-day per Capita Energy Intake and Promotion of Household Meeting REI for Energy by Region: Philippines, 2015



Source: Fourth Quarter 2017 Social Weather Report Dec. 8-16, 2017, National Survey

Table 7.6. Total Hunger, By Area, % of Households, 2010-2017

		NCR	BAL. LUZ	VIS	MIN			NCR	BAL. LUZ	VIS	MIN			NCR	BAL. LUZ	VIS	MIN
Mar	10	17.3	20.9	21.2	24.0	Mar	13	21.7	14.7	15.0	29.2	Apr	16	14.0	12.3	10.3	19.0
Jun	10	22.0	18.3	21.0	26.0	Jun	13	23.7	26.0	21.0	17.0	Jun	16	17.0	15.3	15.7	13.3
Sep	10	20.3	14.7	15.3	16.3	Sep	13	24.3	16.7	11.0	22.3	Sep	16	7.3	11.7	13.0	8.3
Nov	10	21.7	18.3	15.3	18.0	Dec	13	24.3	15.7	17.5	19.7	Dec	16	13.0	15.0	16.7	10.0
Ave.	10	20.3	18.1	18.2	21.1	Ave.	13	23.5	18.3	16.1	22.1	Ave.	16	12.8	13.6	13.9	12.7
Mar	11	20.7	25.0	14.7	16.7	Mar	14	12.0	20.0	16.7	18.0	Mar	17	12.0	11.3	13.7	11.7
Jun	11	13.0	9.7	21.0	21.7	Jun	14	15.5	14.7	14.7	21.3	Jun	17	11.3	8.3	8.7	11.3
Sep	11	23.0	28.3	15.3	13.0	Sep	14	22.0	24.3	18.7	20.3	Sep	17	11.7	13.8	9.7	9.7
Dec	11	22.0	23.0	25.0	19.7	Dec	14	14.7	18.3	16.4	17.3	Dec	17	14.7	17.7	13.3	15.3
Ave.	11	19.7	21.5	19.0	17.8	Ave.	14	16.0	19.3	16.6	19.2	Ave.	17	16.0	12.8	11.4	12.0
Mar	12	24.3	28.0	10.3	26.7	Mar	15	12.7	14.3	11.0	14.3						
May	12	16.0	14.3	17.3	28.3	Jun	15	18.3	10.7	11.7	14.3						
Aug	12	26.0	16.0	17.3	30.3	Sep	15	18.3	14.7	9.3	21.7						
Dec	12	25.3	12.7	13.3	20.0	Dec	15	17.0	9.7	11.3	13.0						
Ave.	12	22.9	17.8	14.6	26.3	Ave.	15	16.6	12.4	10.8	15.8						

Source: Fourth Quarter 2017 Social Weather Report Dec. 8-16, 2017, National Survey

In 2015, only 69.2 percent (estimated 2.2 million) of households in Metro Manila did not meet their caloric requirements. This level is slightly higher than the national percentage of 69.0 percent. While this level is lower than that recorded in 2013 (72.0%) it is higher than the level reported in 2008 (67.7%).

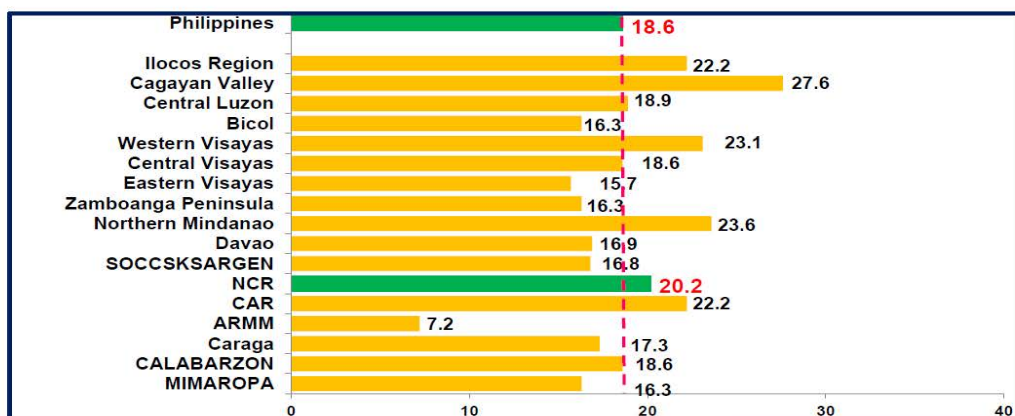
Poor Infant and Young Child Feeding Practices. Breastfeeding is the most effective strategy to improve child survival. Breastmilk is the most complete food for infants during their first six months of life.

The percentage of women who initiated breastfeeding within an hour after delivery has increased from 61.8 percent in 2011 to 83.0 percent in 2013. These results are higher than the national prevalence of 51.9 percent and 77.1 percent in 2011 and 2015 respectively.

However, the percentage of infants 5 months old who are exclusively breastfed has decreased from 44.5 percent in 2011 to 31.8 percent in 2015 (2015 NNS). This is lower than 17 points than the national percentage of 48.8 percent, far lower than the IYCF program target of 70 percent. By the sixth month, the child should receive proper complementary foods in addition to breastmilk, but, only 20.2 percent of infants 6-23 months old received the minimum acceptable diet (MAD). The mean duration of breastfeeding in the region remains at 7 months (*Figure 7.18* and *Figure 7.19*).

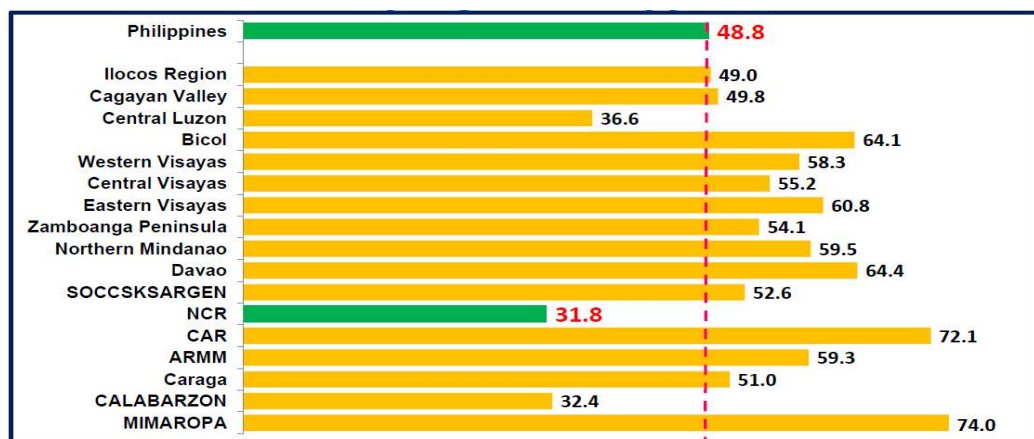
MAD is based on the minimum frequency of feeds and diet diversity or consumption of foods from four groups of a group of seven groups that include grains, roots and tubers; legumes and nuts; dairy products; meat, fish, poultry and liver/organ meats' eggs, vitamin-A rich fruits and vegetables and other fruit and vegetables.

Figure 7.18. Percentage of Children 6-23 Months Meeting the Minimum Acceptable Diet by Region: Philippines, 2015



Source: 2016 Regional Dissemination Forum FNRI-DOST

Figure 7.19. Percentage Distribution of Exclusive Breastfed Children 0-5 Months by Region: Philippines, 2015



Source: 2016 Regional Dissemination Forum FNRI-DOST

Basic Education

The 1987 Philippine Constitution guarantees the right to education of every Filipino. It was further emphasized in RA 9155 or the Governance of Basic Education Act of 2001 along with RA 6655 or the Free Secondary Education Act. These laws reaffirm the policy of the country to protect and promote the rights of all Filipinos by providing children free and compulsory education in elementary and high school level.

Increasing gap between children of school age in school versus children of school age who are not in school. Despite the effort of the government to provide education for all, with the provision of free tuition fee in all levels, the access to education remains to be a major challenge particularly in the elementary level.

As of SY 2015-2016, there are over 1,744,280 children of school age (5-11 yrs. old). The table shows that there is a need to immediately address the increasing gap between children of

school age 5-11 in school over those children of school age who are not in school. In the elementary level alone, about 15.80 percent or 275,565 children are not in school. Meaning 62 percent or 171,742 are out-of-school.

In the secondary level, we have about 32.46 percent or 242,121 children who are not in schools. About 77 percent of these figures are Out of School Youths (OSYs). This data tells us that we need to intensify our programs that address the needs of children under difficult situations. Also, it is important to understand why these learners are not in school so that appropriate interventions can be provided (*Table 7.7* and *Table 7.8*).

In SY 2015-2016, there were 25,599 dropouts for elementary and 22,489 for secondary. By looking at the key stages of education, it was found out that 65.80 percent or 16,830 dropouts come from

the primary level (Kinder to Grade 3) and about 57 percent in the secondary come from Grade 7 and 8. There are many reasons why children dropout of school. Some are due to relocation of families to other places, children are working together with their parents to earn a living, health problems and the like (*Table 7.9*).

Technical Vocational Education and Training

At the center of focus are program interventions directed to intensify program delivery to special groups and to bring down TVET programs at the grassroots or barangay level. Likewise, promote vertical mobility of graduates through scaling up of TVET qualifications and further enhance worker productivity and in tune with the global competitiveness strategy.

Table 7.7. Elementary Level

INDICATOR	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016
Enrolment	1,249,328	1,246,884	1,248,208	1,225,680	1,199,337
NER	97.16%	94.42%	92.54%	89.67%	88.13%
GER	83.61%	101.81%	105.66%	102.01%	99.37%

Source: DepEd-NCR

Table 7.8. Secondary Level

INDICATOR	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016
Enrolment	674,237	680,656	690,032	684,934	671,839
NER	57.07%	77.00%	76.30%	72.21%	75.49%
GER	73.11%	96.30%	94.71%	91.95%	90.22%

Source: DepEd-NCR

Table 7.9. Number of Dropouts SY 2015-2016

LEVEL	NO. OF DROPOUTS	LEVEL	NO. OF DROPOUTS
Kinder	4,562	Grade 7	6,522
Grade 1	5,822	Grade 8	6,333
Grade 2	3,402	Grade 9	5,896
Grade 3	3,064	Grade 10	3,738
Grade 4	3,107		
Grade 5	3,122		
Grade 6	2,520		

Source: DepEd—NCR

In the course of realizing these interventions, a number of challenges remains to be addressed including emerging ones. The present regional challenges include but are not limited to:

Quality-Related Issues

Job-Skills Mismatch. The rapid change in technology and long turnaround time for qualifications review (every three years) becomes difficult for TVET program providers to cope with the supply of graduates that will be at par with the industry requirements. This leads to a widening gap in the quality of supply of skills with that of the industry demand and further contributing to job-skills mismatches. With NCR, at the center of TVET innovation it will have to come up with adequate resource support in partnership with industry to support emerging industries as well as a radical shift in policy and programs for more dualized or enterprise based program.

Satisfying the Requirements of the Key Growth Areas. While the NCR is mainly dominated by the service sector, it will have to address the Build, Build, Build thrust of the new administration which has huge infrastructure investment in the construction industry sector that will be located in the NCR. Apart from government spending in the construction industry, the private sector also has expanded investment in the construction of hotels, condominiums, malls and housing requirements of the growing populace. This should be translated into more program offerings in construction related qualifications. Dearth in skills demand for masonry, carpentry, rigging etc. are already observed.

Cyber Services/BPO-IT sector companies proliferate in the NCR and are expected to expand in the next six years. With this trend, the TVET sector in the region should continue to update its program offerings to sustain the talent requirement of this sector.

Graduates in IT related courses are also supplied under the senior high school tech voc track, TESDA will have to gradually refocus on shifting its scholarship grants towards higher level competencies in the IT sector if Levels 1 and 2 IT qualifications will be mainly supplied by the Senior High School program providers.

In the tourism industry sector, a similar trend is observed, popularity of this sector among the youths significantly increase the supply under the Senior tech voc track. The NCR being the one of the centers of conventions in the country and the major routes of tourists will translate into a robust demand for tourism related qualifications and the focus of Technical Vocational Institutions (TVIs) program offering should take into consideration the complementation with the Senior High school providers.

On Equity and Access

Reintegration Program. Around 62 percent of OFWs annually find jobs in tech voc related occupations and those with finished contracts or those who encountered problems abroad or repatriated may opt to go into retraining or retooling. The requirements of this sector should be taken into account and where possible developed for higher level qualifications (i.e. to fill demand for supervisors, technicians or technologists) or as potential assessors or trainers. The returning OFWs with industry experience may be able to help boost the TVET sector with potential trainers or assessors.

Other New Issues

Low Productivity Occupations. While the underemployed in Metro Manila maybe low in terms of percentages the work force count is enormous as it has the biggest employment among all Regions. The bulk of the underemployed in NCR consists mainly of laborers and unskilled workers which

have steadily grown in the past years.

Programs that would target the provision of basic skills and competencies that would open opportunities for them at higher value added must be pursued.

Full Implementation of K-12. With the increasing number of Senior high school students taking the tech voc track, the region has to prepare the sector to transit to higher level of program qualifications.

Expansion programs of the Region's TVET capacity for assessment will have to be in place.

ASEAN Integration. The NCR-TVET continuous support to the National Office on the ASEAN Integration in the adoption of Mutual Recognition Agreements. Focus will be on the Hotel and Restaurant Industry and other Tourism related areas as the sector is one of the leading Key Growth Areas of the ASEAN Region.

Existing TVIs likewise must be prepared to be competitive with possible incoming foreign schools and institutions in the coming years.

Higher Education

The National Capital Region has the biggest share in terms of number of higher education institutions and enrolment in the country. Its higher education system is a large mixed public-private system consisting of 337 Higher Education Institutions (HEIs) (of highly uneven quality), excluding SUC satellite campuses.

With this number, the Higher Education in the region has several persistent challenges:

Proliferation of Higher Education Institutions (HEIs) and programs, both private and public. There are 337 HEIs in

the NCR or about 17 percent of the total number of HEIs in the country. Majority or 90 percent of these HEIs are privately owned.

The biggest concentrations of HEIs are in Quezon City and City of Manila with 99 and 72 HEIs, respectively. About 50 percent of HEIs in the region is in these two cities while the remaining 50 percent are in the fifteen other cities and municipality, each having 27 HEIs and below. However, half of these HEIs have less than 500 students. Because of so many HEIs offering almost the same programs, they resorted to open admission for incoming students.

Out of the 4,064 programs offered in the different HEIs in the region during Academic Year 2017-2018, the top five most common programs offered by HEIs are the following: Business and Related Programs with 19.8 percent share; Humanities, Social Sciences and Communications with 14.6 percent share; Information Technology with 10.5 percent share; Teacher Education with 8.5 percent share; and Medical and Allied Health with 7.6 percent share.

Among the regions in the country, NCR has the most number of enrolment, majority of which are in the private HEIs. Out of the 531,376 student enrolment data gathered during Academic Year 2016-2017 in NCR, 449,465 students or 84.68 percent of which are enrolled in private HEIs. With these number, the top five heavily populated disciplines are the following: Business and related programs with 32.5 percent enrolment share; Engineering and Technology with 13.4 percent share; Information Technology with 12.5 percent share; Humanities, Social Sciences and Communications with 9.7 percent share; and Medical and Allied Health with 8.8 percent student enrolment. Although Teacher Education is one of the top 5 most common programs, it only has 4 percent enrolment share.

Programs offered by HEIs in the region are primarily market driven, those that are either in-demand or low cost on capital. Business and related programs remains to be the top program offered because it is the most subscribed program in terms of number of enrolment and with minimal laboratory and technical requirements. Although Engineering and Technology is one of the most sought programs, it is not in the top five most common programs offered because of the heavy requirement on laboratories that needs bigger capital outlay. On the contrary, Teacher Education is one of the most common programs offered, however, it is not included in the list of top five most popular programs that is appealing to the students.

Low quality programs that led to skills-job mismatch, low productivity in research and development, and a deficient science and innovation culture. Most of the HEIs in the region are complying only with the minimum requirements set by the commission on Higher Education (CHED) in offering degree programs. Some of the programs have already phased-out due to non-compliance with the new CHED policies, standards and guidelines.

Out of the 1,706 private HEIs in the country, only 75 HEIs have autonomous and deregulated status. Majority of these HEIs are in NCR with 26 HEIs, (23 are autonomous and 3 are deregulated) which comprises 7.7 percent of the total number of private HEIs in the region. In terms of programs awarded as Centers of Excellence and Centers of Development during AY 2016-2017, majority are also in the NCR with 100 programs, or 50 percent, out of the 199 programs nationwide for Centers of Excellence, while 49 programs or 21 percent, out of the 233 programs nationwide for Centers of Development. However, these numbers are only 3.67 percent of the 4,064 programs offered in the region.

In terms of voluntary accreditation, only 862 programs or 21 percent of the total number of programs in the region is accredited by the different accrediting agencies.

In relation to the performance in the national licensure examinations, there have been some modest gains over the past few years as reported by the Professional Regulation Commission (PRC). The performance of our graduates and percentage passing in professional licensure examinations across all disciplines increased from 39.29 percent in 2011 to 58.59 percent in 2015 (first time takers), 35.92 percent in 2011 to 39.29 percent in 2015 (all takers). However, the performance of takers from SUCs yielded higher results than the national passing rate while the private HEIs have lower. This is problematic in the region since, 70 percent of the total enrolment is in the private HEIs.

Hard to reach deserving poor and disadvantaged students to avail student financial assistance programs. There are 274,721 students or 51.7 percent participation rate of underprivileged students in higher education in the region from the total enrolment. Out of the 531,376 student enrolment data gathered during Academic Year 2016-2017, only 72,189 students have availed of the different Student Financial Assistance Programs (StuFAPs) offered by the Commission. This is only about 26 percent of the participation rate of underprivileged students in the region.

Nonetheless, not all of the slots allotted for the region are availed particularly in the Tulong Dunong program due to several reasons such as lack of qualified applicants, failure to meet the documentary requirements and failure to meet the deadline of submission, among others.

Strategic Framework

To ensure human capital development under the pillar Pagbabago, there is a need to warrant care in all stages in life, health and nutrition from womb to tomb, lasting learning opportunities for all, continuing free basic education and improvement in higher and technical education in order to increase employability and productivity as well as global competitiveness.

Targets

Table 7.10 shows the indicators and corresponding targets for improving human capital development outcomes.

Table 7.10. Plan Targets to Accelerate Human Capital Development

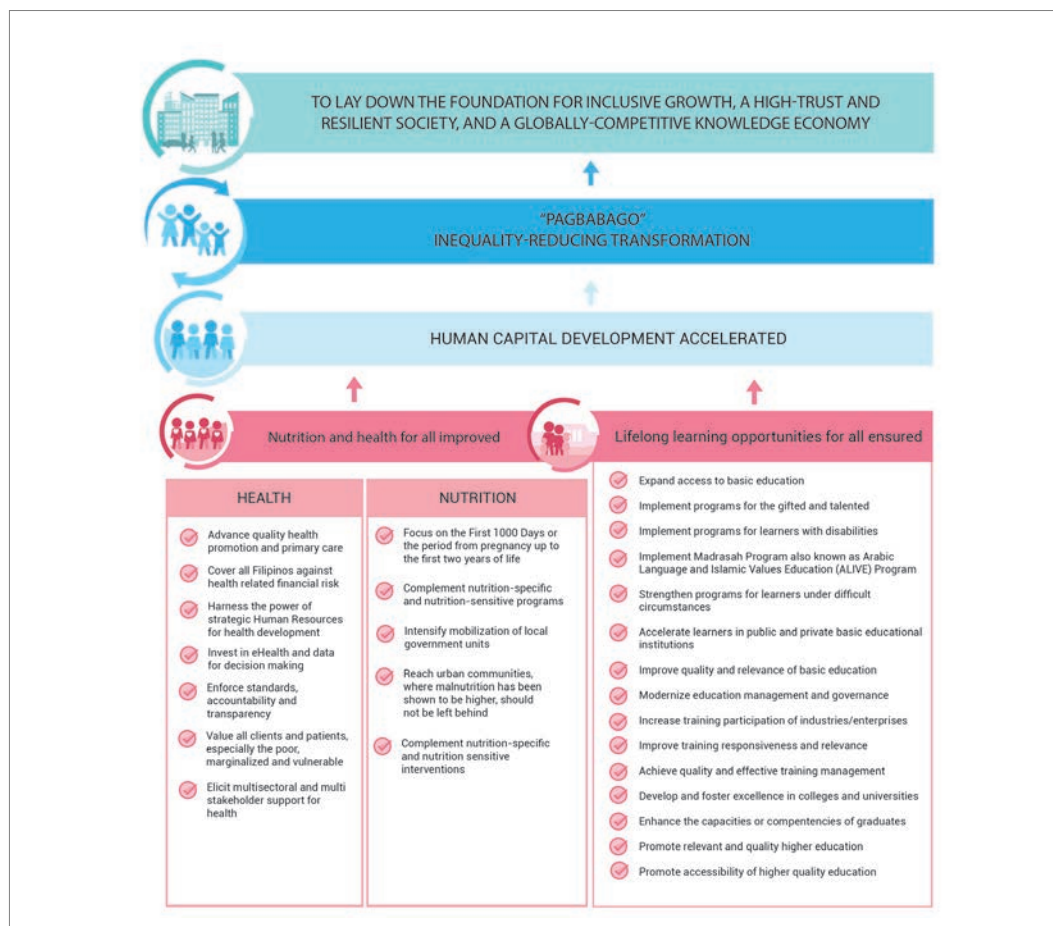
INDICATORS	BASELINE DATA YEAR	VALUE	END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
HEALTH AND NUTRITION				
Life Expectancy at birth, years	2010-2015	72.2	75 (2020-2025)	DOH
Male	2015	68.8	71.26	
Female	2015	75.6	77.54	
Maternal mortality ratio per 100,000 live births	2015	60.3	45	DOH
Neonatal mortality rate per 1,000 live births	2015	9.8	7.1	DOH
Infant mortality rate per 1,000 live births	2015	15.4	11.6	DOH
Under 5 mortality rate per 1,000 live births	2015	20.3	14.5	DOH
Premature mortality attributed to cardio vascular disease, cancer, diabetes, and chronic respiratory diseases (in rate)	2015	220.8	216.5	DOH
Tuberculosis incidence per 100,000 population	2015	175.7	145.5	DOH
Malaria prevalence per 100,000 population	2015	0	0	DOH
New HIV infections (newly diagnosed cases/year)	2015	<1	<1	DOH
Modern Contraceptive Prevalence Rate	2015	44.3	65	DOH
Proportion of households with diets that meet the energy requirements	2013	30.8	36.0	NNC
Prevalence of stunted children under five years old (%)	2015	24.9	15.9	NNC
Prevalence (in percent) of wasted children				NNC
Under five years old	2015	6.4	4.9	NNC
6 – 10 years old	2015	7.5	4.9	NNC
Vitamin A deficiency				
Prevalence (%) of children 6 months to 5 years old with vitamin A deficiency (low to deficient serum retinol)	2013	27.3	14.9	NNC
Anemia: Median urinary iodine concentration, mcg/L				
Children 6-12 years old	2013	220	≥100	NNC
Pregnant women	2013	157	≥150	NNC
Lactating women	2013	98	≥100	NNC

INDICATORS	BASELINE YEAR	DATA VALUE	END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
Percent with urinary iodine concentration <50mcg/L				
Children 6-12 years old	2013	8.9	8.8	NNC
Lactating women	2013	27.4	19.9	NNC
Prevalence (in percent) of overweight				
Under five years old	2015	6.0	5.0	NNC
6 – 10 years old	2013	17.9	8.5	NNC
Adolescents	2013	16.1	4.9	NNC
Adults	2013	37.7	28.0	NNC
Proportion of nutritionally-at-risk pregnant women	2015	20.8	16.8	NNC
Reduce the prevalence of low birthweight	2015	21.4	16.6	NNC
Increase the percentage of infants 5 month old who are exclusively breastfed	2015	31.8	40.3	NNC
Increase the percentage of children 6-23 months old meeting the minimum acceptable diet	2015	20.2	24.4	NNC
Increase the percentage of households with diets that meet the energy requirements	2015	30.8	36.0	NNC
Health manpower to population ratio within the standard (MD, nurses, midwife)				
Proportion of births attended by health professional	2015	94.7%	95	DOH
Proportion of births delivered in facility increased	2015	92.1%	95	DOH
BASIC EDUCATION				
Mean Years of Schooling				
Functional Literacy Rate				
Net enrolment rate				
Kindergarten				
Elementary	2015-16	88.113%	Increased	DepEd
Secondary	2015-16	75.49%	Increased	DepEd
Completion/cohort survival rate				
Kindergarten				
Elementary	2015-16	85.25%		
Secondary	2015-16	78.75%		
Drop-out rate				
Elementary	2015-16	1.75%	Decreased	DepEd
Secondary	2015-16	3.36%	Decreased	DepEd
Achievement rate				
Achievement rate in elementary				
Proportion of students moving towards mastered, closely approximating mastery, and mastered				
Proportion of students moving towards mastery				
Achievement rate in junior high school				
Proportion of students closely approximating mastery				
Proportion of students at low mastery				

INDICATORS	BEGINNING DATA YEAR	VALUE	END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
HIGHER EDUCATION				
Number of Higher Education Institutions (HEIs)	2015-16	337		CHED
Average Faculty-Student Ratio	2015-16	1:37*	1:25 (standard)	CHED
Number of higher education enrollees	2015-16	531,376		CHED
Number of higher education graduates				CHED
Licensure Examination Regional Passing Rate				PRC
TECHNICAL VOCATIONAL EDUCATION				
No. of TVET enrollees	2017	297,479		TESDA NCR
No. of TVET graduates	2017	257,424		TESDA NCR
No. of TVET graduates certified	2017	408,552		TESDA NCR
No. of TVET graduates employed	2017	159,473		TESDA NCR
Rate of TVET certification	2017			TESDA NCR
Rate of TVET employment				

Strategies

Figure 7.20. Strategic Framework to Accelerate Human Capital Development, 2017-2022





Health

Philippine Health Agenda Framework. Guided by the commitment to the Sustainable Development Goals and Ambisyon Natin 2040, the Department of Health formulated the Philippine Health Agenda (PHA) 2016-2022 adopting “All for Health towards Health for All” as the rallying point to realize the vision of a Healthy Philippines by 2022.

The PHA aims to 1) ensure the best health outcomes for all, without socio-economic, ethnic, gender and geographic disparities; 2) promote health and deliver health care through means that respect, value and empower clients and patients as they interact with the health system and 3) protect all families especially the poor, marginalized, and vulnerable against the high costs of healthcare.

The PHA provides for 3 Guarantees:

Guarantee 1: All Life Stages and Triple Burden of Disease – refers to population and individual level interventions for all life stages that promote health and wellness, prevent and treat the triple burden of disease, delay complications, facilitate rehabilitation and provide palliation.

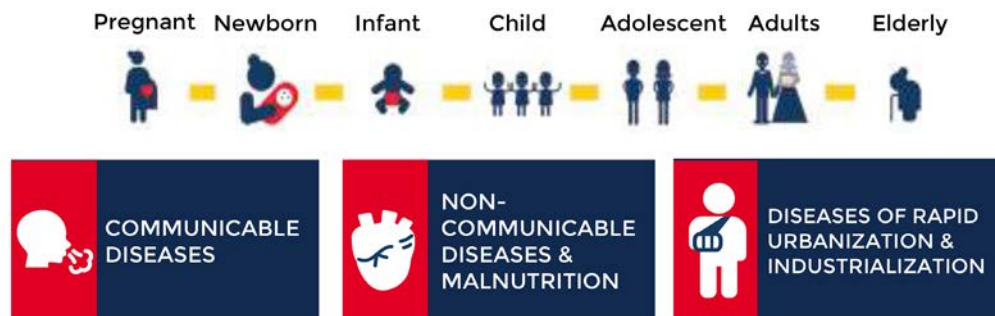
All life stages would cover services for pregnant women, children, adolescents, adults and older persons.

Triple burden of disease pertains to the communicable and tropical diseases that continue to be a burden 2) increasing challenge of non communicable diseases such as cancer, diabetes, heart disease, and their risk factors like obesity, smoking, poor diet, sedentary lifestyle, as well as malnutrition; and 3) problems related to globalization, urbanization and industrialization like injuries, substance use and abuse, mental illness, pandemics travel medicine and health consequences of climate change.

Guarantee 2: is access to health interventions through functional Service Delivery Networks (SDNs) that:

- Consist of primary care networks linked to Level 3 hospitals and specialty care
- Ensure well-equipped and fully staffed network of health facilities, either static or mobile, that are close to the people
- Render services that are compliant with clinical practice guidelines, available 24/7 and are resilient during disasters
- Practice gate keeping and utilize telemedicine to expand access to specialty services

Guarantee 1: All Life Stages & Triple Burden of Disease



Guarantee 3: is financial freedom when accessing these interventions through Universal Health Insurance such that the:

- National Health Insurance Program (NHIP) enrolls 100 percent of Filipinos where formal sector premiums are paid through payroll and non formal sector premiums are paid by tax subsidies
- NHIP's support value is 100 percent (or zero co-payment) for the poor and those admitted in basic accommodation; and a predictable (fixed co-payment for those admitted in non basic, private accommodation
- NHIP covers a comprehensive range of services and becomes the main revenue source of public health care providers

The strategies needed to fulfil the guarantee may be summarized by the word "ACHIEVE"

- Advance quality, health promotion and primary care
- Cover all Filipinos against health related financial risk
- Harness the power of strategic Human Resources for health development
- Invest in eHealth and data for decision making
- Enforce standards, accountability and transparency
- Value all clients and patients, especially the poor, marginalized and vulnerable
- Elicit multisectoral and multi stakeholders support for health

Nutrition

Focus on the First 1000 Days or the period from pregnancy up to the first two years of life. This is the period when health and nutritional care is optimum because this is the period when the body needs such the most. If this period is missed, all shortfalls in physical and mental development cannot be recovered. The irreversible effects of not intervening in this critical period will have an impact not only on the health status of the children but also on their future performance in schools and on productivity as adults.

Complement nutrition-specific and nutrition-sensitive programs. This strategic thrust recognizes that malnutrition has immediate, underlying, and basic causes, which should be addressed to achieve targeted nutritional outcomes. Thus, there is a need to implement and deliver nutrition-specific and sensitive interventions. Nutrition specific interventions are programs that address the immediate causes of malnutrition such as inadequate food intake and nutrient intake, care giving and parenting practices, and low burden of infectious diseases. Nutrition sensitive programs are non-nutrition programs that are tweaked to produce nutritional outcomes.

Intensify mobilization of local government units. There will be intensified mobilization of LGUs particularly on provinces with high levels of child stunting to ensure that PPAN programs are implemented and actively reduce malnutrition to targeted levels.

Reach the urban communities, where malnutrition has been shown to be higher, should not be left behind.

Complement nutrition-specific and nutrition sensitive interventions to address the immediate causes of malnutrition as well as its underlying and basic causes. As LGUs are charged with the delivery of services, including those related to nutrition, the national government is charged with creating an enabling environment through appropriate policies and continuous capacity building of various stakeholders.

Basic Education

Expand access to basic education. The Kindergarten Education Law through Republic Act 10157, made Kindergarten the compulsory and mandatory entry stage to basic education. This Act provides that all five (5)-year old children shall be given equal

opportunities for Kindergarten Education to effectively promote their physical, social, emotional and intellectual development, including values formation so they will be ready for school, since Kindergarten is the transition period from informal to formal literacy (Grades 1-12) and considering that age five (5) is within the critical years where positive experiences must be nurtured to ascertain school readiness.

Secondary Education as third stage of compulsory basic education consists of four (4) years of junior high school education and two (2) years of senior high school education with entrant age of twelve (12) and sixteen (16) years old, respectively.

Section 8 of the said Act includes the implementation of programs designed to address the physical, intellectual, psychosocial, and cultural needs of learners, which shall include, but shall not be limited to, the following:

Implement programs for the gifted and talented. These shall refer to comprehensive programs for the gifted and talented learners in all levels of basic education.



Implement programs for learners with disabilities. These shall refer to the comprehensive programs designed for learners with disabilities which may be home-, school-, center- or community-based.

Implement Madrasah Program also known as Arabic Language and Islamic Values Education (ALIVE) Program. This shall refer to the comprehensive program using the Madrasah curriculum prescribed by the DepEd, in coordination with the Commission on Muslim Filipinos, for Muslim learners in public and private schools.

Strengthen programs for Learners under Difficult Circumstances. This shall refer to the timely and responsive programs for learners under difficult circumstances, such as, but not limited to: geographic isolation; chronic illness; displacement due to armed conflict, urban resettlement, or disasters; child abuse and child labor practices.

Accelerate learners in public and private basic educational institutions. Acceleration of learners in public and private basic educational institutions shall be allowed, consistent with DepEd rules and regulations.

Improve quality and relevance of basic education. To improve the quality and relevance of basic education, the DepEd NCR as well as the 16 Schools Division Offices, continues to implement programs and projects that would help improve quality education, to with:

- Professional Learning Community or Learning Action Cell – this is a regular professional meeting of teachers, a venue where best practices or issues in the classroom are discussed, peer coaching/mentoring happen, new strategies are introduced, etc. all geared towards improving instruction and delivery; consequently, improving

student's performance.

- Continuous Improvement (CI) Program – is a result based intervention, basically it follows a step-by-step procedure in deciding how a particular classroom issue or challenge shall be addressed; hence it is informed decision making;
- Research and Development – similar with CI, it addresses classroom issues or challenges through informed decision making, however, this follows the usual research format while action research is most often done for long term results. This is often funded through the Basic Education Research Fund or the in some cases, the local government provides incentive for those who wish to undertake research which could be related to access, quality or governance of basic education.

Modernize education management and governance

Technical Vocational Education

The Regional Technical Education and Skills Development Plan of TESDA-NCR 2011-2016 (RTESDP 2011-2016), is the spatial dimension of the National Technical Education and Skills Development Plan 2011-2016. It essentially adopts and applies the objectives and strategies enunciated by the National Plan (NTESDP) 2011-2016. The variations basically lie on the magnitude clientele and specific policies and programs formulated and pursued to implement the strategies, as these are suited to the local conditions, TESDA capacity and social environment prevailing within the time period covered by the Plan.

The Plan envisioned supporting the Regional and National goals of inclusive growth through decent and productive growth via wage employment and support to entrepreneurship development of a Filipino workforce that is ready to take on the challenges of the 21st Century.

To realize the above, the Plan set five (5) objectives and twenty-six (26) strategies that would translate the NTESDP's three (3) major objectives of: 1) increase training participation; 2) improve training responsiveness and relevance; and 3) achieve effective training management.

There are basic quantitative measures which can be used to gauge the extent of Plan implementation. In the case of the TESDA-NCR, it adopted the following criteria for assessment covering the 2011-2016 Plan performance:

Increase training participation of industries / enterprises. With the TESDA-NCR's implementation of the 2011-2016 Plan strategies aimed at strengthening industry-government collaboration, the total enrolment output for Enterprise Based Training (EBT) in the past three (3) years, as against the output in 2011-2013, has generally increased.

Improve training responsiveness and relevance. Improving the responsiveness and relevance of TVET is basically done through the accurate analysis and interpretation of TVET relevant labor market information. Labor market information like expected future labor demand can certainly help improve the fit between the TVET graduates and available jobs. These measures are the employment rates and utilization rates.

Achieve quality and effective training management. For the TESDA-NCR to perform well, the TESDA Regional and District Offices must be strong internally in terms of personnel, systems and processes to effectively govern. The Regional Office and its six (6) District Offices pursued ISO Certification in 2013.

Higher Education

Develop and foster excellence in colleges and universities. The Commission has identified ways of fostering excellence in higher education such as 1) the harmonization of Key Academic programs with International Standards such as Engineering, IT and Maritime programs; 2) developing niche areas in food production and security, environment disaster risk reduction and response, climate change and energy, marine resources, biodiversity and conservation, smart analytics and engineering innovations and health systems to name a few; 3) establish research and innovation infrastructure in Philippine HEIs and maintain international partnerships such as the PCARI project; 4) increase the number of identified Centers of Excellence and Centers of Development for curricular programs and awarding of Autonomous and Deregulated status to deserving Higher Education Institutions; 5) identification of HEIs as college, professional institution and university through horizontal typology and to motivate schools to be the best it can be per classification; 6) entice schools to conduct Institutional Sustainability assessment to check its institutional soundness based on 5 key result areas (KRAS); 7) strive for full membership in the Washington and Seoul Accord. Continuous reforms in the maritime education and improve Global ranking of HEIs in the region.

Enhance the capacities or competencies of graduates. The Commission has issued several memoranda addressing this concern such as CMO 46 s. 2012 known as "Policy Standard to Enhance Quality assurance in Philippine Higher Education through an Outcome based and Typology based Quality Assurance and CMO 20 s. 2013 known as "General Education Curriculum: Holistic Understanding, Intellectual and Civic Competences".

Other program addressing this challenge is the K to 12 Transition program which includes: Scholarships for graduate studies, instruction, research and sectoral engagement grants and Institutional Development and Innovation grants.

Quality is further assured through the integration of the Philippine Qualifications Framework and its competencies and the ASEAN Qualification Reference Framework to the enhancement and implementation of OBE-Based Curriculum for Higher Education programs. Ensuring a higher SUC budget and the continuous policy on the closing and phase-out of substandard programs.

Promote relevant and quality higher education. In ensuring the promotion of relevant and quality higher education:

All higher education programs should be evaluated and monitored regularly to ensure compliance with the minimum standards contained in the policies, standards and guidelines for each program set by the Commission of Higher Education as recommended by its technical panel members from both the academe and industry. With the current development in the education system in the country, the policies, standards and guidelines for all curricular programs should be revised to become compliant with the outcomes based education and with the implementation of K to 12 program;

Priority programs in the higher education are set to ensure continuous supply of graduates for key industries and sectors. In the national level, these programs are clustered into Information Technology, Agriculture and related fields, Teacher Education, Science and Math, Engineering, Health Sciences, Arts and Humanities Social and Behavioral Sciences, Business Administration and related courses, Architecture, Maritime and Communication. In addition to these

programs, there are also programs which are identified as priority programs in the regional level.

Continuous promotion of quality assurance mechanisms are a) conducted thru granting of autonomous and deregulated status to HIEs with exemplary performance; b) provision of benefits and incentives with exemplary performance; provision of benefits and incentives to HEIs undergoing voluntary accreditation, and c) identification of Centers of Excellence and Centers of development in various degree programs.

Another developmental approach to quality assurance being implemented is thru Typology and Institutional Sustainability Assessment (ISA). Typology is the mechanism used by CHED in classifying HEIs in the country. Horizontal typology specifically aims to classify institutions into three categories whether it be a college, professional institution or university while vertical typology is the classification of HEIs according to the three elements of quality: 1) alignment and consistency of the learning environment with the institution's vision, mission and goals; 2) demonstration of exceptional learning and service outcomes; and 3) the development of a culture of quality. The first element is related to the horizontal type of HEI (college, professional institution or university) while the last two are related to level of program excellence (manifested thru accreditation, COE/COD and international certifications) and institutional quality (through institutional accreditation, Institutional Sustainability Assessment). These classifications aim to equalize opportunities and identify specific challenges and concerns unique within each institutional category. Proper classification will also help an HEI to focus on its mission-vision and goals and be able to realize it. Hence, quality assurance is no longer a one size fits all quality assurance that HEIs should all aspire to become autonomous or deregulated HEI.

ISA on the other hand, is quality assurance mechanism made available to schools for free by the government to assess the institutional sustainability of the institution in five key results area: 1) governance and management; 2) quality of teaching and learning; 3) quality of professional exposure, research and creative work/innovation; 4) support for students; and 5) relations with the community. ISA looks into the internal QA systems of an HEI and determines whether these systems lead to the attainment of the desired quality of outcomes.

The K to 12 Transition programs of the Commission is a five year program which commenced in Academic Year 2016-2017 to AY 2020-2021. It provides opportunity to improve the quality of higher education in the country by upgrading and updating the qualifications of teaching and non-teaching staff through scholarships and grants such as Scholarships for Graduate Studies (SGS) (both local and abroad), Professional Advancement and Post Doctoral Studies, Sectoral Engagement Grants, and Institutional Development and Innovation Grants (IDIG), and Discovery-Applied Research and Extension for Trans/Inter-disciplinary Opportunities (DARE TO).

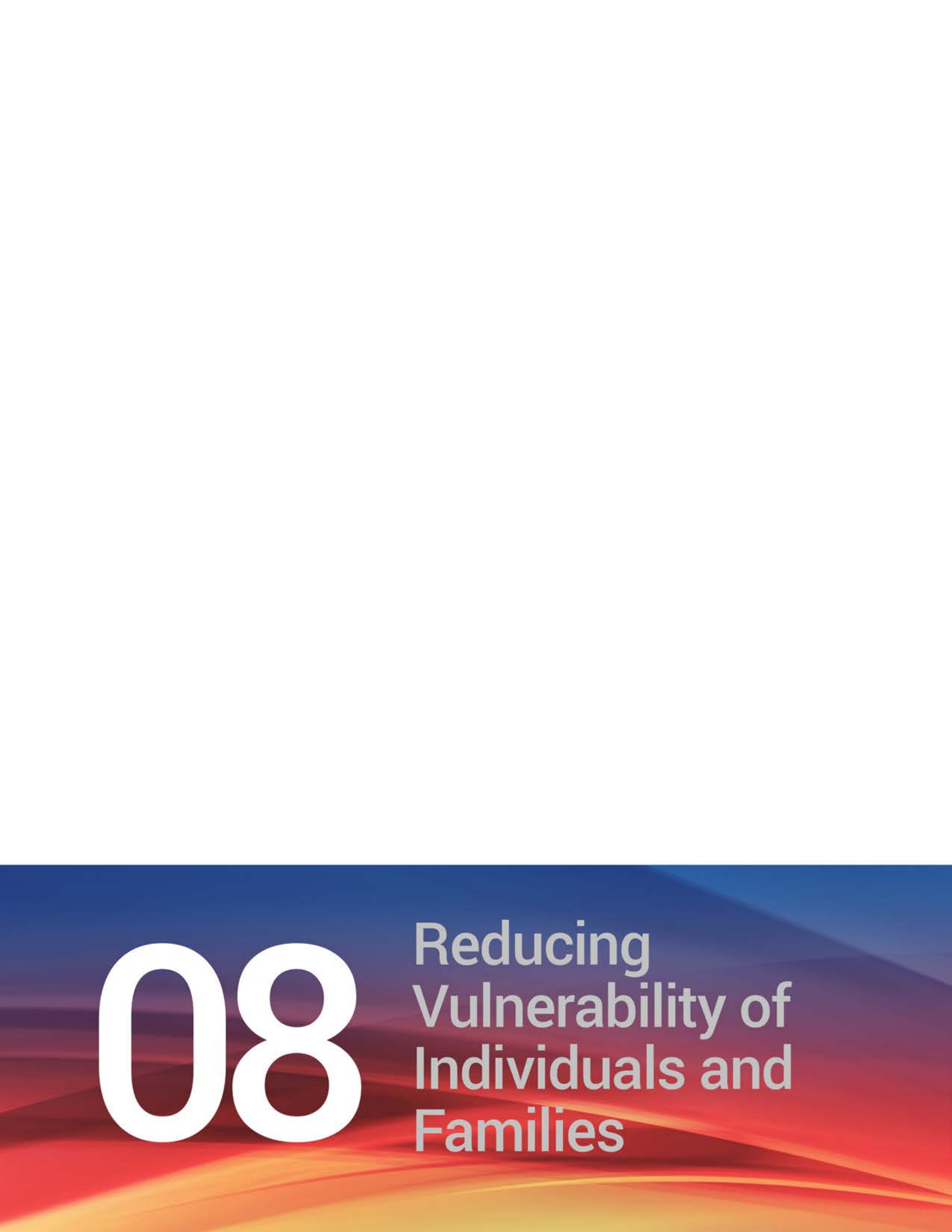
Both displaced and deloaded teaching and non-teaching personnel can avail of these grants during the transition period to K to 12 program having no college students to cater their services. At present, there are a total of 1,069 teaching and non-teaching

personnel of HEIs in NCR given various grants under the K to 12 transition program.

Promote accessibility of higher quality education. The Philippine government through CHED, is also exerting efforts to ensure that quality higher education is accessible to the poor but deserving students through its Student Financial Assistance Programs (StuFAPs) namely: Scholarships (merit-based); Students Loans; and Grants-in-Aid (need-based).

Other student financial assistance programs being implemented by the Commission include the Expanded Students' Grant-in-Aid Program for Poverty alleviation (ESGP-PA), the PAMANA Study Grant Program, Iskolar and Bayan, and scholarship grants for children for sugarcane workers and other small famers. Several landmark legislations were also passed into the law such as R.A. 10931 otherwise known as the Free Tuition Law for State Universities and Colleges which is being implemented beginning Academic Year 2017-2018 and R.A. 10687 otherwise known as Unified Student Financial Assistance System for Tertiary Education Act or UNIFAST which will be implemented beginning AY 2018-2019.

For NCR, the bulk of these grantees are enrolled in the programs offered by 16 Local Universities and Colleges (LUCs) and 8 State Universities and Colleges (SUCs). With the implementation of the Free Tuition and Other School Fees in the SUCs and LUCs, thousands of students will be benefited.



08

Reducing
Vulnerability of
Individuals and
Families

Reducing Vulnerability of Individuals and Families

Another main concern of Metro Manila, similar to the country as a whole, is the vulnerability of its individuals and families to different forms of risks given the fast and unpredictable changing needs in our environment. Risks are pervasive and are of different types which require different kinds of response.

Metro Manila as the Capital of the Philippines is also the 2nd most populous region among 18 Administrative Regions in the country having 12,877,253 or 12.75 percent of the total 100,981,437 population (2015 PSA Census of Population) which puts great constraint in the provision of socio-economic services and interventions in the metropolis. Vulnerability of people to Life Cycle Risks (hunger, illness, disability, etc.); Economic Risks (unemployment, low and irregular income, etc.); Environmental and Natural Risks (floods, fire, bombing, earthquakes, etc); and Social and Governance Risks (crime, corruption, political instability, etc) are prevalent.

The exposure to risks may be seen as one of the many dimensions of poverty. Individuals, households, and communities are exposed to multiple risks from different sources. The poor households are typically more exposed to risk and least protected from it. They have less access to risk management instruments that can allow them to deal and adapt with risks (SPVAM, DSWD 2014).

Vulnerability of people due to the social and/or economic conditions is prevalent in Metro Manila. This can be attributed to the high population density of the area which puts great pressure in the provision of socioeconomic services in the metropolis.

In addition to these conditions, it can often be threatened by the environmental vulnerability of Metro Manila. The country lies on the western rim of the Pacific as well as within the Pacific Ring of Fire¹, thereby subjecting it to natural hazards such as storms, typhoons, floods, earthquakes, volcanic eruptions, droughts, and other related calamities.

The Philippines placed 14th among more than 190 countries around the world that have suffered the most extreme weather events such as flooding and storms over the past 20 years.² With this, Metro Manila is affected by the said events that aggravate its vulnerability, as multiplied by factors such as population density and current living conditions.

Being highly vulnerable to these risks, the Metropolis must maximize its approach and responses to ensure collective and transformative social protection towards

¹ Pacific Ring of Fire is an area in the basin of the Pacific Ocean wherein earthquakes and tsunamis occur frequently.

² 2013 Germanwatch Global Climate Risk Index

resiliency of its people. Specifically, this chapter will discuss high vulnerabilities and resiliency of the region focusing on the four components of Social Protection (Social Welfare, Labor Market, Social Insurance and Social Safety Nets).

Assessment and Challenges

Exposure to Environmental Risks: Disasters (Natural and Human-Induced Hazards) and Climate Change Impacts

Metro Manila is greatly susceptible to disaster and climate change. The following are the major considerations of this issue:

High exposure to natural and human-induced disasters. Metro Manila is highly vulnerable to flood and earthquake. Flooding occurs due to several factors, while exposure to earthquake or ground-shaking is high as the metropolis is situated along the West Valley Fault. It also experiences severe wind and tropical cyclones since it is located along the Western Pacific Basin. An average of 20 typhoons enter the Philippine Area of Responsibility every year, of which four to six typhoons pass the vicinity of Metro Manila that bring flooding.

Flooding in Metro Manila, especially in low-lying areas, is caused by high intensity and/or long duration of rainfall or storm waters which result to surface run-off and overflowing of rivers, primarily because of the insufficiency of the modified natural channels, canals, and drainage system. Other major factors aggravating flooding are the present condition of the waterways and aquifers where there are obstructions and constrictions brought about by the proliferation of informal settlements, illegal structures, and indiscriminate dumping of garbage.

On the other hand, the country experiences an average of five earthquakes per day,³ in turn resulting to ground shaking, ground rupture, liquefaction and lateral spreading, landslides and tsunamis. Because of this complex tectonic situation, Luzon Island — in which Metro Manila is located — shows high seismic activity.

Aside from natural hazards, there are also human-induced disasters, such as fire, bombing, stampede/mobs, demolitions and the like, that pose great threat to a highly populated Metro Manila.

Given these hazards, there are potentially immense dangers in the region especially on damages to properties and infrastructure, loss of labor hours and income, cause of epidemics and widespread diseases and results in substantial loss of life. In addition, disasters disrupt economic activities which affect the overall growth and development in the region.

Issues on Disaster Preparedness, Prevention and Recovery. In the aftermath of the 2009 typhoons experienced in the country,⁴ different agencies and actors in the government drastically became active in promoting climate change and disaster risk reduction. Various efforts were made to reduce the region's risk and vulnerability to natural disasters through laws, policies and other government issuances. In order to enhance capacities to reduce and manage disaster risks as well as increase resilience against devastation and damages brought about by natural disasters and climate change, various projects and programs have

³Source: Philippine Institute of Volcanology and Seismology (PHILVOLCS)

⁴Namely typhoon Ondoy (international name: Ketsana) and Pepeng (international name: Parma)

also been undertaken by different agencies and stakeholders.

With certain disaster management efforts put into place, achieving positive results and impacts on the lives of the people remain to be a constant challenge.

Insufficient knowledge/awareness on disaster preparedness and mitigation. The Report of the United Nations International Strategy for Disaster Reduction (UNISDR) noted that the high number of casualties caused by Tropical Storm Washi (Typhoon Sendong) in 2011 could be due to lack of awareness of the risk and proper action of residents in affected areas, as they have not experienced floods of such magnitude in the past. It is concurred that informed people, through information dissemination and community disaster preparedness, would be properly prepared in times of disaster.⁵

In times of disasters and climate events, financial constraints and absence of information about occurrence of these events limit the effectivity of the initiatives and interventions available. The government should ensure that households have access to adequate and timely information.⁶

Massive Economic Damages. The aftermath of any disaster is never an easy one. There are different assistance and aids that the government extends especially for the victims directly affected by the disaster. At the national level, disasters greatly burden the government funds with an average of ₱15 billion in annual direct damages, which in turn negatively impacts the poverty reduction efforts.⁷

Moreover, the government spends huge amount of budget for rescue and relief operations whenever disaster strikes in

Metro Manila. On 26 September 2009, Metro Manila was hit by Tropical Storm Ondoy (International name Ketsana), which was considered as one of the worst recorded calamities in the region, that brought rains equivalent to 600 millimeters in just a matter of 12 hours. With this, Metro Manila incurred serious damages and loses which affected 123,221 persons and ₱1.3 million⁸ total cost of infrastructure damages including school buildings and health facilities.

In 2014, the estimated cost of damaged properties during the onslaught of Super Typhoon Glenda in Metro Manila reached ₱100.6 million and affected 8,817 families. Another major calamity which affected Metro Manila was the 2012 August Habagat disaster wherein the total damages amounted to almost ₱1 billion (US\$ 23.8 million).⁹

Inadequate Evacuation Centers and/or Disaster-Proof Areas and Structures/Facilities/Equipment. Lack of identified evacuation centers is the main issue in Metro Manila. In practice, when calamities strike, public school buildings and covered courts are being utilized to shelter the affected families in an extended period of time. This practice is being discouraged by experts and authorities as reflected in DEPED-DSWD-DILG-DOH Joint Circular No. 1, since these structures are designed only for school activities and not for temporary shelter. There is also insufficient supply of basic utilities, such as water, electricity, and sanitation facilities for evacuees as they are not designed for this purpose. This practice significantly disrupts the schedule of classes of schools.

Need for Quick Response for Rehabilitation and Recovery. As observed from the major calamities previously experienced by the country such as during Ondoy, Yolanda, and Sendong, government

⁵ UNISDR (2011) Philippines tragedy underlines need for improved risk management. <https://www.unisdr.org/archive/24261>

⁶ Danilo Israel and Roehlano Briones (2014). Disaster, Poverty, and Coping Strategies: The Framework and Empirical Evidence

⁷ Official Gazette. Disaster Preparedness. <http://www.gov.ph/aquino-administration/disaster-preparedness/>

⁸ http://nap.psa.gov.ph/headlines/statsspeak/2009/110909_rav_mrsr_typhoons.asp

⁹ World Bank, 2013. The Socio-Economic costs of Informal Settlements

response on disaster response, rehabilitation and recovery has been considered lacking and unhurried. There were instances that a week after the calamity, many communities remained inaccessible due to road blockages and had not received the required assistance.

Complexity of the Institutional Arrangement of Disaster-Related Agencies. The Metro Manila Disaster Risk Reduction and Management Council (MMDRRMC) is the main coordinating body that manages the operations before, during, and after disasters in Metro Manila. However, because of the impact of previous extreme weather events that impacted the region, there was a rise in the response of various individuals, government, and other organizations. At the community level, it was seen that adaptation or risk reduction was hindered by the insufficient collaboration among various stakeholders. Thus, there is a need to have a concerted effort of ongoing and future endeavors when it comes to disasters to be able to achieve optimum coordination, monitoring, response and reporting.

In addition, there are agencies which provide services in relation to disasters. For instance, MMDA and LGUs provide immediate clean-up, rescue, relief and rehabilitation operations in affected areas in Metro Manila in times of disaster. Coordination and proper communication with other agencies must be in place for a more effective and efficient provision of assistance during disasters.

Need to Mainstream Disaster Risk Reduction and Climate Change Adaptation (DRR/CCA) in the Comprehensive Land Use Plans (CLUPs) and Zoning Ordinances (ZOs). The CLUP is an effective instrument in addressing

DRR and CCA concerns. As indicated in the HLURB Supplemental Guidelines, land use planning is a cost-effective and proactive approach in managing current and future risks considering the high costs of structural measures to address unplanned spatial development. It serves as a framework to guide in the preparation of various local level plans to implement DRR and CCA.¹⁰

As a major thrust in mitigating the effect of disaster and other calamities, the government enacted Republic Act 10121 series of 2010 or the Philippine Disaster Risk Reduction and Management Act. The aim is to streamline the Disaster Risk Reduction/Climate Change Act (DRR/CCA) in the formulation and updating of the Comprehensive Land Use Plans (CLUPs) of the 17 LGUs in Metro Manila. Of the 17 LGUs, only six (6) cities like Caloocan, Las Piñas, Makati, Pasay, Pasig, and Quezon City have updated their CLUPs as of 2017.

The HLURB led the initiative to disaster proof the CLUPs of LGUs by integrating DRR and CCA approaches in the plan.¹¹ It was recognized by the HLURB as crucial that LGUs prepare climate and disaster risk-sensitive CLUPs/ZOs that control the allocation of land use to minimize or prevent the exposure and vulnerability of population, infrastructure, economic activities and the environment.

It is expected that through anticipatory interventions the population would be safer, the economy more resilient, and basic services and infrastructure more vigorous. In the process, substantially reducing resources for disaster response and post disaster recovery and rehabilitation.

Disasters also entail large amount of resources for structural repairs to

¹⁰ Housing and Land Use Regulatory Board. (2015) Supplemental Guidelines on Mainstreaming Climate Change and Disaster Risks in the Comprehensive Land Use Plan. http://hlurb.gov.ph/wp-content/uploads/services/lgu/CLUP/HLURB_Supplemental_Guidelines.pdf

¹¹ As indicated in HLURB's on Mainstreaming Climate Change and Disaster Risks in the Comprehensive Land Use Plan

buildings and recovery or replacement of damaged equipment needed to restore daily operations. These impacts can range from short and even medium to long term effects. Regrettably, the disaster risk reduction and management framework of government has not been effectively translated into sectoral or business plans. Thus, they are highly vulnerable and have weak adaptability.¹²

Exposure to Social and Governance Risks

Individuals in various facets of society in Metro Manila have inherent vulnerabilities, who are even placed in a worse position due to several risk factors, such as social, economic and environmental.

There are sectors in the society that are inherently vulnerable to social risks: those in poverty, children, women, persons with disabilities, indigenous peoples, and older persons.

Poverty remains as a major concern. Metro Manila has the highest annual per capita poverty and food thresholds among all regions in the country due to high cost of living. It continues to increase which correlates to the increase of poverty and subsistence incidence. NCR's Annual Per Capita Poverty Threshold¹³ registered from ₱20,344.00 (2012) to ₱25,007.00 (2015), which is higher than the national

poverty threshold of ₱21,753.00 (2015). While the NCR's Annual Per Capita Food Threshold¹⁴ in CY 2015 is ₱17,462.00 which is higher than the national food threshold of ₱15,189.00. This is higher than the threshold of ₱14,206.00 (2012) and ₱13,426.00 (2009).¹⁵

The NCR data for poverty and subsistence incidence¹⁶ is lowest in the country. In terms of the regional poverty incidence among families, it registers 2.7 in 2015, as compared to the national poverty incidence of 19.7 in 2015. As regards to subsistence incidence,¹⁷ there is no significant increase from years 2012 to 2015 as compared to 0.1 percent increase between the years 2009 and 2012.

Most Vulnerable Groups

Children.¹⁸ They are inherently vulnerable as they are exposed to different risks and are highly dependent on others to survive. Those identified at risk are children in need of special protection (CNSP)¹⁹ or children at risk (CAR), such as abandoned, neglected, street children, children in conflict with the law, sexually-exploited, among others.

In 2012, the total number of reported Children in Need of Special Protection from the 13 LGUs is 4,262 which is lower than the previous year. Majority of the CNSP cases were street children followed by voluntary surrendered and abandoned children.

¹² Marife Ballesteros and Sonny Domingo. Philippine Institute for Development Studies. Building Philippine SMEs Resilience to Natural Disasters. Discussion Paper Series No. 2015-20 (Revised). http://dirp3.pids.gov.ph/webportal/CDN/PUBLICATIONS/pidsdps1520_rev.pdf

¹³ Poverty Threshold or poverty line is the minimum level of income required by an individual to meet the basic non-food needs such as clothing, housing, transportation, health, and education expenses, among others.

¹⁴ Food threshold is the minimum income required by an individual to meet the basic food needs and satisfy the nutritional requirements set by the Food and Nutrition Research Institute (FNRI), while remaining economically and socially productive.

¹⁵ Source: CY2016 Regional SPDR of NCR from DSWD

¹⁶ Poverty Incidence is the proportion of families/individuals with per capita income/expenditure less than the per capita poverty threshold to the total number of families/individuals.

¹⁷ Subsistence Incidence is the proportion of families/individuals with per capita income/expenditure less than the per capita food threshold to the total number of families/ individuals.

¹⁸ A Child is a person below 18 years of age, or a person over 18 years of age but is unable to fully care for himself/herself or protect himself/herself from abuse, neglect, cruelty, exploitation or discrimination because of a physical or mental disability or condition. – RA 9523

¹⁹ DSWD AO 24 series of 2011: Guidelines on Adopting the Use of Modified Social Stress Model (MSSM) as a Tool in Managing Children in Need of Special Protection (CNSP)

These poor children experience hunger that they seek refuge in the streets to beg for food or money to survive a day. Most likely, they become victims of violence given their powerlessness if not used for crimes by unscrupulous adults and syndicates.

The condition of the street children in Metro Manila is one of the phenomenal situations in any developing urban areas not only in the Philippines but also in other countries. However, the description and perception on street children vary among different countries and researchers on the said social issues. Department of Social Welfare and Development (DSWD) Administrative Order No. 08, series of 2009 or the Standards for Community Based Service for Street Children defines street children as those children who spend at least four (4) hours in the streets or public places and engage in any street activities such as begging, scavenging, vending etc.

Based on the 2001 Research Study of the United Nations International Children's Emergency Fund (UNICEF) and the National Protection for Street Children, there are 246,011 street children in the Philippines though only 45,000 to 50,000 were highly visible on street of major cities. The study claims that street children is estimated to be three percent (3%) of the population aged between 0-17 years old. Of these number, 20 percent are estimated to be highly visible. Around 11,346 streetchildren are in Metro Manila. Four (4) LGUs were identified with the most number of highly visible street children, Manila with 3,266 children followed by Quezon City with 2,867 children, Caloocan with 1,530 children and Pasay City with 1,420 children.

In November 2010, the DSWD-NCR in collaboration with 17 LGUs and Civil Society Organizations (CSOs) conducted a rapid appraisal to determine the number of street families and street children in Metro Manila covering the areas in Metro North,

Metro South, Metro East, Quezon City, and Manila. The study showed that there are approximately 5,086 people living in the streets. Of this number, 3,072 are children and 2,014 are unattached adults. The study further estimated a total number of 657 families in the streets.

Further, based on the results of Reach-Out operations in 17 LGUs from January to September 2016, a total of 4,445 children at risk were reached out, wherein 3,302 are children on the streets (community-based) and 1,143 are children of the street (street-based). The LGU of Manila has the most number of reached out cases followed by Quezon City.

Women. There are still some who view women as weak and with limited capacity and they were treated as such in all aspects of life.

Attaining equality between women and men and eliminating all forms of discrimination against women are fundamental human rights and United Nations values. Women around the world nevertheless regularly suffer violations of their human rights throughout their lives, and realizing women's human rights has not always been a priority. Achieving equality between women and men requires a comprehensive understanding of the ways in which women experience discrimination and are denied equality so as to develop appropriate strategies to eliminate such discrimination.

The United Nations has a long history of addressing women's human rights and much progress has been made in securing women's rights across the world in recent decades. However, important gaps remain and women's realities are constantly changing, with new manifestations of discrimination against them regularly emerging. Some groups of women face additional forms of discrimination based on their age, ethnicity, nationality, religion,

health status, marital status, education, disability and socioeconomic status, among other grounds. These intersecting forms of discrimination must be taken into account when developing measures and responses to combat discrimination against women.

In terms of violence against women, the Department of Interior and Local Government (DILG)-NCR in 2013 reported that the number of violence against women (VAW) cases in the Philippines as reported by the Philippine National Police (PNP) increases by 49.4 percent from 2012.²⁰ In NCR, there are a total of 819 VAW cases in 2013 based on the report from DILG. Further, the number of VAW cases served in DSWD NCR Center and Residential Care Facilities increased from 397 in 2013 to 657 in 2016.

Persons with Disability (PWDs). Disability is closely linked to poverty as one may cause the other. In 2010, the PSA reported that NCR is the 2nd among the 17 regions with highest number of PWDs having 67,000. In 2015, the DSWD-NCR conducted the 2nd round of Listahanan assessment wherein 74,028 individuals (poor and non-poor) claimed that they have disabilities based on the Proxy Means Test (PMT) result.

There have been laws enacted and services set to protect the PWDs. However, the efforts of the government to help improve the lives of the said sector remained far from being realized. Varied issues experienced by the sector includes the lack of opportunities for employment, lack of livelihood projects limited only for PWD, discrimination in employment and in school, non-compliance of some public and private establishments to Batas Pambansa 344 or An Act to Enhance the Mobility of Disabled Persons by Requiring Certain Buildings, Institutions, Establishments and Public Utilities to Install Facilities and Other Devices or most commonly known as the “Accessibility

Law,” and unavailability of comprehensive, accurate and reliable data of PWDs.

Indigenous People (IPs). There are instances that IPs make their way to Metro Manila in the hopes of finding economic opportunities and more income in the city. In some cases, they were those found as vagrants on the streets since infiltrating the work force and housing settlements are more difficult in Metro Manila.

Result of the DSWD-NCR Listahanan 2 data noted that there are about 438 Poor households who claimed that they belong or who have lineage of Indigenous people. Out of the 17 LGUs, the following are the top 5 with the highest turnouts: Manila (207), Quezon City (78) Caloocan (47), Taguig (39) and Pasig (19).

Older Persons. They are particularly vulnerable since they are physically more fragile and have limited income sources. NCR senior citizens population constitute of 13.08 percent or 999,600 of the total national population of senior citizens. Out of 999,600, 13.32 percent or 454,800 are male senior citizens and 12.89 percent or 544,800 are female senior citizens.²¹ Senior Citizens (SCs) population contributed 7.76 percent or 999,600 to the total 12,877,253 household population as recorded during the Census of Population in year 2015.

SCs are particularly vulnerable since they are physically more fragile and have limited income resources. Their opportunities to participate in the nation building activities are also at risk due to their condition hence, may limit their access to services.

During the assessment of the Philippine Plan of Action for Senior Citizens (PPASC) 2012-2016, among the challenges, issues and concerns identified were: absence of national sectoral agency which will monitor

²⁰ <http://www.pcw.gov.ph/statistics/201405/statistics-violence-against-filipino-women>

²¹ Source: CY 2016 Regional SPDR of NCR from DSWD

the implementation of PPASC; lack of Regional Inter-Agency Committee for Senior Citizens (RIAC-SC) Functionality; lack of representation of senior citizens during sectoral plan formulation in the city and barangay level; there is no common and centralized template used for data banking of SC; program used for databanking varies in different LGUs resulting to non-uniformity, duplication of services and inaccurate and bloated number of SC; no clear mechanism in issuance, replacement and cancellation of Office of Senior Citizens Affairs (OSCA) ID, no permanent positions/limited number of OSCA personnel with plantilla position; and proliferation of privately run (business oriented) group homes/residential care facilities for unattended, neglected, abandoned and homeless senior citizens which has no regulation from any agency.

Social Protection (SP) mechanisms were put in place but these remain inadequate.

Programs like conditional cash transfers, social security systems, and income and livelihood support have been existent but have limited coverage and benefits.

Social Protection (SP) constitutes policies and programs that seek to reduce poverty and vulnerability to risks and enhance the social status and rights of the marginalized by promoting and protecting livelihood and employment, protecting against hazards and sudden loss of income, and improving people's capacity to manage risks." (National Economic and Development Authority - Social Development Committee Resolution No. 1, Series of 2007).

This definition harmonized existing SP definitions and clearly identified the four (4) SP components: Labor Market Intervention, Social Insurance, Social Welfare and Social Safety Nets wherein priority strategies in the PDP 2011-2016 are also anchored with. These include strategic interventions that promote not just growth but inclusive growth.

The government has existing SP mechanisms for vulnerable sectors but the coverage and benefits are limited.

All 17 LGUs were trained and provided with the developed Social Protection Handbook and the Social Protection - Vulnerability and Adaptation Manual (SP-VAM), a useful tool for providing a social protection lens in the regular course of planning, budgeting, implementing, monitoring and evaluating inclusive interventions for the underprivileged and to assist these localities in mainstreaming social protection in local development planning.

However, programs like CCT, social insurance, social pension, among others are not yet sufficient due to its limited coverage, benefits and insufficient funding.

Exposure to Economic Risks

There is also a need to focus on identified sectors which face economic risks, namely Overseas Filipinos and their families, workers in the informal sector, workers in the formal sector, and female workers.

The underlying factors that affect the risks for the said sectors are the limitations on employment opportunity due to biases against women, elderly, displaced and distressed workers, and workers with special concerns, as well as the incidence of non-compliance to general labor standards, social welfare protection, occupational safety and health standards and potential loss of decent and productive employment due to illegitimate contractualization.

Overseas Filipinos and their families: Effects of Migration in the Labor Market. Issues on job and skills mismatch, inadequate employment opportunities and low salary have consistently been identified as the push factors for Filipinos to leave and

work overseas. Records show that NCR is the 2nd top regional source of overseas Filipino workers in 2016. It can also be noted that medical workers specifically nursing professionals, as well as skilled and semi-skilled workers in manufacturing and civil works are part of the region's top skill categories deployment.

These are the skills which are critical in promoting growth and development considering the government's strategy of massive infrastructure development as well as rehabilitation efforts. These big construction projects for which increased government expenditure has been allocated for the next 5 years will result to increased demand for construction related manufactures and transport equipment, among others.

The overseas out-migration of professionals and skilled technical workers resulting to non-availability of qualified manpower supply in the domestic labor market could well be a major factor which can affect the timely implementation of civil works and other important infrastructure projects deemed critical in resolving the gigantic problems of the metropolis such as lack of housing, and worsening traffic situation. The Build, Build, Build strategy is also seen as requirement for poverty alleviation with the end goal of inclusive growth.

On the other hand, just like the national picture, the household service worker category is the top skill category deployed from NCR comprising almost 30 percent of total workers deployed.

This sector which is composed of more than 90 percent women are deemed vulnerable in view of the nature of the work environment and in many instances, the host countries' lack of legal framework for their protection from abuse. Most of these workers are college graduates and

usually forced to work abroad in foreign households due to lack of employment opportunities and low salary in the country. Despite government efforts to discourage them from leaving, Filipino household service workers, caregivers and nannies have always been in great demand overseas due to their endearing qualities/ qualifications such as industriousness, trustworthiness, good education, facility in the English language, adaptability and caring attitude. This is coupled with the availability of working age women who find it difficult to find employment locally thus, are willing to forget their education/ qualifications for an overseas job.

Because of the prevailing vulnerabilities and lack of adequate legal mechanisms to protect them in the jobsite, thus resulting to cases of maltreatment, exploitation and abuse of workers' rights and welfare, there is need to provide domestic workers employment opportunities locally where they can utilize their education and skills and in the process be able to have the option to choose not to leave the country. Where they will not be subjected to abusive situations.

Workers in the Informal Sector. Central to achieving the outcome of accessibility of social protection and safe work for all is the implementation of social protection interventions to reduce vulnerabilities of informal workers and their families against risks, such as loss of income, unemployment, and sickness, particularly during times of crisis. Philippine Statistics Authority (PSA) defines the informal sector to include unincorporated household enterprises, informal own-account enterprises, and enterprises of informal employment basically owned and operated by own-account workers, either individually or in partnership with members of the same or other households.²²

²² Source: Department of Labor and Employment news <https://www.dole.gov.ph/news/view/2113a>

Workers in the formal sector. One main issue for workers in the formal sector is contractualization or labor-only contracting, outsourcing work due to a strike, and the so-called “555” or “endo.” “Endo” is the practice of repeated hiring of workers on five-month contracts so employers don’t need to regularize them on the sixth month.²³

Female workers. Women in the work force have more vulnerabilities than men in terms of pregnancy and maternity, aside from other aspects such as gender segregation and bias, violence, and harassment. It is necessary to enable an environment for women in the workforce in facilitation of the entry of women workers and promoting work-life balance.

Strategic Framework

Following the Framework from the PDP 2017-2022, this RDP chapter will contribute in achieving universal and transformative social protection, focusing on the strategies on mitigating environmental, social and governance risks as well as economic risks (Figure 8.1).

Targets

Table 8.1 shows the plan indicators and corresponding targets. It aims to reduce each type of risk and enable people to manage individual, economic and environmental risks.

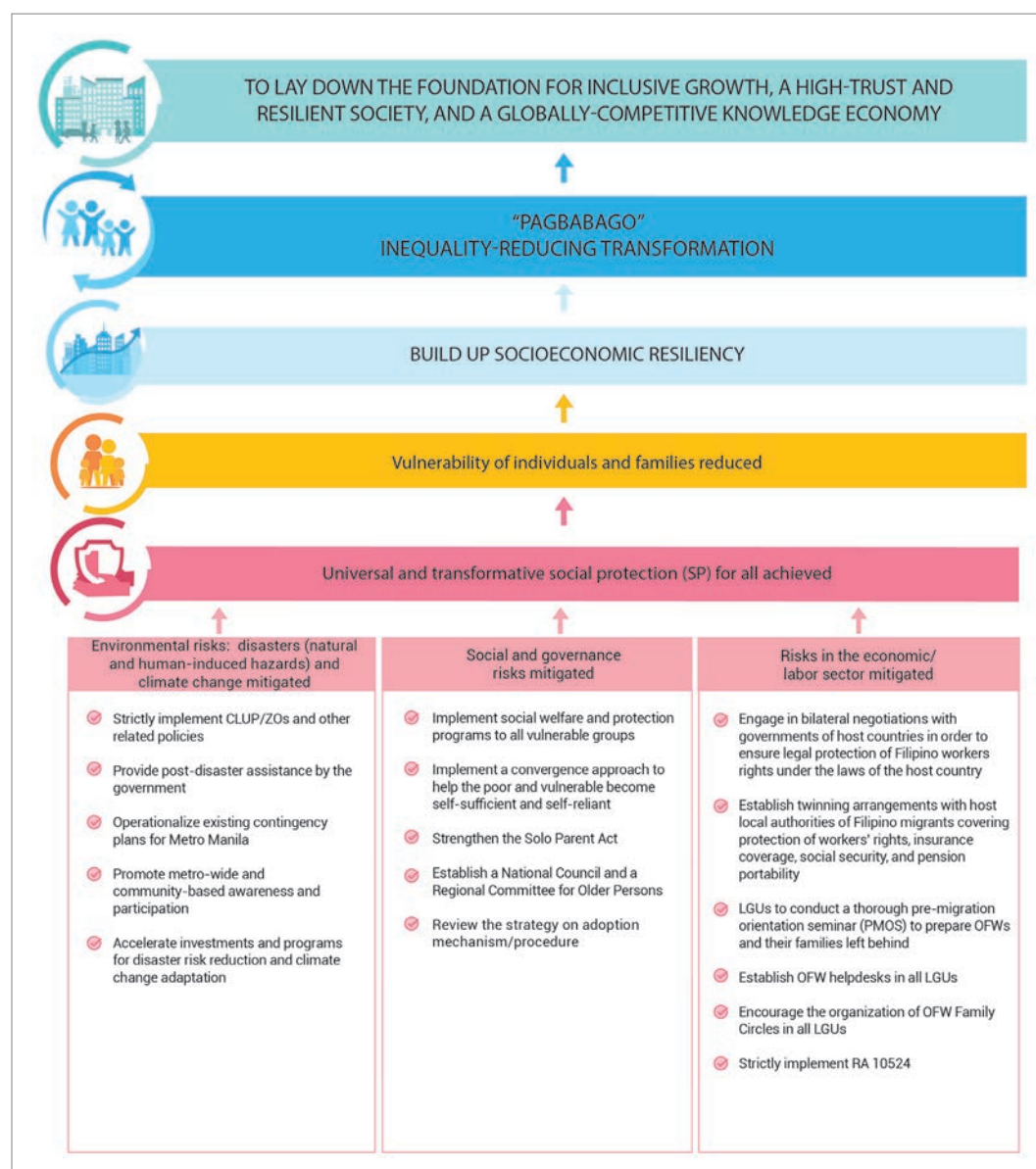
Table 8.1. Plan Targets to Reduce Vulnerability of Individuals and Families

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
ENVIRONMENTAL AND NATURAL HAZARDS				
Proportion of families affected by natural and human-induced calamities provided with relief assistance (%)				
Emergency shelter assistance provided increased				
Number of deaths due to disaster per 100,000 population				
Number of missing persons due to disaster per 100,000 population decreased				
SOCIAL AND GOVERNANCE RISKS				
Reduction of Extreme Poverty				
Poverty incidence among families (%)	2015	2.7		
Poverty incidence among population (%)	2015	3.9		
Subsistence incidence among families (%)	2015	0.4		
Subsistence incidence among population (%)	2015	0.7		

²³ Source: cnnphilippines.com/news/2017/07/04/DOLE-admits-failure-to-end-contractualization.html

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Improving Access to Social Protection Programs				
Percentage of population covered by PhilHealth insurance				
PhilHealth coverage of poor households (%)				
PhilHealth coverage of OFWs				
Social pension coverage of indigent senior citizens (%)				
Improving Access to Basic Social Services				
Increase in number of poor households covered by CCT				
Proportion of poor households covered by 4Ps (%)				
Disadvantaged groups with protective services:				
Children in need of special protection				
Women in difficult circumstances				
Older persons				
PWDs				
Victims of illegal recruitment assisted				
Victims of human trafficking assisted				
Number of scholarship provided under various programs:				
Students Grants-in-Aid Program for Poverty Alleviation				
IPs				
OFWs				
Number of Beneficiaries trained under GAD-related programs				
Increasing Opportunities for Livelihood and Microfinance				
Number of beneficiaries provided with livelihood assistance:				
Sustainable Livelihood Program (SLP)/4Ps				
DOLE Integrated Livelihood and Emergency Employment Program (DILEEP)				
Workers affected by economic crisis and natural/man-made disasters, assisted thru emergency employment program				
Youth provided with bridging employment (SPES and GIP)				
ECONOMIC RISK				
No. of children aged 5-17 years engaged in child labor				
No. of children aged 5-17 years engaged in child labor and working hazardous environment				
Percentage of SSS regular members over the total number of workers employed by private sector				
Percentage of employed in the government covered by GSIS				
Percentage of OFW membership to OWWA over total documented OFWs increased				
Percentage of OFWs covered by social security increased				

Figure 8.1. Strategic Framework to Reduce Vulnerability of Individuals and Families, 2017-2022



Strategies

To mitigate environmental risks: disasters (natural and human-induced) and climate-change impacts

Strictly implement CLUP/ZOs and other related policies. The risk-sensitive or responsive land use guidelines and zoning ordinances must be strictly adopted and enforced by each LGU to successfully mitigate future adverse impacts from natural hazards. The government must also enforce seismic building codes to be followed by establishments and developments. Given the high risk of earthquakes, with particular risk along well-identified seismic faults in Metro Manila, it is imperative to follow seismic building codes.

Provide post-disaster assistance by the government.

Post-disaster response and strengthening inter-agency linkages of LGUs and MMDRRMC with other CC/DRRM agencies. The government from various levels must strengthen their disaster preparation and response activities with other agencies and organizations. Under Republic Act 10121 or the Philippine Disaster Risk Reduction and Management Act of 2010, the LGUs are the first responders in the event of natural and man-made calamities. They are in-charge of rescue operations, provision of immediate relief assistance, and management of evacuation centers. Other agencies are tasked to give augmentation, such as the DSWD which also provides the needed basic services through distribution of family food packs and standby funds.

For Metro Manila, the Metro Manila Disaster Risk Reduction and Management Council (MMDRRMC) will carry out coordination, integration, supervision, monitoring, and evaluation functions. The national

government, through the National Disaster Risk Reduction and Management Council (NDRRMC) will also provide assistance as needed and given the magnitude of the effect of the calamity.

Financial assistance. The financial assistance agencies such as the Home Development Mutual Fund or the Pag-IBIG Fund, Government Service Insurance System (GSIS), and Social Security System (SSS) also offer Calamity Loans to those affected by disasters. In addition, the establishment of the People's Survival Fund, as indicated in Republic Act 10174, aims to provide long term financing to enable the government to address climate change.

In addition, there is also a need to address the existing limitation in the current system of disaster risk financing. Usually, budgetary sources do not represent proactive efforts to mitigate the expected damages and risk from natural disasters. It merely reflects post disaster relief and rehabilitation expenditures. As provided in RA 10121, budget should also reflect disaster reduction strategies.

Psycho-social and mental health intervention. Post-disaster psychological assistance to those directly and indirectly affected is the responsibility of DSWD and DOH. The aim is to restore them to a psychologically sound, safe and secure state to return to normal functions after the disaster.

PWD, women, and child-friendly spaces in evacuation centers. The establishment of PWD, women and child friendly spaces is also integrated in evacuation camp which aims to respond to the effects of displacements and vulnerability of women and children in natural and man-made situations.

Livelihood assistance / emergency employment. For those vulnerable workers and their families affected by disasters, the Department of Labor and Employment

(DOLE) also provides immediate livelihood support through emergency employment program. This enables vulnerable households which have lost their livelihood to endure the critical period through wages and social protection benefits in order to minimize dependency on relief aid and promote self-recovery of those affected.

Operationalize existing contingency plans for Metro Manila. The Oplan Metro Yakal Plus has been a step-up of the previous contingency plan of the MMDRRC on the event of a 7.2 magnitude earthquake in NCR. It details the protocols and procedures that would be activated right after the earthquake.

In addition, MMDA has the Metro Manila Inclement Weather Emergency Preparedness and Response Plan or “Oplan Unos”. It is designed to respond to life- and property-threatening situations caused by storms and disasters, and specifically addresses flood mitigation; traffic management; search, rescue and evacuation; debris removal and road clearing; hazard abatement; emergency transport and heavy equipment; medical assistance, relief and sanitation; and emergency engineering and construction.

Moreover, RA 10121 also states that there must be a comprehensive Local Disaster Risk Reduction and Management Plan (LDRRMP) for each localities to integrate disaster risk reduction efforts in the national, regional and provincial framework.

Promote metro-wide and community-based awareness and participation. All sectors of the metropolis must be aware and educated on the possible occurrence of disasters including actions that must be done during and after said event. It is vital to exercise knowledge, innovation, and

education to build an environment of safety and resilience at all levels. The key to disaster reduction is knowledge and understanding of vulnerability measures coupled with commitment to act.²⁴ Empowering communities to reduce their own risks is an important feature of any meaningful risk reduction program. Community-based and participatory approaches that engage communities in decision-making, implementation, and monitoring ensure the alignment of investments and promote accountability and transparency.

A major component to strengthen resiliency of the community and the people in Metro Manila would require the cooperation of all stakeholders, both in the government sector, the private sector as well as in the community. Considering that the people themselves bears the major brunt of disasters, it is essential that they remain always aware, vigilant, organized and prepared to cope with any eventualities.

In the spirit of volunteerism and community service, there are also efforts to promote and form rescue volunteers to augment the manpower resources in times of disasters and emergencies. They are trained with life-saving skills and properly briefed for proper response.

Accelerate investments and programs for disaster risk reduction and climate change adaptation. Since the vulnerability of Metro Manila is largely a function of its unplanned urbanization, environmental degradation, and inadequate infrastructure related measures, there is an urgent need to reduce vulnerabilities associated with natural hazards through a combination of structural and non-structural measures. The government must accelerate their investment in disaster risk reduction programs, equipment, and infrastructure to aid in the preparation and increase the

²⁴ Dr. Liwayway Vilorio. (2014). Barangay Government Disaster Preparedness: The Case of Typhoon Sendong Affected Iligan Communities. http://philippinesociology.com/wp-content/uploads/2014/08/VILORIA_BARANGAY-GOVERNMENT-DISASTER-PREPAREDNESS_COMPLETE.pdf

capacity of the people in recovery and adaptation, such as the following:

- Improve the monitoring system for Climate Change and Disaster Risk Reduction Management implementation (e.i. financing and effectiveness of programs and projects).
- Establish comprehensive and socio-economic monitoring systems to track set indicators for vulnerability reduction and resilience development.
- Generate and cultivate knowledge on disaster risk management along with climate change in a systematic way. Protocols on data sharing and knowledge management must be established among relevant users and producers of information.
- Strengthen the national and local capacities for monitoring, forecasting, hazard identification, early warning, and risk evaluation and enhance the capability of Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) in providing services to protect the people and environment and to ensure economic security against natural hazards since PAGASA is the frontline government agency for meteorological observation.

To mitigate social and governance risks:

Implement social welfare and protection programs to all vulnerable groups (children, older persons, and persons with disabilities). Social welfare programs help ensure income security and access to basic services for poor and vulnerable populations. Key sectors that may serve as entry points for social assistance programs include education, finance, health and public sector management.

Social assistance interventions may include:

- Cash transfers (conditional and unconditional)
- Social pensions
- In-kind transfers, such as food-for-work, school feeding programs, etc.
- Asset transfers
- Scholarships

The vulnerable sectors can avail of programs and services from the government under the DSWD which provides social protection benefits to the poor and marginalized sectors of society. Said services rendered by the agency include educational, burial, medical and transportation services under their Assistance to Individuals under Crisis Situation (AICS), Sustainable Livelihood Program (SLP) and Pantawid Pamilyang Pilipino Program (4Ps).²⁵ DSWD as an agency that caters to the poor and vulnerable sectors of society is guided by its core values of prompt and compassionate and corrupt free service to the people.²⁶

Children

- Intensify implementation of social protection programs and ensure functionality of Barangay Council for the Protection of Children (BCPCs)
- Strengthen capacities of LGUs as primary duty bearers to reduce risk and vulnerabilities of street children
- Enhancement and operationalization of activity centers for children in Metro Manila
- Improvement in the management and operation of Center/Residential Care Facilities of LGUs and DSWD

²⁵ Asian Development Bank: <https://www.adb.org/themes/social-development/social-protection/social-assistance>

²⁶ Source: DSWD showcases social protection programs for poor in 'Kilos Sambayanan' campaign <http://www.pna.gov.ph/articles/1001093>

Older Persons

- Strengthen implementation of Social pension through on time pay-out, quarterly consultation with LGU implementers and Senior Citizen organizations and explore other mechanisms or service provider to deliver the cash grants to senior citizens
- Continuous lobbying with local government executives to gain support to the following:

- Enactment of supportive ordinance at the local level
- Appropriation of funds for implementation of programs and services for SCs
- Replication of Home Care Support Services for SCs, Intergenerational Program for SCs and Children and Sheltered Workshop for Older Persons and PWDs

- Lobby the following with DSWD Central Office:
 - Advocate/lobby for the approval of the Creation of Office/Council that will serve as the National Office to overlook the implementation of Philippine Action Plan for Senior Citizens
 - Review existing policies of the Social Pension Program for Indigent Senior Citizens to consider senior citizens with limited pension from other institution
 - Formulate guidelines on regulating private (business oriented center/ residential care facilities with DTI and DOH
 - Conceptualize Social Technology Project that will address emergent needs of Senior Citizens (e.g. center that will cater senior citizens from middle income family that wanted to house their SCs to a safe and good environment while they are working; database that are link and connected

to all LGUs in the Philippines to avoid duplicity of registration PWDs

- Enhance Database of Social Pension Program which should be linked with SSS, GSIS and PVAO
- Monitor implementation of other National Government Agencies' compliance related to the sector

Persons with Disability

For DSWD-NCR

- Continuous lobbying and advocacy with the local government executives to gain support to the following:
 - Appropriation of funds for implementation of programs and services for PWDs.
 - Establishment of Persons with Disability Affairs Office (PDAOs) and Local Committee on Disability Affairs
- Strengthening of the Regional Committee on Disability Affairs particularly in the implementation of the Regional Inter-agency Plan, policy and program development and data sharing.

For DSWD-CO/NCDA

- Review existing policies and programs and mainstream concerns of PWDs in the Pantawid Pamilya Pilipino Program eg. Children with disabilities
- Review the existing Magna Carta for PWDs and consider proposing amendments to include provision of additional government assistance "disability pension" to PWDs
- Review and implement the Guidelines of the MIFF
- Conceptualize Social Technology project that will address emergent needs of PWDs
- Monitor implementation of other National Government Agencies compliance to Social Welfare

Development (SWD) Laws related to the sector.

Implement a convergence approach to help the poor and vulnerable become self-sufficient and self-reliant — convergence of government programs that promote good well-being of individuals and families. There is a need to converge the efforts of the government agencies and institutions as DSWD cannot do it alone. The convergence will be anchored on key elements such as unified targeting system, synchronized implementation of social preparation and mobilization activities, harmonized engagement of the LGUS, coordinated capability building, harmonized monitoring and reporting and enhanced partnership with the private institutions.

DSWD will further intensify its various programs in order to address poverty and improve the quality of life of the marginalized using a convergence strategy to achieve its commitment to serve the poor, vulnerable and disadvantaged sector through its social protection program.

Strengthen the Solo Parent Act. Strengthening the existing law in promoting and protecting the rights of solo parents, which is considered as one of the vulnerable sector in the country has to be prioritized by the government as it is seen to be an effective mechanism to promote and protect the rights of the solo parents.

To advocate for support of the passage of the bill, the following activities are being initiated by DSWD: Discussion of the draft bill in regular activities where stakeholders and duty-bearers are involved i.e. SWD Forum, Regional Inter-Agency Committee on Filipino Family (RIAC-FF) and Regional Inter-Agency Committee on Anti-Trafficking and Violence Against Women (RIACAT-VAWC) Meetings, Celebration of International Day of Families and National Filipino Week, Social Welfare and

Development Forum, Advocacy Forum, Consultation Dialogue, Legislator's Briefing and Area-Based Standards Network (ABSNET) Meeting.

Establish a National Council and Regional Committee for Older Persons. This is one of the priority bills of DSWD filed at the 17th Congress that seeks to institutionalize the National Commission for Senior Citizens (NCSC), which shall have the best interest of our country's senior citizens. It aims to create the National Commission for Senior Citizen and Regional Committee for Older Persons.

Our Filipino senior citizens have contributed immensely to the growth and progress of the country. As they reach the twilight of their lives, it is our responsibility as a society to ensure their rights and privileges, and that proper accommodation is given to them. (SB 674)

As a national agency, the NCSC will formulate and implement policies, plans, and programs that promote senior rights and privileges or address issues plaguing the sector. With sectors such as the youth, women, and PWDs having a specialized body catering to their constituents, it is about time that our Filipino seniors are awarded their own commission as well as to ensure that no Filipino gets left behind. (SB 674)

Review the strategy on adoption mechanism / procedure. DSWD, in collaboration with partner-government agencies and child caring and placement institutions, will lead in the continuous call for legal adoption advocacy in the Philippines. Yearly, the Adoption Consciousness Celebration (ACC) is being conducted particularly on the month of February. Likewise, several advocacy activities through the Adoption Family Support are also regularly conducted.

To mitigate risks in the economic/ labor sector

Engage in bilateral negotiations with governments of host countries in order to ensure legal protection of Filipino workers rights under the laws of the host country.

The government, through the OWWA, implements several programs for overseas Filipino workers and their families such as social benefit, educational and training assistance, workers welfare assistance program, repatriation program, and reintegration program.

Establish twinning arrangements with host local authorities of Filipino migrants covering protection of workers' rights, insurance coverage, social security, and pension portability.

LGUs to conduct a thorough pre-migration orientation seminar (PMOS) to prepare OFWs and their families left behind with the adjustment of having a family member living and/or working overseas and to encourage both the OFW and households which are recipients of OFW remittances to save and put their money in more productive

use such as entrepreneurial/business activities utilizing government programs such as those offered by SSS, Philhealth and Pag-ibig as well as the Philippine Financial Freedom Campaign Platform Peso Sense which make use of social media to reach out with savings and investment messages for Overseas Filipinos and their families.

Establish OFW helpdesks in all LGUs as mandated by Republic Act No. 10022; LGUs are also encouraged to scale up service provision to migrants and their families through the establishment of Migrant Resource Center to provide social and economic services in all phases of migration and pathways to return and reintegrate towards ensuring a successful migration experience.

Encourage the organization of OFW Family Circles in all LGUs towards creating the environment for migrants and their families to have their own support network that will also serve as a bridge with local authorities.

Strictly implement Republic Act No. 10524 – at least 1% of all positions in government agencies and private companies will be reserved for persons with disability.

09

Building Safe and
Secure Communities

Building Safe and Secure Communities

Redeveloping underutilized land in Metro Manila, if unlocked, the government will gain around US\$200 billion, and eventually provide opportunities to address the current housing needs through high density housing, large scale in-situ redevelopment and other alternative uses.¹

Moreover, the increasing population in Metro Manila has major effects on urban areas especially the inevitable proliferation of informal settlements or slum dwellings which contributed significantly in producing negative environment and health impacts. These informal settlers living in danger areas are vulnerable and exposed to different disasters, such as flooding.

The incidence of informal settlements in the Philippines can be traced to low-income, weak urban planning, lack of serviceable land, shortage of affordable socialized housing and dysfunctional legal system. The proliferation of informal settlements is a phenomena associated with big cities and expanded urban centers. Habitat III Philippine Report 2016 indicated that there are about 1.5 million Informal Settler Families (ISFs) nationwide or about 15 percent of the Philippine total population, 39 percent of whom are located in Metro Manila. The same study projected the growth of ISFs at 3.4 percent per year. It is for this reason that access to affordable housing remains a development challenge.

In 2015, according to PSA survey, a family of five in Metro Manila has an average annual income of ₱425,000.00 which is still the highest among the 17 regions in the country.

In addition, in 2009, the average monthly rental in housing or apartment in Metro Manila can be classified according to its rental class ranges: less than ₱2,000.00; from ₱2,000.00 to ₱5,999.00; ₱6,000.00 to ₱9,999.00; ₱10,000.00 to ₱19,999.00; ₱20,000.00 to ₱29,999.00; ₱30,000.00 and more. It can be noted that on the following rental classifications, majority of families which is around 50 percent of more than 2.4 million families in Metro Manila is renting from ₱2,000.00 to ₱5,999.00.

Assessment and Challenges

Over the years, Housing and Urban Development Coordinating Council (HUDCC) together with the Key Shelter Agencies have been fulfilling its mission of facilitating access to a variety of housing options that are decent, affordable and responsive to the diverse and changing needs of homeless and underprivileged Filipino families. However, the housing sector has identified several challenges, in

¹ The Metro Manila Greenprint 2030, 2013. Building a Vision

Table 9.1. ₱50 Billion Oplan LIKAS Housing Program for ISFs In Metro Manila (As of September 2016)

PARTICULARS	IN-CITY	OFF-CITY	TOTAL
NHA Housing Production	16,748	84,462	101,210
Completed	6,898	73,075	79,973
On-going	7,700	7,011	14,711
To be Started	2,150	4,376	6,526
SHFC High Density Housing Program	9,248	10,615	19,863
Completed	2,257	1,255	3,512
On-going	3,711	7,980	11,691
To be Started	3,280	1,430	4,710
TOTAL	25,996	95,077	121,073

Source: HUDCC

addition to rapid urbanization and poverty. These are the factors that contribute to the growing housing problem in the country, to include the following:

Proliferation of Informal Settler Families (ISFs). There is a significant increase in the number of informal settlers based on the 2011 census which reached to 584,425 families.² These figures are distributed according to the location of ISFs: i) private lands with 279,966 families; ii) government lands with 158,647; iii) government infrastructure project areas recorded at 21,132; iv) danger areas³ with 104,219; and v) other areas with 20,461 families. Of the 104,219 families living in danger areas, about 65,723 families are located along waterways.

In response to the continued proliferation of ISFs situated in danger areas (waterways, transmission lines in Metro Manila), the national government implemented the Oplan-Lumikas para Iwas Kalamidad at Sakit (Oplan-LIKAS) Program for the period 2011-2016 (*Table 9.1*). It is the pre-emptive evacuation and relocation of ISFs

along danger areas. Said program allocated ₱50 billion which aims to relocate around 104,219 ISFs in Metro Manila.

As of September 2016, a total of 79,973 housing units has already been constructed by the National Housing Authority (NHA) while 3,512 housing units have been constructed by the Social Housing Finance Corporation (SHFC) under their High Density Housing Program (HDHP). It should be noted that of the total constructed housing units for the ISFs in Metro Manila, 74,330 were off-city while 9,155 were in-city, majority of which are located in the cities of Quezon, Pasay, Manila, Caloocan, Navotas and Valenzuela. For the in-city, the average cost of socialized housing per unit is pegged at ₱450,000.00 while off-city the cost per housing unit averages from ₱215,000.00-₱280,000.00.

It is apparent that the housing program for the same period which was to provide in-city or on-site resettlement sites was not realized based on the accomplishments of NHA and SHFC.

²NHA, LGUs

³Danger Areas mean esteros, railroad tracks, garbage dumps, riverbanks, shorelines, waterways, and other public places such as sidewalks, roads, parks, and playgrounds (RA 7279)

Table 9.2. Housing Accomplishment in Metro Manila, 2011-2015

PROGRAM	AGENCY	ANNUAL ACCOMPLISHMENT (NUMBER OF FAMILIES ASSISTED/BENEFITTED)					TOTAL
		2011	2012	2013	2014	2015	
DIRECT HOUSING PROGRAM							
In-City	National Housing Authority (NHA)	0	0	660	3,244	1,522	5,426
Off-City		5,093	779	1,040	4,831	4,781	16,524
Community Mortgage Program	Social Housing Finance Corporation (SHFC)	6,521	2,487	4,293	3,848	4,569	21,718
High Density Housing Program		0	0	212	5,436	8,270	13,918
End-User Financing	Home Development Mutual Fund (HDMF)	3,507	2,897	3,037	3,601	2,903	15,945
TOTAL		15,121	6,163	9,242	20,960	22,045	75,531
INDIRECT HOUSING PROGRAM							
Residential	Housing and Land Use Regulatory Board (HLURB)	45,798	106,383	65,996	58,942	61,849	338,968
Non-Residential		34,589	4,559	2,079	33,110	35,048	109,385
Abot Kaya Pabahay Fund	Home Guaranty Corporation (HGC)	4,263	4,398	4,500	4,991	5,424	23,576
Retail Guaranty Program		31,441	30,162	43,848	24,441	36,558	166,450
TOTAL		116,091	145,502	116,423	121,484	138,879	638,379

Source: NHA, SHFC, HDMF, HLURB, HGC

For the past five (5) years in 2011-2015, the total housing need in Metro Manila was estimated at 1,737,039 housing units.

For the period 2011-2015 (*Table 9.2*), for the direct housing program under the program of the NHA, SHFC, Home Development Mutual Fund (Pag-IBIG) were able to provide housing units for the 75,531 families. On the other hand, indirect housing program which is composed of Housing and Land Use Regulatory Board (HLURB) and Home Guaranty Corporation (HGC) were able to provide housing assistance to 638,379 families. These accomplishments were very low to the total target housing units in Metro Manila for the same period.

Limited appropriations for housing. Historically, government annual appropriations for the housing sector only accounts to less than 0.5 percent of the GAA and 0.12 percent of our GDP, which is one

of the lowest in South East Asia according to the World Bank.

Limited supply of government land for residential use. According the the Department of Environment and Natural Resources (DENR), the only available land for housing and urban land uses are the remaining untitled alienable and disposable lands of 0.54 million hectares or 4 percent of the estimated total land area of the country (300,000 square kilometers). While the total land requirement to address the 6.57 million housing need in the country is 42,057 hectares at an average lot size of 64 square meter per housing unit.

Low affordability of targeted beneficiaries. Based on the 2015 Family Income and Expenditures Survey, a Filipino family can only allocate 12.2 percent of their income for rent/rental value and house repair and maintenance, which is 23 percent lower

than the 35 percent income requirement under the end-user financing program of the Home Development Mutual Fund (HDMF) for loans more than ₱1.25 million and 17.8 percent lower than the 30 percent income requirement for loans up to ₱1.25 million.

Adverse impact of climate change. For the last five years, there is a total of 1.8 million housing units need to be either repaired or completely replaced due to the calamities that hit the country.

Low housing production. The current level of housing production represents a small fraction of the estimated housing need perpetuating the formation of informal settlements and slums in major urban centers. The government shelter program for the last six years, has only provided housing assistance to 168,535 families per year on the average. This only constitutes 16 percent of the estimated average annual housing need of more than 1 million.

Lack of affordable land for the socialized housing program within Metro Manila. This recurring challenge prompted the government to shift its focus by relocating majority of ISFs outside Metro Manila (*Table*

9.1). Though the government has introduced the construction of High Density Housing or Low Rise Buildings within Metro Manila, the number of housing allocated was not enough wherein only 9,155 housing units (both NHA and SHFC) were completed and 11,411 with on-going construction for the period of 2011-2015.

Housing Needs. While the population in Metro Manila is constantly at the annual average growth rate of 1.58 percent for the period 2010-2015, the demand for housing grows in enormous numbers both for the low-income and middle-income families. This translates to the housing demand in Metro Manila which reaches 1,737,039 units in 2016.

Delay in the implementation of the projects attributed to the following:

- Government Policy on Voluntary Relocation vs. Regular Relocation;
- Objection of Receiving LGUs to absorb ISFs residing outside their cities/ municipalities;
- Delayed issuance of Bureau of Internal Revenue (BIR) ruling on Capital Gains Tax exemption for developers;
- Slow payment of financial assistance;
- Titling for High Density Housing.

Strategic Framework

Targets

Table 9.3. Plan Targets to Build Safe and Secure Communities

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Total housing need	2011-2015	1,737,039		HUDCC
Number of families provided with direct and indirect housing assistance				HUDCC
Direct				
• SHFC	2011-2015	13,918		SHFC
• Pag-IBIG	2011-2015	15,945	52,680	Pag-IBIG
• NHA	2011-2015	16,524		NHA
Indirect	2011-2015	482,083		HLURB, HGC
Number of Informal Settler Families			Decreased	NHA
Private lands	2011	279,966		NHA
Government lands	2011	158,647		NHA
Government infrastructure project areas	2011	21,132		NHA
Danger areas	2011	104,219		NHA
Other areas	2011	20,461		NHA
Number of socialized housing target to housing needs				HUDCC
Number of LGUs with approved Local Shelter Plan				DILG

Figure 9.1. Strategic Framework to Build Safe and Secure Communities, 2017-2022



Strategies

In general, the housing sector aims to expand access to affordable, adequate, safe and secure shelter in well-planned communities expanded:

Develop integrated neighborhoods and sustainable communities particularly for low-income households. Provision of housing services particularly to low income households undertaken in an integrated manner, focusing on the development of neighborhoods and communities rather than just housing. To address the proliferation of informal settlements in the city, planning for the provision of housing should consider the costs (as well as the benefits) of providing vertical housing within the city vis-à-vis off-city resettlement. Promotion of high-density mass housing and vertical developments for socialized housing are advocated to encourage in-city development, emphasizing that such resettlement approach is more economical given the principle of maximum retention and minimum dislocation of beneficiaries who should not be displaced from access to opportunities and services (e.g., livelihood, schools, hospitals, etc.). The upscaling of SHFC's HDHP which was able to serve 13,918 ISFs in a span of 2.5 years, should be pursued as a successful model. The implementation of HDHP has received interests from LGUs outside Metro Manila for possible replication in their area. Hence, continued funding for housing program/project innovations should ensure that the ISF communities' shelter concerns are being addressed by the National Government.

It is essential to ensure the provision of necessary infrastructure support and basic services (including access to livelihood and employment opportunities) to address the needs of resettled ISFs. Relevant government agencies (e.g., DSWD for social development, DepEd/TESDA/CHED for education, DOLE for employment

opportunities, and the Department of Public Works and Highways (DPWH) and Department of Transportation (DOTr) for infrastructure and transportation networks) should be involved to ensure that other basic needs that are key to the success of any sustainable and liveable community are adequately satisfied. The physical infrastructures and their location must ensure that housing and human settlements comply with disaster risk reduction and management (DRRM) and climate-change adaptation (CCA) requirements to mitigate risks and vulnerability.

Intensify implementation of alternatives and innovative solutions in addressing the housing need of the lower income classes and the vulnerable sectors. Solutions such as public rental housing, mixed-income/mixed-use housing development, housing microfinance initiatives and possible non-mortgage shelter program are reiterated to increase/enhance housing affordability and better cater to underserved housing beneficiaries more sustainably. Moreover, public rental housing and housing microfinance may address the issue of low occupancy rates in socialized housing and unwillingness of ISFs to voluntarily relocate primarily due to the low affordability of housing units. Such interventions are proposed in recognition of the need to cater the homeless, poor and underprivileged as most of them are unable to meet the standards necessary to own a socialized housing unit.

Partnership with LGUs/NGAs by offering their employees / constituents / members with housing units will eventually influence project design, specifications and housing packages.

The National Informal Settlements Upgrading Strategy (NISUS) that aims to be institutionalized under the National Urban Development and Housing Framework (NU DHF) as a ten-year program to improve

the lives of ISFs nationwide is the entry point to integrate these proposed solutions. The NISUS envisions and targets the provision of secure tenure to at least one million ISFs by 2025.

The national and LGUs shall prioritize the establishment of safe and secure settlements for fisherfolks: i) who are empowered to serve as stewards for ensuring sustainable use of coastal and marine resources; and ii) who are enabled to add value to the marine fishery for employment and livelihood.

Strengthen decentralization of housing and urban development interventions.

LGUs shall be the forefront of housing and urban development with the support of the national government. The decentralization of housing and development efforts, will be reinforced especially on local shelter planning, comprehensive land use planning with a ridge-to-reef approach, land acquisition and development, curbing proliferations of urban settlers, implementation of Regional Resettlement Action Plan (RRAP) in pursuit of SDGs in coordination with NGAs.

Adopt viable land acquisition approaches and fast-track the conduct of inventory of lands for socialized housing development.

The importance of adopting land acquisition approaches aside from land ownership (such as usufruct, long-term lease, and land banking, among others) is also emphasized to address the perennial problem of identifying suitable land for socialized housing particularly in urban areas while also addressing affordability. Along with this, fast-tracking the inventory of lands and cadastral surveys are crucial to hasten the process of identifying land areas for housing projects.

Further, the government should facilitate the identification of idle and non-performing government lands for socialized housing as directed under Memorandum Circular No.

87, series of 2015 which directed all NGAs and instrumentalities, including GOCCs, to submit an inventory of their respective idle lands and created an inter-agency task force to identify lands and sites for socialized housing. This inventory should be integrated into a national land, ISF and geohazards information system that would enable government to map out government properties, forfeited properties, hazard areas and locations of informal settlers as the platform for developing city or municipality wide housing projects.

The limited supply of lands for socialized housing necessitates a reassessment of the use of government properties as well as the location of government offices with the objective of unlocking the potential of government lands in pursuing urban renewal and rural development.

Mainstream program convergence budgeting in housing and resettlement, and innovative housing finance modalities.

A programmatic and convergent approach to planning and budgeting for housing and resettlement interventions must be mainstreamed in the national budget process. This would ensure that shelter, utilities, infrastructure and community facilities to be implemented by various agencies (e.g., NHA, DEPED, DOH, LWUA, NEA, DSWD) will be identified early in the budget planning process. Additional budgeting for community facilities and infrastructure will be determined annually based on the site selected as in-city and near-city relocations are expected to reduce costs since most community facilities would already be available in the area. The budget estimate should likewise factor in the proposed policy on an income-based subsidy scheme wherein those with less income will receive higher subsidy from government to bridge the gap between housing costs and varying income levels of families. The income-based subsidy scheme shall be integrated into the National

Resettlement Policy Framework. The Key Shelter Agencies (KSAs) shall likewise pursue securing additional financing from the private sector through Public-Private Partnerships (PPPs) wherein government's share would be in the form of government-owned lands under a sale, usufruct or long-term lease arrangement.

Support HUDCC key program of Oplan Likas for "20 in 2022" through venture with developers to refurbish foreclosed properties and offer it to low-income members at a discount and sale of existing acquired assets to NHA or LGUs as inventory for relocation of low-income families.

Strengthen partnerships with stakeholders.

As a cross-cutting strategy, the sector will continue strengthening its partnerships with stakeholders, especially LGUs, and ensuring effective and efficient delivery of housing interventions. This is essential in ensuring linkage of local shelter plans with regional and national development plans, encouraging PPPs for housing projects and improving compliance of developers to balanced housing development.

Conduct of Fund Briefing for employers regarding latest Fund updates, Provident Benefit and Home Lending programs, as well as, One-on-one consultation regarding Fund-related queries.

Adopt a Community-Driven Development (CDD) approach in shelter provision towards safe and secure communities. This approach may result in better occupancy rates and improvement of housing loan amortization collection efficiency and better estate management since beneficiaries are involved in the entire process, from site selection to the design of their homes and the construction of necessary community facilities.

This method can be best implemented through Memorandum of Understanding

(MOU) signing and ground breaking of government projects such as Pag-IBIG, SHFC, NHA, among others.

Mainstreaming the community-driven or people's plan approach also requires sufficient funding to provide capacity-building and technical support activities (e.g. training on collection of association dues, addressing electricity and water theft, other estate management issues) for communities to ensure that they can be effective participants of the programs.

Cultural aspects, gender-responsiveness, and accessibility must also be considered in the provision of housing for the different groups. For instance, mosques or spaces for congregational prayer may be included as one of the amenities in housing projects for Muslim communities. Further, housing designs must also consider the reproductive and productive activities of both women and men, as well as ensure that housing facilities are elderly and PWD-friendly.

Government must also undertake home financing literacy campaigns to raise the awareness level of the underserved on how to manage their finances as well as the available safety nets for times of financial distress.

Strengthen housing as a platform to reduce poverty and improve social outcomes.

Housing programs will be linked to other social development programs. It will help maximize the "multiplier effect" of the provision of housing units as a means to reduce poverty, generate jobs and employment, and spur downstream economic activities.

Conduct of briefing and loan counselling to members which includes unserved/underserved to provide awareness relevant to housing in support to HUDCC's Home Financing Literacy Campaign.

Implement BALAI (Building Adequate, Liveable, Affordable and Inclusive Filipino Communities), a nationwide housing program that aims to improve government efforts in addressing Filipinos' housing needs, as well as engage participation of the private sector.

HUDCC has set its Program Thrusts for fiscal year 2018, in tight collaboration with the Key Shelter Agencies (KSAs) working together as one team under a single banner program BALAI, or “building adequate, liveable, affordable and inclusive Filipino communities, with the following thrusts:

1. Develop synergy through a complementation of the various housing programs offered by the KSAs from planning, housing production, financing and regulations. With this complementation, there is a chance to increase the government's shelter agencies' impact in the housing sector.
2. From the average 172,000 annual housing production in the past six years, the KSAs is aiming to accelerate housing production by 45% to hit the target of building 250,000 units to house low-income families for FY 2018. This is especially crucial with the government's BUILD BUILD BUILD, in anticipation that more housing needs from those who will be potentially displaced, and may require resettlement assistance from the shelter agencies. In order to accomplish this, the need of partnerships should be strengthened with private developers, non-government organizations (NGOS), civil society organizations (CSOs), communities and other groups from the private sector. Because admittedly, the government cannot deliver this alone.
3. Continue to develop an array of affordable home financing schemes being offered by SHFC and HDMF to bridge the gap in the demand and the supply in the housing market brought about by the low-income household's inability to afford even the socialized housing supply in the market. It is good to note that the program is currently

on the right track, with the approval of the 3% interest rate for housing loans of minimum wage earners enrolled in the Pag-IBIG fund, which is the lowest rate in the market. This was also complemented by the 5.5% interest rate for the regular housing loans under the fund.

4. Anti-red tape which is one of the primary concerns in the housing development is the very long and tedious process in accomplishing the required permits and clearances from the different agencies of the government prior to the implementation of housing development projects. On the average, the estimated pre-construction processing time for the private sector is one to four years due to red tape. In response, KSAs are working on the design of an online housing portal where all required transactions across agencies can be undertaken, speed up the flow of the process and minimize face-to-face contact that creates opportunities for corruption. KSAs are also facilitating government-to-government (G2G) arrangements to speed up the processes and, thereby, reduce the burden of cost to housing beneficiaries.
5. One of the primordial thrust in the sector is to ensure that the houses to be produced are disaster-resilient and that they should be built in safe and hazard-free areas. This means that the houses should be compliant with the national building code and they should not be located in areas prone to floods, landslides and other geo hazards or sitting in an earthquake fault that may imperil the families' lives.
6. Under the vision of BALAI FILIPINO, the securitization and guaranty programs of the NHMFC and the HGC will be scaled-up, respectively, of the housing developments being implemented by the private sector. The securitization will recycle the invested capital to housing back to the market and the guaranty program will boost the confidence of the developers in the

housing market and encourage them to produce more.

7. Related to drawing in private sector participation in addressing the country's massive housing needs, it will ensure the strict compliance of the private developers to the balanced housing requirement. Based on RA 10884 otherwise known as the "Balanced Housing Development Program Amendments", subdivision developers and condominium developers are required to allocate 15% and 5% of the total land or cost of their projects to building socialized housing projects, respectively. This will increase the affordable housing stock available to low-income families.
8. Focus on developing and establishing an updated and central repository of data on lands and informal settler families (ISFs) to hasten the identification of available lands suitable for housing and to prevent inclusion and exclusion errors among the intended beneficiaries. In line with this, close coordination is essential with the league of municipalities and barangays in the updating and verification of housing statistics.

Integrate planning collaboration with various stakeholders such as government infrastructure agencies, Local Government Units, social services providers, utility providers and other stakeholders

Relocate and resettle affected ISFs using the Community Initiative Approach (CIA) for In-City, near city and on-site relocation.

Adapt alternative housing modes i.e., Public Rental Housing for ISFs affected by Waterways, Gov't Infra, etc., New township, and Mixed-Use / Mixed-Income

Upgrade government lands proclaimed as socialized housing sites

Implement High Impact housing projects in partnership with Private Sector

Enhance resettlement assistance program for LGUs

Implement housing program for low-income employees of government agencies and local government units

Implement Land Assembly Program, forward planning and new business development

PART IV

INCREASING GROWTH POTENTIAL (“PATULOY NA PAG-UNLAD”)



10

Prospecting for a
Demographic
Dividend

Prospecting for a Demographic Dividend

Economists' perspective have focused mainly on population growth. They tend to overlook the changes within populations, particularly the changes in age distributions which are equally important as population growth. It is imperative to look into a critical dimension of the population which is the age structure and corresponding changes to it as these are essential in noting the trend of workforce and dependency. Likewise, development should be viewed broadly factoring in the complex relation between economic and human development and the effects of the changing population age structure.

When a country is comprised of mostly young population below 15 years old, the degree of young dependency is relative to that population belonging to the working age group. Similarly, the number of elderly population would notably increase as population ages thereby increasing their dependence to the working ages. On the other hand, when people in the working ages comprise a relatively large proportion in the population, the workforce can generate increasingly higher production and income for the country and its population. This is indicative that such country has made a demographic transition, resulting to what is conventionally referred to as demographic dividend.¹

Each age group in the population tend to behave differently. As such, they provide distinct economic consequences like the young require intensive investment in health and education, prime-age adults supply labor and savings, and the aged require healthcare and retirement income. When the relative size of each group in the population changes, so does the intensity of the economic behaviors.²

Countries with relatively high proportion of working age population compared to dependent populations have the possibility of benefitting from a "demographic dividend" or "demographic bonus."

As such, increase in economic productivity among the working population can accelerate economic growth.³ When older working population remains economically productive and continues to earn income and assets, this is the possibility window for the second demographic dividend. The second dividend begins after the first has ended and continues indefinitely (Lee and Mason 2003).

However, it should be noted that demographic transition directly influences the age structure. When the transition from high fertility and high mortality transcends to low levels of fertility that approximates that of mortality, a demographic dividend is possible.

¹ Philippine Development Plan 2017-2022, Chapter 13-1

² Bloom, D., D. Canning and J. Sevilla, "The Demographic Dividend: A New Perspective on the Economic Consequences of Population Change," RAND, 2003

³ Commission on Population, "Is Demographic Dividend an Imminent Prospect in the Philippines?," Policy Brief, 2016

A lower fertility affects for decrease in dependency ratio and increase in the working age population. Increase in the proportion of working age will yield increase in per capita output while decline in dependency ratios can facilitate more savings and investments. Considerations would have to be made, of course, on the constant labor participation rate and output per capita, current wage rates, available economic opportunities, and quality of labor force are just among those which may critically affect the human resource utilization and productivity.

The rapid changes in population age structure in many countries have created an opportunity to experience rapid economic growth over a long period of period, provided the right policies in the labor market, health, governance and economy are in place. Demographic transition links the evolving age structure and economic development where there is change from a level of high fertility and high mortality to one of low fertility and low mortality.

According to a publication of Bloom, Canning and Sevilla, demographic dividend is delivered through a number of mechanisms to which labor supply, saving and human capital are considered most important. First, the demographic transition affects labor supply in such a way that when the regular and inevitable aging of the baby-boom generation is between 15 and 64, it is more likely to be working, thus lowering the ratio of dependents to non-dependents. The number of people who would like to work (labor supply) therefore gets bigger and, provided the labor market can absorb the larger numbers of workers, per capita production increases. Also, they reflected that women are more likely to enter the workforce as family size declines where the effect is magnified with increase in life expectancy among adult women. Yet, they suggested that an upward adjustment of the retirement age could mitigate the labor force effects of population aging.⁴

Secondly, they stressed that demographic transition encourages the growth of savings, thus improving a country's prospects for investment and growth. The young and the old consume more than they produce, whereas working-age people tend to have a higher level of economic output and also a higher level of savings. When people are less likely to be investing in children, more savings are provided in preparation for retirement to maintain standard of living.

Lastly, they presented that the demographic transition has significant effects on investments in human capital, though with the least tangible yet significant effects. The demographic transition begins with changes in mortality that result in a population that lives longer and stays healthier. A longer life expectancy provide fundamental changes in the way that people live, shifting attitudes about education, family, retirement, the role of women and work.

NCR is considered as the largest conurbation (urban agglomeration) in the country, it is composed of the 16 cities and one municipality with the City of Manila as its regional capital.

⁴ Bloom, D., D. Canning and J. Sevilla, "The Demographic Dividend: A New Perspective on the Economic Consequences of Population Change," RAND, 2003

Generally, the country benefitting from demographic window of opportunity is possible considering its relatively young population which would create the demographic dividend that can boost economic growth. However, demographic transition in the country is slow as trend in fertility rates remain high with the latest Total Fertility Rates (TFR) of 3 based on the 2013 NDHS, still one child higher compared to the replacement level of 2.1 children. Moreover, unemployment rates create an additional challenge in reaping the benefits.

For NCR, there is seemingly a greater chance of reaping the benefits of demographic dividend sooner than other regions in the country provided that the challenges on fertility rates reduction and improvements in employment rates will be addressed. TFR should continue to decline or be sustained with corresponding increases in contraceptive prevalence rate particularly among the poorest households. Increasing incidences of teen pregnancies, however, may impede in this aspect.

Concomitant to this, high unemployment and underemployment rates among young workers should likewise be dealt with. The region should look into the flow of young workers to the labor force from school where greater challenges are faced in terms of unemployment and low income. Boosting for the achievement of demographic dividend would entail improvements in current employment and average income among young workers where appropriate labor government policies should be in place. Other considerations would include more women entering the labor force, better health status and highly skilled workforce, and improved support policy and environment can propel its attainment.

Essentially, given that the high proportion of the working age population who is healthy and highly productive with the appropriate skills, the region may be able to reap the fruits of the demographic dividend.

Assessment and Challenges

Increasing Population Size. Over the years, the total population of NCR had significantly increased having been among the top three most populous regions in the country. *Figure 10.1* shows the trend in population growth over the years. As of 2015, NCR's population has grown to 12.88 million.

In 2015, the Census of Population and Housing recorded a total of 12.88 million Filipinos living in the NCR, an increase of about 1.02 million in a span of 5 years from its 2010 figure of 11.86 million as shown in (*Table 10.1*). During the period 2010 to 2015, the population increased on the average by 1.58 percent annually, a decline from the growth rate recorded for the period

2000-2010 to which the population grew a rate of 1.78 percent (*Table 10.2*).

About half of the total population of the region was occupied by three cities namely, Caloocan (1.58 million); Manila (1.78 million); and Quezon City (2.94 million). Meanwhile, the municipality of Pateros together with the cities of San Juan and Navotas comprised the areas with the smallest population size (*Figure 10.2*).

Between 2010-2015 period, Taguig City posted the highest population growth rate across 17 localities in the region. This was followed by the cities of Mandaluyong, Paranaque and Pasig with annual population

Table 10.1. Population of NCR: 2000, 2010, 2015

CENSUS YEAR	CENSUS REFERENCE YEAR	POPULATION
2000	May 1, 2000	9,932,560
2010	May 1, 2010	11,855,975
2015	August 1, 2015	12,877,253

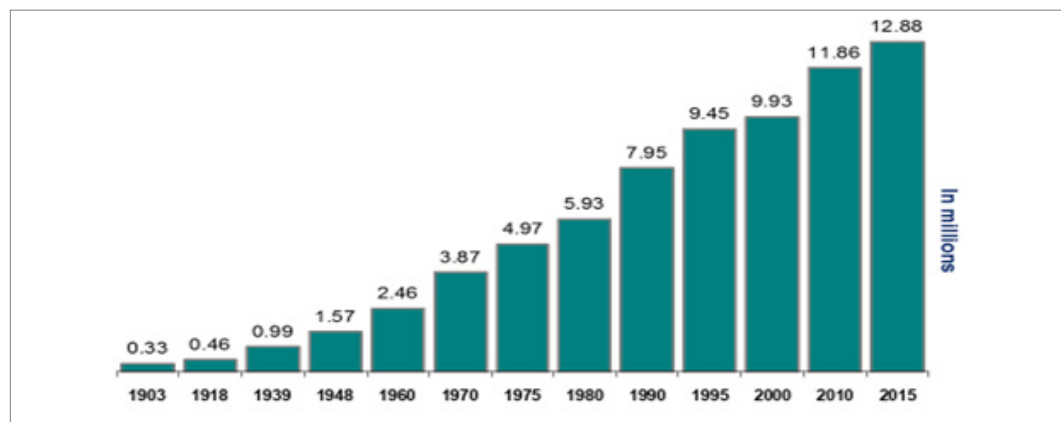
Source: Philippine Statistics Authority

Table 10.2. Annual Population Growth Rate, NCR: 2000, 2010 and 2015

REFERENCE PERIOD	POPULATION GROWTH RATE
2000-2010	1.78
2010-2015	1.58
2000-2015	1.72

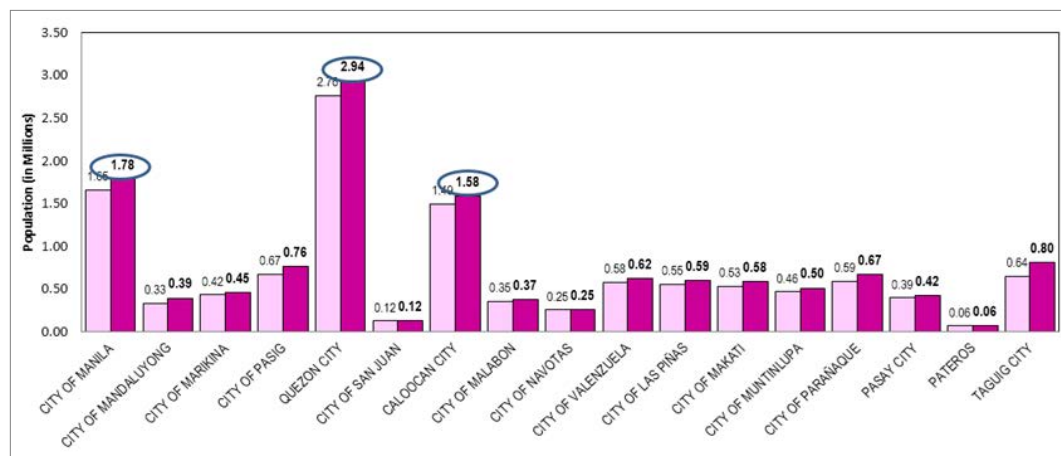
Source: Philippine Statistics Authority

Figure 10.1. NCR Population Census Years 1903 -2015



Source: Census of Population & Housing, PSA

Figure 10.2. NCR Population by City/Municipality, 2010 & 2015



Source: Philippine Statistics Authority

Table 10.3. Annual Population Growth Rate by City/Municipality, NCR (Based on the 2000, 2010, and 2015 Censuses)

CITY/MUNICIPALITY	POPULATION GROWTH RATE	
	2010-2015	2000-2010
City of Manila	1.43	0.44
City of Mandaluyong	3.12	1.67
City of Marikina	1.16	0.81
City of Pasig	2.31	2.86
Quezon City	1.17	2.42
City of San Juan	0.12	0.31
Caloocan City	1.18	2.37
City of Malabon	0.65	0.42
City of Navotas	0.03	0.78
City of Valenzuela	1.45	1.71
City of Las Piñas	1.22	1.57
City of Makati	1.85	1.16
City of Muntinlupa	1.78	1.95
City of Parañaque	2.39	2.72
Pasay City	1.12	1.02
Pateros	-0.09	1.12
Taguig City	4.32	3.2

Source: Philippine Statistics Authority

growth rates of 3.12, 2.39 and 2.31, respectively. Meanwhile, the Municipality of Pateros had incurred a negative population growth rate for the period 2010-2015 (*Table 10.3*).

Of the 1,706 barangays in the region, Barangay 176 from the Caloocan City topped the list of the most populated barangays in the region and even across the country in 2010 and 2015 with about a quarter of a million in size (*Table 10.4*).

Densely Populated Areas. With a total land area of approximately 636 square kilometers, the population density in the NCR in 2015 was posted at 20,263 persons per square kilometers. This represent an increase of 1,622 persons per square kilometer (8.7%) from population density of 18,641 person per square kilometer in 2010. In 2000, there were 15,617 persons residing in every square kilometer of land.

Relatively young population. NCR population has remained relatively young (*Figure 10.3*). Compared to the country's age-sex composition in 2015, the population in NCR had its bigger share of the working age group who were relatively young. In terms of sex composition, there were more

Table 10.4. Most Populated Barangays in NCR: 2010 & 2015

BARANGAY	CITY/MUNICIPALITY	POPULATION	
		2010	2015
Barangay 176	Caloocan City	243,890	246,515
Commonwealth	Quezon City	186,543	198,285
Batasan Hills	Quezon City	150,764	161,409
Pinagbuhatan	Pasig City	126,503	151,979
Payatas	Quezon City	119,053	130,333
Poblacion	Muntinlupa City	103,104	115,387
Holy Spirit	Quezon City	101,385	110,447
Barangay 178	Caloocan City	97,068	107,596
Pasong Tamo	Quezon City	85,954	103,100

Source: Philippine Statistics Authority

females in the region compared to that of the country. Sex ratio in NCR posted 97 males for every 100 females. Meanwhile, based on the PSA data, life expectancy at birth (2010-2015) for male is 69 years, while for female is 76 years. The crude death rate was posted at 6.1/1000 population in 2014.

In 2015, the young dependents (0 to 14 years) comprised 27.3 (2010:29.1) percent of the household population in NCR while the old dependents (65 years and over) posted a share of 3.9 (3.4) percent. The working-age population (15 to 64 years) accounted for the remaining 68.7 (67.5) percent. The age-sex composition of the Philippines and NCR in 2010 is shown in *Figure 10.3*. Evidently, NCR indicates a

“youth bulge” or significant proportions of young people aged 20-34 years.

In *Figure 10.4*, the age-sex composition of the region in 2015 was compared to its projected age-sex composition for 2030 with the condition that the TFR had remained at 2.3. The resulting age-sex distribution is indicative of a larger working age population - a prospect for demographic dividend.

The overall dependency ratio of the region in 2015 was 46, which indicates that for every 100 working-age population, there were about 46 dependents (40 young dependents and 6 old dependents). This ratio is lower than the dependency ratio in

Figure 10.3. Age-Sex Composition of Philippines and NCR: 2010

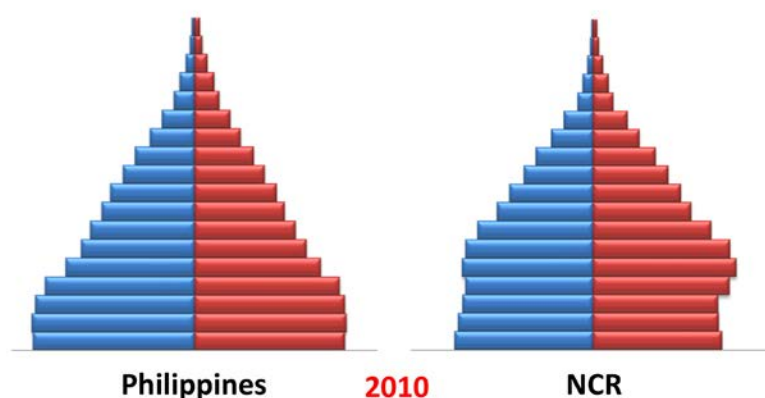
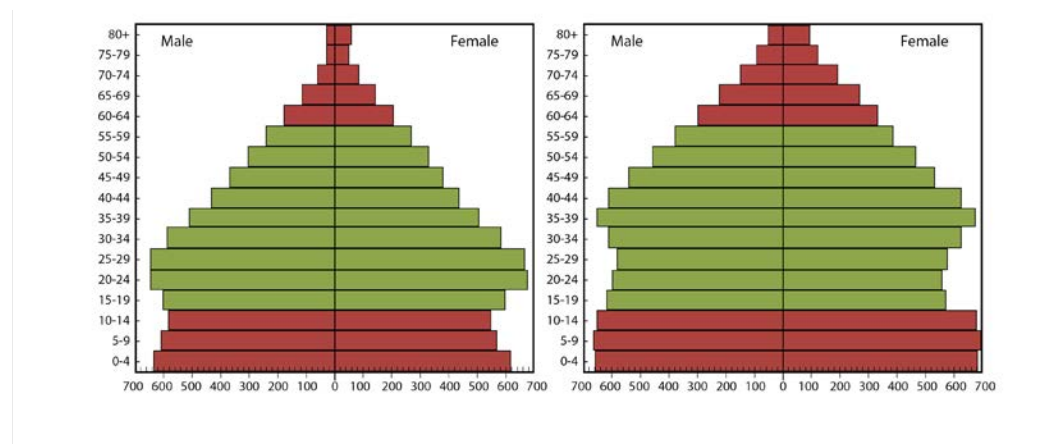


Figure 10.4. Comparative Age-Sex Composition of NCR: 2015 and 2030



2010, which was recorded at 48 dependents per 100 working-age population (43 young dependents and 5 old dependents).

Nine cities had lower an overall dependency ratio lower than the regional figure. For every 100 working-age population of their respective household population, the City of Marikina, City of Pasig, and City of Muntinlupa each had 47 dependents. Moreover, the City of Parañaque had 46 dependents, Pasay City had 45 dependents, and City of Mandaluyong had 44 dependents. Both the City of San Juan and the City of Makati recorded the lowest overall dependency ratio of 40.

Low fertility level yet low in modern contraceptive prevalence. TFR in the region was accounted at 2.3 percent based on the 2013 National Demographic and Health Survey (NDHS). However, wanted fertility rate was posted at 1.7.

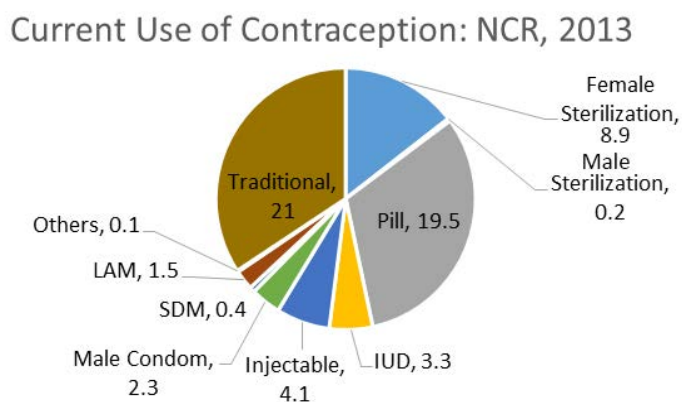
The 2013 NDHS Report also revealed that of the 16,155 women, age 15-49 surveyed, one in five married women and one in four uneducated married women has unmet need for FP compared with one in six wealthy and highly educated married woman which can be a contributing factor for high unmet need.

Although some of the married women who want to stop having children or want to delay the next pregnancy are already using family planning, still around 18 percent of these women are not using any contraceptive method and who are at risk of unwanted pregnancy and have unmet need for family planning. Based on the NDHS results, NCR reported unmet need for spacing at 5.5 percent while 8.6 percent for limiting.

The 2013 NDHS Report also indicated that contraceptive prevalence of NCR at 60.1 percent of which 40.1 pertains to any modern method while 21.0 percent on traditional method (*Figure 10.5*). Meanwhile, based on FHSIS report of 2014, CPR registered 42.18 percent with Quezon City, Navotas, Valenzuela, Muntinlupa, Mandaluyong and Makati reporting CPR levels higher than the regional levels. Essentially, low contraceptive use is often linked to the poor and less educated.

Low level of men's involvement in Responsible Parenthood and Family Planning (RFPF). The improvement of maternal and child health is among the concerns of the RFPF program. One of the identified reasons for non-use of contraceptive method is due to husband's opposition. As such, men's involvement in

Figure 10.5. Current Contraceptive Use by Method, NCR: 2013



Source: 2013 NDHS

actively supporting RFPF practices including anti-violence promotions are important areas to look into to lift the status of women and improve maternal child health and nutrition.

High proportion of births among unwed mothers. According to the 2013 NDHS Report, women in NCR were expected to have around 2.3 children during her childbearing years (i.e. total fertility rate). Data further revealed that 24.3 percent of women age 15-24 in the region have begun childbearing.

Based on the report of the PSA on live births, there were about 1.75 million births registered in 2014 in the country. This equates to average of 4,791 babies born daily in 2014 or about 200 births per hour. Data also revealed that NCR accounted for 13.7 percent of the total births in the country in same year. 94.8% (227,389) of births delivered in NCR were medically attended with.

In the same report, more than half (880,524 or 50.3%) of the total births recorded in 2014 were born by unwed mothers. The three regions that recorded the highest number of illegitimate children born in 2014 are NCR (148,660) followed by Region IV-A (139,773) and Region III (95,214). The proportion of illegitimate babies born in 2014 in NCR is about 62.0 percent.

Increasing Out-of-School Youth. According to the results of the 2013 Functional Literacy, Education and Mass Media Survey (FLEMMS), one in every 10 or about 4 million Filipino children and youth in the country were out-of-school in 2013. Out-of-school children is defined in the FLEMMS as persons aged 6 to 14 years who are not attending school while out-of-school youth are persons aged 15 to 24 years who are not attending school, have not finished any college or post-secondary course, and are not working.

The 2013 FLEMMS results also showed that of the nearly 4 million out-of-school children and youth, 22.9 percent have entered into union or marriage. Another 19.2 percent cited insufficient family income to send child to school as the reason for not attending school (this refers to all educational expenses other than tuition fee), while 19.1 percent lack the interest in attending school.

Among females, 4 out of every 10 have already entered marriage while 3 in every 10 males have no interest in attending school. The proportion of out-of-school children and youth whose family income was not sufficient to send child to school was 22.7 percent and 17.0 percent, among males and females, respectively.

In NCR, there are 4.56 million school age children and youth aged 6 to 24 years old. The proportion of these groups who are out of school is about 10.3 percent. With the identified reasons for these sector's drop-out, they are most vulnerable to STI/HIV/AID, teen pregnancy, and engagement to non-sexual risky behaviors.

Prevalence of Risky Behaviors among NCR Youth. The number of young people aged 15-24 years old in the country reached 18M in 2010, an increase of 3 million from the 2000 figure. NCR's youth accounted for 2.4 million or 14.3 percent of the country's youth population. In 2015, NCR youth increased to 2.58 million representing 20 percent of the region's total population.

Meanwhile, Metro Manila has the highest proportion of youth with college or better education (34.9) while one in five is already working. Moreover, according to the 2013 Young Adult Fertility and Sexuality Study (YAFSS), the proportion of those in live-in arrangement is about one in 5.

Engagement in too early sexual activities.

Across all regions, NCR posted the highest in the prevalence of premarital sex among the youths. Based on the study, about two in 5 have engaged in pre-marital sex of which 7 out of 10 were unprotected against pregnancy and/or STI. It also revealed other risky sexual activities NCR youth engage in making them vulnerable to impending risks to their health and development.

Increasing number of adolescent mothers.

Teenage fertility (aged 15-19 years) in NCR almost tripled in the past decade. YAFSS report revealed that NCR ranks third across all regions in teenage fertility (17.7 percent), next to CAR and Region II. Of those young women aged 15-19 years, around 13.9 percent of them have begun childbearing while 3.8 were pregnant with first child.

Meanwhile, the results of 2013 NDHS reported that about one in ten young Filipino women age 15-19 had begun childbearing: 8 percent are already mothers and another 2 percent are pregnant with their first child.

Engagement in unprotected sexual activities. According to the YAFSS Study, nearly 7 in 10 NCR youth's first premarital sex was unprotected against early pregnancy and/or STI/HIV.

According to WHO, the Philippines has the fastest growing HIV in the world, while other neighbors in Asia like India, Myanmar, Papua New Guinea and Thailand have decreased its cases.

Young people now make up the majority of new human immunodeficiency virus (HIV) diagnoses in the Philippines. Youth aged 15 to 24 accounted for an estimated 62 percent of all new HIV infections, according to the National Youth Commission (NYC). There are an estimated 29 new HIV infections reported daily. More than half of these infections – about 19 – are from the youth sector.

Meanwhile, based on the HIV/AIDS surveillance report of the Epidemiology Bureau of the DOH, there were 9,264 reported cases of HIV infection in the country from January to December 2016 which bring the total number to 39,622 (since January 1984). The numbers are on an upward trend. In 2008, only one person was getting infected daily. This figure jumped to 26 per day in 2016. The age group with the biggest proportion of cases has become younger: from 2001 to 2005, it was 35-49 years old; and starting 2006 to 2016, it was 25-34 years.

Out of the 9,264 new cases from January to December 2016 alone, 22 cases were recorded among 14 years old and below 2,625 cases were recorded among 15 to 24 years old and 4,921 recorded cases among 25 to 34 years old while 35 to 49 years old were only about 1,510 cases.

NCR recorded the most number of HIV reported cases, with 16,899 or 43.0 percent of the total of cases distribution from 1984 to 2016. In 2016, there were 3,744 new HIV cases reported.

Engagement in sex with multiple partners.

According to YAFSS, 13 percent of youth in NCR have engaged in casual sex or sexual activities outside a romantic relationship. Around 8 percent of males have disclosed that they had experienced having sex with other males, while 11 percent of youth who have ever been married, including those who are living-in status, have experienced extramarital sex.

Prevalence of violence among young boys and girls.

According to the report of the Violence Against Children in East Asia and the Pacific regarding A Regional Review and Synthesis of Findings Evidence Strengthening Child Protection Systems (Series, No. 4 UNICEF EAPRO, 2014), violence against children have profound and damaging consequences in childhood,

adolescence and throughout adulthood, its lifelong effects includes: adverse health, social, and economic consequences exist for those affected by violence, including mental and physical health conditions; increased health and other risk behaviors; exposure to further violence; disability from physical injury; reduced health-related quality of life; lower educational attainment; and lower levels of adult economic well-being (Gilbert et al., 2009; Fang et al., 2012).

In NCR alone, violence is not limited among an adult inflicting violence to young boys and girls. According to YAFS4 results, NCR had the largest proportion of youth who experienced physical violence within 12 months based wherein they have been the aggressor (34.5%) and a victim (29.5%).

Engagement of young boys and girls in non-sexual risky behaviors The YAFSS showed only a slight decline in the number of young people who are current smokers, from 20.9 percent in 2002 – when the third YAFSS was conducted to 19.7 percent in 2013. However, NCR recorded the highest proportion of youth smokers at 26.6 percent. In relation to alcohol and drug abuse among adolescents, only national data is currently available.

Alcohol drinkers among adolescents reflected a significant decline from 41 percent to 37 percent, while a substantial decline was found in drug use, with only 4 percent of young Filipinos admitting to the act as compared to almost 11 percent in 2002.

Maternal and Infant deaths due to unintended pregnancies. From the World Health Organization (WHO) data, about 830 women die from pregnancy or childbirth-related complications around the world every day. Accordingly, it was estimated that in 2015, roughly 303,000 women died during and following pregnancy and childbirth. Almost all of

these deaths occurred in low-resource settings, and most could have been prevented.

In Metro Manila, in the 2012 data of DOH Philippine Health Statistics, 153 women/mothers have died due to complications related to pregnancy occurring in the course of labor, delivery and the puerperium, with the following breakdown; 15 for aged 15-19; 28 for aged 20-24; 39 for aged 25-29; 28 for aged 30-34.

Based on the 2013 NDHS report, about 98.7 percent of pregnancies in NCR received antenatal care from skilled health provider to which 90.9 percent of birth deliveries were also provided by a skilled health professional. Moreover, around 82 percent of the deliveries were held in a health facility. However, the 2014 FHSIS report of the DOH-NCRO indicated that maternal mortality ratio in the region was at 69 per 100,000 livebirths while infant mortality rate was 16 per 1,000 livebirths. Meanwhile, data for under-five mortality was available only in 2013 at 22 per 1,000 livebirths.

The non-use of any family planning contraceptives during the reproductive stage could result to unintended pregnancy. The 2013 NDHS found that 28 percent of pregnancies were mistimed and 11 percent were not wanted at all. These are very common in the Philippines that often lead to abortion and other reproductive complications. About 1.43 million pregnancies each year—nearly half of all pregnancies in the Philippines—are unintended. Furthermore, six in 10 Filipino women say they have experienced an unintended pregnancy at some point in their lives that increases the risk in their lives and health. And worse, some 54% of women who have ended an unintended pregnancy by abortion were not using any family planning method when they conceived.

Urban migration of labor force. Internal migration of people to the region remains as one of the major source of population growth in Metro Manila. Based on the 2010 Census of Population and Housing (CPH), around 284,000 individuals migrated to Metro Manila, from which, 200,000 are in-city migrants.

In NCR, the total population ages 15 years and over reached 8.78 million based on the 2016 Annual Labor Force Survey. Of these, 63.1 percent belong to the labor force. Of those belonging to the labor force, only 93.4 percent are employed. Employment rate in NCR is considered one of the lowest across regions. Thus, actual dependency ratio would be much higher since not everyone in the working age population are actually engaged in productive labor.

The influx of labor force, complemented by students from the neighboring regions are temporary movements that comprised the region's daytime population.

Magnitude of urban poverty. In the Philippines, the annual per capita food threshold in 2015 was set at ₱15,189 with NCR posting a higher food threshold of

₱17,462. On the other hand, the annual per capita poverty threshold (which includes basic non-food needs such as clothing, housing, transportation, health, and education expenses) was set at ₱21,753 for the country while NCR was pegged at ₱25,007.

Given this, poverty incidence among Filipinos in 2015 was estimated at 21.6 percent, lower than the poverty incidence of 25.2 percent recorded in 2012. Meanwhile, subsistence incidence, or the proportion of those whose incomes fall below the food threshold, was estimated at 8.1 percent in 2015, lower than the 10.4 figure posted in 2012.

For Metro Manila, poverty incidence among population in 2015 was posted at 3.9 percent, similar to 2012 figure. Subsistence incidence, however, was recorded at 0.7 percent.

Thus, in terms of magnitude, data showed an estimate of 3.75 million poor families or about 21.93 million poor individuals in the country. From this figure, Metro Manila holds its share of poor families at 2.1 percent (80,246) while the magnitude of poor population was estimated at 2.3 percent (494,630) of that of the country.

Strategic Framework

To contribute to the achievement of the Philippine Development Plan goal of laying a strong foundation for inclusive growth, a high-trust society, and a globally-competitive knowledge economy, the region should be able to maximize its potential growth. The region should be able to ascertain its contribution in the phenomenon of demographic transition, also keyed as the demographic window of opportunity, where low levels of fertility and mortality will be able to shift the population age structure to a point that the working age

population comprises the bulk. As the shift near towards this significant precondition may be earlier than the rest of the regions in the country, the region should aim towards maximizing the potential dividend for economic growth.

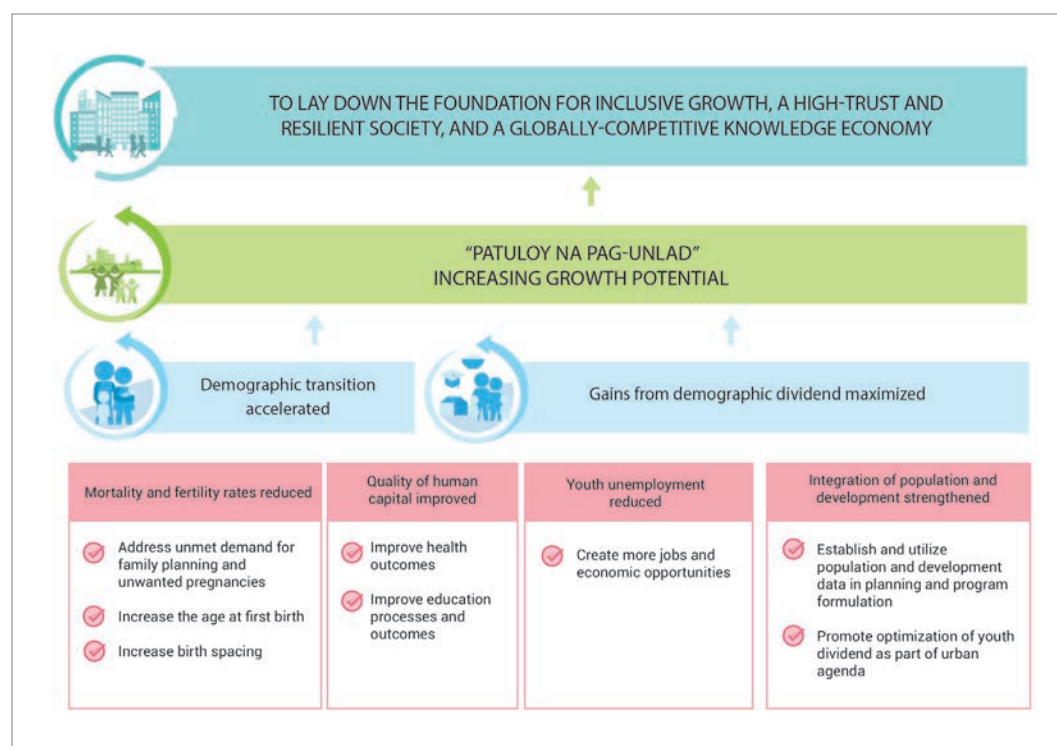
Targets

Table 10.5 shows the indicators and targets for the plan period 2017-2022 to maximize the demographic dividend.

Table 10.5. Plan Targets to Maximize the Demographic Dividend

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Population growth rate	2010-2015	1.58		
Increased life expectancy (in years)				
Male	2010-2015	68.8	Increased	
Female	2010-2015	75.6	Increased	
Reduced crude death rate per 1,000 population	2010-2015	5.1	Decreased	PSA
Decrease total fertility rate (children)	2013	2.3	2.1	NDHS
Increased contraceptive prevalence rate for modern FP (%)	2017	39.9	65	NDHS
Decreased unmet need for FP (%)	2017	12.3	Decreased	
Reduced proportion of adolescents aged 15-19 years old who have begun childbearing (%)	2013	17.7	9	YAFS
Decreased maternal mortality ratio per 100,000 live births	2013	76	70	FHSIS
Under-five mortality per 1,000 live births	2013	22.17	Decreased	FHSIS
Number of new HIV infections (newly diagnosed) decreased	2016	3,744	Decreased	

Figure 10.6. Strategic Framework to Maximize the Demographic Dividend



Strategies

In speeding up demographic transition and maximizing the potential gains from the demographic dividend, the following strategies will be pursued:

To reduce mortality and fertility rates

The government will continue to attain for a sustained universal health care program to contribute to the reduction of mortality rates among infants, young children and women. Meanwhile, the country's population management program and reproductive health policies and programs will be strengthened to effect a decrease in fertility level. The government and key stakeholders in the region shall work within the key strategy that will assist and enable couples and individuals to achieve their desired family size within the context of responsible parenthood and informed choice in order to achieve population outcomes that are facilitative of economic growth.

The government will assist couples and individuals in achieving their desired fertility intention as part of the promotion for responsible parenthood. For couples with fewer children, this equates to more savings from the expenses related to health, education, and other basic needs required for the total well-being of children. Interventions to reduce early and unintended pregnancies will positively impact on its economic implications. Financial inclusion strategies such as the efficient delivery of microfinance and micro-insurance products and services will be pursued as well. *See Chapter Six.*

Address unmet demand for family planning and unwanted pregnancies. Pursuant to Executive Order No. 12 on attaining and sustaining zero unmet need for modern family planning, the government and key stakeholders will work collectively

towards this particularly through the full and aggressive implementation of the Responsible Parenthood and Reproductive Health Act (RA 10354), the Magna Carta of Women (RA 9710), the National Population Policy (PD 79, s. 1972) and other relevant national interventions that promote the reproductive rights of Filipinos.

Demand-generation strategies and family planning services will be intensified with priorities given to the poor and vulnerable population groups. Advocacy for increased local government counterpart resources and funding as well as the establishment of service delivery networks will also be pursued to ensure improved access to the wide range of family planning methods and quality maternal and adolescent health care. Civil society and private sector engagements will be encouraged and tapped to augment efforts in the promotion of universal access to modern family planning methods. Men's involvement will also be fostered to support efforts for the reduction in unmet need for family planning and unintended pregnancies.

Increase the age at first birth. The region shall adapt strategies focusing on promotion of age-appropriate and comprehensive sexuality education among adolescents and pre-adolescents. This approach will contribute to increasing the age at first birth and empower young people to exercise responsible sexuality to avoid or prevent early and unintended pregnancies. Reproductive health education among young people will be pursued in different modalities in schools and communities. Social media, online networks and electronic platforms may be engaged for responsible sexuality education.

In compliance with the reproductive health law, age-appropriate reproductive health services will be provided to sexually-active adolescents and those who had their first birth in order to prevent unwanted

and repeated early pregnancies. The establishment of information and service delivery networks for adolescent health and development at the local level will be further promoted to ensure a continuum of care for young people. Likewise, the role of parents, teachers, and adults in guiding adolescent children toward responsible sexuality will be strengthened through institutionalized capacity-building. Partnerships with civil society organizations and youth organizations will be strengthened for a more coordinated implementation of adolescent health and development interventions.

Increase birth spacing. Lowering desired number of children with proper birth spacing of three to five years will be promoted to reduce fertility levels among women as part of the integrated promotion of reproductive health and family planning interventions. This will contribute to the increase in women's productive participation in the labor force and in human resource formation activities.

To improve quality of human capital

Quality human capital resource is essential for optimizing the increasing labor force for more productive economic output. It requires investments and improvements in health, education, and total wellbeing of the human capital. *See Chapter 7.*

Improve health outcomes. Increased investments in health initiatives contribute to human capital formation or development by ensuring that children grow healthy and educated, that members of the labor force are productive, and members of the family can enjoy the fruits of their labor. The government will therefore: (a) guarantee services that will provide care at all life stages; (b) ensure the accessibility of these services in functional delivery networks;

and (c) sustainably finance these services through universal health insurance.

Improve education processes and outcomes.

Maximizing the potential gains from a demographic transition requires educated and skilled labor force. It is key that reforms in the basic and higher education curriculum will be able to adapt to current realities and needs. The K to 12 curriculum integrates lessons on sexuality education and essential life skills to equip students to make informed choices and decisions on issues that affect their wellbeing. The new general education curriculum in higher education aims to develop learners' intellectual competencies, personal and civic competencies, and practical skills. Substantive investments to improve educational services and facilities are entailed along this end.

Likewise, interventions to address issues on education will also dovetail with initiatives to improve school participation especially among girls to prevent young and unintended teen pregnancy and delay marriage. Women who marry later tend to have fewer children than those who marry at a young age.

To reduce youth unemployment

Create more jobs and economic opportunities. Optimizing the use of a substantive number and quality human resource through job creation in various sectors will be facilitated by the government in collaboration with the private sector. The establishment of amore favorable climate for expanded trade and investment that is environmentally sound; greater investment in human resource development; development of democratic institutions; and good governance will have to be realized and supported. Policies and program initiatives will be made for the creation of productive jobs as well as for transfer of relevant skills and modern technology.

Attuned to the administration's thrust to create more jobs, the government will pursue initiatives through major employment-promoting strategies such as employment generation, employment preservation, employment facilitation, and employment enhancement. Thus, programs supporting employment generation have been pursued to directly or indirectly create new employment opportunities in the domestic labor. Employment programs for the youth will also include support to young parents as part of social protection. Improving the living wage of young workers, especially those with young children to support, will be prioritized.

To strengthen integration of population and development (PopDev)

Poverty and serious social and economic inequities have a significant influence on, and are in turn influenced by, demographic parameters such as population growth, structure and distribution. To ensure that demographic dynamics particularly those related to demographic dividend are considered in sustainable economic and other development initiatives, the integration of population factors in all development initiatives will be promoted.

Establish and utilize population and development data in planning and program formulation. Demographic data needed for planning and formulating appropriate interventions to optimize demographic dividend will be strengthened. The use of population data will be promoted particularly in planning interventions in health, education, employment, infrastructure, and economic development. Development strategies will reflect the short-, medium-, and long-term implications of, and consequences for, population dynamics as well as patterns

of production and consumption. Specific strategies will include the following:

- Capacity building of regional and local government institutions in establishing and utilizing a demographic and socioeconomic database in development planning. This includes strengthening the local survey and information systems for demographic and socioeconomic data to be conducted by local population and planning offices. The survey and information systems will support local development planning and program development initiatives.
- Improving the knowledge base on population and development interrelationships through continuing research and studies, including those related to daytime population and population movement.
- Building public awareness and political commitment to address population issues that have implications on development concerns.

Promote optimization of youth dividend as part of urban agenda. The prospect of demographic dividend is more pronounced in the NCR where the economic opportunities thrive enticing more youth workers to move and work in the metropolis. As a highly urbanized region, its pull factor is economically-motivated.

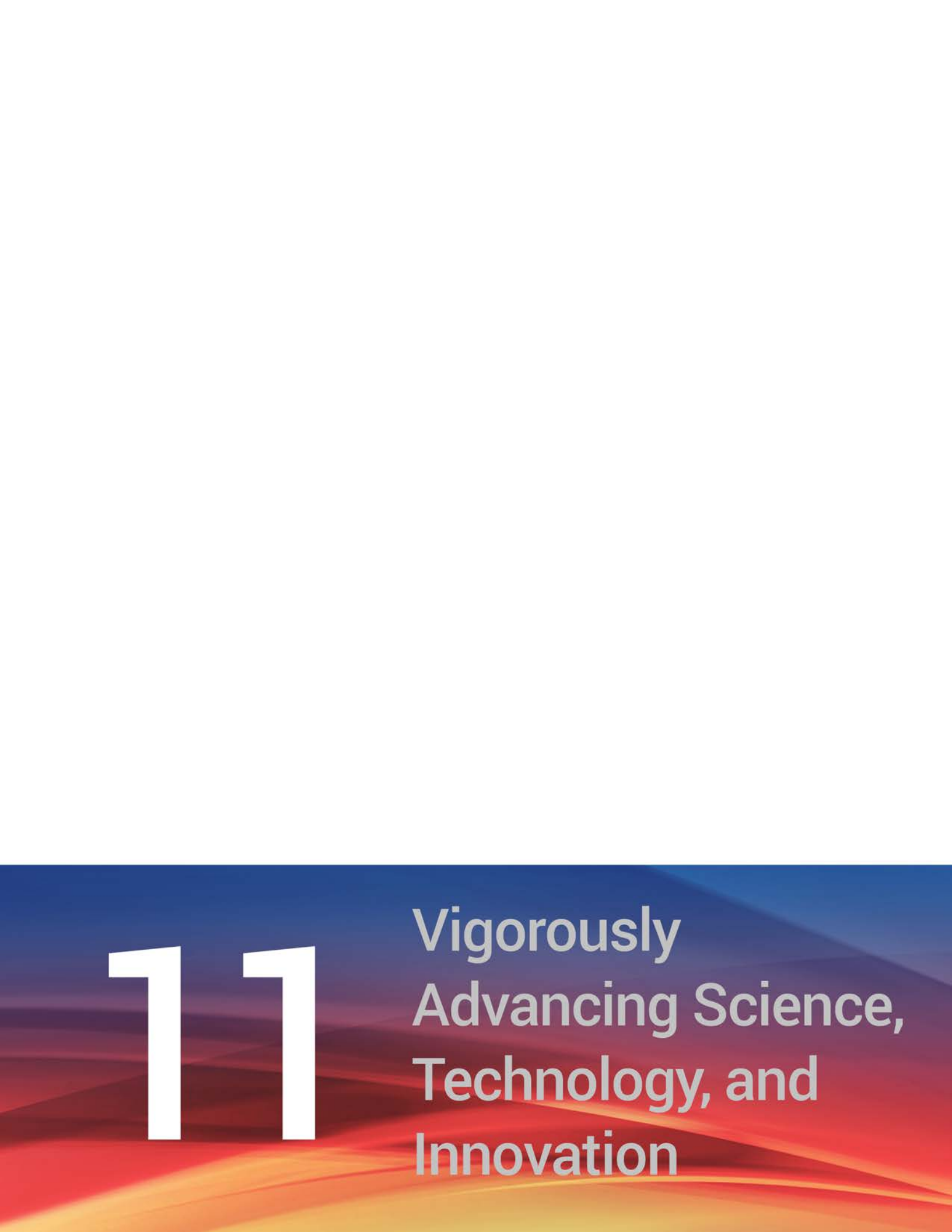
Initiatives to optimize demographic dividend entails improving the capacities of the localities to employ the bulk of human resources that will enter the region. This involves a review of urban management initiatives to integrate strategies for the optimization of demographic dividend as an urban agenda. Demographic interventions will also be developed to address its effects within the region and contiguous areas.

Legislative Agenda

Legislative support to maximize demographic dividend, which is anchored on the national legislative agenda of the Philippine Development Plan, will be essential for the effective implementation of the strategies.

Table 10.6. Legislative Agenda to Maximize Demographic Dividend, 2017-2022

LEGISLATIVE AGENDA	RATIONALE
Institutionalization of Local Population and Development Policy	There is a need to have a cohesive policy in addressing population and development at the local level.
National Policy on the Prevention of Teenage Pregnancy	The alarming levels of teenage pregnancy requires a comprehensive policy to address the problem at all levels of government.
Strengthening of the National Population and Development Policy	A comprehensive policy for population and development will address the fragmentation of government efforts and initiatives for population and development. It will ensure that population and development is mainstreamed.



11

Vigorously
Advancing Science,
Technology, and
Innovation

Vigorously Advancing Science, Technology, and Innovation

The application of Science, Technology, and Innovation (STI) plays an important role in economic growth and development.

In 2017, Metro Manila ranked 19th most dynamic city among 134 major business hubs across the globe in the fourth annual City Momentum Index (CMI) released by global property consulting firm Jones Lang Lasalle (JLL). Appearing for the first time in the Global Top 30, Metro Manila was cited to have a 'particularly dynamic office market supported by a growing and globally competitive BPO sector'.

Furthermore, JLL described Metro Manila as one of the 'Emerging Megacities', along with Delhi, Mumbai, and Chennai, which are 'leveraging their dynamic labour markets to drive momentum' and whose 'growth in domestic corporations and foreign direct investment (FDI) into areas such as business process outsourcing (BPO) is helping fuel one of the fastest economic growth among the CMI cities'. However, it was pointed out that these cities face 'significant infrastructure and quality of life issues, with high levels of inequality, congestion and pollution hindered by weak city governance'.

As such, there is a need to increase the region's potential growth through STI, which will pave the way to the Metro Manila Greenprint 2030 vision of building a globally competitive knowledge economy, providing world-class services, and establishing Metro Manila as a smart metropolis.

Assessment and Challenges

Weak Science Technology and Innovation Culture. The Philippine government believes that achieving *AmBisyon 2040* is based on a solid foundation for inclusive growth, a high-trust society, and a globally-competitive knowledge economy by anchoring its development thrusts on *Malasakit*, *Pagbabago*, and *Patuloy na Pag-unlad*. A globally competitive-knowledge economy will only be possible on a society relying firmly on science, technology and innovation.

Studies have shown that a strong science, technology and innovation culture should be imbedded in well-established

social institutions and networks of the government, industry, academia and civil society. When all these sectors act together, they drive one another as collaborative, synergistic, and innovative catalysts of social and economic growth. The government provides the necessary infrastructure, investment and enabling environment; the academe generates highly skilled human capital in the fields of science, technology, engineering, agriculture and mathematics; while together, the academe and private and government research and development institutions provide a pool of multi-disciplinary experts for research, knowledge creation and technology development, and

knowledge and technology transfer. The industry sector, meanwhile, provides the economic engine by capitalizing on their inherent risk-taking characteristic, and investing on and utilizing the knowledge, processes, products and technologies developed by the academia and the research and development institutions. The successful collaboration among these innovation actors, however, is contingent on a high level of cooperation, coordination, information sharing and mutual trust (RTI International, 2014; Wilson, 2012).

The 2017 Global Innovation Index (GII), co-published by “*Institut Européen d’Administration des Affaires*” (INSEAD) or European Institute of Business Administration, Cornell University and the World Intellectual Property Office, ranked the Philippines 73rd out of 127 countries, one rank higher than its 2016 74th finish. However, the country lagged behind its ASEAN neighbours, with Singapore (ranked 7th), Malaysia (ranked 37th), Thailand (ranked 51st) and Brunei (ranked 71st). Among the seven pillars of the GII, science and technology in the Philippines played a big role in human capital and research, infrastructure, business sophistication, knowledge and technology

outputs, and creative outputs. The most significant progress was in Business Sophistication, where the ranking rose from 113th place in 2014 to 45th place in 2017. Business Sophistication Pillar is measured in terms of knowledge workers, innovation linkages, and knowledge absorption, one that was indicated as our strength. Still, the government recognizes that creating innovation linkages for the academe and the market remains a challenge.

Meanwhile, the World Economic Forum Global Competitiveness Report of 2016 to 2017 (*Table 11.1*) showed that the Philippines ranked 62nd out of 138 countries in the innovation pillar (12th measurement pillar of global competitiveness index), trailing its ASEAN neighbours of Singapore, Malaysia, Thailand and Indonesia. The Philippines ranked low in terms of quality of scientific research institution, industry-university collaboration, availability of scientists and engineers, Patent Cooperation Treaty (PCT) patent applications, and government procurement of advanced technologies.

These point to a weak science, technology and innovation culture brought about by the following factors:

Table 11.1. Global Competitiveness Report, 2016-2017

ASEAN COUNTRY (2017 RANKING)	PILLAR 12: INNOVATION	12.01: CAPACITY FOR INNOVATION	12.02: QUALITY OF SCIENTIFIC RESEARCH INSTITUTION	12.03: COMPANY SPENDING ON R&D	12.04: INDUSTRY-UNIVERSITY COLLABORATION ON R&D	12.05: GOVERNMENT PROCUREMENT OF ADVANCED TECHNOLOGIS	12.06: AVAILABILITY OF SCIENTISTS AND ENGINEERS	12.07: PCT PATENT APPLICATIONS
Singapore	9	20	10	15	7	4	9	13
Malaysia	22	13	23	8	11	3	7	36
Thailand	54	70	56	46	41	65	57	70
Indonesia	31	32	41	26	28	12	38	99
Philippines	62	41	72	44	61	74	78	86
Bruenei	78	100	79	102	84	41	80	53
Vietnam	73	79	98	49	79	27	84	95

Source: World Economic Forum

Low Research & Development Expenditure.

The Gross Domestic Product (GDP) in the Philippines was worth ₱14.48 billion in 2016. Based on the latest data released by the Philippine Statistics Authority (PSA), National Capital Region (NCR) continued to have the largest share of the country's GDP at 38.1 percent in the same year.

Looking at the country's Gross R&D Expenditure (*Table 11.2*), there was a significant growth from 2009 to 2013, an increment of 30 percent from 2009 to 2011, and 40 percent from 2011 to 2013. Accordingly, the R&D Expenditure of NCR doubled in 2011 and increased by 9.17 percent (₱8.07 billion) in 2013, which was 51 percent of the country's Gross R&D Expenditure. This supports the fact that research institutions are

concentrated in NCR.

Evidently, the progress in the region's R&D expenditure is attributed to the increased spending of public higher education institutions (HEIs) and government, plus the huge share of private industries despite its abrupt decrease in spending from 2011 to 2013. The public sector has now become the biggest R&D spender replacing the private sectors.

The government through the Department of Science and Technology (DOST), Commission on Higher Education (CHED) and other agencies continue to expand its coverage of research, development and innovation programs, primarily on S&T human resource development,

Table 11.2. R&D Expenditure in the Philippines and NCR, 2003 - 2013

INDICATOR	2003	2005	2007	2009	2011	2013
PHILIPPINES						
GDP (in current prices/ in million pesos)	4,316,402	5,444,038	6,648,619	7,678,917	9,708,332	11,548,191
R&D Expenditures (in current prices/ in million pesos)	5,910	6,363	7,556	8,779	11,384	15,915
RDE/ GDP (%)	0.14	0.12	0.11	0.11	0.12	0.14
NATIONAL CAPITAL REGION						
Regional GDP (in current prices/ in million pesos)	1,556,808	2,015,929	2,478,589	2,871,470	3,461,448	4,294,857
R&D Expenditures (in current prices/ in million pesos)	2,198	3,378	3,548	3,845	7,391	8,069
RDE/ RDGP (%)	0.14	0.17	0.14	0.13	0.21	0.19
Public R&D Expenditures (in current prices/ in million pesos)	790	1,172	1,326	1,375	3,061	5,400
% to total RDE	36%	35%	37%	36%	41%	67%
Government	597	615	926	879	1,108	2,335
Public HEIs	193	557	400	496	1,953	3,066
Private R&D Expenditures (in current prices/ in million pesos)	1,408	2,206	2,221	2,470	4,330	2,668
% to total RDE	64%	65%	63%	64%	59%	33%
Private HEIs	88	195	255	255	548	447
Private Non-Profit Institution	78	43	53	60	29	110
Private Industry	1,243	1,967	1,913	2,155	3,753	2,111

Source: Philippine Statistical Authority and Department of Science and Technology

R&D capacity building, improvement of industry competitiveness, health, food and nutrition, renewable energy, climate and environmental sciences, biotechnology, and DOST new programs on space technology, ICT development and artificial intelligence. Consequently, public HEIs continuously take advantage of the government's and other institutions' research grants. The priority R&D spending for public HEIs and government was on exploration and utilization of natural resources. On the other hand, Private Industries spent more on developing new products and services prompted by the pressure of globalization and competition in the international market.

Research activities from private HEIs was found to be consistently low among the sectors, contributed only 7.4 percent and 5.5 percent respectively in 2011 and 2013 total R&D expenditure in the region.

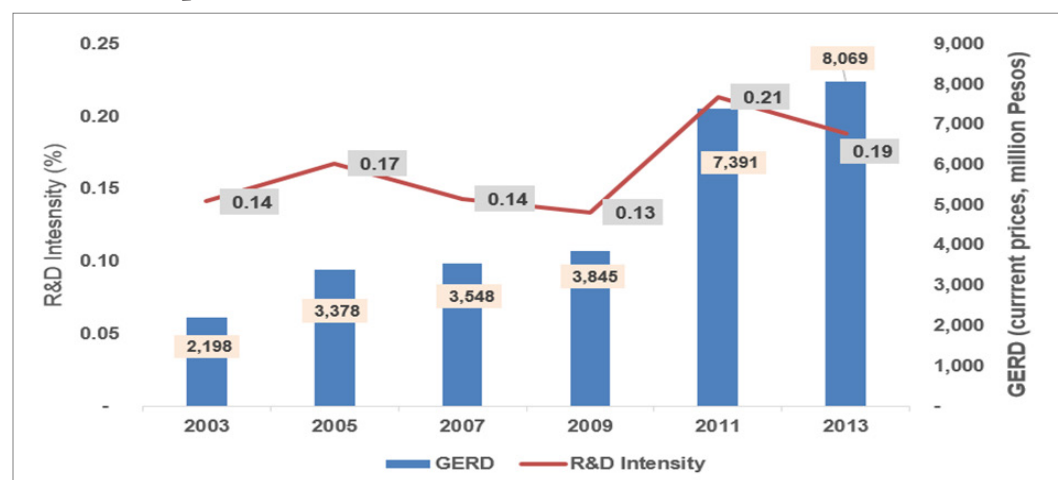
NCR's gross expenditures on R&D (GERD) as percentage of the gross regional domestic product was pegged at 0.21 percent and 0.19 percent in 2011 and 2013 respectively. These were also the highest within the 10-year period (*Figure 11.1*). However, this trend in R&D expenditures still implies

underinvestment based on the United Nations Educational Scientific and Cultural Organization's (UNESCO) recommendation of one percent of GDP. Further, this is lower than the R&D expenditures of neighbouring ASEAN countries such as Singapore (2% in 2012), Malaysia (1.13% in 2012), and Thailand (0.39% in 2011) but equal to that of Vietnam (0.19% in 2011).

Limited S&T Human Capital and Inadequate STI Infrastructure. From the R&D Survey conducted by DOST, R&D personnel are categorized into researchers, technicians and auxiliary personnel. In 2013, a total of 13,107 or 36 percent of the country's R&D personnel are based in the NCR, in which 69 percent (9,067) are researchers, scientists and engineers (RSE). Remarkably, 66 percent (8,698) of the total R&D personnel in the region are in private industries (*Table 11.3*).

According to CHED data as of June 2017, there are 317 private HEIs and 28 public HEIs in the region. However, only 2,197 (17%) of the total R&D personnel are involved in research activities. This low ratio of R&D personnel per HEI can be attributed to the lack of funding and the

Figure 11.1. R&D Intensity (R&D expenditures as % of Gross Domestic Product) and Gross Expenditure on R&D in NCR, 2003- 2013



Source: Department of Science and Technology

restrictive HR policies of confining faculty to teaching functions. There is also difficulty in increasing opportunities and retaining S&T human capital because of the inadequate merit system of rewards and incentives resulting to brain drain.

A study conducted by the USAID and RTI International titled Science, Technology, Research and Innovation for Development (USAID-STRIDE), Philippine Innovation Ecosystem Assessment noted that the quality of STEM-related trainings is acceptable by global standards. However, the study also noted that STEM (Science, Technology, Engineering and Mathematics) graduates are perceived as lacking exposure to current best practices and to the creative possibilities of technology which translates to the need for further training for technical employees hired in multinationals.

Although there is a high number of potential and competent R&D personnel in the region, opportunities are limited by lack of adequate or state-of-the-art laboratories (equipment & expertise), related facilities (library access to international technical publications and internet connectivity) and availability of sufficient private or public funding. This was validated by the USAID-STRIDE report which noted that “public institutions failed to provide equipment

packages for young researchers particularly those returning from PHDs abroad with more advanced research agendas.” (RTI International, 2014).

Further, many local companies that would hire technical personnel cannot compete with the salaries offered by overseas employers or the BPO. At the same time, Filipino scientist trained abroad choose to remain abroad where there is higher demand and higher pay, and access to superior laboratories, equipment and expertise (RTI International, 2014).

Given the crucial role and responsibilities of HEIs in: inculcating science, technology and innovation as development tools; and, as generators of knowledge and skilled manpower, there is a need to closely align undergraduate, graduate and post-graduate curriculum with actual industry needs and intensively invest on necessary infrastructures and improve the research capabilities of HEIs faculty, research staff and graduate students.

Weak Linkages among Players in the STI Ecosystem. The Philippines ranked 59th out of 127 countries in the Global Innovation Index 2017 in terms of University-Industry Research Collaboration while the Global Competitiveness Report 2017 of the World

Table 11.3. R&D Personnel in the Philippines and NCR by Sector, 2013

SECTOR	PHILIPPINES		NCR	
	R&D PERSONNEL	RSE	R&D PERSONNEL	RSE
Government	3,774	2,965	2,019	1,486
Public HEIs	7,646	7,144	1,406	1,286
Private HEIs	2,543	2,364	791	748
Private Non-Profit Institution	227	179	193	154
Private Industry	22,327	13,843	8,698	5,393
Total	36,517	26,495 (72%)	13,107	9,067 (69%)
RSE per million population		270		

Source: Department of Science and Technology

Table 11.4. Sources of information and cooperation for innovation active firms

INFORMATION SOURCE	SIZE OF ENTERPRISE (PERCENTAGE OF ALL ENTERPRISES)					
	MSME		LARGE		ALL	
	NATIONAL	QUEZON CITY	NATIONAL	QUEZON CITY	NATIONAL	QUEZON CITY
1. Internal						
a. Within your establishment or enterprise	67%	52%	75%	75%	71%	61%
2. Market Source						
a. Suppliers of equipment, materials, components, or software	50%	41%	49%	50%	49%	45%
b. Clients or customer	65%	59%	68%	65%	66%	61%
c. Competitors or other enterprise in your sector	40%	34%	35%	40%	38%	37%
d. Consultants, commercial laboratories, or private R&D institutes	21%	31%	22%	20%	21%	27%
3. Institutional Source						
a. Universities or other higher education institutions	11%	21%	9%	10%	10%	16%
b. Government or public research institutes	8%	17%	6%	5%	7%	12%
4. Other Sources						
a. Conferences, trade fairs, exhibitions	28%	34%	15%	25%	22%	31%
b. Scientific journals and trade/ technical publications	19%	17%	15%	25%	17%	20%
c. Professional and industry associations	18%	21%	14%	15%	16%	18%

Source: Philippine Institute for Development Studies

Economic Forum ranked the Philippines 61st out of 138 countries in terms of the same parameter. This implies a weak collaboration between the academic institutions, which are the generators of knowledge and technology, and the industry, which are the knowledge and technology takers.

Further, a study conducted by DOST and Philippine Institute of Development Studies (PIDS) through the funding support of International Development Research Center (IDRC) which measured the innovative activities of firms in selected pilot sites namely Quezon City, Cebu City, Davao City and Laguna (PEZA) showed that knowledge networks are weak and that firms do not access the technical assistance from the government, research institutions and the academe.

In particular, only 21 percent and 17 percent of the innovation active firms in Quezon City listed universities or HEIs and government or public research institutes as their sources of information and cooperation respectively

(Table 11.4).

Moreover, R&D thrusts and outputs of academe and government RDIs are not aligned with the needs of the industry sector. Most of the publicly funded R&D outputs are not commercialized for lack of ready end-users or adoptors. There is also lack of information and access among the academe and industries in finding the right collaborative platform and partners. There is a need for closer and stronger collaboration among government, academe and industry to ensure the relevant and sustainable impact of STI on society and the economy. Further, researchers from the academe are often more concerned on publishing research results.

Need to Improve the Technology Absorptive Capacity and Technology Proactivity among Micro, Small and Medium Enterprises (MSMEs). Of the total establishments in the country, 99.5 percent are MSMEs, in which 21.2 percent or 190,166 MSMEs are operational in the NCR

(DTI, 2015). From 2003-2017, the DOST Small Enterprise Technology Upgrading Program (SETUP) assisted 308 MSMEs through technology transfer and equipment upgrading, and 13,704 MSMEs through technical consultancies and trainings. This minimal percentage of assisted MSMEs invokes the need to increase STI awareness.

SETUP is one of the flagship programs of the DOST to enhance the productivity and competitiveness of MSMEs through technological innovations and upgrading of products and processes towards progress and growth in the countryside.

SETUP supports MSMEs by providing: (1) seed fund for technology acquisition, (2) needed equipment and equipment upgrading, (3) technical trainings and consultancy services, (4) packaging and label design, (5) database information systems, and (6) support for establishment of product standards, including testing, and calibration of equipment.

Assisted MSMEs were found to be highly dependent on government's interventions to maintain and implement continuous quality improvements and innovation because of inadequate technical know-how, resources and financial capability. There is a need to translate technologies developed by RDIs into appropriate technologies using locally designed, fabricated and cost efficient equipment.

Government RDIs and academic institutions should develop their technologies side by side with the required equipment prototypes (with accompanying design for scale-ups) to facilitate the transfer of the technology and eventual commercialization.

Cooperation among MSMEs should be explored in co-dependent manufacturing utilizing intermediate products amongst themselves and the establishment of support industries to supply their requirements.

Thus industry clustering, sectoral linkages and marketing networks should be established amongst them.

Advantages

The prime educational institutions (public and private) and government R&D institutions are located in the region. 18 percent of the HEIs in the country are located in the NCR. Moreover, 25 percent of the top 259 HEIs in the country, as ranked by the Spanish National Research Council, are located in the NCR. This makes the region a possible locus for knowledge and technology transfer not just for the needs in the region but in the whole country as well. However, a huge disparity in capabilities exists among these institutions in terms of expertise and facilities.

A complementary research collaboration approach among these institutions is necessary to address the gaps and overcome the disparities. This could be done in cooperation with the manufacturing industries within the region and its immediate suburbs.

The NCR is the central trading hub for most raw materials, supplies and other resources making it an ideal site for collaborative R&D.

Disadvantages

Traffic and housing congestion limits the mobility and convenience of STI practitioners/ providers thus preventing an effective and efficient generation, transfer and utilization of technologies. Improved and efficient mass transport system coupled with stricter vehicle regulation would address the limited mobility and man-hours wasted on traffic gridlocks. Smarter urban planning and zoning could provide affordable housing for transients and semi-permanent residents thru the creation of dormitories with full amenities and public

lounging areas (digital libraries and internet hubs with free wi-fi connection).

With the high concentration of competent STI manpower in the region, the environment for the conduct of STI activities becomes highly competitive with restrictive

opportunities given the limited available funding resources and facilities to the great disadvantage of the development process and the potential end-users/adoptors of technology. This in turn contributes to the brain drain problem.

Strategic Framework

Following the framework of PDP 2017-2022, this RDP chapter will contribute to achieve the societal goal of establishing the "foundation for inclusive growth, a high-trust and resilient society and a globally competitive knowledge economy", thus achieving the long term vision of *Matatag, Maginhawa at Panatag na Buhay*. R&D endeavours and other S&T based projects

will focus on: Promoting and Accelerating Technology Adoption and Stimulating Innovation.

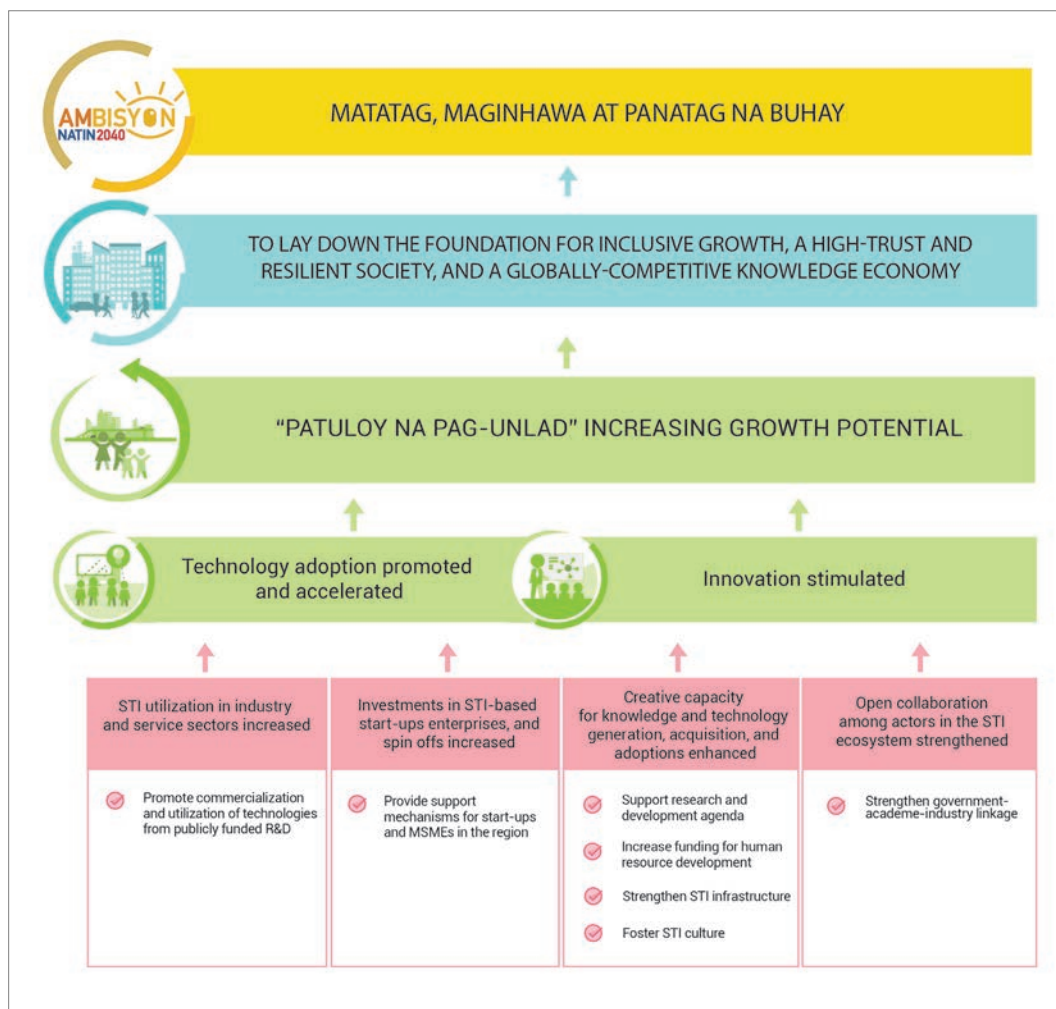
Targets

Over the next six (6) years, the government will aim to achieve the targets indicated in the following table (*Table 11.5*).

Table 11.5. Plan Targets to Leverage Science, Technology, and Innovation

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	Year	VALUE		
Improved rankings among ASEAN Global Competitiveness Report	2016 - 2017	62nd	Improved	
RSE per million population	2013	270 (nat'l)	Improved	DOST
Number of R&D personnel	2013	13,107	Improved	DOST
Number of researchers, scientists, and engineers	2013	9,067	Improved	DOST
R&D expenditure as a proportion of GDP	2013	0.19	Improved	DOST
Number of patents applied	2016	123	Increasing	IPOPIL
Number of patents registered	2016	18	Increasing	IPOPIL
Number of utility model applied	2016	279	Increasing	IPOPIL
Number of utility model registered	2016	252	Increasing	IPOPIL
Number of industrial design applied	2016	552	Increasing	IPOPIL
Number of industrial design registered	2016	378	Increasing	IPOPIL

Figure 11.2. Strategic Framework to Vigorously Advance Science, Technology, and Innovation 2017-2022



Strategies

By 2022, the regional development thrusts shall employ the following strategies:

To increase STI utilization in industry and service sectors

Promote commercialization and utilization of technologies from publicly funded R&D.

The government will intensify its efforts in promoting publicly funded R&D outputs to increase commercialization and utilization of technologies. Adoption of appropriate S&T interventions will be increased to enhance industry productivity and competitiveness.

Development of industry priority areas or industry clusters, sectoral linkages, marketing and promotions networks will be fostered to undertake effective production and marketing strategies. This can also be done through the extensive use of quad-media, conduct of regular technology-investment fora, S&T caravan and fairs.

To increase investments in STI-based startups, enterprises and spin-offs

Provide support mechanisms for startups and MSMEs in the region. Policy and regulatory environment should be improved, simplified and tailored for MSMEs to encourage investment and market expansion.

Most start up enterprises and operational MSMEs are equipped with obsolete and inefficient machineries due to lack of funding. Strengthened implementation and promotion of the DOST's Small Enterprise Technology Upgrading Program will increase the number of assisted MSMEs adopting technological innovations to upgrade products, processes and operations.

The government shall provide more platforms for technology commercialization such as Technology Business Incubators and Innovation Hubs, and mechanisms for productivity and competitiveness such as technical consultancy services (technology needs assessment, food safety assessment, energy audit, resource efficient and cleaner production) and packaging and labelling assistance.

Also, an example is the Food Innovation Facility (FIF), housed in Diliman, Quezon City at the UP College of Home Economics, which is primarily for the development of product prototypes, process optimization and the limited production of samples for preliminary market testing. FIF also serves as instructional facility for students, industry & other government agency personnel and private individuals.

To enhance creative capacity for knowledge and technology generation acquisition and adoption

Support research and development agenda.

Regional R&D Agenda will be formulated and institutionalized thru the Regional Research, Development & Innovation (RRDI) Committee of the RDC to ensure that all researches, innovations and other S&T based projects are relevant with the regional development thrusts. Research priorities identified shall address the pressing needs of the various sectors and the identified priority industry sectors in the NCR. There shall be sufficient funding support for the implementation of R&D projects aligned with the Regional R&D Agenda.

Increase funding for human resource development. By 2025, the government aims to achieve the UNESCO's recommended 380 RSE per million population. Though a high number of RSE is represented by NCR,

the government shall continue capacitating R&D personnel to be more competent and globally competitive. This can be done through continuous support and funding to ongoing S&T scholarships and providing specialized trainings.

The DOST's Grants-In-Aid (GIA) program aims to harness the country's scientific and technological capabilities to spur and attain a sustainable economic growth and development.

The GIA program provides grants for the implementation of programs/projects identified in the current DOST priorities and thrusts such as (1) funding of research and development, promotion, technology transfer and utilization, human resources development, information dissemination, advocacy, and linkages; (2) pilot testing of new or scaled-up invention and potential application in Metro Manila; (3) enhancement of community based institutions and improvement of quality of life in disadvantaged areas through application of S&T based interventions and various capacity building activities under CEST program; and (4) enhancement of instructional and engineering capacities and facilities of SUCs to facilitate R&D activities through equipment and facility upgrading.

Furthermore, the government should create more conducive work environment, promote and provide more incentives for research as career. Raising the rewards-incentives scale in R&D systems will create a more appealing opportunities to restrain migration of highly trained and competent R&D personnel. Support to the Balik Scientist program shall be intensified to tap the expertise of overseas Filipino scientists in providing scientific solutions and technological advancement in the region. Their international linkages could be an advantage for collaborative undertakings.

Strengthen STI infrastructure. Niche Centers in the Region for Research and Development (NICER) will be established with support funding from DOST. These centers shall be aligned with the R&D priority agenda crafted by the RRDI Committee of the RDC. Further, they shall function under the spirit of collaboration with other HEIs and sectoral institutions in the region.

Upgrading of existing R&D facilities and establishment of R&D center aligned with the industry needs of the region by capacitating HEIs. This will promote conduct of collaborative research, and technology adoption among industries to accelerate competitiveness.

The Science for Change Program aims to accelerate the R&D and Human Resource Development programs of DOST thru (1) Niche Centers in the Regions for R&D (NICER) Program, (2) R&D Leadership (RDLead) Program, (3) Collaborative R&D to Leverage PH Economy (CRADLE) for RDIs and Industry Program, (4) Business Innovation through S&T (BIST) for Industry Program. The program aims to capacitate Higher Education Institutions in the region to undertake quality researches that will bridge the gaps and challenges of regional socio-economic development.

Adoption of the Open Laboratory and Library Policy among the academic institutions and government RDIs could hurdle lack of facilities and access to current information. Such access could be restricted to R&D projects screened and approved by a committee panel comprised of representatives from these institutions.

The government shall also establish, equip and promote the utilization of product development centers, material and products testing facilities, disaster risk reduction laboratories, innovation hubs, shared service facilities, and other similar mechanisms.

Foster STI culture. A culture of inventiveness and creativity among children and young adults shall be promoted through integration of the Invent School Program into the curriculum of primary and secondary educational institutions. Hence, students will be encouraged to pursue STEM courses. The government should intensify its support programs such as scholarship, financial assistance on research studies and specialized trainings.

There should be regular conduct of technology - investment fora, technology pitching, and other promotional activities to increase public awareness and appreciation on STI.

Moreover, the government should also provide avenues on raising awareness on the importance of intellectual property protection, providing assistance and consultancies on intellectual property filing.

To strengthen open collaboration among players in the STI ecosystem

Strengthen government-academe-industry linkage. To engage R&D personnel in HEIs and foster collaborative R&D activities with the government and industries, career advancement programs of HRDs should put more credit on technologies commercialized than the number of publications derived from R&D.

A closer and stronger collaboration among the three sectors can be attained by invigorating a wide sectoral inclusive R&D consortium thru the supervision of RRDI committee of the RDC. This will ensure that R&D activities will be undertaken in partnership with private industries so that R&D outputs will be adopted and commercialized by industries. Interagency collaboration will promote commensurate sharing of resources, compensation and benefits among members.

The government should create avenues on collaborative opportunities between the universities and the industry to pose possible commercial and market use of their patent portfolios.

The DOST's Collaborative Research and Development to Leverage Philippine Economy (CRADLE) program for R&D funding which aims to have a synergistic relationship between the industry and the academe by having the private sector identify the problem and the HEI provide the research and development for the technological solution shall be one of the platforms for the increased linkage.

Likewise, a database of R&D projects and initiatives of government and HEIs shall be maintained to capacitate the RRDI committee of the RDC in setting policy directions to stimulate an innovation driven economy in the region.

PART V

FOUNDATIONS FOR SUSTAINABLE DEVELOPMENT



12

Ensuring Security,
Public Order, and
Safety

Ensuring Security, Public Order, and Safety

Ensuring the security, peace and order situation in Metro Manila is among the top priorities of the government because of its impact on economic growth and development not just of the region but of the entire country. The impact is evident in the areas of local and foreign investments, tourism and employment opportunities. The region's stable situation in ensuring security will lead to social order notwithstanding differences on cultural beliefs and political ideologies of the people.

Metro Manila aims to be a safe, drug-free and crime-free metropolis. In order to achieve this, law enforcement agencies are harmonizing their efforts in maintaining public safety, and peace and order in Metro Manila.

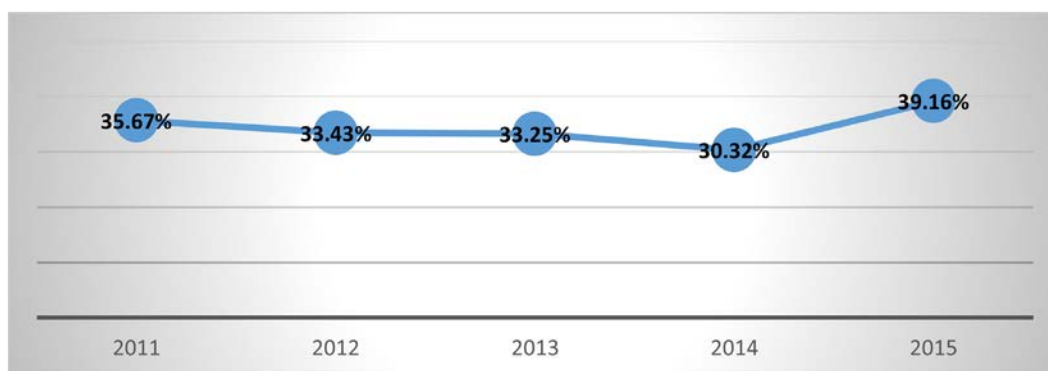
Assessment and Challenges

Low crime solution efficiency. Despite the increase in the crime solution efficiency in index crimes from 2011-2015 in the region, the efficiency rate in solving crime is still low but can be considered a good mark (Figure 12.1).

Insufficient police personnel. As of June 2016, there are 29,584 personnel from the four (4) different police districts,

as well as regional headquarters and other special Philippine National Police (PNP) units distributed all over the metropolis. There are 3,696 personnel in Northern Police District (NPD), 2,764 in Eastern Police District (EPD), 4,339 in Manila Police District (MPD), 4,835 in Quezon City Police District (QCPD) and 5,371 in Southern Police District (SPD). There are 685 in the regional headquarters, 1,037 in

Figure 12.1. Crime Solution Efficiency Rate (Index Crime): National Capital Region, 2011-2015



Regional Personnel Holding Accounting Unit (RPHAU), 95 in PPSDU, and 4,342 for those undergoing Public Safety Field Training Program (PSFTP) (*Table 12.1*).

Table 12.1. Number of Police Personnel in Metro Manila, 2016

HEAD-QUARTERS	NUMBER OF PNP PERSONNEL	POLICE TO POPULATION RATIO
NPD	3,696	1:740
EPD	2,764	1:592
MPD	4,339	1:380
SPD	5,371	1:669
QCPD	4,835	1:593
RPSB	2,420	-
RPHAU	1,037	-
PPSDU	95	-
PO1 (under-going PSFTP)	4,342	-
RHQ	685	-
TOTAL	29,584	1:475

Source: NCRPO

In terms of police to population ratio within the region, in 2016, the ratio was recorded at 1:475 which is within the standard of 1:500. However, when comparing per district, it should be noted that four (4) districts fall below the standard ratio i.e., EPD has the ratio of 1:592, QCPD has the ratio of 1:593, SPD has the ratio of 1:669, and NPD has 1:740. Only MPD has the ratio of 1:380, way above the standard.

Inadequate proper crime prevention and analysis equipment. Despite having achieved the firearm to police officer ratio of 1:1 in

the previous years, the lack of proper crime prevention and analysis equipment and state-of-the-art technology is a major challenge in the police force. The police force particularly in the different districts can be more efficient in crime prevention and solution if armed with proper equipment, such as closed circuit television (CCTV) cameras, dashboard cameras and body cameras and monitors as well as digital maps with satellite image overlays. To better serve and protect the community, the NCR police force also needs additional prowler cars, motorcycles, radio equipment and ammunition, among others, that would help improve their performance in their war against crime.

Fire incidences are increasing despite the efforts on fire prevention and suppression.

With the implementation of Intensified Verification Inspection (IVI), there was a remarkable increase in the number of inspection conducted on establishments for the period of 2015-2016, with 157,526 and 252,266, respectively, in the region (*Table 12.2*). This move was in response to the ill-fated Kentex fire incident in Valenzuela City wherein establishments that failed to secure Fire Safety Inspection Certificate were issued either a Notice to Comply, Notice to Correct Violations, Abatement Orders, and Order of Stoppage of Operations.

Given the number of Business Permits and Licensing Office (BPLO) registered business establishments, it is worth noting that only

Table 12.2. Comparative Data of the Inspection Conducted Fire Safety Inspection Certificate (FSIC) Issuances

	2012	2013	2014	2015	2016
Total Inspection Conducted	77,052	80,099	87,936	157,526	252,266
Establishment Inspected and Issued FSIC for Occupancy	8,307	7,797	8,500	8,723	9,535
Establishment Inspected and Issued FSIC for Business (including renewal)	54,132	62,985	70,793	111,627	222,472
Notice to Comply/Notice to Correct Violations/Abatement Orders/Order of Stoppage of Operations	14,613	9,317	8,643	37,176	20,259

*Figures are based on the annual accomplishment report of the Fire Safety Enforcement Branch (FSEB)

Table 12.3. Status of All Establishment Inspected

TAB	CITY/MUNICIPALITY	NOT EXISTING				TOTAL	NO. OF ESTABLISHMENT		TOTAL
		CLOSED	CANNOT BE LOCATED	TRANSFERRED	NOT OPERATIONAL		RE-SCHEDULED	REFUSED TO ENTRY	
FIRE DISTRICT I									
A	City of Manila	4,024	1,469	756	131	6,380	0	0	0
FIRE DISTRICT II									
B	Caloocan City	143	104	36	82	365	71	45	116
C	Malabon City	308	1,154	16	4	1,482	0	9	9
D	Navotas City	72	415	21	2	510	0	11	11
E	Valenzuela City	319	492	0	203	1,014	0	0	0
SUBTOTAL		842	2,165	73	291	3,371	71	65	136
FIRE DISTRICT III									
F	Pasay City	0	571	47	938	1,556	0	0	0
G	Makati City	678	5,185	950	3	6,816	2,368	19	2,387
H	Parañaque	830	1,417	131	1	2,379	0	25	25
I	Las Piñas City	793	0	37	340	1,170	0	0	0
J	Muntinlupa City	2,116	212	0	0	2,328	0	160	160
SUBTOTAL		4,417	7,385	1,165	1,282	14,249	2,368	204	2,572
FIRE DISTRICT IV									
K	Marikina City	48	832	81	293	1,254	0	40	40
L	Pasig City	2,693	1,721	613	30	5,057	1	5	6
M	Pateros	177	78	13	22	290	8	0	8
N	Taguig City	223	0	143	0	366	0	0	0
O	Mandaluyong City	104	546	37	23	710	0	104	104
P	San Juan City	670	125	0	0	795	0	0	0
SUBTOTAL		3,915	3,302	887	368	8,472	9	149	158
FIRE DISTRICT V									
Q	Quezon City	1,814	4,981	499	27	7,321	1,044	2,783	3,827
TOTAL		15,012	19,302	3,380	2,099	39,793	3,492	3,201	6,693

**Figures based on the IVI/Catch Up Plan Accomplishment Report of CY 2015*

few were inspected. For instance in 2014, only 87,936 out of 292,619 total BPLO establishments in the region were inspected (Table 12.2 and Table 12.4).

A significant factor that tends to bloat the BPLO figures are the so-called ambulatory businesses wherein they possess business permits without securing Fire Safety Inspection Certificate (FSIC) from the BFP. Thousands of this type of businesses are included in the registered establishments in the entire NCR.

To remedy the situation, there should be continuous move to conduct inspection of all establishments and at the same time intensifying the fire safety information drives in fire prone areas even those beyond the jurisdiction of inspection such as informal settlers and other depressed areas. Another effort is to train more Fire Safety Inspectors through the conduct of Competency Seminars to strengthen the Inspectorate Team.

Table 12.4. Comparison of Business Permits and Licensing Office Registered Establishment to Actual Inspection Conducted (Updated BPLO-Registered Establishments)

DISTRICT	NUMBER OF ESTABLISHMENTS BASE FROM BPLO (2014)	NUMBER OF ESTABLISHMENTS BASE FROM BPLO (2015)	NUMBER OF ESTABLISHMENTS BASE FROM BPLO (2016)
Fire District I			
Manila	52,681	39,226	53,590
Fire District II			
Caloocan	7,909	7,909	8,203
Malabon	6,660	6,632	6,899
Navotas	3,300	3,586	3,372
Valenzuela	14,288	14,438	15,693
	32,157	32,505	34,167
Fire District III			
Pasay	11,115	11,115	12,198
Makati	31,736	34,840	35,933
Parañaque	16,191	13,255	17,072
Las Pinas	12,075	11,413	11,679
Muntinlupa	11,688	13,255	13,761
	83,805	83,828	90,643
Fire District IV			
Marikina	15,387	16,677	17,402
Pasig	22,479	26,629	26,695
Pateros	1,456	1,410	1,417
Taguig	7,188	8,715	9,693
Mandaluyong	11,157	12,992	15,511
San Juan	6,804	7,058	10,590
	64,471	73,481	81,308
Fire District V			
Quezon City	60,505	60,505	69,204
TOTAL	292,619	289,555	328,912

High Incidence of Fire. Even with the intense information drive conducted by the government to the communities, fire incidence is still increasing at 3,424 in 2012 to 5,121 in 2016. For the period 2012-2016, the estimated amount of property damage reached at ₱2,974,363,909.44 with total deaths of 442 and 1,551 injuries (Table 12.5).

Although there was a significant reduction in the number of deaths and amount of property damaged in 2016 there was an increase in the overall fire incidence in the

region and a slight increase in the number of injuries as compared to 2015. This can be attributed to the severity of the fires that occurred within 2016 as compared to the previous year. It should be noted that Quezon City has the highest fire incidences and amount of property damages among the 17 Local Government Units (LGUs) which reached an estimated amount of ₱123,420,964.86 (Table 12.6).

In terms of types of occupancies involved, residential has the most number of

Table 12.5. 5-Year Fire Incident Statistics 2012 – 2016 (As of 31 December 2016)

YEAR	NUMBER OF FIRE INCIDENTS	ESTIMATED AMOUNT OF PROPERTY DAMAGE (IN PHP)	NUMBER OF DEATHS	NUMBER OF INJURIES
2012	3,424	494,877,630.00	75	268
2013	3,691	812,410,623.00	68	296
2014	4,563	458,958,300.00	75	302
2015	4,374	606,826,885.00	157	339
2016	5,121	601,290,471.44	67	346
TOTAL	21,173	2,974,363,909.44	442	1,551
AVERAGE	4,235	594,872,781.88	88	310

Source: BFP-NCR

Table 12.6. Number of Fire Incidents, Deaths, Injuries, and Amount of Damages by City/ Municipality, NCR 2016

AREA	FIRE INCIDENCE	DEATHS	INJURIES	AMOUNT OF DAMAGE (IN PHP)
METRO MANILA (NCR)	5,121	67	346	601,290,471.44
CALOOCAN CITY	418	1	32	29,280,600.00
LAS PIÑAS CITY	65	4	20	84,646,376.00
MAKATI CITY	363	1	28	10,414,300.00
MALABON CITY	128	1	9	6,565,200.00
MANDALUYONG CITY	134	2	6	16,237,000.00
MANILA CITY	770	9	95	112,829,200.00
MARIKINA CITY	265	9	1	1,395,250.00
MUNTINLUPA CITY	252	3	19	53,037,450.00
NAVOTAS CITY	78	-	6	2,698,800.00
PARAÑAQUE CITY	220	5	10	14,604,074.00
PASAY CITY	142	6	6	4,198,200.00
PASIG CITY	219	12	17	73,365,000.00
PATEROS	42	-	4	2,606,700.00
QUEZON CITY	1,246	9	76	123,420,964.86
SAN JUAN CITY	54	0	0	2,071,700.00
TAGUIG CITY	265	5	16	31,762,429.00
VALENZUELA CITY	460	0	1	32,157,227.58

fire incidences with 1,465 followed by mercantile with 153 and industrial with 45 establishments. Motor vehicle has 281 fire incidences in 2016. The high incidence of residential fires can be attributed to informal settlers with houses commonly made from light materials and was further aggravated by difficulty of access of fire trucks because of narrow roads and presence of road obstructions.

As far as causes of fires are concerned, based on available records in 2016, fire incidence was mostly caused by electrical connections with 2,730, followed by lighted cigarette butts with 666, and electrical appliances with 392.

Limitations in the Fire Fighting Operational Equipment and Personnel.

BFP-NCR has a total of 131 fire stations (*Table 12.7*); 21 Central Fire Stations, and 110 fire sub-stations. Out of 131 fire stations, 123 of which are occupying under MOA/usufruct agreement or owned by LGU/private owners/other agencies. Only 13 fire stations are owned by the BFP, and two (2) of the total number of fire stations were occupying a donated lot. In addition, there were two (2) stations which have been activated only on 2016.

It can be noted that the NCR has 257 firetrucks (*Table 12.8*), 76 of which are LGU-owned, while 181 are BFP-owned. Of the total 257, 146 of which are serviceable while 111 are either unserviceable, unrepairable or beyond economic repair. Based on standard, there should be one firetruck for every 28,000 population. Hence, a population of 12,877,253 in NCR needed at least 460 firetrucks. Thus, Metro Manila is

still needing an additional of 314 serviceable firetrucks, assuming all are serviceable, which is not the case.

In order to protect and able to respond quickly in times of fire incidents, the government needs an additional 3,207 firemen to augment the current shortage in the number of fire personnel of 3,232 in the region. According to the standard ratio, there should be one (1) firefighter in every 2,000 individuals (*Table 12.9*).

In terms of PPE, the ideal is 1 personnel should have 1 complete set of PPE. However, it can be inferred from the table that there is a significant shortage of PPE. For instance, for the number of 2,700 operational personnel in 2016, there is only 1,270 available boots in the inventory (*Table 12.10*). Moreover, the BFP lacks the necessary number of operational equipment such as fire trucks, rescue vehicles, as well as protective firefighting gears to properly

Table 12.7. Number of Fire Stations, NCR 2016

NUMBER OF FIRE STATIONS							
	MUNICIPALITY	CITIES	MUNICIPALITY/CITIES				TOTAL NUMBER OF FIRE STATIONS
			W/	W/O	CENTRAL	SUB-STATION	
DISTRICT I	0	1	5	0	5	9	14
DISTRICT II	0	4	4	0	4	30	34
DISTRICT III	0	5	5	0	5	27	32
DISTRICT IV	1	5	6	0	6	20	26
DISTRICT V	0	1	1	0	1	24	25
TOTAL	1	16	21	0	21	110	131

Table 12.8. Number and Status of Fire Trucks, 2016

NUMBER AND STATUS OF FIRE TRUCKS									
	SERVICEABLE		UNSERVICEABLE		UNREPAIRABLE		BEYOND ECONOMIC REPAIR (BER)		TOTAL NUMBER OF FIRE TRUCKS
	BFP	LGU	BFP	LGU	BFP	LGU	BFP	LGU	
DISTRICT I	12	4	9	2	0	0	8	3	38
DISTRICT II	19	14	5	2	5	0	8	0	53
DISTRICT III	18	20	11	3	2	1	12	0	67
DISTRICT IV	22	11	6	5	2	0	12	1	59
DISTRICT V	18	8	5	1	5	1	2	0	40
TOTAL	89	57	36	13	14	2	42	4	257

Table 12.9. Number of Fire Fighters, 2016

DISTRICT	POPULATION	OFFICERS	NON-OFFICERS	NUP	IDEAL NUMBER OF FIRE PERSONNEL	ACTUAL NUMBER OF FIRE PERSONNEL	SHORTAGE OF FIRE PERSONNEL
ORD		23	252	5		280	
District I	1,780,148	28	395	42	890	465	426
District II	2,819,388	32	468	22	1,410	522	887
District III	2,758,349	58	630	28	1,379	716	663
District IV	2,583,252	48	603	14	1,292	665	626
District V	2,936,116	28	532	24	1,468	584	884
TOTAL	12,877,253	217	2,880	135	6,439	3,232	3,207

perform its functions. On 2016, the fire fighters were able to perform their duties even with the meager number of personal protective equipment (PPEs). Their performance will certainly reach optimum level if PPE is adequately and individually assigned to each of them.

Table 12.10. Status of the Personnel Protective Gears (Inventory of Personal Protective Equipment as of 31 December 2016)

PARTICULARS	EXISTING	SHORTAGE
Boots	1,270	1,430
Gloves	1,075	1,625
Trousers	631	2,068
Coat	669	2,031
Helmet	1,274	1,426
Breathing Apparatus	350	230

*Number of Operational Personnel: 2,700

Standard: 1 Personnel in the Operations: 1 Set of PPE

High jail congestion rate. As of second quarter of 2017, the NCR accounts the largest jail population in the country with a 32,593 Persons Deprived of Liberty (PDL) or 23.59 percent of the total number of clients temporarily detained in the Bureau of Jail Management and Penology (BJMP) facilities.

For the past 5 years from 2012-2017, the jail population catered by BJMP-NCR has been relatively increasing. In 2015, there was a considerable increase of 21,258 detainees in the NCR jails due to the nationwide implementation of the PNP OPLAN Lambat-Sibat – a strategic Operational Plan against criminality. Despite the effort of BJMP's program for decongestion, the jail population further aggravated in 2016 which accounted to 24,320 detainees when President Duterte announced the intensified campaign against illegal drugs. The overall programs of the government showed an overflow of 27,914 inmates in the total 43 jail facilities in the region.

Out of the 43 jail facilities in the region, 10 of which has highest congestion rates ranging from 1,000% to almost 4,000 rate (Table 12.11). These jail facilities were located in the cities of Navotas, Caloocan, Muntinlupa, Quezon, Taguig, San Juan and Las Piñas. It is also noted that the average congestion rate is 553.33 percent in Metro Manila, which is way above the ideal standard.

All jails in the NCR are overcrowded and evidently way beyond the ideal United Nations standard of 1 inmate for every 4.7 square meter jail space. The data presented show the necessity to either expand jail facilities in the region or improve the efficiency of court case resolutions or both.

Table 12.11. Top 10 Most Congested Jails, 2017

JAIL	IDEAL CAPACITY	ACTUAL CAPACITY	CONGESTION RATE
Navotas City Jail MD	23	911	3,795.29%
Caloocan City Jail MD	101	2,042	1,927.33%
Muntinlupa City Jail MD	46	922	1,904.72%
Quezon City Jail FD	56	1,033	1,747.45%
San Mateo Municipal Jail MD	22	393	1,657.64%
Taguig City Jail MD	128	1,794	1,305.30%
Las Piñas City Jail MD	67	885	1,220.48%
Muntinlupa City Jail FD	13	156	1,122.00%
San Juan City Jail MD	22	253	1,062.59%
Quezon City Jail MD	280	3,247	1,058.54%

Logistics, utilities and other administrative concerns which has a negative effect on the rehabilitation of inmates:

Health Concerns among Inmates and BJMP Personnel. Jail congestion, aggravates the situation in the jail facilities and impact negatively on the inmates' health and limitation of basic and special medicines.

Top health concern in most NCR jails is the prevalence of inmates with pulmonary diseases such as Tuberculosis (TB). According to medical experts, one (1) infected person can easily infect up to 20. In a congested jail, there is a higher probability of spreading such disease among the inmates as well as the BJMP personnel due to poor ventilation.

In addition, it also poses high risk of spreading the disease due to limited space for isolation for suspected TB patients. Apart from pulmonary diseases, other diseases that are most common to detainees are skin diseases such as scabies, boil, fungal infection, and among others.

Shortage of Water and Electricity Supply in the Jail Facilities. Most of NCR jail facilities are still way below the ideal structure design enough to give ample

supply of water and electricity to its jails. In the case of the Manila City Jail Female Dormitory facility which was built to house 139 is currently utilized by 1,177 detainees. The water pipes are no longer appropriate to sustain the water supply needs of the residents leading to hygiene problems of detainees. On the other hand, series of power interruptions has been experienced by some of NCR Jails due to high demand of electricity supply. Such interruptions pose security risks such as escape and other jail disturbances.

Shortage in the Subsistence Allowance for Food Services. There was a sudden increase of jail population which started from the second quarter of 2016. The noted increase was enormous as compared to the previous year's gradual increases that has been stable and within the safe range of projections of about 5 to 10 percent. This upshot had substantial impact on BJMP management, particularly on the part of the Wardens who are in direct supervision of inmates within their areas of responsibility. The upsurge on jail population from the third quarter 2016 has exhausted the supposed Prisoner's Subsistence Allowance intended to be utilized until end of the year 2016. Similar incident is foreseen this 2017. As per evaluation, the allotment given by the national government will only last

Table 12.12. Personnel to Inmate Ratio

YEAR	JAIL POP	JAIL PERSONNEL	RATIO
2011	18,369	1,767	1:10
2012	18,524	1,872	1:10
2013	18,363	1,889	1:10
2014	19,294	1,995	1:10
2015	21,258	2,049	1:10
2016	24,320	2,114	1:12
2 ND QUARTER 2017	32,593	1,925	1:17

until August or September. The BJMP top management is continuously coordinating with the Department of Budget Management (DBM) and the Office of the President to address this matter. Apart from this, BJMP-NCR is likewise facing a challenge on rice shortage since the National Food Authority (NFA) can no longer sufficiently provide for the demand.

Lack of Human Resources. In 2017, there is a total of 43 jails in NCR consisting of male and female district, city, municipal, extension and attached jails. As of the 2nd quarter of 2017, the total population of detainees in NCR are around 32,593 which are manned by a total of 1,925 jail personnel (*Table 12.12*). Since the establishment of the BJMP, its manpower is far from the ideal number of personnel required for easy governing of its regional and field unit administrative and operational functions. For the past five (5) years, the annual average custodial personnel to inmate ratio had been consistent 1:10. However, in 2016

and 2017, it rose to 1:12 and 1:17, which is significantly higher than the standard ratio set by the United Nations which is 1:7 for three shift.

The manpower requirement of the BJMP is widely based on the functions the duty personnel will be rendering. As per ideal personnel distribution set forth by the national headquarters, 71 percent should render custodial function, 8 percent should render escorting function, 6 percent should render Inmate's Welfare and Development (IWD) function and lastly 15 percent of the entire manpower should render administrative function (*Table 12.13*). The lack of human resources has been addressed by the management through personnel's multi-assumptions of duties. A suppose custodial personnel usually does other tasks such as escorting, IWD and administrative jobs. The disadvantage of this current set-up is over-work that can lead to less effectiveness and efficiency of personnel.

Table 12.13. BJMP Manpower Requirement by Duty Function

PERSONNEL DISTRIBUTION		
FUNCTION	REQUIRED PERCENTAGE (IDEAL DISTRIBUTION)	IDEAL RATIO (PERSONNEL PER INMATE)
CUSTODIAL	71%	1 personnel : 7 inmates
ESCORT	8%	1 personnel + 1 : 1 inmate
IWD	6%	
ADMIN	15%	
	100%	

Table 12.14. Logistical Capability of the BJMP–NCR as of May 2016

LOGISTICAL ITEM	NEED	ON HAND	VARIANCE	READINESS
Prisoners' Van	83	51	32	61.45%
Prisoners' Bus	3	3	0	100.00%
Short Firearms	2,056	1,715	341	83.41%
Long Firearms	2,535	311	2,224	12.27%
Batons	2,056	1,429	627	69.05%
Handcuffs	11,312	2,487	8,825	21.99%
Handheld Radios	195	100	95	51.28%
CCTV Systems	228	167	95	73.25%

Source: BJMP-NCR

Logistical Support. To successfully deliver daily operations of a jail facility, the logistics should be not just available but should be sufficient in number. Unfortunately, both the transportation and equipment capability have not yet reached the desired percentage. Three (3) buses were provided by the national government and two (2) buses by the BJMP. These were distributed to NCR's biggest jails (in terms of jail population) namely Caloocan City Jail, Manila City Jail, and Quezon City Jail. Nonetheless, as assessed (in *Table 12.14*) BJMP-NCR is only 61.45 percent in terms of its lift capability. This goes to show that many of NCR facilities are still in need of additional prisoners' van, bus or staff car. There is an increasing demand on vehicle requirement as the jail population rises. In Manila City Jail for example, out of 4,599 detainees (May 2017), a total of 5,698 hearings has been accounted in a month. That means that there is an average of 190 inmates for court hearings per day. Wardens that were

not able to send inmates for hearing are usually issued with subpoena or being fined or charged in contempt by courts.

Another adverse effect, in some cases, are the few instances wherein either personal car of personnel or public transport are utilized in transporting inmates. This happens especially during emergency cases wherein inmates are needed to be brought to hospital for immediate medical attention.

As to equipment prerequisites, it reveals the current operational readiness of the bureau. Over-all, NCR jails are in need of more equipment to efficiently manage jail operations. For instance, the NCR needs to acquire 84 metal detectors, wherein there is only 2 actual metal detectors installed, as wells as 42 walkthrough detectors (*Table 12.15*). Similar problem is being encountered by other jails in different regions. Gradually, the national government is providing funds for procurement of

Table 12.15. Equipment Distribution in NCR Jails as of May 2016

EQUIPMENT		IDEAL	ACTUAL	READINESS
Metal Detector		84	2	2.4%
Walkthrough Detector		42	0	0%
Base Radio		42	2	4.8%
Other Equipment	CCTV Cameras	-	228	-
	Computers	-	167	-
	Printers	-	63	-

equipment. However, it will take a while to fulfill what is ideal. For now, all Equipment Outlay Fund are centrally managed and are being handled by the national headquarters. As to its distribution, judgment is based on prioritization.

Implementation of Inmates Welfare and Development (IWD) Programs. Apart from safekeeping of inmates, the BJMP caters different developmental programs for its clients. Inmates Welfare and Development Programs includes provision of basic needs, spiritual services, livelihood services, educational services, guidance and counselling, physical fitness & recreational services and Therapeutic Community Modality Program. The intention of this program is to holistically develop inmates' attitude, knowledge and skills in order for them to regain self-worth. All-in-all, the IWD is viewed to be effective as evidenced by the improved discipline and maintained peace and order inside facilities. Still, the following are the evaluated limitations of IWD program implementation:

Limited space for various activities of inmates. Apart from the activities hosted by BJMP-NCR, Jail IWD Personnel works on inviting different service providers and stakeholders. At times, the supposed number of PDLs to participate in an activity are lessened due to the limited space. At present, only 23 out of NCR's 43 jails have an established Multi-Purpose Hall/Activity Center.

Willingness of the client. All IWD Programs are open for all PDLs. Upon commitment, a detainee's background is assessed so as to categorize what program

will be most appropriate for him/her as an individual. However, the willingness of the client to engage himself/herself contributes to the success of the programs being introduced. Common personal reasons for resistance are the detainees' current emotional state, maturity level, sense of responsibility, laziness and pride.

Sustainability of programs. The key factors that affect the sustainability of IWD Programs are the undetermined length of stay of PDLs and the availability of resources (manpower & logistics).

Availability of separate detention facility specific for drug dependent inmates. Treatment for detained drug dependents should be different from the regular PDLs. Ideally, drug dependents should be separated and implementation of Therapeutic Modality Program and behavioral shaping tools should be undertaken.

Absence of Aftercare Program . There is a high rate of recidivism in all NCR jails. In Quezon City Jail alone, there is a substantial 28 percent rate of recidivist. As per jail's record, in every ten (10) released PDLs, three (3) of which will become a recidivist. One comprehended factor of recidivism is the society's stigma on previously detained person. Undeniably, there is a negative connotation, regardless if the person previously detained was proven innocent more so to those who were convicted guilty of charges. As of 2018, the BJMP does not have a firm policy or program for after care of its clients. In other countries, there are half-way houses (community-based detention) which prepares PDLs who are about to be released.

Strategic Framework

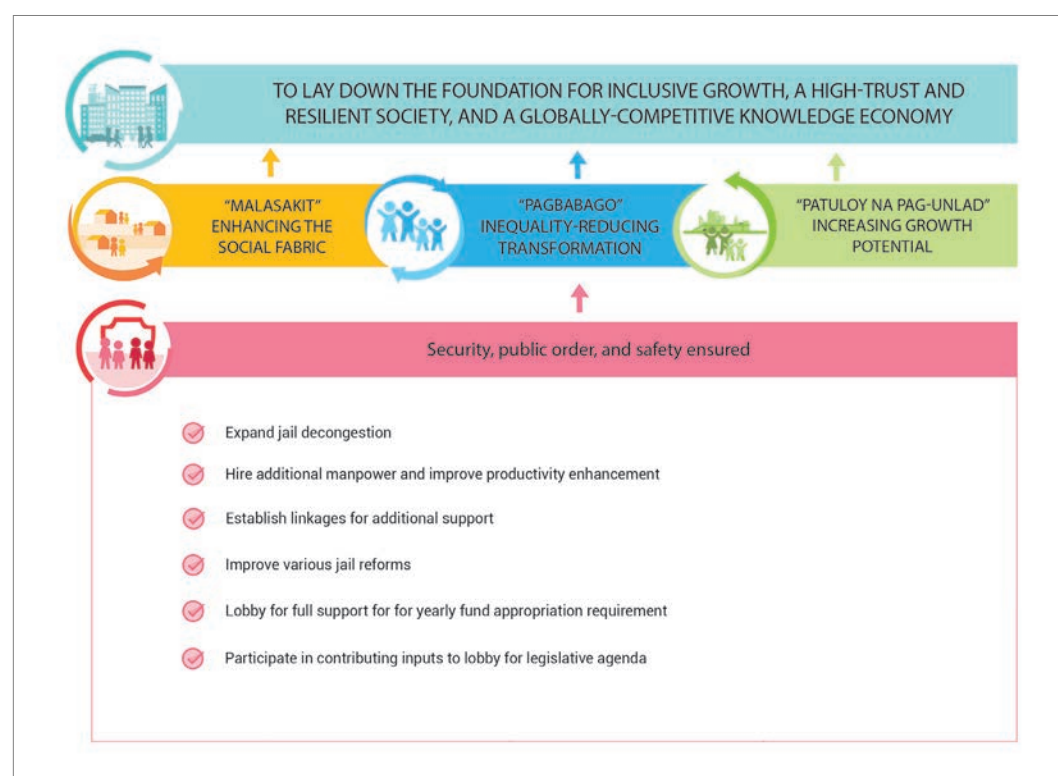
Targets

Table 12.16 shows the indicators and corresponding targets for the security, public order and safety.

Table 12.16. Plan Targets to Ensure Security, Public Order, and Safety

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Crime Solution Efficiency Rate in Index Crimes	2015	39.16%		NCRPO
Crime Solution Efficiency Rate in Non-Index Crimes				
Police to Population Ratio	2016	1:475	1:500 per district	NCRPO
Number of fire incidences	2016	5,121	Decreased	BFP
Firetruck to Population Ratio	2015	1:50,106	1:28,000	BFP
Firefighter to Population Ratio	2015	1:3,984	1:2,000	BFP
BJMP Personnel to Inmate Ratio	2015	1:10	1:7	BJMP
Occupancy/congestion Rate of Prisons and Detention Centers				

Figure 12.2. Ensuring Security, Public Order, and Safety, 2017-2022



Strategies

The following strategies identified under the subsector outcomes will help address the challenges enumerated:

To expand jail decongestion. The effort to improve the living condition of PDLs under BJMP care are directed to three (3) aspects namely construction of jails, release of qualified inmates and intensified paralegal services. These are the strategies to address overcrowding in jails. The most effective but the hardest to achieve due to budget constraints is the construction of new facility. The top management of BJMP-NCR, with wardens of specific jails being the major role player, lobbies possible improvements and/or construction of new facility. Although the BJMP provided with national fund, the status of lot ownership per jail hinders the approval of allocation for its construction, repair and improvements. As per guidance of the Commission on Audit, construction or improvements should be assigned to those BJMP owned facilities. A constant coordination with counterpart LGUs is being undertaken to address this concern. The record shows that only 10 out of the 43 NCR jail facilities are BJMP owned (*Table 12.17*).

Also, the release of qualified inmates is strictly implemented and monitored. The foremost rule observed by BJMP is the 100 percent release of qualified inmates through different modes (*Table 12.18*). The granting of Good Conduct Time Allowance by wardens likewise contributed to minimizing the stay of PDLs in jails.

Moreover, in line with professionalization of jail services, all jails have a designated paralegal officers who are either graduates of Bachelor of Laws, Political Science or professional criminologists. This qualification standard is set by the Legal Service Unit primarily to ensure job

efficiency. The paralegal officers provide for legal assistance to BJMP detainees, in order to fast track their cases and contribute to decongest jails.

To hire additional manpower and improve productivity enhancement. As what was presented on the preceding paragraphs, the BJMP, not just the NCR, was overwhelmed with the unexpected increase of commitments nationwide. It resulted to a higher personnel-inmate imbalance. In addressing this issue, the BJMP continues its yearly recruitment process in accordance to the Department of Budget and Management (DBM)'s approved allotment for salary. Proportion to the huge number of PDLs under NCR's care, it is usually NCR who has the highest quota for recruits.

To limit the utilization of manpower, the BJMP also developed electronic systems such as the National Inmates Monitoring System (NIMS), Human Resource Monitoring Information System (HRMIS) and Fugitive Inmates Monitoring System (FIMS). These Jail Management Information Systems lessened the required personnel to track data and monitor the same as it is now saved and input in one database.

To augment the lack of manpower, the bureau also accepts assistance from the Local Government Units (LGUs), On-the-Job Trainees (OJT) to support in the following inmate's welfare activities i.e., gate personnel, researchers, and enhancement productivity.

To establish linkages for additional support. Most LGUs are responsive to the needs of their constituents under the BJMP care. LGUs have been a good partner of BJMP in its virtue for advancements. Constant coordination is established to convey the needs of PDLs and maintain the assistances

Table 12.17. Status of Ownership of NCR Jail Facilities

NAME OF JAILS	REMARKS	REFERENCE/AUTHORITY
Caloocal CJ	NHA	Letter from the NHA
Las Piñas CJ	LGU owned	None
Las Piñas CJFD	LGU owned	None
Makati CJ	LGU owned	MOA/Deed of USUFRUCT
Makati CJFD	LGU owned	MOA/Deed of USUFRUCT
Malabon CJ	BJMP owned	None
Malabon CJFD	BJMP owned	None
Mandaluyong CJ	LGU owned	Presidential Declaration
Mandaluyong CJFD	LGU owned	Presidential Declaration
Manila CJ	HGC	Presidential Declaration
Manila CJFD	HGC	Presidential Declaration
Manila CJ Annex	BJMP owned	Presidential Declaration
Manila CJ	BJMP owned	Presidential Declaration
Marikina CJ	LGU owned	MOA/Deed of USUFRUCT
Marikina CJFD	LGU owned	MOA/Deed of USUFRUCT
MMDJ	BJMP owned	Presidential Declaration
SICA I	BJMP owned	Presidential Declaration
SICA II	BJMP owned	Presidential Declaration
Muntinlupa CJ	LGU owned	MOA/Deed of USUFRUCT
Muntinlupa CJFD	LGU owned	MOA/Deed of USUFRUCT
Navotas CJ	LGU owned	None
Parañaque CJ	LGU owned	MOA/Deed of USUFRUCT
Parañaque CJFD	LGU owned	MOA/Deed of USUFRUCT
Pasay CJ	LGU owned	None
Pasay CJFD	LGU owned	None
Pasig CJ	LGU owned	None
Pasig CJFD	LGU owned	None
Pateros MJ	LGU owned	None
Quezon CJ	LGU owned	MOA/Deed of USUFRUCT
Quezon CJFD	LGU owned	MOA/Deed of USUFRUCT
Quezon CJ Annex	BJMP owned	Presidential Declaration
Rodriguez MJ	LGU owned	MOA/Deed of USUFRUCT
San Juan CJ	LGU owned	MOA/Deed of USUFRUCT
San Juan CJFD	LGU owned	MOA/Deed of USUFRUCT
San Mateo CJ	LGU owned	MOA/Deed of USUFRUCT
San Mateo CJFD	LGU owned	MOA/Deed of USUFRUCT
Taguig CJ	BJMP owned	None
Taguig CJFD	BJMP owned	None
Valenzuela CJ	LGU owned	None
Valenzuela CJFD	LGU owned	None

given to improve their current living condition. These supports are usually in terms of rehabilitation supplies, prisoners' subsistence allowance, constructions and facilities improvements. Aside from this, LGUs also extends support for the operational necessities of jails. These comes in various ways, from personnel allowance, office supplies, gas allocations, utilities expenses, and even janitorial services.

Similarly, BJMP-NCR work on establishing good rapport with counterpart agencies of the government, non-government organizations, different service providers and stakeholders. As a result, activities like medical and dental missions, feeding programs, book donations, skills enhancement training, and livelihood programs, are regularly facilitated to most NCR Jails. These pro-active move for partnerships even gave rise to several jail improvements and provision of equipment. In addition, this assistance help mitigate the inadequacies of NCR's lift and capability.

To improve various jail reforms. Taking into consideration the development of inmates, IWD programs are improved to provide the clients their individual needs while they are under the care of the BJMP. The utilization of modern technology is

one way, as "E-dalaw" (online visitation) was initiated in Quezon City Jail. This gives opportunity to the inmates who came from different regions to see their loved ones without spending much money and time for travel. This is also a good way of decreasing number of daily visitors which will further result to the reduction of jail security issues. Another improvement is the participation of the bureau in local trade fairs and exhibits. In participating in such activities, we are exposing the products of the inmates in the community. Further, we are currently improving the marketing of the products with the aid of social media and internet. We are also considering of engaging in partnerships with service providers and stakeholders to give our clients more opportunities while they are under our custody. As of now, the Alternative Learning System (ALS) is offering only educational advancement to elementary and high school undergraduates; we are presently looking for a possibility of having a college degree courses be offered to our clients in coordination with the local universities and Department of Education.

Every person has individual needs, likewise with our clients – inmates. In order to effectively implement development and reformation, we should address their individual needs. Different programs must

Table 12.18. Number of Released Inmates thru Application of Penal Laws

RELEASE OF QUALIFIED INMATES CY 2017					
MODE OF RELEASES	NUMBER OF RELEASES				
	1 ST QUARTER	2 ND QUARTER	3 RD QUARTER	4 TH QUARTER	TOTAL
Released on Recognizance (Republic Act 6036)	34	128	34	37	233
Probation & Parole (Presidential Decree 968)	61	70	57	59	247
Juvenile Justice and Welfare Law (Republic Act 9344)	0	0	0	0	0
Released on Bail	1,196	1,594	1,192	1,664	5,893
Served Sentenced	843	1,109	1,097	1,283	4,332
Case Dismissed/Acquitted	1,281	1,592	2,021	2,661	7,555
Art 29, RPC (Good Conduct Time Allowance)	183	224	238	134	779
Total	3,598	4,717	4,639	5,838	18,792

be applied to different type of offenders. For now, as part of the IWD, BJMP-NCR incorporated the Therapeutic Community Modality Program where its philosophies are anchored to positive values promoting self-realization to initiate and sustain inward transformation.

With the campaign of the present administration, Greyhound Operations and OPLAN Galugad are regularly conducted by BJMP officers and sometimes in coordination with other government agencies in consonance with the BJMP-OPLAN RODY. Contraband control operations are intensified for the inmates to turn away from their old ways and focus on individual development.

The drug related cases has drastically increased from 12,822 cases of 2nd quarter of 2016 to 23,325 cases for the same quarter of 2017. That is 81.91 percent escalation. Crimes against property almost doubled and crimes against person remains the least with the total of 3,211 cases for the 2nd quarter.

During the commencement of the anti-illegal drugs and anti-criminality campaign of the present administration, there was a drastic increased of drug-related cases. This requires a great demand of proper drug rehabilitation facility which is 67.12 percent or 94,109 inmates (May 2017) incarcerated in the BJMP nationwide came from the NCR.

As to the rehabilitation of drug offenders, the bureau is currently proposing a construction of a drug rehabilitation facility, where Therapeutic Community Modality Program (TCMP) will be religiously implemented for drug dependent clients. Once implemented and efficiently handled, it will help greatly in the development and reformation, and further lessen recidivism, proper programs must be addressed to specific offenders.

To lobby for full support for yearly fund appropriation requirement.

To participate in contributing inputs to lobby for legislative agenda.

13

Accelerating
Infrastructure
Development

Accelerating Infrastructure Development

Infrastructure is one of the driving forces that supports the socio-economic activities of the country towards development. It sustains and guides urban and rural development, integrates diverse communities in our country and links our nation to the global community. Thus, the state of infrastructure, including the way it enhances and complements the natural environment, provides a physical measure of national and local development and, ultimately, of the quality of life.

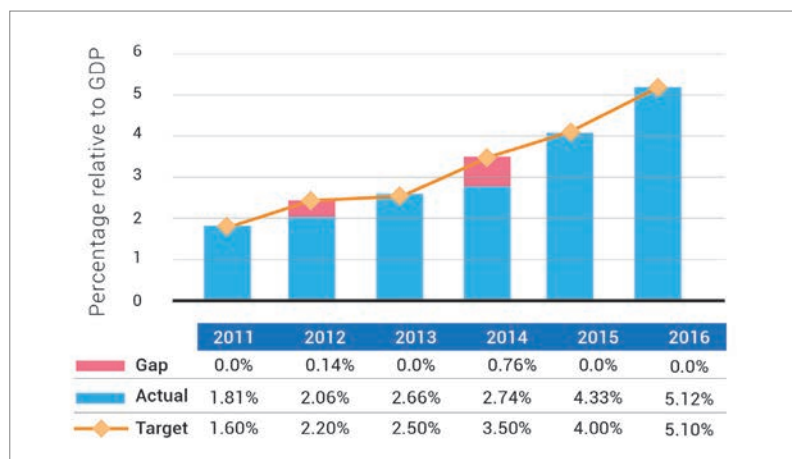
The National Capital Region (NCR), as a premier region requires strategic investments for support infrastructure to further promote growth and development in consonance with the 0-10 Socioeconomic Agenda and the country's Long Term Vision, *AmBisyon Natin 2040* consistent with the massive Build Build Build Program of the current Administration.

Assessment and Challenges

Infrastructure development is one of the top most priorities of the government, particularly in Metro Manila over the past five years. This is very evident since the government is continuously increasing its spending in infrastructure within this period, though the government fell short of the target in 2012 and 2014 (*Figure 13.1*).

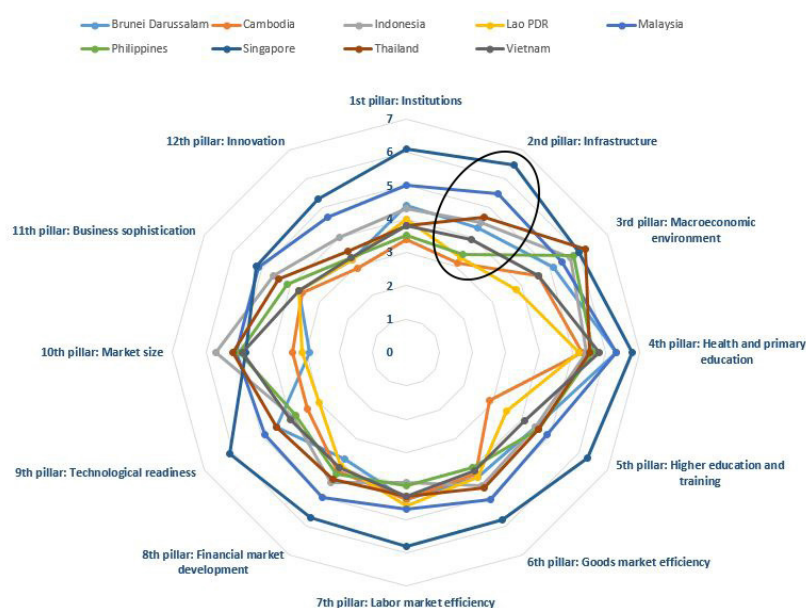
Despite of this remarkable increase in public spending on infrastructure development, the country still struggles behind with the other ASEAN Member States (AMS) in terms of delivery of quality infrastructure, based on the Global Competitiveness Report 2017-2018 of The World Economic Forum (WEF) as shown in *Figure 13.2*.

Figure 13.1. Public Spending in Infrastructure



Source: National Economic and Development Authority (2017). *Philippine Development Plan 2017-2022*

Figure 13.2. The Global Competitiveness Index in ASEAN-9 Countries



On the said report, the Philippines, along with Cambodia and Lao PDR, is surpassed by other AMS in the Infrastructure Pillar (and its components) of the Global Competitiveness Index (GCI). This GCI pillar combines eight (8) indicators on quality and availability of land, air, water, and electricity and Information and Communications Technology (ICT) infrastructure and provides an overall measurement for infrastructure development among the economies around the world (Table 13.1). Further, the inadequacy of infrastructure supply is

identified as one of the top three problematic factors for doing business in the Philippines, based from the results of the Executive Opinion Survey 2017 which is included in the same WEF report.¹

Delays in the implementation of infrastructure projects are often attributed to this sluggish infrastructure quality performance. Mostly, these delays are due to the difficulty in the acquisition of lots required for the road right-of-way, utility and informal settler families (ISF) relocation as well as on the tedious process in the issuance

Table 13.1. Global Infrastructure Quality Rankings of ASEAN-9 Countries

	QUALITY OF OVERALL INFRA-STRUCTURE	QUALITY OF ROADS	QUALITY OF RAILROAD INFRA-STRUCTURE	QUALITY OF PORT INFRA-STRUCTURE	QUALITY OF AIR TRANSPORT INFRA-STRUCTURE	QUALITY OF ELECTRICAL SUPPLY	MOBILE-CELLULAR TELEPHONE SUBSCRIPTIONS	FIXED-TELEPHONE LINES
Brunei Darussalam	51	33	-	74	63	53	61	62
Cambodia	99	99	94	81	106	106	52	115
Indonesia	68	64	30	72	51	86	18	104
Lao PDR	83	94	-	127	101	75	131	60
Malaysia	21	23	14	20	21	36	28	71
Philippines	113 (9 ^a)	104 (9 ^a)	91 (6 ^a)	114 (8 ^a)	124 (9 ^a)	92 (8 ^a)	88 (8 ^a)	111
Singapore	2	2	4	2	1	3	23	27
Thailand	67	59	72	63	39	57	5	91
Vietnam	89	92	59	82	103	90	44	96

¹World Economic Forum (2017). The Global Competitiveness Report 2017-2018.

of project requirements such as clearances and permits. Aside from the delays, physical connectivity to support trade and tourism activities among and within urban centers also remains as a challenge.

Among the reforms that will drive the 0-10 Socioeconomic Agenda of the administration is the acceleration of infrastructure and the development of industries that will yield robust growth, create jobs and uplift the lives of Filipinos (*Table 13.2*).

Table 13.2. Programs and Projects on Infrastructure

PROJECT NAME	DESCRIPTION	IMPLEMENTING OFFICE
Metro Manila Integrated Transport System	Establishment of integrated transport terminals that are par with international standards to provide interconnections between different transport modes and services at the following proposed locations: South – FTI Compound along SLEX, Taguig City (5.57 has.) Southwest – PRA lot along Coastal Road, Parañaque City (2.8 has.) North Terminal – selection of terminal location sites: a) NFA, Valenzuela, Brgy. Malanday; b) Caloocan PNR Depot; c) Balintawak (original site); d) Balintawak (opposite GHC); e) Balintawak (GHC); and e) Pandacan Oil Depot	Department of Transportation (DOTr)
MRT 3 Capacity Expansion Project	Purchase of LRV vehicles; Upgrade traction power substations and signalling system; Installation of additional stabling and catenary system in the depot; and Improvement of depot maintenance equipment and workshops.	Department of Transportation (DOTr)
Metro Rail Transit Line 7 (MRT 7)	The project involves the implementation of an integrated transportation system consist approximately 23-km Metro Rail Transit Line 7 from North Avenue Station in EDSA to San Jose Del Monte in Bulacan, Intermodal Transportation Terminal (ITT), and 22-km Highway from Bocaue interchange of NLEX to the ITT.	Department of Transportation (DOTr)
North-South Railway Project (South Line)	Involves the design and construction of a standard gauge 56-km commuter line between Tutuban and Calamba and long haul railway between Tutuban and Legaspi including long haul operations on the branch line between Calamba and Batangas and extension between Legaspi and Matnog.	Department of Transportation (DOTr)
North-South Railway Project (Tutuban-Malolos Commuted Line)	Construction of approximately 38 kilometers elevated commuter train line from Tutuban, Manila to Malolos, Bulacan, procurement of rolling stock (104 cars), and Installation of Electro-Mechanics System	Department of Transportation (DOTr)
Metro Manila Bus Rapid Transit Line 1	Establishment of a 12.3 km bus system from Philcoa, Quezon City to Manila City Hall, passing along Quezon Avenue and España Boulevard	Department of Transportation (DOTr)
NAIA Development Projects	The project will improve, upgrade and enhance the operational efficiency of all existing terminals of the Ninoy Aquino International Airport (NAIA) covering landside and airside to meet the International Civil Aviation Organization (ICAO) standards and develop the main gateway airport of the Philippines.	Department of Transportation (DOTr)
Premium Point-to-Point (P2P) Buses	The operation of high quality bus services such as the Premium Point-to-Point (P2P) buses	Department of Transportation (DOTr)
Rationalization of Public Transport Services	The rationalization of public transport services consistent with the Omnibus Franchising Guidelines (OFG) that will address the operation of "colorum" public utility vehicles	Department of Transportation (DOTr)
Establishment of the Public Transport Information and Management Center (PTIMC)	The establishment of the Public Transport Information and Management Center (PTIMC) that will manage operation of services that will correct "out of line" operations	Department of Transportation (DOTr)
Enforcement of Motor Vehicles Inspection System (MVIS)	To ensure operation of safe and road-worthy vehicles, Motor Vehicles Inspection System (MVIS) tests prior to the renewal of registration shall be strictly enforced	Department of Transportation (DOTr)
Driver's Academy	The Driver's Academy which aims to upgrade the competence and skills as well as enhance the character of Public Utility Vehicle drivers, will be operated by the LTRFB to promote disciplined public transport drivers	Department of Transportation (DOTr)
North-South Commuter Rail Project	37-km North-South Commuter Rail Project from Tutuban to Malolos, Bulacan, the Rehabilitation of the PNR South Line	Philippine National Railway (PNR)
Mega Manila Subway Project	25-km Mega Manila Subway Project	Department of Transportation (DOTr)

PROJECT NAME	DESCRIPTION	IMPLEMENTING OFFICE
NAIA Redevelopment, Improvement, and Capacity Expansion Project	Construction of Rapid Exit Taxiways (RET) at Runway 06/24 Improvement and Rehabilitation of NAIA Terminal 2 NAIA Rehabilitation, Capacity Maximization, and Expansion Construction of parking area and public transport hub at Terminal 3 Rehabilitation and expansion of existing toilets and re-orientation of the location of PWD at Terminals 1, 2 & 3 Purchase of two (2) units of Aircraft Rescue and Fire Fighting Major Foam Vehicles (6 x 6) Detailed Engineering Design and Construction Supervision for the Repair and Upgrading of Taxiway Detailed Engineering Design and Construction Supervision for the Repair and Overlay of Runway 13/31 including Interconnecting Taxiway and Proposed Repair and Upgrading of Runway 13 Extension Detailed Engineering Design and Construction Supervision for the Repair and Overlay of Runway 06/24 Upgrading of 9 PBB including Layout of New Power Feeder Line Cable at NAIA Terminal 1 Transfer of General Aviation to Sangley (Passengers, Airlines, Concessionaries) Airport Slotting/Limitation of Air Traffic to 40 Flights per hour	Department of Transportation (DOTr)
Improvement of Clark International Airport	Clark International Airport New Passenger Terminal Building Procurement of secondary surveillance radar, upgrading of Airport Ground Lighting System (AGLS)	Department of Transportation (DOTr)
Flood Management in Metro Manila Phase I	The Master Plan (M/P) for Flood Management in Metro Manila and Surrounding Areas was assisted by the World Bank through funds from the Australian Government amounting to P23,464.75 Million. The M/P includes structural and non-structural measures towards sustainable flood management in Metro Manila and surrounding areas. The project aims to reduce flood risk to people and property in the flood prone areas of Greater Metropolitan Manila in the following approach: Component 1: Modernization of Drainage Areas Component 2: Minimizing Solid Waste in Waterways Component 3: Participatory Housing and Resettlement Component 4: Project Management Support and Coordination	Department of Public Works and Highways, UPMO-Flood Control Management Cluster
Data Collection Survey on Parañaque Spillway in Metro Manila	The project aims to prepare a comprehensive flood management plan for the entire Laguna de Bay coastal area and to conduct a pre-feasibility study on the Parañaque Spillway to reduce the high water level of Laguna de Bay.	Department of Public Works and Highways, UPMO-Flood Control Management Cluster
Feasibility Study and Preparation of Detailed Engineering Design of the Flood Protection Works in Marikina River, Development of Comprehensive Flood Forecasting System and Monitoring & Early Warning System and Development of Institutional Arrangement for Sustainable Flood Management of Greater Metro Manila Area Flood Management Project	The overall project objectives are to provide sustainable flood management project and resilience in the Greater Metro Manila Area; to have comprehensive technical and engineering studies, and social and environmental studies on the identified priority structural measures of flood control (Upper Marikina Dam) in the target area; and to come up with the feasibility study and detailed engineering design of the selected option. The total cost of the project is P134,388,310.810 under the assistance of the World Bank.	Department of Public Works and Highways, UPMO-Flood Control Management Cluster

PROJECT NAME	DESCRIPTION	IMPLEMENTING OFFICE
Metro Manila Flood Management Project Phase I: Component 2, Minimizing Solid Waste in Waterways under World Bank Project	The project is Phase 1 of the implementation of the Metro Manila Flood Management Master Plan (P 350 billion), approved by the government in 2012. The Master plan aims to reduce vulnerability and to strengthen resilience against floods. It specifically addresses the structural measures to improve urban drainage in Metro Manila. It will be achieved through an integrated set of interventions to modernize existing pumping stations to accommodate urban expansion; increase short-term water retention capacity in the drainage areas; reduce the volume of indiscriminately dumped solid waste into waterways; and support community-driven resettlement of Informal Settler Families (ISFs) encroaching in easements for drains and waterways linked to the project pumping stations to safer in-city locations.	Metropolitan Manila Development Authority and Department of Public Works and Highways
Flood Mitigation Project in the East of Mangahan Floodway Area	The purpose of the project is to solve the flooding in the eastern side of Mangahan Floodway and to complete the drainage system of North Laguna Lakeshore Area.	Department of Public Works and Highways, UPMO-Flood Control Management Cluster
Improvement of Existing Drainage System	The objective of the project is to divert floodwater of the Cainta River into the Buli Creek to discharge directly through Buli, Cainta and Taytay river into the Laguna Lake through the improved Mangahan diversion channel with the design scale of 30-year return period.	Department of Public Works and Highways, UPMO-Flood Control Management Cluster
Flood Control Plans	The project scale is proposed to protect the area against floods of less than a 10-year return period, since the major waterways in the area are assumed to be urban drainage channels. The following measures are considered applicable for formulation of flood control plans: Construction of lakeshore dike to protect the lakeshore area in the east side of Mangahan; Channel improvement for Buli, Cainta, Bongad and Bonguod to confine and carry the design flood towards the Laguna Lake; Improvement of Mangahan Diversion Channel with floodgates to prevent any backwater effects; Construction of pumping stations to discharge rainwater confined inside the diking area; and Construction of retarding ponds in the upstream reaches of the waterway (middlestream of Bulao Creek)	Department of Public Works and Highways, UPMO-Flood Control Management Cluster
Pasig-Marikina River Channel Improvement Project, Phase IV	The project is the continuation of channel improvement works of Phase III for the Upper Marikina River from Rosario Bridge to Marikina Bridge including the construction of Marikina Control Gate Structure. The objective of the projects is to decrease flood inundation, reduce flood damages including affected population and eliminate flooding. The estimated project cost is P26.80 Billion.	Department of Public Works and Highways, UPMO-Flood Control Management Cluster

The government has also sought private sectors participation through the creation of Public-Private Partnership (PPP) Center in 2011 to address the gaps in infrastructure; fast track and ensure efficient and effective implementation; financing; and operations and maintenance of major infrastructure projects such as elevated expressways, rail expansion and extension, integrated bus terminal, modernization of flood control facilities, school and health facilities, disaster resilient housing, boosting energy efficiency among others. These has led to the realization of the large-scale infrastructure projects in the region, such as Muntinlupa-Cavite Expressway (Daang Hari-SLEX Link Road), NAIA Expressway and NLEX-North Harbor Link Segment 9. As of 15 September 2017, 37 projects amounting to ₱438.65 billion are already included on the center's PPP Project Pipeline, in which 15 of these projects, worth ₱310.51 billion, are already awarded.² However, the operation of PPP projects is still based on the Build-Operate-Transfer (BOT) law issued in 1990, hence, there is a need to revisit the said law.

Transport

An integrated and effective transportation system is important to build further the multi-centered spatial structure enhancing connectivity among and within existing and new urban and economic centers in the Greater Metro Manila Area (GMMA) to include Regions III and IV.

Several master plans and roadmaps were already crafted in the past to expand the road network and ensure that the implementation of the government priority projects is harmonized and well-coordinated. However, what is important now is the rapid implementation of the projects identified in the existing master plans and roadmaps, complete with feasibility studies and detailed

engineering design.

Road-Based Transport

The region national road has an aggregate length of 1,159.01 kilometers, which accounts for 3.54 percent of the total road length of 32,770.27 kilometers nationwide having the shortest length among the 17 regions (*Figure 13.3*). In terms of road density, NCR has the highest road density among the regions with 187.08 kilometers of national road per square kilometers.³

With regards to the local roads in the NCR, there is a total of 6,753.637 kilometers based on the submission of the 16 local government units except Taguig City to the MMDA (*Table 13.3*).

Table 13.3. Metro Manila Local Roads

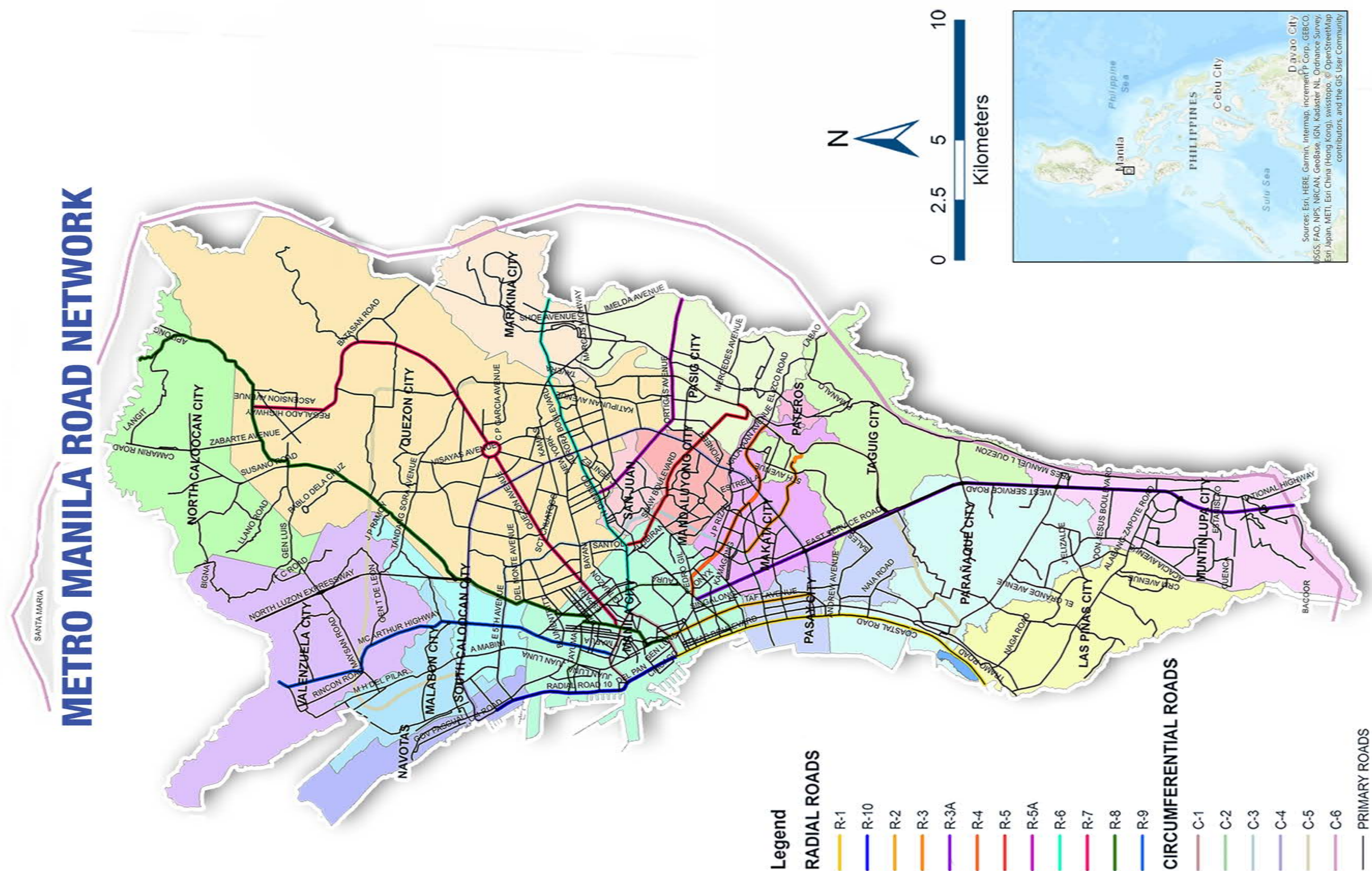
	Local Government Unit	Local Road Length, km
1	Caloocan	1,000.570
2	Las Piñas	172.517
3	Makati	332.621
4	Malabon	119.212
5	Manila	510.442
6	Mandaluyong	80.93
7	Marikina	449.535
8	Muntinlupa	348.765
9	Navotas	50.015
10	Parañaque	203.599
11	Pasay	372.51
12	Pasig	353.419
13	Pateros	23.551
14	Quezon City	2,327.159
15	San Juan	81.994
16	Taguig	280.013
17	Valenzuela	326.798
	TOTAL (km)	7,033.650

Note: Caloocan - data from CLUP

² PPP Center (2017). The Philippine PPP Program presentation.

³ DPWH Atlas 2016 (Road and Bridge Information Application)

Figure 13.3. Metro Manila Road Network Map



Furthermore, 289 permanent bridges in the national roads are recorded across the region having a total length of 27,252 linear meters, of which 276 bridges (25,734 linear meters) and 13 bridges (1,518 linear meters) are made of concrete and steel, respectively.

Despite the increasing trend of the newly reclassified local roads into national roads, as shown in *Table 13.4*, there is a need to re-evaluate the total roads (local and national) in the NCR to determine the inadequacy to accommodate the growing number of motorists.

The road based transport faces the following challenges:

Longevity of road and bridges affected by overloading. The DPWH reached the maximum target of 100 percent paved national roads for NCR among the regions in the country for 2016. Out of 1,159.01 km of national roads, 62.44 percent (723.69 kilometers) is in good condition, 22.54 percent (261.27 kilometers) in fair condition, 6.6 percent (76.50 kilometers) in poor condition, 1.42 percent (16.55 kilometers) in bad condition while 7 percent (81 kilometers) of the total road network were not assessed with an average

International Roughness Index (IRI) of 4.69.⁴

With regards to the overall condition of the bridges in the region, out of 289 bridges, 49.48 percent (143 bridges) are in good condition, 43.94 percent (127 bridges) in fair condition, 4.15 percent (12 bridges) in poor condition, 1.73 percent (5 bridges) in bad condition and the remaining 0.70 percent (2 bridges) are for further assessment (*Table 13.5*). The figures show that majority of the bridges are in good to fair state but requires routine or major maintenance, while the bridges with poor to bad condition is due to the defects that affect the performance and structural integrity of the structures which requires upgrading or replacement.⁵

One of the major causes of poor road quality and durability is overloaded trucks traversing along the road network. The damage done to the already poor road quality has resulted to rapid deterioration of facilities and road accidents. The said road network has already been maximized to the fullest and proved to be insufficient for the current volume of vehicles.

High volume of registered vehicles. In 2016, NCR has the largest number of

Table 13.4. Summary of NCR Newly Reclassified Local Roads into National Roads

SUMMARY OF NCR NEWLY RECLASSIFIED LOCAL ROADS INTO NATIONAL ROADS									
Year	2009	2010	2011	2012	2013	2014	2015	2016	Total
No. of Roads	3	3	0	0	7	11	2	14	40
Length (Km)	13.38	11.04	0	0	12.76	5.56	6.96	5.04	54.74

Source: DPWH—ATLAS 2016

Table 13.5. National Road and Bridges Condition

	GOOD	FAIR	POOR	BAD	NO ASSESSMENT	GRAND TOTAL
Roads length, km	723.69	261.27	76.50	16.55	81.00	1,159.01
Number of Bridges	143	127	12	5	2	289

Source: DPWH—ATLAS 2016

⁴ibid

⁵ibid

registered vehicles among all the regions accounting for 2.4 million (26 percent) from the total number of registered vehicles in the country of 9.25 million.

As evident in *Table 13.6*, NCR is experiencing rapid motorization resulting to more road-users adding to the volume of traffic being experienced by every commuter in the region.

The historical data of Motor Vehicle (MV) Registration in the NCR (2011-2016) reveals that motorcycles, utility vehicles and cars are the dominant vehicle type in terms of the number of vehicle registration. In 2016, the motorcycles accounts for around 39.6 percent of the total MV registrations in NCR, while Utility Vehicles and Cars registered around 25.5 percent and 19.48 percent, respectively. These three motor vehicle types accounts for almost 85 percent of the total motor vehicle registration for NCR in 2016.

In terms of vehicle classification, private vehicles dominate the volume of registration covering almost 92 percent of the total registration of NCR in 2016. Based on vehicle registration statistics for NCR, it can be surmised that the increase in motorization may be attributed to the prominent number of private motor vehicle ownership in the area. It is important to note that in terms of

passenger occupancy, private vehicles are much lower than public vehicles.

According to JICA's study entitled the "Roadmap for Transport Infrastructure Development for Metro Manila and Its Surrounding Areas (Regions III and IV-A)" estimated economic losses of at least ₱2.4 billion per day (in 2012 prices) or at least 4.6 percent of GDP annually due to traffic congestion in a "Do Nothing scenario or "without projects".⁶ The World Bank and Asian Development Bank (ADB) also participated in the creation of informative studies specifically addressing domestic transport infrastructure needs. Preliminary analysis in the study showed that the average low-income household has to spend at least 20 percent of its total income on transport costs. Said study also concluded that without intervention, the traffic demand will likely increase by 13 percent by 2030 and transport costs will be 2.5 times higher.

Weak implementation of traffic laws and regulations. Transport-concerned agencies such as MMDA, Department of Transportation (DOTr) and Department of Public Works and Highways (DPWH) with the LGUs joined forces to improve vehicular traffic on the road. However, despite efforts, traffic condition is significantly affected by several factors:

Table 13.6. Registered Motor Vehicles, 2011 to 2016

	2011	2012	2013	2014	2015	2016
Philippines	7,138,942	7,463,393	7,690,038	8,081,224	8,706,607	9,251,565
NCR	2,014,750	2,032,390	2,101,148	2,222,370	2,317,204	2,405,122
Cars	446,106	450,189	452,959	462,880	477,320	468,520
Utility Vehicles (UV)	575,614	566,083	578,732	589,627	643,686	613,440
Sports Utility Vehicles (SUV)	156,188	167,105	185,623	210,151	214,327	243,935
Trucks	72,121	72,009	78,521	75,168	83,735	90,225
Buses	13,345	13,007	13,989	12,221	15,278	14,500
Motorcycles (MC)/Tricycles (TC)	734,465	745,777	773,291	853,089	861,594	951,231
Trailers	16,911	18,220	18,033	19,234	21,264	125,821

Source: LTO-DOTr

⁶JICA Study

1. Undisciplined driving behavior resulting to road accidents. In 2016, the region experienced an annual average of 299 accidents per day, in which one (1) is fatal, 45 are non-fatal and 253 accidents involve damaged to properties.⁷
2. Obstructions on the road such as illegal parking of vehicles, illegal structures/ encroachments, among others.
3. Use of old and dilapidated public utility vehicles that are still plying along the major thoroughfares in Metro Manila.
4. Presence of colorum and out-of-line operation of vehicles plying along the roads
5. Construction of infrastructure projects and delays in the implementation.
6. There are variation in some of the traffic ordinances of 17 LGUs and MMDA which significantly affects traffic management.

Rail-Based Transport

The Philippines ranked 9th among countries in Asia based on rail transport network which significantly affects the service efficiency of rail-based transport system (Table 13.7).

Insufficient capacity of railway systems.

Railway system is another mode of mass transport system popularly used by commuters. It addresses the growing passenger demand and eases the traffic congestion along major thoroughfares in the metropolis. According to the World Economic Forum-Global Competitiveness Report (WEF-GCR) 2015-2016, the quality of our rail infrastructure network ranks 84th out of 140 countries.

Currently, there are four (4) mass transit systems operating in the region, namely: LRT 1, LRT 2, MRT 3, and PNR (Table 13.8). The original LRT Line 1 has 18 stations with a route length of 15 km from Monumento, Caloocan City to Baclaran and started full operations in 1985. An extension of 5.4 kilometers was completed in 2010, thus, closing the loop with MRT3 at the intersection of EDSA and North Avenue. The Megatren or LRT Line 2 which started full operations in 2005 has a route length of 13.8 km with 11 stations from Santolan, Pasig City to Avenida, Manila. These 2 lines which are under the jurisdiction of LRTA have a combined daily ridership of about 700,000 passengers.

The Metrostar, or commonly called MRT 3, has a route length of 16.9 kilometers with 13 stations from North Avenue, Quezon City

Table 13.7. Comparison of Rail Transport Network in Asia

COUNTRY	RAILWAY (KM)	ELECTRIFIED (KM)	AREA PER TRACK KM (KM ² /KM)	POPULATION PER TRACK KM
China	127,000	80,000	77	11,048
Japan	27,182	16,700	16	5,451
Indonesia	8,529	550	223	27,853
South Korea	4,078	2,522	19	9,348
Thailand	4,507	107	126	16,084
Vietnam	3,147		105	27,765
Malaysia	1,833	207	178	15,324
Cambodia	650		279	20,609
Philippines	531	52	626	196,270
Singapore	161		4	28,682
Brunei	13		444	30,692
Laos	4		59,200	1,557,550

Source: Department of Transportation

⁷MMDA – MMARAS Annual Report 2016

Table 13.8. Metro Manila Mass Transit System

	NO. OF STATIONS	LENGTH (KM)	DAILY PASSENGERS
A. Existing Railways			
1. LRT 1	18	20.4	500,000
2. LRT 2 or Megatren	11	13.8	200,000
3. MRT 3 or Metrostar	13	16.9	500,000
4. PNR	24	56	40,000
B. Ongoing Construction			
1. MRT 7	14	23	350,000
2. LRT Cavite Ext.	8	12	200,000
3. LRT 2 Ext.	2	4.2	100,000

Source: Department of Transportation

to Taft Avenue, Pasay City. The trains carry about 500,000 passengers daily or about 42% more than its maximum 350,000-passenger capacity, hence, passenger congestion is one of the inconvenience in the daily operations of the MRT 3.

In the past years, the MRT 3 has been experiencing several technical problems and glitches involving derailment incidents, electrical and signaling problems, abrupt stopping, etc. resulting in the temporary shutdown of operations. The DOTr, in its desire to improve train operations, has embarked on the MRT 3 Capacity Expansion (Capex) Projects I and II. Capex I involves the procurement of 48 additional light rail vehicles (LRV) while Capex II is the upgrade of ancillary systems. These projects have encountered delays in the implementation due to procurement and technical concerns, among others.

The planned construction of the Common Station at the intersection of EDSA and North Avenue which will house the LRT 1, MRT 3, and the future MRT 7 stations has been delayed, due partly to the tug-of-war among stakeholders on its final location. Recent favorable developments, however, indicate start of construction within 2017.

Having started operations in November 1982 with the inaugural run of the 195-km Manila-Dagupan railroad, the PNR Line is one of the oldest railways in the world. It is currently operating the commuter line with 24 stations and a 56-kilometer route length

from Tutuban, Manila to Calamba, Laguna. Daily weekday ridership is only about 40,000 passengers which is way below the patronage during PNR's golden years in the 1960's. For the past years, the PNR has encountered serious accidents and technical problems involving rail derailment, collisions with road-based vehicles, etc. resulting in deaths and stoppage of operations. Furthermore, the railroad right-of-way which has been cleared of informal settlers by the national government over a decade ago is being encroached on again.

On the possible side, the government has embarked on expanding the railway network in the metropolis. On 2017, the MRT 7 started construction of the 23-km line from North Avenue, Quezon City to San Jose Del Monte, Bulacan. The initial ridership of the system when completed in 2020 will be 350,000 daily.

Not to be outdone, the 12-km LRT 1 Cavite Extension broke ground early 2017. Once completed, the LRT 1 will have a total length of 32.4 km catering to about 700,000 passengers.

On the eastern side of Metro Manila, the 4.2-kilometer LRT 2 Extension is about to be completed in one (1) year, bringing the total length of LRT 2 to 18 kilometers.

Air Transport

The region's aviation service is unable to meet the expected demand. Air transport is one of the keystone hubs transport system promoting tourism in the country. With the unprecedented traffic growth in the last decade, the Ninoy Aquino International Airport (NAIA) terminals and runways have been facing airside and landside congestion that leads to delays, discomfort, safety concerns and deteriorating service level of passengers. In 2016, NAIA has generated about 39,564,972 air passengers. DOTr is utilizing other airports such as the Diosdado Macapagal International Airport or the Clark International Airport in Pampanga to address the growing passenger demand that is currently experienced in NAIA.

Water Transport

Infrastructure quality and operational efficiency of Pasig River Ferry System still needs to be improved. Ports are essential part in the Philippine transport network as connecting hubs for the 3,219 navigational waters.⁸ The Port of Manila is the largest and the premier international shipping gateway in the country. Congestion inside the port area and its adjacent roads continues to be a challenge for the transport and logistics. The increasing volume of cargoes being processed at the Port of Manila resulted in high volume of truck container traffic in the metropolis.

At its peak, the ferry had 17 stations and 2 lines. The first line was the Pasig River Line which stretched from Plaza Mexico in Intramuros, Manila to Nagpayong station in Pasig City and the second line was the Marikina River Line which served the Guadalupe station in Makati City up to Santa Elena station in Marikina City.

After a year of poor traffic, the number of passengers picked up to the point that Nautical Transport Services Inc. (NTSI) considered purchasing more boats. This service was also promoted by the Pasig River Rehabilitation Commission (PRRC) to highlight the importance of the environment to the people of Manila.

To augment the decongestion efforts of vehicular traffic on major thoroughfares in Metro Manila, the operation of Pasig River Ferry Service System was resumed in 2014 through the joint efforts of the MMDA, DOTr and PRRC. The MMDA operates 12 ferry stations and utilizes 11 boats which benefitted a total of 105,475 passengers in 2016, an increase of 17 percent ridership from 2015 data of 89,933, a manifestation that the Pasig River can serve as an alternative transport corridor for commuters in Metro Manila. However, the ridership of the ferry was affected due to the ongoing rehabilitation of the stations and less operational boats owing mainly to engine breakdowns as well as the presence of water hyacinth and garbage along the river which threaten the operation of the ferry.

The DOTr has commissioned a consultant to conduct a study on the possible establishment of a ferry route system along the Manila Bay Coast (to cover the areas of Cavite, Bulacan and Bataan) and Laguna Lake (to cover the areas of Laguna and Rizal) and assess the impact of interconnecting these with the existing Pasig Ferry Service. The DOTr engage the services of CPCS Transcom Ltd., through the PPP Center in the preparation of a Business Case Report for the Manila Bay-Pasig River-Laguna Lake (MAPALLA) Ferry System Project in order to establish the most appropriate implementation strategy for the project and also to explore its possible implementation through PPP.

⁸The Report the Philippines 2015 by Oxford Business Group

Water Resources

The MMDA, in its effort to improve the quality of living in Metro Manila, is closely coordinating with the 17 LGUs, national government agencies and the private sector for the implementation of programs, projects and activities that will improve water resources management.

Water Supply

Wide Water Crisis Remains a Threat in the Region/Limited Source of Water for Metro Manila Households. Metro Manila faces a water crisis unless the government finds a source to supplement its single source of water supply which is Umiray–Angat–Ipo or Angat reservoir system, which will not be able to meet the needs of its more than 12.8 million residents four years from now. Its replenishment is reliant on the amount of rain during the rainy season. With its limited capacity, water crisis is always experienced at the onset of the dry season and during prolonged dry spells. Moreover, this 49-year-old Angat reservoir system happens to sit on the West Valley Fault in Bulacan province.

The main sources of Metro Manila's water supply are the Angat, Ipo, and La Mesa dams. The water from these dams are then processed by the La Mesa and Balara Treatment Plants, which convert it from a raw state to clean and potable water.

The La Mesa Water Treatment Plant serves the western half of Metro Manila which is 60 percent, while the remaining 40 percent of the eastern half supplies by the Balara Water Treatment Plant.

When both Balara and La Mesa Treatment Plants are operation, the total processing capacity will be 4,000 mld (million liters per day). The available supply from the control distribution system is not enough to meet the total demand of the service areas particularly during the summer months. Some areas are to schedule water rationing when insufficient supply and low water pressure becomes more frequent. Projected average water demand by 2021 will be 4,136 mld, thus, the need to find other sources of water.

As an alternative source to the Angat Dam, the Putatan Treatment Plant⁹ in Muntinlupa City is using water from Laguna Lake.¹⁰ The facility is producing 100 mld of water per day. The plant supplies potable water to Alabang, Bayanan, Poblacion, Putatan, Tunasan and New Alabang.

Sewerage and Sanitation

Provision of sewerage and sanitation services remains to be a challenge. In compliance to the Supreme Court Mandamus on the full implementation of the Operational Plan for the Manila Bay Coastal Strategy for the rehabilitation, restoration, and conservation, Metropolitan Waterworks and Sewerage System (MWSS) together with Manila Water Company Inc. (MWCI)¹¹ and Maynilad Water Services Inc. (MWSI)¹² have aggressively pursued the acceleration of the sewerage service provisions in Metro Manila. These services target to abate the domestic water pollution in the major river systems which drain into Manila Bay.

However, provision of sewerage and sanitation

⁹It is the largest membrane-based water treatment plant in the Philippines using large-scale microfiltration and reverse osmosis.

¹⁰The largest lake in the Philippines and the third-largest freshwater lake in South East Asia located east of Metro Manila between the provinces of Laguna to the south and Rizal to the north. It has a surface area of 911–949 km², with an average depth of about 2.8 meters and an elevation of about 1 meter above sea level.

¹¹A company that holds the exclusive right to provide water and wastewater services to the East Zone of Metro Manila, under a Concession Agreement entered into between the company and MWSS.

¹²The water and wastewater services provider in the West Zone of the greater Metro Manila area under a Concession Agreement entered into between the company and MWSS

services in Metro Manila encountered numerous challenges, summarized into five (5) major categories: a) Land Availability, b) Stakeholder Endorsement, c) Cooperation with Government Agencies, d) Social Acceptability and lastly e) Operational Challenges.

Land availability is one of the major challenges in building treatment facilities. Treatment plants and lift stations require large areas of land. Unfortunately, Metro Manila is the most populated metropolis in the country. Large available open spaces are scarce, and minimal available land is typically inhabited by informal settlers.

Drainage and Flood Management

The drainage system in Metro Manila is a network composed of drainage laterals, drainage mains and interceptors, and open waterways such as creeks and rivers. The drainage network provides the linkage for the passage of floodwater to ensure the continuity of flow, from the time the rain falls to the ground until it reaches its final destination – the Manila Bay. Any hindrance within this network may cause the disruption in the flow of water when not properly addressed and will eventually result to flooding.

The drainage system has a total length of 1,615 kilometers, 897 kilometers of which are for drainage laterals along major thoroughfares with varying sizes of reinforced concrete pipe culverts from 24” to 60” diameter; 59 kilometers are for drainage mains, outfalls and interceptors; and 659 kilometers for 273 open waterways such as rivers and creeks/esteros.

However, other existing drainage laterals are still using 18” and 24” diameter pipes or not yet completely replaced/upgraded to a bigger ones to accommodate a higher

return period of flooding. In the previous studies, the existing drainage is insufficient and/or deteriorated and cannot lodge heavy, concentrated rainfall which results to flooding.

In 2012, the Metro Manila Flood Management Master Plan was approved by the government. With the implementation of the said plan, urban drainage in Metro Manila will reduce vulnerability and strengthen resilience against flood.

For improved delivery of basic infrastructure, the following challenges must be addressed:

Presence of informal settler families (ISFs) along waterways. On June 2016, there are 32,082 informal settler families living along the waterways in Metro Manila.¹³ They aggravate the problem of deteriorating water quality as they indiscriminately dump all forms of domestic waste in the water system as well as lack of access to clean water and poor sanitation. Furthermore, the presence of ISFs were certainly contribute in the exposure to natural and manmade disasters, particularly to the dangers of floods or typhoons and health risks due to the congestion and unsanitary environment.

Relatively poor condition of waterways/estero. Narrowing and swallowing due to siltation and garbage, and poor quality of water makes it not suitable for swimming and other forms of water recreation.

Severe flooding caused by clogged drainage systems due to solid wastes and heavy siltation. This problem has led to severe flooding despite moderate rains causing inconvenience to people in low-lying areas. Flooding in major roads also results to heavy traffic congestion.

Soil erosion due to the adverse effects caused by the deforestation of the Marikina Watershed and Sierra Madre

¹³ LGUs consolidated by MMDA in June 2016

Mountains. The deforestation in the upper catchment (head waters) of Marikina River watershed which stretches from the Sierra Madre Mountains has resulted to soil erosion occurring at a fast alarming rate that accumulates in the lower reaches of the river system.

Reduced absorptive capacity of soil due to rapid urbanization and development that results in increased flash flood events. As the premier urban center, Metro Manila has built-up status which reduces the availability of open spaces which is important for rain absorption.

Energy

Metro Manila, touted to be 100 percent electrified¹⁴ is highly dependent on the efficient distribution and reliable supply of power crucial to attaining sustained socio-economic growth. Considering that the distribution facilities are predominantly overhead installations, its reliability is affected during severe weather conditions, often resulting in the disruptions of the socio-economic activities in the region.

The demand for electricity (power generation) in Metro Manila has increased by 32.50 percent from 2011 (22,725GWh) to September 2016 (30,013GWh). Distribution in energy sales by zone are as follows: the highest is in the commercial sector at 39.4 percent, followed by the residential sector at 31.30 percent and the industrial sector at 29.00 percent. Distribution in consumer sales by zone are as follows: the highest rate was reflected in the residential sector at 91.70 percent, the commercial sector at 2.5 percent and the industrial sector at 1.70 percent.¹⁵

In 2015, the Manila Electric Company

(Meralco)¹⁶ provides electricity to more than 2.8 million customers in Metro Manila. Based on the average retail rate of ₱8.09/kwh, electricity cost in Metro Manila is very expensive making the cost of doing business relatively higher particularly for small and medium enterprises. The biggest component of electricity cost is the generation charge at ₱3.70/kwh or about 54 percent of the total rate, while transmission and distribution charges are about 33 percent and taxes and universal charges contributed another 13 percent. The members of the low and middle-income classes are partially shielded from increases in power prices through a lifeline subsidy system being implemented by Meralco. Households consuming 100kwh or less receive discounts ranging from 20 percent to 100 percent. These households constitute around 40 percent of the Meralco residential customers.

The Philippine electricity prices is still the highest among the power rates in the ASEAN region (*Figure 13.4*). A major reason for these higher prices is the absence of government subsidies. As of August 2017, data shows that the Philippines has a higher residential and commercial power rates than Thailand, Indonesia, Malaysia, Vietnam and even the highly developed Singapore. However, it is noted that the country is charging the true cost of power as envisioned by the Electric Power Industry Reform Act (EPIRA) law while other countries are still subsidizing their power which make their rates lower.

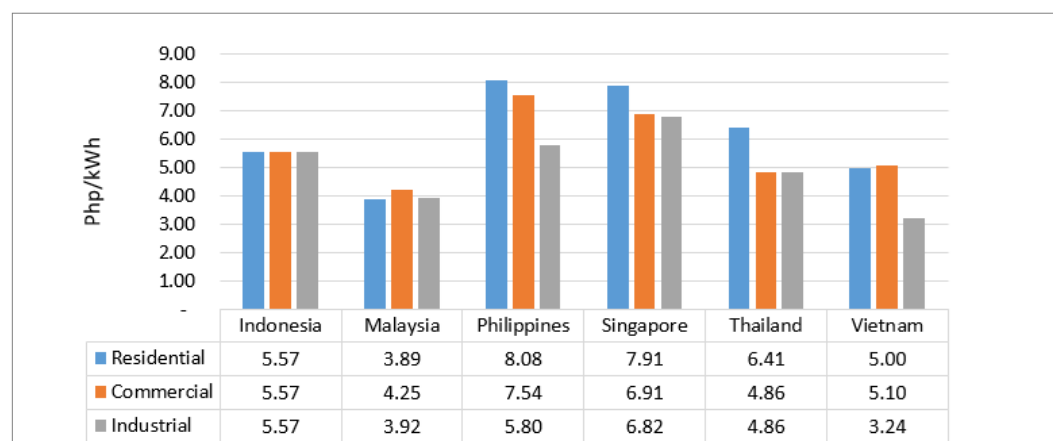
In addition, taxes, fees and other charges are also levied on the power industry sectors composed of the generation, transmission, and distribution levels which constitute a portion on electricity rates in the Philippines.

¹⁴ Meralco Report May 2016

¹⁵ Meralco Report September 2016

¹⁶ The largest electric power distribution company in the Philippines covering 36 cities and 75 municipalities, including Metro Manila

Figure 13.4. Average Electricity Rates in Selected ASEAN Countries, August 2017



Sources:

Indonesia - Perusahaan Listrik Negara, Effective since January 2015; Malaysia - Tenaga Nasional Berhad, Effective since January 2014; Singapore - Singapore Power, 3rd Quarter 2017 Rates; Philippines - MERALCO, August 2017; Thailand - Metropolitan Electricity Authority, July 2017; Vietnam - Ministry of Industry and Trade, Effective since March 2015

The following are the challenges:

The need to enhance investment climate to encourage greater private sector participation. The country faces major challenge in securing energy investment. Potential investors are faced with the challenge of lengthy permitting and licensing processes that makes the business climate unattractive. This discourages private sector participation and detrimental to our aspiration for competitiveness. This also impacts economic expansion as inadequate energy infrastructure hampers economic development.

The need for congressional support for the passage of energy-related bills (e.g. energy efficiency and conservation, etc.). The passage of pending energy-related bills will provide regulatory/policy framework and enabling environment to support the development of energy industry. For instance, the energy efficiency and conservation bill will promote energy efficiency and conservation through various means, including fiscal and non-fiscal incentives. For Liquefied Petroleum Gas

(LPG), the establishment of a regulatory framework for the importation, refining, refilling, transportation, distribution and marketing of LPG, among others.

The need to harmonize the energy policies/laws with related government agencies. In order to have a timely implementation of energy projects, there is a need to harmonize the different policies, rules or procedures of related government agencies involved in the permitting process. A simplified procedure is expected to expedite processes that cause delay in the implementation of energy projects.

Social acceptability of energy resources or technologies. In the communities where energy projects are implemented, normally social acceptance issues persists. This is also a major challenge as local permits could also be difficult to acquire without the approval of local authorities. Better understanding of the benefits/advantages that these new technologies could bring to the community could facilitate the development of new policy to support the deployment of energy projects.

Information and Communications Technology (ICT) Infrastructure

More than being a social networking¹⁷ and selfie capital¹⁸ of the world, NCR has made significant contributions in ushering the usage of information and communications technologies or ICTs¹⁹ in spurring region's socio-economic growth. The success of Information Technology-Business Process Management (IT-BPM) industry is a testament for this contribution of ICT on the region's growth. In 2016, the IT-BPM industry garnered \$ 22.9 billion in revenue. This industry has also provided 1.15 million direct employment and generated additional 3.68 million indirect and induced employment to Filipinos.²⁰ This remarkable feat of the IT-BPM industry has made Metro Manila reached the 4th place out of 100 cities surveyed in the recent Tholons Services Globalization City Index 2017 report²¹ (Table 13.9). NCR also houses several IT-BPM hubs particularly in Cities of Manila, Makati, Mandaluyong, Parañaque, Pasig, Taguig and Quezon City, which provides voice-based services as well as non-voice and complex process outsourcing services.

Despite this significant contribution of IT-BPM industry to the ICT sector,²² the country's ICT landscape faces challenges, particularly in policy and regulatory environment, business and innovation environment, infrastructure and affordability pillars of the Networked Readiness Index,

Table 13.9. Tholons Services Globalization City Index 2017

RANK	SUPER CITIES
1	Bangalore
2	Mumbai
3	Delhi (NCR)
4	Manila (NCR)
5	Hyderabad
6	Sao Paolo
7	Dublin
8	Krakow
9	Chennai
10	Buenos Aires
11	Pune
12	Cebu City
13	Santiago
14	San Jose
15	Johannesburg
16	Singapore
17	Prague
18	Dubai
19	Toronto
20	Kuala Lumpur

according to the Global Information Technology Report (GITR) 2016 of World Economic Forum (Figure 13.5).

Aside from the GITR 2016, other reputable bodies detailed the country's status on other areas of ICT such as telecommunications, e-government, and cybersecurity, among others (Table 13.10).

Telecommunications

During the past decade, the cellular mobile telephone system (CMTS) is the primary means of the Filipinos in accessing telecommunication services. Due to recent developments in ICT, broadband internet

¹⁷ Hootsuite and We Are Social Ltd. (2017). Digital in 2017. <https://www.slideshare.net/wearesocialsg/digital-in-2017-global-overview>

¹⁸ TIME (2014). The Selfiest Cities in the World. <http://time.com/selfies-cities-world-rankings/>

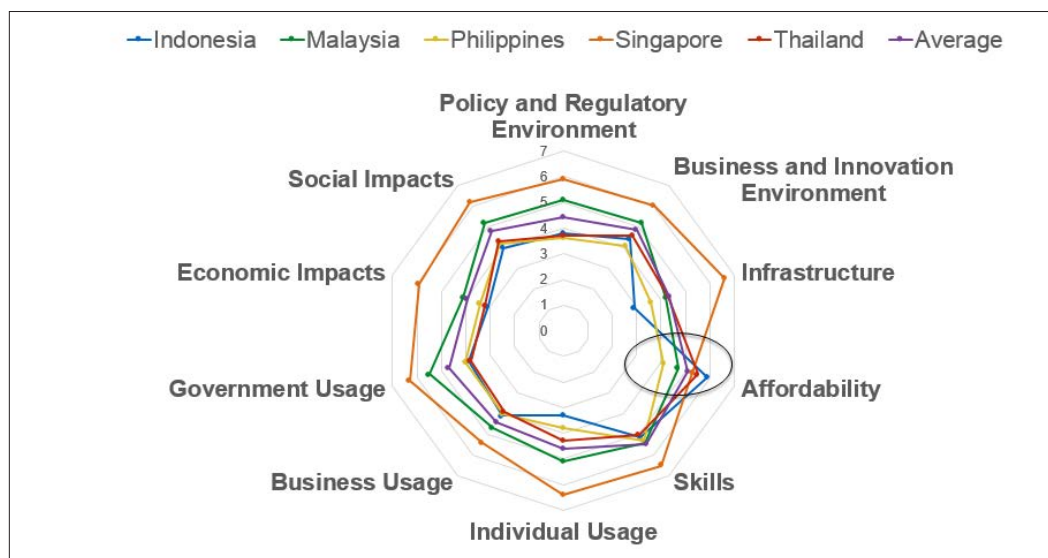
¹⁹ Information and Communications Technology or ICT shall mean the totality of electronic means to access, create, collect, store, process, receive, transmit, present and disseminate information.

²⁰ Manila Bulletin (2017). IT-BPM Sector Expects Sustained Growth to Continue. <https://business.mb.com.ph/2017/02/04/it-bpm-sector-expects-sustained-growth-to-continue/>

²¹ Tholons (2017). Tholons Services Globalization Index 2017: Disruption by Digital. <http://www.tholons.com/TholonsTop100/pdf/TSGI%20Press%20Release%202017.pdf>

²² ICT sector shall mean those engaged in providing goods and services intended to fulfill or enable the function of information processing and communication by electronic means. The ICT sector includes telecommunications and broadcast information operators, ICT equipment manufacturers, multimedia content developers, and providers, ICT solution providers, internet service providers, ICT training institutions, software developers and ICT-Enabled Services (ICT-ES) providers.

Figure 13.5 The Network Readiness Index in ASEAN-5 Countries



Source: World Economic Forum and INSEAD (2016). The Global Information Technology Report 2016: Innovating in the Digital Economy. http://www3.weforum.org/docs/GITR2016/GITR_2016_full%20report_final.pdf

Table 13.10. The Philippine ICT Sector Standings in Key International ICT Indicators versus Other ASEAN-5 Countries

ICT INDICATOR	PHILIPPINES	OTHER ASEAN-5 COUNTRIES
ICT Development Index (IDI) ^a	Though the IDI score increased from 4.52 to 4.67, the Philippines' rank slipped down from 100 th place in 2016 to 101 st place in 2017; second lowest among ASEAN-5 countries	Indonesia: 111 Malaysia: 63 Singapore: 18 Thailand: 78
Networked Readiness Index (NRI) ^b	Fell down from 56 th place in 2015 to 76 th place in 2016; lowest among ASEAN-5 countries	Indonesia: 73 Malaysia: 31 Singapore: 1 Thailand: 62
e-Government Development Index (EGDI) ^c	Improved from 95 th place in 2014 to 71 st place in 2016; ranked 3 rd among ASEAN-5 countries	Indonesia: 116 Malaysia: 60 Singapore: 4 Thailand: 77
Global Cybersecurity Index (GCI) ^d	Though the GCI score increased from 0.353 to 0.594, the Philippines' rank slipped down from 17 th place in 2015 to 37 th place in 2017; second lowest among ASEAN-5 countries	Indonesia: 70 Malaysia: 3 Singapore: 1 Thailand: 20

a: Developed by International Telecommunication Union (ITU), ICT Development Index (IDI) is a unique benchmark of the level of ICT development in countries across the world. IDI combines eleven indicators on ICT access, use and skills, capturing key aspects of ICT development in one measure that allows for comparisons to be made between countries over time.

b: Networked Readiness Index (NRI) is a composite indicator made up of four main categories (subindexes), 10 subcategories (pillars), and 53 individual indicators distributed across the different pillars. NRI is a joint undertaking of World Economic Forum (WEF) and INSEAD.

c: E-Government Development Index (EGDI) is a composite index based on the weighted average of the three normalized indices, namely Telecommunications Infrastructure Index (TII), Human Capital Index (HCI) and Online Service Index (OSI). Jointly developed by United Nations Department of Economic and Social Affairs (UNDESA), Division for Public Administration and Development Management (DPADM) and many external experts, EGDI measures the readiness and capacity of national administrations in using ICT as a means to deliver public services.

d: Global Cybersecurity Index (GCI) is a composite index combining 25 indicators into one benchmark measure to monitor and compare the level of ITU Member States cybersecurity commitment with regard to the five pillars (Legal, Technical, Organizational, Capacity Building, and Cooperation) identified by the High-Level Experts Group and endorsed by the ITU Global Cybersecurity Agenda (GCA).

gradually replaces CMTS in accessing aforesaid services.

Despite of this development, the broadband internet of the country, particularly in Metro Manila, lags behind its peers in ASEAN-5 in terms of cost, coverage and speed (*Table 13.11*). To address this challenges, President Rodrigo Roa Duterte instructed the DICT to develop a National Broadband Plan (NBP) to accelerate deployment of fiber optic cables and wireless technologies to improve internet speed.²³

Now on its implementation, NBP²⁴ lays the foundation for developing the Domestic Wideband Information Network (DWIN) which comprises of the following but not limited to: (1) Construction of the Luzon Bypass Infrastructure; (2) Provision of Satellite Broadband Services; (3) Operationalization of Domestic Fiber Optic Backbone through National Grid Corporation of the Philippines (NGCP) facilities; (4) Construction of

Middle and Last Mile Facilities and (5) Deployment of Access Points in government sites and establishments. Also, the NBP targets to formulate and modernize policies and regulations related to ICT in order to facilitate entry of a new major player in the telecommunications market fostering competition, introduces networking as the backbone of operation of intelligent cities, and employ ICT systems and technologies for informed decision making and urban planning.

Broadcasting

In the pursuit of providing better quality of TV transmission and reception, the government, through the National Telecommunications Commission (NTC) issued the Memorandum Circular (MC) No. 05-11-2013,²⁵ a landmark regulation which provides Integrated Services Digital Broadcasting-Terrestrial (ISDB-T), as a sole standard in the delivery of Digital Terrestrial Television (DTT) in the country. MC No. 07-

Table 13.11. The Philippine Telecommunication Industry Performance vs Asia and the Pacific and World Averages

INDICATOR	VALUE	ASIA AND THE PACIFIC	WORLD
Fixed-telephone subscriptions per 100 inhabitants	3.7	10.0	13.6
Mobile-cellular subscriptions per 100 inhabitants	109.2	98.9	101.5
Fixed-broadband internet subscriptions per 100 inhabitants	5.5	11.3	12.4
Mobile-broadband subscriptions per 100 inhabitants	46.3	47.4	52.2
Fixed-broadband price (1GB), in % GNI per capita	7.1	14.5 (4.62 in ASEAN-5)	13.9
Mobile-broadband price (1GB), in % GNI per	2.1	5.4 (1.4 in ASEAN-5)	6.8
CMTS coverage in NCR, in % cities/municipalities ^a	100%	-	-
Broadband internet coverage in NCR, in % cities/municipalities ^b	100%	-	-
Average Download Speed, in Mbps ^c	5.5	-	7.2
Individuals using internet, %	55.5	41.5	45.9
Households with internet access, %	39.1	45.5	51.5
International internet bandwidth per user, in kbps/user	43.4	48.0	74.5

a National Telecommunications Commission 2016 data

b Hootsuite and We Are Social Ltd. (2017). Digital in 2017.

c Akamai (2017). Akamai's State of Internet – Connectivity Report Q1 2017. <https://www.akamai.com/fr/fr/multimedia/documents/state-of-the-internet/q1-2017-state-of-the-internet-connectivity-report.pdf>

Source: International Telecommunication Union (2017). *Measuring Information Society Report 2017*. https://www.itu.int/en/ITU-D/Statistics/Documents/publications/misr2017/MISR2017_Volume2.pdf

²³ Office of the President (2016). The State of the Nation Address 2016. <http://www.officialgazette.gov.ph/2016/07/26/the-2016-state-of-the-nation-address/>

²⁴ Department of Information and Communications Technology (2017). National Broadband Plan: Building Infrastructures for a Digital Nation. <http://www.dict.gov.ph/wp-content/uploads/2017/09/2017.08.09-National-Broadband-Plan.pdf>

²⁵ National Telecommunications Commission (2013). MC No. 05-11-2013. Standard for Digital Terrestrial Television (DTT) Broadcast Service. <http://ntc.gov.ph/wpcontent/uploads/2015/10/LawsRulesRegulations/MemoCirculars/MC2013/MC-05-11-2013.pdf>

12-2014, which provides rules and regulation for the Digital Terrestrial Television Broadcast Service was issued a year later.²⁶

To address policy, regulatory, and technical issues involved in the country's migration to Digital TV, the Department of Information and Communications Technology (DICT) published the Framework of the DTT Broadcasting (DTTB) Migration Plan²⁷ last 13-14 February 2017. Such plan includes the utilization of Early Warning Broadcast System (EWBS) feature of ISDB-T in facilitating the delivery of vital information in times of disaster.

NCR is home to the major broadcasting networks in the country, serving around 3,095,766²⁸ TV households. Currently, fourteen (14) broadcasters already operates on digital TV channels in Mega Manila but on a simulcast basis (*Table 13.12*).

E-Government

To sustain the country's unprecedented socioeconomic growth, the government, through the Information and Communications Technology Office (now DICT) developed the e-Government Masterplan 2013-2016²⁹ which aims to integrate, harmonize and ensure interoperability of ICT systems and processes. To institutionalize this reform, the Medium-Term Information and Communications Technology Harmonization Initiative (MITHI) was conceptualized. MITHI Steering Committee then was formed to operationalize the initiative.

Meanwhile, Integrated Government Philippines (iGovPhil) Program was formed for the execution of the basic infrastructure and shared services needed across government. iGovPhil, which is launched on June 2012, seeks to achieve a higher level

Table 13.12. List of Digital TV Channels in Mega Manila

CALLSIGN	UHF CHANNEL	BROADCASTER	STATUS
DZKB-TV	19	Radio Philippines Network (CNN Philippines)	Test Broadcast*
DWCP-TV	22	Solar Entertainment Corporation (via Southern Broadcasting Network)	Test Broadcast
DZTV-TV	26	Intercontinental Broadcasting Corporation	Test Broadcast
DZBB-TV	27	GMA Network, Inc.	Test Broadcast
DZRJ-TV	30	Rajah Broadcasting Network	Test Broadcast
DWKC-TV	32	Broadcast Enterprises and Affiliated Media	Test Broadcast
DZOZ-TV	33	ZOE Broadcasting Network	Fully migrated from analog to digital on February 28, 2017
DWAQ-TV	38	Progressive Broadcasting Corporation	Test Broadcast
DWBP-TV	40	Sonshine Media Network International	Test Broadcast (Low Power Signal)
DWGT-TV	42	People's Television Network	Test Broadcast (Low Power Signal)
DWWX-TV	43	ABS-CBN Corporation	Test Broadcast
DWVN-TV	44	Gateway UHF Broadcasting	Test Broadcast
DZCE-TV	49	Christian Era Broadcasting Service International	Test Broadcast
DWET-TV	51	TV 5 Network	Test Broadcast

* Returned on-air last 10 January 2017 after its brief status for maintenance and upgrades according to DTV Pilipinas

Source: Wikipedia (2018). List of Digital Television Stations in the Philippines. [https://en.wikipedia.org/wiki/List_of_digital_television_stations_in_the_Philippines#Mega_Manila_\(Metro_Manila,_Cavite,_Bulacan,_Laguna,_Rizal\)](https://en.wikipedia.org/wiki/List_of_digital_television_stations_in_the_Philippines#Mega_Manila_(Metro_Manila,_Cavite,_Bulacan,_Laguna,_Rizal))

²⁶ National Telecommunications Commission (2013). MC No. 07-12-2014. Rules and Regulation for the Digital Terrestrial Television (DTT) Broadcast Service. http://region7.ntc.gov.ph/images/LawsRulesAndRegulations/MC/Broadcast/MC_07-12-2014_Rules_and_Regulations_for_Digital_Terrestrial_Television.pdf

²⁷ Department of Information and Communications Technology (2017). Framework of the DTTB Migration Plan. http://www.dict.gov.ph/wp-content/uploads/2017/10/PHL-Framework-for-the-DTTB-Migration-Plan_V1-3-1.pdf

²⁸ PSA, 2015

²⁹ Department of Science and Technology – Information and Communications Technology Office (2013). E-government Masterplan 2013-2016. http://i.gov.ph/wp-content/uploads/2015/07/eGovMasterPlan_Final-Version.pdf

of e-governance or the application of ICT to rationalize government operations and improve the delivery of goods and services to people.³⁰ Currently iGovPhil was able to implement the following high-impact e-government systems:

1. Government Network (GovNet) - connects 15 agencies in Cebu City and 160 national government agencies in the NCR located in Manila, Quezon City, Pasay, Pasig and Taguig. This success paved way to implement GovNet in other regions, aptly called Regional GovNet, which have connected 317 agencies in Tuguegarao, Iloilo, Pampanga, Baguio, Legazpi, Palo-Leyte and Butuan.
2. National Government Data Center (NGDC)
3. Government Cloud (GovCloud)
4. Government Web Hosting Service (Administrative Order 39)
5. National Government Portal (NGP) – which aims to unify all Government-to-Government (G2G), Government-to-Citizen (G2C), and Government-to-Business (G2B) services in one venue. Mandated by the President during his first SONA last 2016. NGP also targets to eliminate queues while transacting with government. Currently, DICT already activated its beta website, gov.ph.

Policy and Regulatory Reforms

In the past years, reforms was already made by the government. Such reforms include the issuance of the following critical laws and regulations:

1. Republic Act (RA) No. 10515: Anti-Cable Television and Cable Internet Tapping Act of 2013
2. RA 10173: Data Privacy Act of 2012

3. RA 10175: Cybercrime Prevention Act of 2012
4. Executive Order No. 89, Series of 2015: Creating the National Cybersecurity Inter-agency Committee
5. RA 10639: The Free Mobile Disaster Alerts Act
6. RA 10844: Creating the Department of Information and Communications Technology (DICT)
7. NTC Memorandum Circular No. 07-08-2015: Rules on the Measurement of Fixed Broadband Internet/Access Service
8. RA 10929: An Act Establishing The Free Internet Access. Program in Public Places In The Country and Appropriating Funds Therefor.

Social Infrastructure

Education

In 2016, the National Capital Region has 2,169 elementary schools, of which 518 are public and 1,651 are private schools. For secondary schools, there are 258 public schools and 829 private schools with a total of 1,087 schools. Among the LGUs, Quezon City, being the largest in terms of size and population, has the highest number of government schools both in elementary with 96 schools and secondary with 447 schools. The DPWH being the agency responsible to construct public infrastructure facilities is responsible in the construction of public schools and its “educational facilities.”³¹

The following are the challenges:

Insufficient school classrooms. In 2013, the classroom-student ratio for primary and secondary public schools is 1:65 and 1:72, respectively which is way beyond the

³⁰ Integrated Government Philippines (2018). About Us. <http://i.gov.ph/about-us/>

³¹ The term “educational facilities” refers to all the physical properties of a school, consisting of the grounds, buildings, and the various facilities within the school and inside the school buildings as describe by Educational Facilities Manual of the DepEd (<http://www.deped.gov.ph/sites/default/files/2010%20Educational%20Facilities%20Manual.pdf>)

Table: 13.13. Student Classroom Ratio by Division of Schools, SY 2009-10 to SY 2012-13

DIVISION	ELEMENTARY-CLASSROOM RATIO				SECONDARY-CLASSROOM RATIO			
	2009-2010	2010-2011	2011-2012	2012-2013	2009-2010	2010-2011	2011-2012	2012-2013
National Capital Region	75.60	66.51	65.74	64.95	78.66	75.76	74.21	72.15
City of Manila	57.60	53.38	52.25	51.98	62.64	53.22	51.01	50.60
City of Mandaluyong	53.01	40.70	41.99	42.03	64.75	53.81	55.28	44.18
City of Marikina	76.78	62.05	61.30	61.24	72.53	71.13	71.74	73.20
City of Pasig	86.72	74.12	72.18	72.92	84.99	79.76	76.97	76.53
Quezon City	86.72	74.12	72.18	72.92	84.99	79.76	76.97	76.53
City of San Juan	42.54	36.42	35.31	34.65	79.46	76.73	77.59	68.98
Caloocan City	106.05	90.60	84.74	86.27	100.17	101.42	102.84	98.03
City of Malabon	93.38	80.04	80.49	77.19	115.16	108.77	118.48	103.97
City of Navotas	90.32	76.62	79.28	72.97	125.66	120.13	117.98	105.11
City of Valenzuela	84.95	79.47	77.95	76.13	93.48	95.60	99.03	92.24
City of Las Piñas	111.42	96.51	93.06	87.34	127.39	135.22	122.23	119.95
City of Makati	55.99	49.27	48.78	46.52	68.06	69.65	66.42	64.61
City of Muntinlupa	73.59	70.26	69.49	68.16	69.81	85.67	73.06	72.49
City of Parañaque	94.99	83.21	95.59	87.56	92.55	88.99	80.84	83.86
Pasay City	56.22	49.00	51.08	51.48	68.58	67.94	66.32	64.54
Taguig City/Pateros	97.58	87.11	81.13	79.35	106.48	103.56	102.00	83.17

Note: Data aggregated by division; Public schools only.

Source: Department of Education, EBEIS

standard ratio of 1:45.³² Although there is a slight improvement from SY 2009-2010 to SY 2012-2013 as shown in *Table 13.13* still, there is a need to construct additional classrooms and other school facilities in the region. The implementation of K-12 program as well as the fast growth rates in the region aggravates the situation.

Relatively poor maintenance of school facilities. Operations and maintenance of school facilities including supplies of water, electrical, telecommunications, security and fire suppression systems to provide comfort and pleasing environment in the school campus is relatively poor. Lack of access to the basic services such as clean water and sanitation facilities affects the learning process, health and nutrition of public school students.

Weak compliance to BP 344 - An Act to Enhance the Mobility of Disabled Persons by Requiring Certain Buildings, Institutions, Establishments and Public Utilities to install Facilities and Other Devices. Another concern

is the accessibility of children with disability in school facilities. Although there are already ramps in some schools, toilets with railings are still insufficient if not lacking.

Unavailability of government land as school building sites. Another constraint is the availability of government land for construction of additional school buildings. Taking into consideration of the limited space of the region, with only 636 sq. km having a population of 12.88 million,³³ land became a luxurious position in the metropolis.

Health

Access to health facility is necessary to have a healthy community. As of 2016, the total number of health facilities in NCR are 477 health centers, which represent 1:27,684 population. In addition, there are 54 lying-in clinics and 56 barangay health stations/satellites/sub-centers.

There are also 164 hospitals located in Metro Manila. Of these, 48 are government-

³² Prescribed under Republic Act 7880

³³ PSA-NCR

owned, operated by both national and local and the rest are privately owned (*Table 13.14*). However, from the existing hospitals operated by the LGUs, three (3) hospitals has been downgraded from level 1 to infirmary category because there were services that are not available in those three hospitals, namely, Pagamutang Bayan ng Malabon, Valenzuela City Emergency Hospital and Caloocan City Medical Center. Further, among all the LGUs, Marikina, Pasig, Taguig, Quezon City, Makati, Mandaluyong, San Juan, Manila, Las Piñas, Muntinlupa, and Pasay City met the standard ratio of 1 bed to 1000 population.

However, for the whole NCR, the bed-population ratio is 1:442 (*Table 13.15*).

In terms of rehabilitation centers, at present, there are eight (8) rehabilitation centers located in the NCR, of which four (4) are government-owned while the other four (4) are being run by the Non-Government Organizations (NGOs). Considering the number of drug dependents surrendered as a result of the ardent campaign against illegal drugs of the present administration, construction of additional drug rehabilitation centers are very much needed.

Table 13.14. Government and Private Hospitals by City and Municipality, 2016

CITY/ MUNICIPALITY	LEVEL OF HEALTH CARE											TOTAL NUMBER OF HOSPITALS
	GOVERNMENT HOSPITAL RETAINED			OTHER LOCAL GOVERNMENT OWNED			TOTAL	PRIVATE HOSPITALS			TOTAL	
	L1	L2	L3	L1	L2	L3		L1	L2	L3		
Malabon	1						1				0	1
Navotas				1			1				0	1
Valenzuela			1				1	1	2	1	4	5
Caloocan		1		1			2	6		1	7	9
Marikina			1				1	5	3		8	9
Pasig			1		1	1	3	9	3	1	13	16
Pateros							0		1		1	1
Taguig				3			3	2	2		4	7
Quezon City			8	2	1	3	14	15	7	9	31	45
Makati						1	1	1	1	1	3	4
Mandaluyong			1		1		2	1		1	2	4
San Juan				1			1	1		1	2	3
Manila			4	5		2	11	5		11	16	27
Las Piñas		1					1	5	1	2	8	9
Muntinlupa			1	1	1		3	3	3		6	9
Parañaque				1			2	4	4	1	9	10
Pasay					1	1	2			2	2	4
TOTAL	1	2	17	15	5	8	48	58	27	31	116	164

Table 13.15. Government and Private Hospital Bed Population Ratio in NCR, 2016

MUNICIPALITIES/ CITIES	POPULATION 2016	NUMBER OF AUTHORIZED BEDS			TOTAL BEDS (GOV'T. & PRIVATE)	BED POP. RATIO
		GOVERNMENT RETAINED	OTHER LOCAL GOV'T. OWNED	PRIVATE		
MALABON	362,836	10	0	0	10	1:36,284
NAVOTAS	261,648	0	50	0	50	1:5,233
VALENZUELA	639,790	100	0	339	439	1:1,457
CALOOCAN CITY	1,722,366	200	82	418	700	1:2,461
MARIKINA	447,099	150	0	370	520	1:860
PASIG	797,054	300	264	940	1,504	1:530
PATEROS	68,663	0	0	58	58	1:1,184
TAGUIG	783,592	0	395	668	1,063	1:737
QUEZON CITY	3,190,989	2,855	2,472	3,411	8,738	1:365
MAKATI	566,015	0	300	695	995	1:569
MANDALUYONG	355,590	4,200	150	203	4,553	1:78
SAN JUAN	123,266	0	150	366	516	1:239
MANILA	1,686,621	1,850	2,634	3,117	7,601	1:222
LAS PIÑAS	608,833	200	0	499	699	1:871
MUNTINLUPA	481,468	50	649	499	1,198	1:402
PARAÑAQUE	692,381	0	50	584	634	1:1,092
PASAY CITY	417,005	0	250	380	630	1:662
Total	13,205,216	9,915	7,446	12,547	29,908	1:442

Source: DOH- NCR

Solid Waste Management (SWM)

Limitation of facilities to accommodate the increasing volume of solid waste in the region. Rapid urbanization has put too much pressure on our land use and on our basic infrastructures. Increasing population and consumption has resulted a growth rate of waste generated of 16.24 percent from 2015 to 2016. This is aggravated by inefficient collection and inadequate disposal facilities in Metro Manila.

The region has three (3) Sanitary Landfills (SLFs) which the MMDA is responsible for the maintenance and operation of SLF while the LGUs are accountable for the collection and disposal of garbage. The

3 SLFs are the Rizal Provincial Sanitary Landfill (RPSLF), Navotas Sanitary Landfill (NSLF) and Quezon City Sanitary Landfill (QCSLF). However, QCSLF has been closed based on the recommendation of the Mines and Geosciences Bureau (MGB) which conducted a technical evaluation of the facility's structural stability. The waste disposal of Quezon City was diverted initially to Navotas SLF via the Pier 18 Marine Loading Station and Rizal Provincial SLF but eventually stopped the use of NSLF to divert all to RPSLF. While sanitary landfill in Carmona, Cavite and San Mateo landfill are need to rehabilitate.

Strategic Framework

As shown in *Table 13.16*, infrastructure development supports all the three (3) pillars and intermediate goals of the plan, as it's vital to enhancing the social fabric, reducing inequality, and increasing the region's growth potential.

Targets

Table 13.16. Plan Targets to Accelerate Infrastructure Development

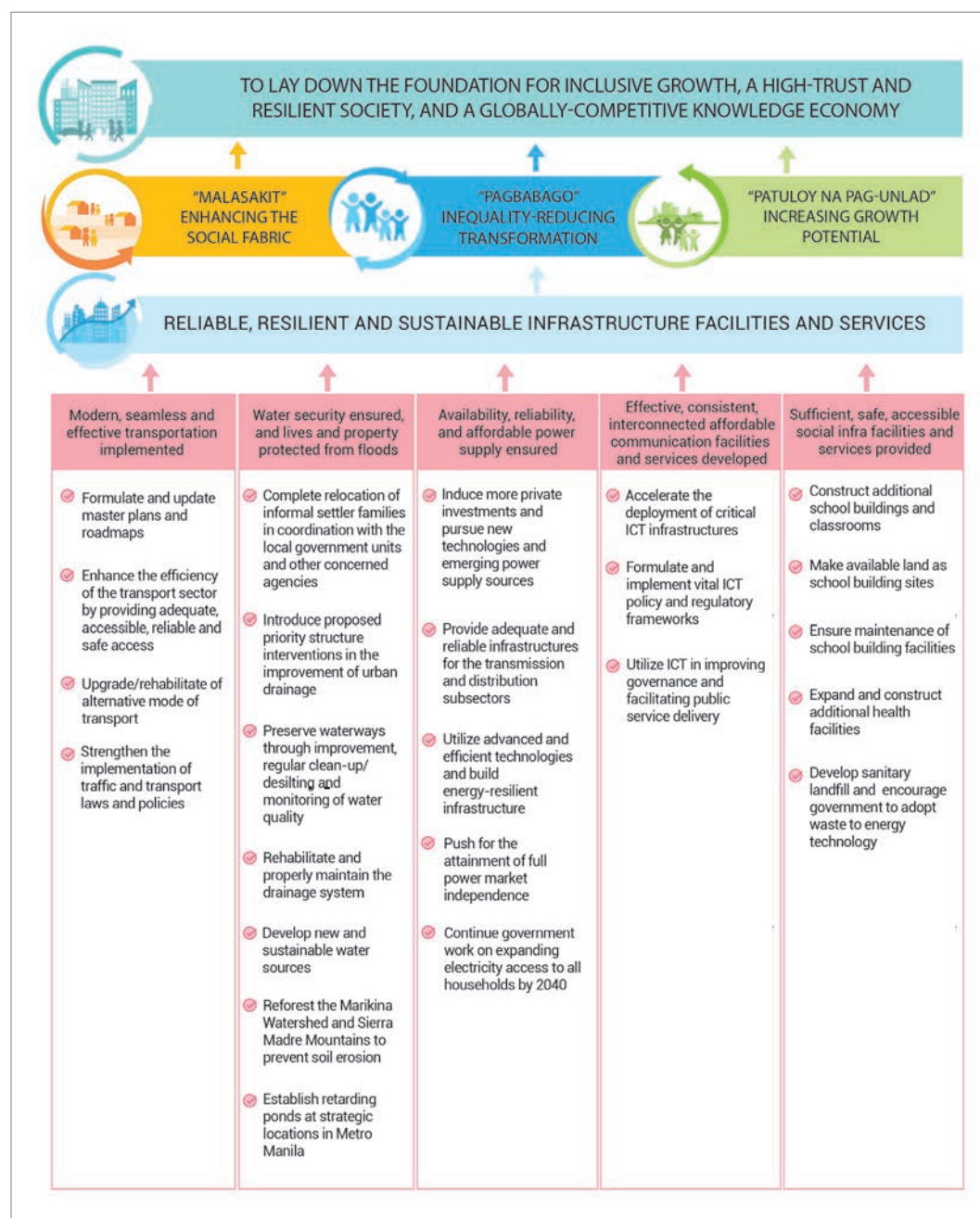
INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Average International Road Roughness Index in national primary roads achieved	2016	4.69	3.0	DPWH
Length of national roads increased (km)	2016	1,159.01	1,204.55	DPWH
Length of existing paved national roads increased (km)	2016	1,159.01	1,204.55	DPWH
Percentage paved of national road to total national road in kilometres	2016	100%	100%	DPWH
Percentage of national roads in good condition increased (%)	2016	62.44%	77.20%	
Length of permanent bridges along national roads increased (lm)	2016	27,252		
Number of permanent bridges along national roads Increased	2016	289		
Percentage of permanent bridges in good condition increased (%)	2016	49.48%	70.00%	DPWH
Percentage of bridges retrofitted	2016	4.51%	4.51%	DPWH
Percentage of bridges widened	2015	1.04%	1.04%	DPWH
Percentage of permanent bridge to total bridge along national roads	2016	7.92%	100%	DPWH
Kilometers of Track Added (LRT/MRT)	2016	51.1	90.3	DOTr
Number of motor vehicles registered	2016	2,405,122	Decreased	LTO
Optimal capacity in train systems achieved, in passengers per square meter (sq. m.)				
Air passenger traffic increased (international and domestic) in number of passengers	2016	39,564,972		DOTr CAAP MIAA
Air cargo traffic increased (international and domestic), in MT	2016	620,283,230		DOTr CAAP MIAA
Number of round-trip international flights increased	2016	103,327		DOTr CAAP
Number of round-trip domestic flights increased	2016	187,868		DOTr CAAP MIAA
No. of Pasig Ferry Stations	2017	12	17	MMDA
No. of passengers via Pasig River Ferry Service	2016	105,475	921,600	MMDA
No. of population with access to safe water supply increased				
Mandaluyong	2016	436,897	436,897	MWCI
Makati	2016	457,710	457,710	MWCI
Marikina	2016	461,880	461,880	MWCI

INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Quezon City	2016	1,083,653	1,083,653	MWCI
Pasig	2016	840,577	840,577	MWCI
Pateros	2016	65,208	65,208	MWCI
San Juan	2016	120,925	120,925	MWCI
Taguig	2016	907,846	907,846	MWCI
Manila	2016	191,221	191,221	MWCI
Parañaque	2016	6,346	6,346	MWCI
No. of population with access to sewerage and sanitation increased				
Sewerage				
Makati	2016	244,737	376,663	MWCI
Mandaluyong	2016	1,522	149,506	MWCI
Manila	2016	41,931	203,737	MWCI
Marikina	2016	172,182	388,577	MWCI
Pasig	2016	72,131	490,081	MWCI
Pateros	2016	-	14,382	MWCI
Quezon City	2016	262,298	500,556	MWCI
San Juan	2016	-	-	MWCI
Taguig	2016	111,196	389,112	MWCI
Parañaque	2016	-	-	MWCI
Sanitation				
Makati	2016	491,242	347,860	MWCI
Mandaluyong	2016	350,822	436,897	MWCI
Manila	2016	193,971	185,484	MWCI
Marikina	2016	452,697	461,880	MWCI
Pasig	2016	714,852	814,687	MWCI
Pateros	2016	68,464	65,208	MWCI
Quezon City	2016	1,110,817	866,922	MWCI
San Juan	2016	129,603	120,925	MWCI
Taguig	2016	687,849	862,453	MWCI
Parañaque	2016	4,610	6,346	MWCI
Total length of drainage system (km)	2016	1,615		
No. of ISFs living along waterways reduced	2016	32,082		
Percentage of HHs with electricity				
Conserved annual amount of electricity and fuel increased, in kilotons oil equivalent				
Classroom to pupil ratio improved				
• Kindergarten			1:25	
• Grades 1-3			1:30	
• Grades 4-6			1:40	
• Junior High School			1:40	
• Senior High School			1:40	
Access to health services improved				
• Number of public hospitals and hospital facilities (i.e. bed) in each LGUs established/provided				
• Number of socialized housing units provided				
• Number of barangays/LGUs with access to Sanitary Land Fills				

Strategies

By 2022, the following subsector outcomes should have been achieved: (a) modern, seamless and effective transportation; (b) water security ensured, lives and property protected from floods; (c) availability, reliability and affordable power supply; (d) effective, consistent, interconnected, affordable communication; and (e) sufficient, safe, accessible and cost effective social infra facilities and services (Figure 13.6).

Figure 13.6. Strategic Framework to Accelerate Infrastructure Development



To implement modern, seamless and effective transportation

Implementation of plans and programs particularly mass transport system projects should be strengthened and made them beneficial to all to encourage commuters to shift to public transport.

Formulate and update master plans and roadmaps to expand road network. Master plans are necessary to ensure that the implementation of the government priority programs/activities/projects is harmonized and well-coordinated. It also, help to identify the location of the roads that need to be develop to link in the tourism sector and business areas. This will also help to lessen the traffic congestion due to the increasing number of registered motor vehicles in the region which affect our economy.

Enhance the efficiency of the transport sector by providing adequate, accessible, reliable and safe access for people and goods across the neighboring regions and other countries. With emphasis on improved connectivity and enhanced mobility, multi-modal transport terminals will be established to improve accessibility and achieve better integration of the economy while respecting the environment. This will include the rehabilitation of ancillary facilities as well as the integration of fare collection systems and stored value cards or similar electronic media to ensure maximum convenience of passengers and easy transfers to different kinds of mode of transport.

Furthermore, continues improvement and rehabilitation of railways with the latest technology will be implemented to encourage passengers in using public transport. The Philippine National Railway including the facilities such as railways will be rehabilitated for the safety of the passengers.

Upgrade/rehabilitate alternative mode of transport like Pasig River Ferry System. The Pasig River Ferry System will be upgraded and additional station will be constructed to encourage the passengers to use the ferry system. Also, it will be upgraded to the latest technology to be used to transport prime commodities and serve as access for emergency response operation during phenomenal calamities such as earthquake.

Strengthen the implementation of traffic and transport laws and policies. The existing laws and policies in transport sector should be revisited and updated to ensure effective enforcement and implementation. Partnership and coordination between stakeholders as well as national and local government units will be strengthened to harmonize policies and programs to address conflicting policies and overlapping mandates across agencies and levels of government which have resulted in conflict and/or duplication of efforts.

To ensure water security, and protect lives and property from floods

Generally, the following strategies must be considered:

1. Complete relocation of informal settler families in coordination with the local government units and other concerned agencies;
2. Introduce proposed priority structure interventions in the improvement of urban drainage in order for the drainage areas to be emptied of flood waters within an acceptable time duration;
3. Preserve waterways through improvement, regular clean-up/desilting and monitoring of water quality suitable for swimming and other forms of water recreation;
4. Rehabilitate and properly maintain the

drainage system to increase its conveyance capacity in draining floodwaters;

5. Develop new and sustainable water sources to meet the water demand of the growing number of households in Metro Manila;
6. Reforest the Marikina Watershed and Sierra Madre Mountains to prevent soil erosion; and
7. Establish retarding ponds at strategic locations in Metro Manila such as parks, playgrounds, gardens and other recreational areas.

To ensure availability, reliability and affordable power supply

It is envisioned that the Philippines obtain a responsive and dynamic energy sector through implementation of policy thrusts and strategic directions geared towards supporting the growing industrialization of the country. In the power sector, full restructuring and reform of the electric power industry by 2040 is envisioned. Focusing on ensuring quality, reliable, affordable and secure supply; expanding access to electricity; and ensuring a transparent and level playing field in the power industry, the following strategies are identified:

1. Induce more private investments and pursue new technologies and emerging power supply sources consistent with the appropriate power mix policy for the generation sector;
2. Provide adequate and reliable infrastructures for the transmission and distribution subsectors to be able to deliver the supply to end-users;
3. Utilize advanced and efficient technologies and build energy-resilient infrastructure to take-off in the short-term up to the long-term planning horizon;
4. Push for the attainment of full power

market independence through the continuous review, issuance and implementation of policy guidelines on Wholesale Electricity Spot Market (WESM) and Retail Competition Open Access (RCOA) for the supply subsector; and

5. Continue government work on expanding electricity access to all households by 2040 through various programs and projects embodied under the electrification roadmap.

To develop effective, consistent, interconnected affordable communication facilities and services

Accelerate the deployment of critical ICT infrastructures. In order to ensure the provision of strategic, reliable, cost-efficient and citizen-centric ICT infrastructure, the NCR-RDC will facilitate the implementation of the plans, programs and projects in the region such as a) National Broadband Plan, b) Digital Terrestrial Television Broadcasting Migration Plan, c) Pipol Konek, d) SECURE GOVNET/Luzon Bypass Infrastructure, e) WiFi Internet Access in Major Thoroughfares (Road Wi-Fi), and f) IT-BPM Roadmap.

Moreover, the NCR-RDC will provide necessary incentives to ensure such facilitation. This will be done by:

1. Mandating all local government units to reduce permit and licensing requirements for putting up ICT infrastructures and standardized fees associated with the aforementioned requirements;
2. Facilitate the acquisition of rights-of-way by coordinating public utility works, provision of space construction of passive infrastructures (e.g. conduits, poles) and promoting co-use or sharing of this facilities; and

3. Maintaining database or information system to support future policy and decision making process.

Formulate and implement vital ICT policy and regulatory frameworks. In order to keep abreast with the fast-changing demand in ICT, reforms on the following critical laws should be pursued: a) Commonwealth Act No. 146 or the Public Service Act; b) Republic Act No. 3846 or the Radio Control Law; c) Republic Act No. 7925 or the Public Telecommunication Policy Act of 1995; and d) Executive Order No. 467, s. 1998 Providing for a National Policy on the Operation and Use of International Satellite Communications in the Country.

Likewise, the masterplans/roadmaps should be developed/updated to provide policy directions for the development of the ICT sector such as a) E-government Masterplan, b) National ICT Ecosystem Framework, and c) National Capital Region ICT Strategic Plan.

Utilize ICT in improving governance and facilitating public service delivery. In furtherance of interoperability and efficiency in the government, NCR-RDC shall facilitate the implementation of the following system among the government agencies and LGUs in the region:

- Authoritative Registries and Government Common Platform
- National Government Portal
- Integrated Business Permits and Licensing System
- Government Operations and Management Platform

To provide sufficient, safe, accessible social infra facilities and services

Education

- Construct additional school buildings and classrooms
- Make available land as school building sites
- Ensure maintenance of school building facilities

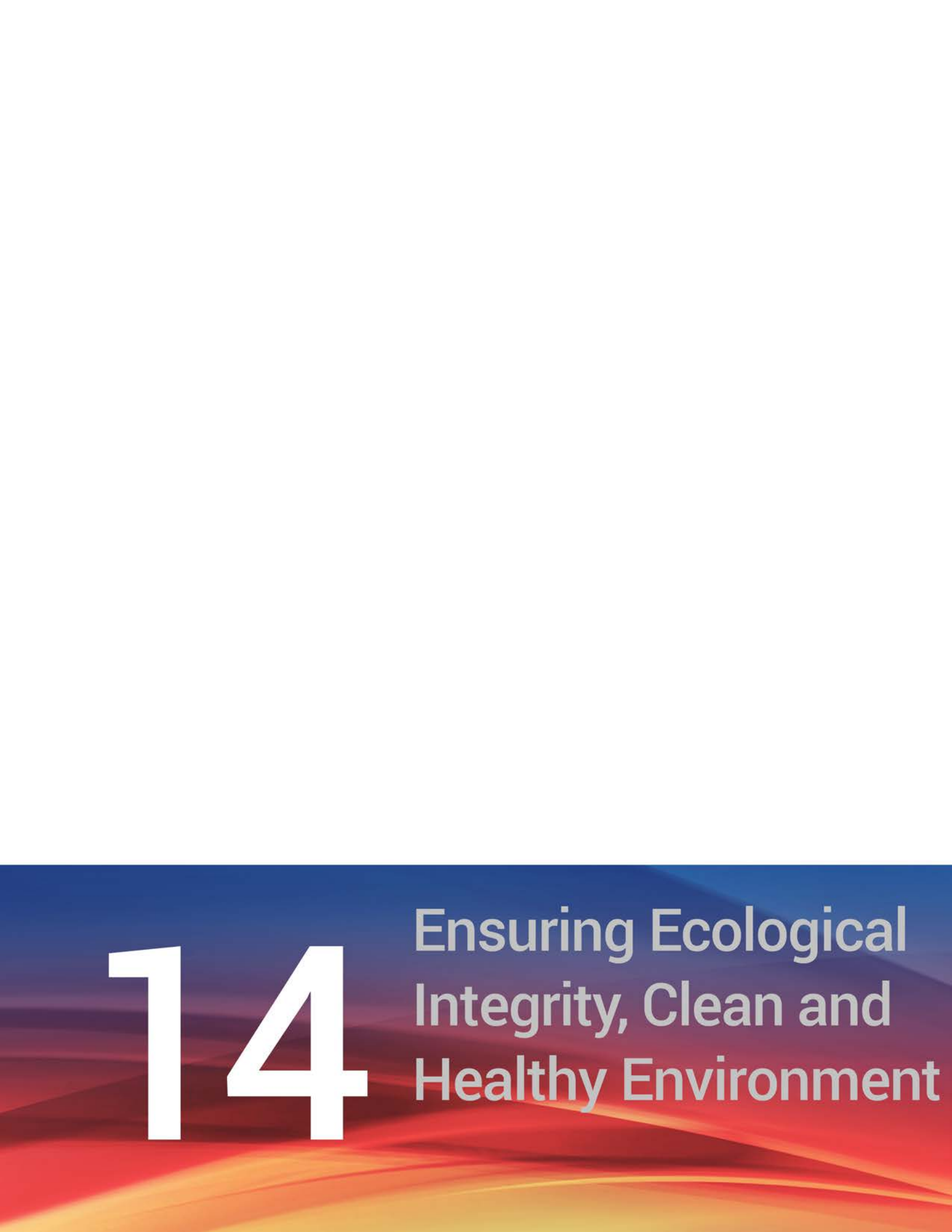
Health

Expand and construct additional health facilities to attain the required standard ratio.

Solid Waste Management

Develop sanitary landfill and encourage government to adopt waste to energy technology. With the increasing volume of waste in Metro Manila and the relatively unsustainable SWM interventions among LGUs, the Government will develop new landfills and push the adoption of the appropriate waste-to-energy (WTE) technologies. WTE technologies will convert wastes into a genuine resource-generating and sustainable renewable energy with minimal residues brought to SLFs, thus, extending the latter's life spans and minimizing land requirements.

The closure and rehabilitation plan for Carmona Sanitary Landfill in Cavite will be pursued as well as the development of San Mateo Landfill to ensure environmental protection and safeguard safety of human health.



14

Ensuring Ecological
Integrity, Clean and
Healthy Environment

Ensuring Ecological Integrity, Clean and Healthy Environment

The environment and natural resources (ENR) desired outcome is to provide the region with more productive endeavors while ensuring sustainable utilization of the ecosystem services. These services are (a) the products obtained from ecosystem such as freshwater, food, fuel, fiber and genetic resources; (b) benefits obtained from regulation of ecosystem processes like regulation of air quality, water, climate and disease; (c) non-material benefits obtained from ecosystems such as spiritual and religious enrichment, recreation and tourism; and (d) services for the production of all ecosystem like habitats for species and maintenance of genetic diversity.

The provision of these basic services, including the maintenance of a healthy and clean environment have put a strain on the limited natural resources of the region due to the demands arising from the development and utilization activities, population expansion, inadequate environmental protection and the global impact of climate change and natural disasters which bring bigger problems like extreme weather conditions. This is reflected in the degraded and deteriorated state of the Manila Bay area, the region's major water outlet, with relatively higher risks of urban decay. On December 18, 2008, by virtue of a writ of continuing mandamus, the Supreme Court ordered "to clean up, rehabilitate, and preserve Manila Bay, and restore and maintain its waters to SB level (Class B sea waters per water classification tables under DENR Administrative Order No. 34,1990 and DAO No. 2016-08) to make them fit for swimming, skin diving, and other forms of contact recreation".

Interrelated environmental concerns such as the quality of air, water, waste management, health and sanitation, and adequacy of land and water resources are some of the challenges facing Metro Manila residents in rehabilitating and restoring degraded natural resources.

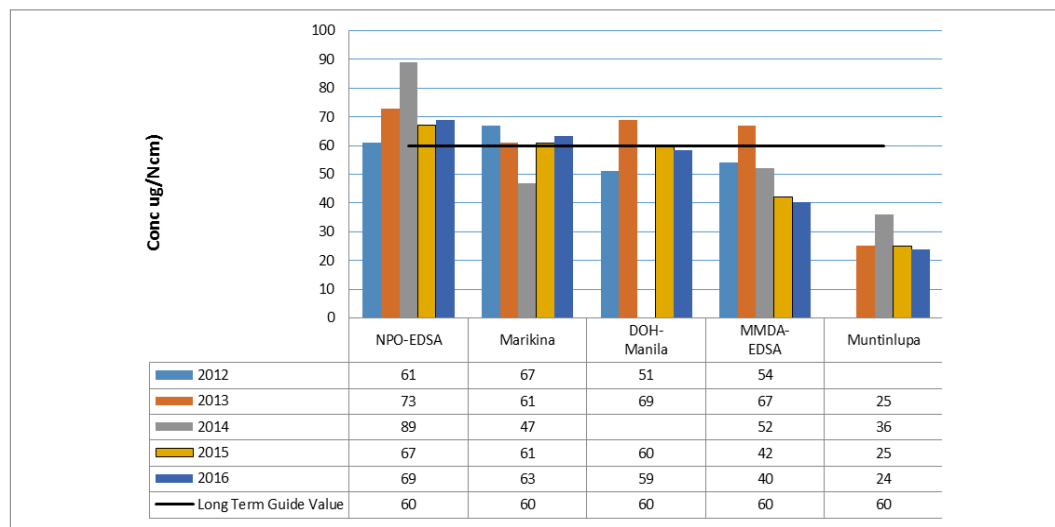
Assessment and Challenges

Poor air quality despite slight improvements recorded. From 2012 to 2016, improvements have been observed in Metro Manila's ambient air quality for particulate matter measuring 10 micrometers in diameter or smaller (PM10) with a long term guide value of 60 µg/Ncm (*Figure 14.1*). Among the five (5) stations measuring PM10, only Muntinlupa station conformed with the standard requirement

of 60 micrograms per normal cubic meter (µg/Ncm) for four (4) years from 2013-2016. While the National Printing Office (NPO) in EDSA, Quezon City remains with the highest data recorded for the past five (5) years which is above the long term guide value.

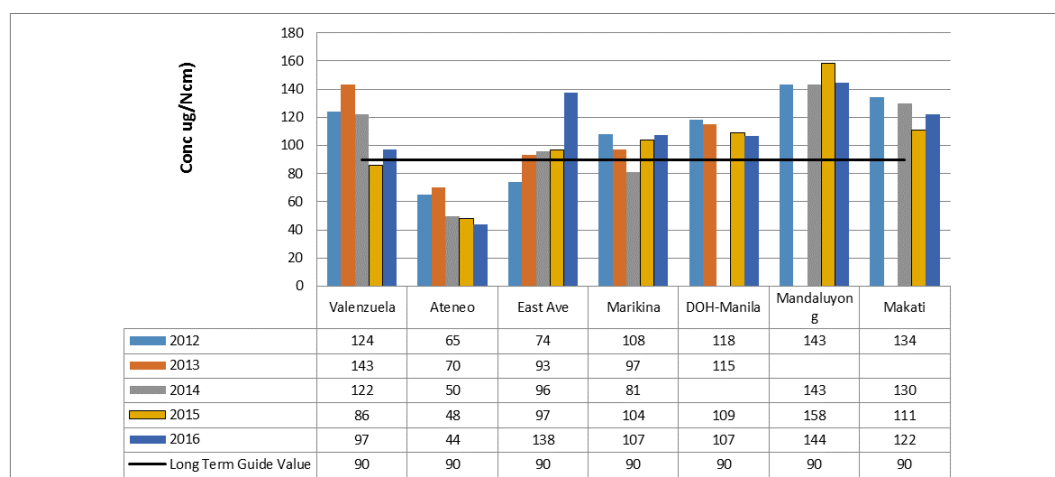
With regards to Total Suspended Particles (TSP), the NCR is operating and

Figure 14.1. PM10 Annual Geometric Mean Comparative Bar Graph (conc. in ug/Ncm) 2012-2016



Source: Environmental Management Bureau—DENR

Figure 14.2. TSP Annual Geometric Mean Comparative Bar Graph (conc. in ug/Ncm) 2012-2016



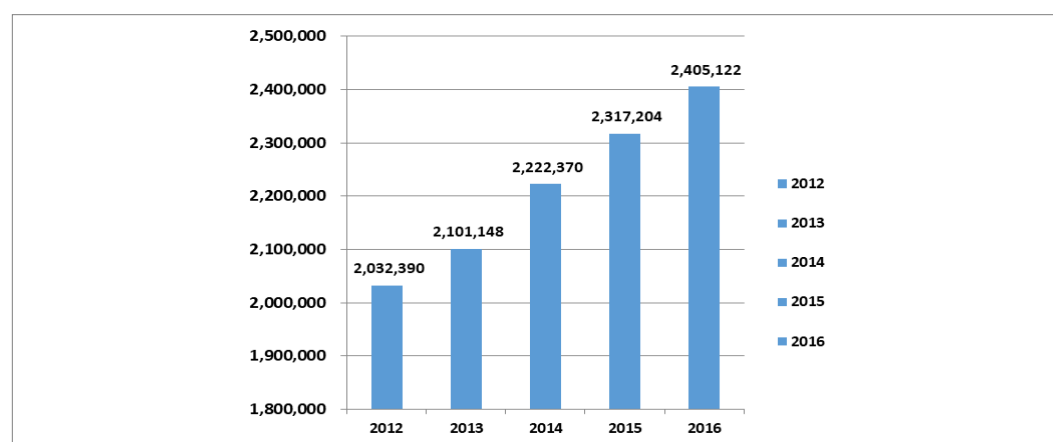
Source: Environmental Management Bureau—DENR

maintaining seven (7) TSP manual air quality monitoring stations within Metro Manila. Out of seven (7) stations measuring TSP only Ateneo station conforms to the requirement of general ambient air of 90 μ g/Ncm (Figure 14.2). Whereas, Mandaluyong City Hall Compound in Maysilo Circle Plainview, Mandaluyong City recorded the highest measurement for the past five

(5) years due to the continuous growth of registered motor vehicles in the road which is the largest contributor of emission in the region, thus, environmental situation in the metropolis is at a critical level.¹ From 2012 to 2016, the number of motor vehicles in Metro Manila increased by about 26 percent, with an average annual increase of 4.36 percent (Figure 14.3). Furthermore, greenhouse gas

¹80% of air pollutants came from the mobile sources, 2009 National Emission Inventory of the Department of Environment and Natural Resources – Environmental Management Bureau (DENR-EMB)

Figure 14.3. NCR Registered Motor Vehicles, 2012-2016



Source: Land Transportation Office

Table 14.1. Manual Air Quality Monitoring Stations in NCR

STATION NO./ID & CLASSIFICATIONS		POLLUTANTS MONITORED	LOCATION/ADDRESS
1	General Ambient	TSP	Pamantasan Lungsod ng Valenzuela, Maysan Road Poblacion, Valenzuela City
2	General Ambient	TSP	Manila Observatory, Loyola Heights, Quezon City
3	Roadside Ambient	TSP	National Ecology Center Compound, East Avenue, Quezon City
4	Roadside Ambient	PM ₁₀	National Printing Office Compound, EDSA Diliman, Quezon City
5	Roadside Ambient	TSP & PM ₁₀	Marikina Sports Complex Compound, Sumulong Highway, Barangay Sto. Niño, Marikina City
6	Roadside Ambient	TSP & PM ₁₀	DOH Compound Rizal Avenue, Sta. Cruz, Manila
7	Roadside Ambient	TSP	Mandaluyong City Hall Compound, Maysilo Circle Plainview, Mandaluyong City
8	Roadside Ambient	TSP	Makati Bureau of Fire Compound, Bel-air, Makati City
9	Roadside Ambient	PM ₁₀	MMDA Compound, Guadalupe, Makati City
10	General Ambient	PM ₁₀	Muntinlupa Bilibid Prison Open Ground, Muntinlupa City

Note: 2 Monitoring Stations each in Sto. Niño, Marikina and Sta. Cruz, Manila

Source: Environmental Management Bureau—DENR

(GHG) emissions from mobile sources were due to decreasing travel speed and increasing travel time which builds up significantly under congested driving conditions.² This attributed to respiratory diseases being the top cause of death in the region.³

Inadequate environmental monitoring equipment. In monitoring the ambient air quality, the region is now facing the challenge in sustaining the continuous monitoring of Air Quality (PM and gases) to comply in the “establishment and operation of appropriate devices, methods, systems and procedures

necessary to monitor, compile and analyze data on ambient air quality” as stated in the Clean Air Act of 1999 (Republic Act 8749) specifically Section 8(b).

The region has a total of 29 monitoring stations located in the different strategic locations in Metro Manila (*Tables 14.1, 14.2 and 14.3*). Of this total, there are 10 manual air quality monitoring stations, three (3) air quality real-time open path – differential optical absorption spectroscopy (DOAS), and 14 PMS-CAMS air quality real-time monitoring stations. Manual monitoring

²Based on an article posted in the Australian Government website of the Department of Sustainability, Environment, Water, Population and Communities entitled “Assessing Vehicle Air Pollution Emissions”, <http://www.environment.gov.au/atmosphere/airquality/publications/vehicle-pollution.html>

³DOH - NCR

Table 14.2. Air Quality Real-Time Monitoring Stations in NCR Open Path-DOAS

STATION NO./ID & CLASSIFICATIONS		POLLUTANTS MONITORED	LOCATION/ADDRESS
1	South Caloocan	PM ₁₀ , PM _{2.5} , CO, NO ₂ , SO ₂ & O ₃	University of the East, 105 Samson Road, Caloocan City
2	Marikina	PM ₁₀ , PM _{2.5} , CO, NO ₂ , SO ₂ & O ₃	Marikina City Environmental Management Office (CEMO), Gil Fernando Ave., Bgy. Sto. Niño, Marikina City
3	Muntinlupa	PM ₁₀ , PM _{2.5} , CO, NO ₂ , SO ₂ & O ₃	Hospital ng Muntinlupa Cpd., Civic Drive Filinvest Ave Corporate City, Alabang, Muntinlupa, 1780 Metro Manila

Source: Environmental Management Bureau—DENR

Table 14.3. Air Quality Real-Time Monitoring Stations in NCR PMS-CAMS

STATION NO./ID & CLASSIFICATIONS		POLLUTANTS MONITORED	LOCATION/ADDRESS
1	North Caloocan	PM10 and PM2.5	Caloocan City Hall Annex Cpd., Caloocan City
2	Malabon	PM10 and PM2.5	Malabon Polytechnic Institute Cpd., Gov. A. Pascual Ave. cor. A. Reyes Street, Malabon City
3	Navotas	PM10 and PM2.5	Navotas City Hall Cpd., M. Naval Street, Navotas City
4	Marikina	PM10 and PM2.5	Marikina Hall Justice open ground, Marikina City
5	San Juan	PM10 and PM2.5	Pinaglabanan Shrine Cpd., Pinaglabanan Street, San Juan City
6	Mandaluyong	PM10 and PM2.5	Plaza Hardin Cpd., Martinez Hi-way, Additional Hills, Mandaluyong City
7	Pasig	PM10 and PM2.5	Country Lodge Cpd., 18 Danny Floro Street, Oranbo, Pasig City
8	Makati*	PM10 and PM2.5	Makati Park and Garden Cpd., Dr. Jose P. Rizal Extension, Makati City
9	Pateros	PM10 and PM2.5	Pateros Elementary School Cpd., P. Herrera Street, San Pedro, Pateros
10	Pasay	PM10 and PM2.5	Philippine Airlines Cpd., Andrew Avenue, Brgy 184 Zone 19, Pasay City
11	Taguig	PM10 and PM2.5	Technical University of the Philippines – Taguig City Cpd., East Service Road, Bicutan, Taguig City
12	Parañaque	PM10 and PM2.5	Don Bosco Brgy. Hall Cpd., Parañaque City
13	Las Piñas	PM10 and PM2.5	Rohm and Hass Warehouse Cpd., CAA Road, Las Piñas City
14	Muntinlupa	PM10 and PM2.5	Muntinlupa Bilibid Prison open ground, Bilibid, Muntinlupa City

* Struck by lightning for replacement

Source: EMB-DENR

stations are high volume samplers which measure either the concentration of TSP or Particulate Matter 10 (PM10) which has been operating for more than five (5) years. Both of these parameters have their own short term and long term guideline value. Short term guideline value covers the period of 24 hours while the long term guide value covers the period for 1 year.

Water bodies in the region remains polluted. One of the major water bodies

in the region is the Pasig River with 27,427 linear meters in length consisting of 47 tributaries with a total length of 146,737.58 linear meters.⁴

The Pasig River Rehabilitation Commission (PRRC) is continuously implementing programs and projects on Pasig River to revive its historically pristine condition, conducive to transport, recreation and tourism. The commission has 13 Pasig River Unified Monitoring System (PRUMS)

⁴ Pasig River Rehabilitation Commission, as of July 3, 2017

Table 14.4. Water Quality Test of Pasig River, 2012-2017

PARAMETER	CLASS C STANDARD	2012	2013	2014	2015	2016	2017
Dissolved Oxygen	5 mg/L	2.79	2.73	2.99	2.69	2.14	1.85
Biochemical Oxygen Demand (BOD)	7 mg/L	24.47	36.13	37.62	56.13	57.77	57.00
Fecal Coliform	200 MPN/100mL	6.77E+09	1.02E+09	1.37E+08	5.68E+09	7.39E+07	5.85E+06
Nitrate	7 mg/L	5.81	5.85	3.50	5.58	6.06	4.38
Phosphate	0.5 mg/L	13.43	7.90	5.46	11.47	10.93	4.86
Total Suspended Solids	80 mg/L	18.81	25.67	62.56	32.85	33.13	31.27
Oil and Grease	2 mg/L	1.53	1.27	3.07	6.21	6.01	3.40
Total Coliform	-	9.14E+08	1.31E+09	2.11E+08	6.19E+09	9.84E+07	-

Source: Pasig River Rehabilitation Commission

located in C6 Bridge, Bambang Bridge, Guadalupe Ferry Station, Lambingan Bridge, Nagtahan Bridge, Jones Bridge, Vargas Bridge, Marikina Bridge, Buayang Bato, Guadalupe Nuevo, Guadalupe Viejo, Sevilla Bridge and Havana Bridge; and six (6) main Pasig Stations located in Napindan, Bambang, Guadalupe Ferry, Lambingan, Nagtahan and Jones to monitor the quality of water in Pasig River. However, some of the parameters used for testing such as dissolved oxygen, biochemical oxygen demand (BOD), fecal coliform, nitrate, phosphate, total suspended solids, and oil and grease were not conformed with the DENR minimum water quality criteria for class C standard of freshwater in the past six (6) years (*Table 14.4*). Highest value of fecal coliform shows that there is a possible contamination of human or animal waste.

Marikina River System. The Laguna Lake Development Authority (LLDA) has been monitoring the water quality of Laguna de Bay and its tributaries to accurately assess the suitability of the lake for all its present uses and to evaluate the impacts of development activities on the lake's water quality that will serve as important criteria for environmental planning and management. There are nine (9) important monitoring stations in Laguna de Bay (LdB) and another 36 monitoring stations in the

tributaries around the lake.

One of the rivers being monitored by LLDA in Metro Manila is Marikina River. This river is connected to the Pasig River near the Napindan Hydraulic Structure. Said Agency's sampling station is situated near Circulo Verde.

Presented in *Table 14.5* are the annual average concentrations of the water quality parameters monitored in Marikina River from 2012 to 2017. The water quality data of the river from 2012 to 2015 were assessed based on compliance to the DENR Class C Water Quality Criteria in DAO No. 34 while for the 2016 to 2017 data, the basis of the assessment is the DENR Water Quality Guidelines in DAO No. 2016-08.

The annual average concentration for dissolved oxygen and biochemical oxygen demand in Marikina River failed the Class C Water Quality Criteria from 2012 to 2015 and the Class C Water Quality Guidelines in 2016 and 2017. Likewise, the annual average inorganic phosphate concentrations from 2012 to 2015 exceeded the 0.4 mg/L Class C Water Quality Criterion although in 2016 conformed with the 0.5 mg/L Class C Water Quality Guidelines. The annual average total coliform counts from 2012 to 2014 were also very high when compared to

Table 14.5. Annual Average Concentrations of Water Quality Parameters Monitored in Marikina River, 2012-2017

PARAMETER	YEAR						DENR CLASS C WATER QUALITY CRITERIA	
	2012*	2013*	2014*	2015*	2016**	2017**	DAO 34	DAO 2016-08
Dissolved Oxygen, mg/L	1.9	1.1	1.8	1.6	1.4	1.0	5	5
Biochemical Oxygen Demand (mg/L)	12	21	28	11	18	16	10	7
Nitrate, mg/L	0.31	0.32	0.03	0.22	0.30	0.17	10	7
Inorganic Phosphate, mg/L	0.42	0.59	1.09	0.49	0.49	0.67	0.4	0.5
Oil and Grease	1.2	1.7	1.3	5.3	2.4	4.4	2	2
Total Coliform, MPN/100ml	2,013,000	2,017,000	8,550,000	No data	3,057,360	2,316,845	5000	-
Notes: *Assessment of the water quality data is based on the Class C Water Quality Criteria in DAO 34								
** Assessment of the water quality data is based on the Class C Water Quality Guidelines in DAO 2016-08								

Source: Laguna Lake Development Authority

the DENR Class C water quality criterion set at 5,000 MPN/100 mL. The results of the assessment could be attributed to the high organic loading in the river from domestic sources. Nevertheless, the annual nitrate concentrations appeared to be lower than the 10 mg/L Class C criterion and 7 mg/L Class C Guidelines. For oil and grease, the 2 mg/L concentration set for Class C criterion and Class C guidelines seemed to be occasionally exceeded that the annual average concentrations met the Criterion from 2012 to 2014 and failed the Guidelines in 2015 to 2017. The very high annual total coliform counts in the river brought about by the poor water quality condition makes the riverside dwellers highly vulnerable to health hazards.

Manila Bay. The NCR has five (5) river systems: Tullahan River System, Pasig-Marikina River System, San Juan River System, Las Piñas-Parañaque River System, and Zapote River System. These are connected to Manila Bay as a catchment area for approximately 17,000 square kilometers that extends far beyond the boundaries of NCR and up to the mountains of other provinces (Manila Bay Site Coordinating Management Office Annual Report, 2017).

After several consultation meetings with the Department of Interior and Local Government – National Capital Region (DILG-NCR) and the 17 LGUs, said river systems were delineated into three (3) major river systems: MANATUTI River System (Malabon-Navotas-Tullahan-Tinajeros Rivers), MUNTIPARLASPIZAP River System (Muntinlupa-Parañaque-Las Piñas-Zapote Rivers), and PAMARISAN River System (Pasig-Marikina-San Juan Rivers) for the area-based management purpose.

Due to several factors, the state of these major river systems in NCR was spoilt and became heavily polluted. This prompted a group called Concerned Residents of Manila Bay to file a complaint before the Regional Trial Court (RTC) of Imus, Cavite against several government agencies and alleged that the deplorable condition of Manila Bay was due to the failure of these agencies to fulfill their respective mandates on the protection of Manila Bay. In 2002, the RTC then rendered a decision ordering the defendant-government agencies to conduct joint clean-up and rehabilitate Manila Bay and restore its waters to SB level fit for swimming, skin diving, and other forms of contact recreation.

Based on various studies, the three (3) major river systems failed to meet the set standards for class C waters. In particular, these major rivers got BOD and DO levels which do not meet the standard ratings. There is also a high presence of phosphate and nitrates as well as coliform in the river systems. This could be accounted for the small cultivated lands in certain areas under these river systems. While industries are reputed to be polluters, according to some studies, it is the domestic waste that contribute most to water pollution. Untreated sewage from houses are directly discharged onto the tributaries and make their way to the rivers and eventually to Manila Bay. Encroachment of both formal and informal settlements has also been contributing to the spoliation of Manila Bay. Similarly, as the PRRC pointed out, “improper waste disposal and the varying seasonal changes” in the flow and volume of water in the rivers act as main culprits of the observed decline in its water quality.

In the NCR, there are five (5) monitoring stations within the Manila Bay namely: 1) Navotas Fish Port Station, 2) Luneta Station, 3) CCP Station, 4) SM Mall of Asia Station, and 5) PEATC Station. Ten years after the Supreme Court issued a writ of continuing mandamus, directing 13 government agencies to clean up, rehabilitate and eventually preserve Manila Bay, significant results have yet to be achieved to make them fit for swimming, skin diving and other forms of recreation.

In spite of various efforts done by the concerned agencies and other government organizations including private sectors, the water quality of Manila Bay remained to be a major issue. The failed water quality tests based on the Standard of DENR DAO 34 (*Table 14.6*) for Class SC and DAO No. 2016-08 for Class SB (*Table 14.7*), is due to the improper dumping of domestic waste and garbage in waterways as well as dumping of untreated wastewater to riverbanks which contribute to the contamination of water bodies. It is therefore inadequate for the intended uses such as fishing grounds, ecotourism and recreational activities, and the like. It will also lead to water-borne diseases and could be detrimental to human health.

Indiscriminate dumping of wastes. Dumping of wastes in Metro Manila remains a major concern due to improper disposal of waste which contributes to environmental problems such as air and water pollution as well as soil contamination.

The region has 689 kilometers of open waterways composed of rivers, creeks and esteros, 879 kilometers drainage laterals along major thoroughfares, and 59 kilometers of drainage mains and interceptors regularly maintained by the Metropolitan Manila Development Authority (MMDA) to avoid deposits of solid wastes. Most of the solid waste dump in the open waterways, drainage main and laterals are from domestic and industrial sources which obstruct the flow of

Table 14.6. Manila Bay “Class SC” Standard for Marine Waters from 2012-2014

PARAMETER	DENR WATER QUALITY CRITERIA	2012	2013	2014
	DAO 34 CLASS SC			
BOD, mg/L	7	14	14	14
Dissolved Oxygen, mg/L	5	3.6	4.2	0.39
Total Suspended Solids, mg/L	-	15	14	16
Total Coliform, MPN/100ml	5,000	4.8x10 ⁵	2.8x10 ⁵	1.2x10 ⁴
Fecal Coliform, MPN/100ml	-	2.3x10 ⁵	7.8x10 ⁴	4.5x10 ⁴

Source: Environmental Management Bureau—DENR

Table 14.7. Manila Bay “Class SB” Standard for Marine Waters from 2015-2016

PARAMETER	DENR WATER QUALITY CRITERIA	2015*	2016
	DAO-2016-08 CLASS SB		
BOD, mg/L	n/a	17	16
Dissolved Oxygen, mg/L	5	3.3	0.99
Total Suspended Solids, mg/L	50	10	12
Total Coliform, MPN/100ml		5x10 ⁶	1.9x10 ⁶
Fecal Coliform, MPN/100ml	100	2.1x10 ⁶	6.4x10 ⁷

Note: *Manila Bay was re-classified as Class SB last March 2015

Source: Environmental Management Bureau—DENR

water.

Part of the clean-up and maintenance operations of the MMDA are the conduct of regular dredging, declogging and desilting of the region’s waterways and drainage networks. Also, the MMDA implements the Estero Blitz Program which aims to reduce maximize conveyance capacity and minimize the inundation of vulnerable areas in Metro Manila. The campaign specifically enjoined communities’ involvement in the dredging, removal and hauling of garbage, spoils, silts and other debris along the waterways. However, there are 362,662 (55%) linear meters length of open waterways not accessible to flood control equipment due to illegal encroachment such as presence of informal settler families or obstruction of buildings, among others.

From the start of the Estero Blitz Program on 8 August 2011 up to 13 October 2017

involving 181 esteros, a total of 50,173.50 cubic meters of garbage/silts were collected and hauled consisting of 6,789 truckloads which is approximately equivalent to 20 Olympic size swimming pool (Table 14.8).

Weak implementation of environment-related laws, policies and regulations. Despite having a complete set of laws that aims to protect the country’s water bodies, to achieve and maintain clean air, to adopt a systematic, comprehensive and ecological solid waste management program, among others, the desired outcomes of our legal environmental governance arrangements are yet to be achieved. The reasons may be the following: (a) the need to amend/ supplement existing laws, policies and regulations to reflect the current values and needs of modern society; (b) conflicting and overlapping roles and mandates of LGUs and NGAs causing implementation gaps and weak policy integration; (c) limited

Table 14.8. Estero Blitz Program

	2011 Aug. 8– Sept 8	2012 May 28– Jun 8	2013 May 28-June 14	2014 April 4-June 6	2015 Feb 17-July 10	2016 Mar 1-July 13	2017 Mar 1-Oct 13	TOTAL
Volume of Garbage and Silts (m³)	7,383	6,401	2,726	7,550	6,636.50	10,858	8,619	50,173.50
No. of Truck Load (trips)	1,150	949	334	1,159	688	1,196	1,313	6,789
No. of Esteros	45	40	20	16	20	16	24	181

Source: Metropolitan Manila Development Authority

commitment of the responsible government agencies/ units; (d) issues of incapacity in the implementation/enforcement; and (e) inability to establish a clear financing scheme. As a result, measuring the region's progress and setting of new targets, as far as environmental quality is concerned, remains a challenge.

An example of which is the implementation of Republic Act 9003 or the “Ecological Solid Waste Management Act of 2000” especially on the 3Rs of waste management. The waste management in Metro Manila remains to be the problem despite compliance of LGUs with the provisions of Republic Act 9003 as well as following the National Solid Waste Management Commission Resolution No. 143, series of 2015, wherein out of the 17 LGUs which submitted their 10-Year Solid Waste Management Plan (SWMP) but only 16 were approved by the NSWMC. The only LGU without an approved SWMP is the Municipality of Pateros.

In 2016, the actual volume of waste disposed in the SLF is about 19,179,142.98 cubic meters with an increase of 16.24 percent from the 2015 actual volume of garbage disposed of 16,499,825 cubic meters. The increase can be due to population increase, “on and off” operation of material recovery facilities (MRFs) and weak implementation of 3Rs of waste management of Reduce, Reuse, Recycle especially at the household level. Finding solution in the increasing volume of waste produced is still a challenge despite the initiatives of LGUs and some non-government and private entities to implement MRFs, source reduction, and the likes.

There is a need to note that as population increases, waste generation increases. In fact about 0.2650 to 1.0032 kilogram per

person per day of waste is generated. Thus, waste generation is a factor of population.

Lack of greenspaces to serve as environmental buffer.⁵ One of the pre-requisites for a healthy urban living is the abundance of greenspaces that are interspersed with the built environment in a city or any urban area. Standard size of greenspace necessary for healthy urban living set by the World Health Organization of the United Nations (WHO-UN) is 9.5 square meters per capita. Greenspaces include road and street corridors, center islands, interchanges, parks, gardens, recreational areas, institutional premises, commercial and industrial compound, residential subdivisions and villages, educational and military reservations, and others that have vegetation such as trees, palms, bamboos, shrubs, ground covers, and other types of plants.⁶

The NCR, being a highly urbanized area has insufficient greenspaces for various social, economic and environmental benefits to society. Based on the land satellite eight (8) imageries of DENR, greenspaces in Metro Manila are estimated to be 12,152.79 hectares (*Table 14.9*).

The city of Manila has very dense residential areas, and commercial and industrial establishments with only very few trees sporadically distributed all over the city. As shown in *Table 14.9*, said city has only 140.4 hectares of estimated green spaces (EGS). Comparing this to their total land area which is 4,176.28 hectares and their population of 1,780,148, Manila lacks a huge area of greenspaces in order to meet the WHO-UN standard of 9.5 square meters per capita. Another is the city of Pasig which has a land area of 3,300.63 hectares but with a 447.66 hectares of estimated greenspaces which is still way below the standard (NCR Urban

⁵ *Environmental Buffer* is a work done on how environment can be purified or at least stabilized by using entities that naturally make environment go back to its original state more or less. It includes some environmental buffers that can be used to reduce air, water and noise pollution.

⁶ DENR-NCR Urban Greening Plan, 2015

Table 14.9. Estimated Green Spaces in Metro Manila based on Classified NDVI using land satellite 8 imageries, March 8, 2014

CITY/MUNICIPALITY	AREA (HA)*	POPULATION (2015)	EGS (HA)**	TGS (HA)	RGS (HA)
Caloocan	5,370.70	1,583,978	1,181.43	1,504.78	323.35
Las Piñas	3,294.19	588,894	867.96	559.45	-308.51
Makati	2,688.61	582,602	599.22	553.47	-45.75
Malabon	1,664.82	365,525	138.87	347.25	208.38
Mandaluyong	1,152.81	386,276	179.28	366.96	187.68
Manila	4,176.28	1,780,148	140.4	1,691.14	1,550.74
Marikina	2,314.37	450,741	515.61	428.20	-87.41
Muntinlupa	3,854.73	504,509	1,720.44	479.28	-1,241.16
Navotas	1,091.81	249,463	18.54	236.99	218.45
Parañaque	4,839.73	665,822	933.75	632.53	-301.22
Pasay	1,807.68	416,522	273.06	395.70	122.64
Pasig	3,300.63	755,300	447.66	717.54	269.88
Pateros	185.15	63,840	23.22	60.65	37.43
Quezon City	16,116.24	2,936,116	3,580.20	2,789.31	-790.89
San Juan	579.20	122,180	24.3	116.07	91.77
Taguig	4,328.17	804,915	854.73	764.67	-90.06
Valenzuela	4,533.31	620,422	654.12	589.40	-64.72
TOTAL	61,298.41	12,877,253	12,152.79	12,233.39	80.60

*Area declared by each City and Municipality

** Estimated Green Spaces (EGS) of Metro Manila based on Classified NDVI using land satellite 8 imageries

Source: Department of Environment and Natural Resources

Greening Plan, 2015). Though the cities in NCR are very active in greening their areas, the current greenspaces are still lacking and are only temporary in nature due to threats of conversion into other more economically profitable uses. These greenspaces will be drastically reduced over time due to fast urban development, such as conversion into subdivisions and condominiums, commercial and industrial purposes. Some of these greenspaces, both public and private such as parks, gardens, and plazas are converted into parking areas, recreational, and commercial establishments.

Though the introduction of any of these improvements on the land may increase its economic value, it may also be detrimental to its intrinsic environmental value. The changes in the urban landscape in the region further aggravate its vulnerability to disaster and impacts of climate change like

typhoon, flooding, storm surge, sea level rise, extreme temperature, water scarcity, and many others. Ultimately, these will affect the ability of urban communities to live safe and may hamper economic progressed of the government.

Due to the rapid urban development in NCR, several greening efforts have already been introduced. An example of which is the establishment of environmental office or unit in some LGUs in Metro Manila such as Manila, Marikina, Makati, Mandaluyong, Taguig, Valenzuela, and Quezon City. These offices are manned with regular staff and personnel who are responsible for the care and maintenance of greenspaces within their jurisdiction. However, other LGUs, though have environmental offices/ units, are still temporary in nature, with no regular budget and no regular manpower, thus, greening efforts may not fully be

implemented. Similarly, some LGUs have not yet established, developed and maintained forest or tree parks of considerable size and number mainly due to unawareness of the law (e.g. Republic Act No. 5752 series of 1969 or “An Act Providing for the Establishment of Municipal and City Forests, Tree Parks, or Watersheds”). Republic Act 5752 mandates every municipality or city to establish, develop and maintain permanent forest, tree parks or watershed within public land owned with an aggregate size equivalent to at least two percent of its entire area and which may be located in one or more places within a city or town (Palijon, 1998).

Weak management of biodiversity protection and conservation. Urbanization causes not only the elimination of native species but also the introduction of exotic and invasive species (Gonzales, 2015). In NCR, being an urban area, also has an ecotourism destinations such as the Las Piñas Parañaque Critical Habitat and Ecotourism Area (LPPCHEA), Navotas Marine Tree Park, and the Quezon City’s La Mesa Watershed Reservation (LMWR) that house various native/endemic/indigenous tree species, as well as migratory and endemic birds that are thriving and usually being sighted within the said habitats.

However, with the continuous threat caused by urbanization, these ecotourism destinations are not spared from the pressing issues on noise pollution; improper solid and liquid waste management; habitat degradation; destructive/illegal fishing; sedimentation and siltation with its various waterways; over-extraction of resources and multiple-use conflict (Navotas ICM Plan). For the LMWR, soil and water conservation problem and siltation from within and outside the area, flora and fauna poaching and over-extraction/collection of other forest products, slash and burn agriculture and other farming practices, application of organic and inorganic

fertilizers, small-scale mining/quarrying, road and other construction activities, and current management schemes threaten the management of LMWR (LMWR 25-Year Integrated Watershed Management Plan, 2014).

Improving urban biodiversity in the remaining mangrove and forestlands in NCR will provide substantial environmental benefits such as helping conserve and reduce energy use, reducing local carbon dioxide levels, improving air quality, and mitigating storm water runoff. Mangrove areas also protect the coast from waves, tidal currents, and typhoons, as well as provide habitats, shelter, breeding sites, and food sources to various groups of fish and other coastal wildlife. The ecological functions of mangroves as land builder and coastline stabilizer are also widely known, thus, these should have to be protected and maintained. Mangroves should also be enhanced and their diversity indices should be improved by planting indigenous species in order to serve as green investments in support to ecologically-sound economic development.

Lack of engagement of private sector in the implementation of climate change and disaster risk reduction measures. Tapping the private sector to get involved in investing in environment and natural resources would surely hasten the programs of the national government. As of 2018, after the enactment of the Republic Act No. 9729 in 2009 and the implementation of National Climate Change Action Plan (NCCAP) 2011-2028, delineating the clear functions of the private sector is still not defined such as providing mechanisms and incentives.

Enticing the support of the private sectors need to be addressed clearly in the policy gap immediately which will benefit the economy in terms of employment generation as well as climate mitigation.

Need to strengthen mainstreaming of Climate Change Adaptation and Disaster Risk Reduction (CCA-DRR) at the local level. Given the Metro Manila's exposure to different types of hazard, several development projects and programs have been implemented to improve the resilience of the region's environment and natural resources to disasters. Different measures were also undertaken to prevent and mitigate the impact of disasters such as (a) mainstreaming CCA and DRR into policies, plans, programs and budget; (b) analysing and assessing the risk and vulnerability as

well as developing of multi-hazard maps; and (c) continuing information, education and communications (IEC) campaigns like capacity buildings, trainings and drills.

However, not all LGUs in the NCR have mainstreamed CCA and DRR measures in their comprehensive land use and development plans (CLUP and CDP) due to lack of capacity to use the available geospatial information and unavailability of technical personnel to prepare the said Plans.

Strategic Framework

Targets

The NCR RDP 2017-2022 targets to increase and upgrade the quality monitoring stations for water and air as well as improve solid waste management which can be linked in the LGUs compliance in RA 9003 to properly assess the environmental quality of the region. Also, it mainstream ecosystem values into local development planning through institutionalization of natural resources monitoring system to conserve and sustain biodiversity and functioning of ecosystem services. Strengthen the management of biodiversity protection and conservation

as well as the establishment of green spaces to attain a healthy urban living. More green spaces within the region can improve the urban environment because it regulates air quality and climate as well as reduce energy consumption by countering the warming effects and recharge the aquifer, and protect streams and creeks from excessive runoff. Furthermore, encourage all LGUs in the region the full implementation of local DRR and CCA plan to increase adaptive capacity and resilience of ecosystems.

Table 14.10. Plan Targets to Ensure Ecological Integrity, Clean, and Healthy Environment

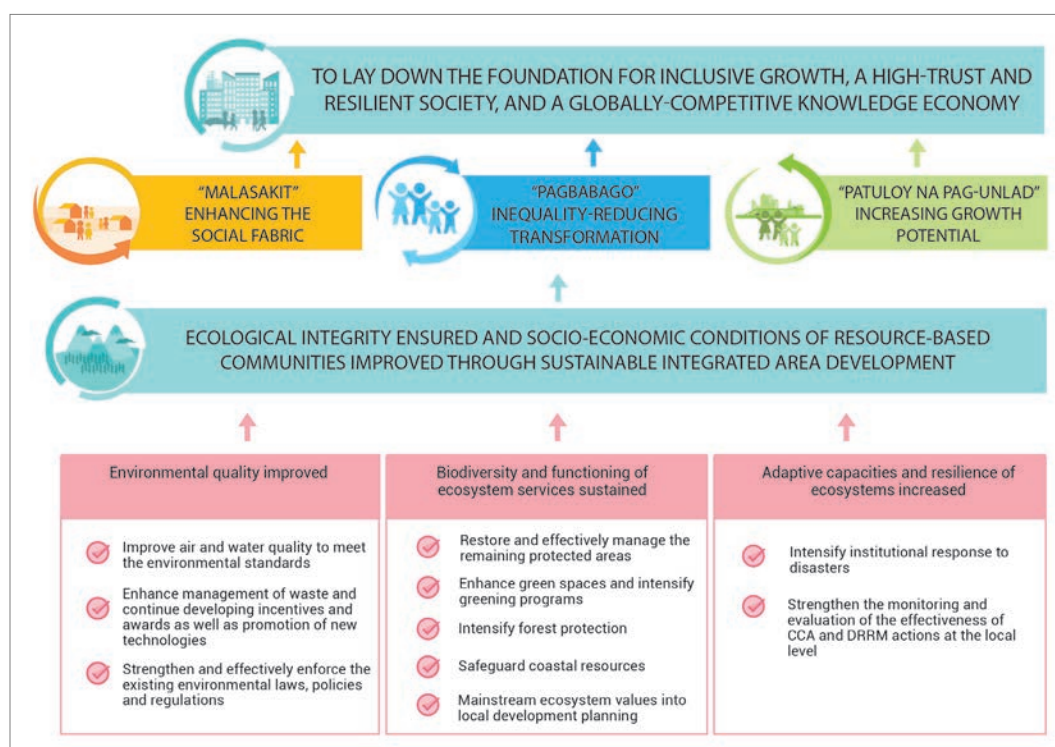
INDICATORS	BASELINE DATA		END OF PLAN TARGET (2022)	RESPONSIBLE AGENCY
	YEAR	VALUE		
Environmental quality improved				
Percentage of air quality monitoring stations within ambient air quality guideline value increased				DENR
Priority water bodies within water quality guidelines				
Pasig River	2017			
BOD	2017	57 mg/L	Reduction in BOD concentration	PRRC, Supreme Court Continuing Mandamus Government Agencies (Manila Bay Clean-Up), and LGU's
DO	2017	1.85 mg/L	Increase in DO concentration	
Fecal Coliform	2017	5.85E+06 MPN/100mL	Reduction in Fecal Coliform	
Manila Bay			Improving	
Solid waste diversion rate increased	2015	48%	80%	DENR
Biodiversity and functioning of ecosystem services sustained				
Increased biodiversity in the remaining protected areas in Metro Manila				
Estimated Green Spaces (EGS) in hectares	2014	12,152.79	Increased	
Adaptive capacity and resilience of ecosystems increased				
Resilience index				
Number of reviewed CC/DRRM enhanced plans increased				

Strategies

Ensuring ecological integrity, clean, and healthy environment will definitely contribute to the attainment of enhancing the social fabric or “*Malasakit*”, reducing inequality or “*Pagbabago*”, and increasing growth potential or “*Patuloy na Pag-unlad*” towards laying down the foundation for inclusive growth, a high-trust and resilient society, and a globally-competitive knowledge economy.

The key outcome under this chapter that should have been achieved by 2022 are the following: (a) improved environmental quality, (b) sustained biodiversity and functioning of ecosystem services, and (c) increased adaptive capacities and resilience of ecosystems.

Figure 14.4. Strategic Framework to Ensure Ecological Integrity, Clean, and Healthy Environment, 2017-2022



To improve environmental quality

In order to ensure ecological integrity, clean and healthy urban environment effective environmental governance will be strengthened through the enforcement of existing laws and policies related to air and water management as well as introduction of new technologies and equipment.

Improve air and water quality to meet the environmental standards.

Air Quality Management. In order to improve the quality of air in Metro Manila and increase percentage of areas within ambient air quality standard, activities to be implemented should focus on the improvement of the monitoring system and tools, strengthen the operationalization of

Airshed Management Governing Board and strengthen the enforcement operations to enforce the air quality standards under Clean Air Act. There should be an approved Airshed Management Action Plan for implementation until 2022. Monitoring the compliance to the standard of the establishments, industries and motor vehicles including the private emission testing centers have to be strictly done by all concerned agencies.

Promotion of environmentally-sustainable transport such as clean fuels and e-vehicles and strengthen the implementation of a centralized, intermodal and integrated bus terminal system at the fringes of Metro Manila will directly contribute in reducing bus traffic volume on the road that has a significant impact of reducing traffic congestion which will directly result in the reduction of air pollutants in the region.

Results of the air quality monitoring have to be publicly disseminated for transparency and accountability of the national and LGUs. Further, air quality status should be the basis of the implementation of enforcement operations and greening programs.

Water Quality Management. In order to improve the quality of water draining to the Manila Bay and increase percentage of water bodies within standard, concerted efforts have to be done focusing on the activities to improve the monitoring of water quality along river system and coastal areas, strengthen the operationalization of Water Quality Monitoring Areas (WQMAs) and its governing boards, strengthen the enforcement operations to ensure that wastewaters from domestic and business/ industrial establishments are within standards, and conduct rehabilitation strategies along river system in Metro Manila like the promotion of the adoption of rivers/creeks by industries/establishments, organizations and communities.

Water quality monitoring stations shall be enhanced by maintaining and establishing the required number of water quality management areas. The improvement of sewage and septage facilities in the urban areas and the development of additional waste water and septage treatment facility in the establishments/ buildings and manufacturing industries shall be implemented. Identify pollutants, including persistent organic pollutants and heavy metals in priority water bodies and determine the appropriate management interventions.

Strengthen the rules and regulations in encroachment along easement of the waterways which contribute blocking of waterways due to presence of illegal structures and uncontrolled throwing of waste along the area resulting water pollution and flooding. For Manila Bay, in order to properly manage and carry out the activities that will lead to the rehabilitation and restoration of Manila Bay, Site Coordinating Offices were created in DENR Regions 3, 4A and NCR. Pursuant to the recommendation of the Supreme Court, the Manila Bay Site Coordinating and Management Committee (MBSCMC) is organizing and adopting an area or river system – based approach. To ensure and tighten coordination and cooperation between all stakeholders, LGUs were grouped according to river system traversing their territorial boundaries. These groups are composed of River Management Committees (RMC) which are comprised of the LGUs. As part also of the management strategies, four (4) thematic clusters namely Liquid Waste Cluster, Solid Waste Cluster, Informal Settler Families Cluster, and Habitat and Resources Cluster were created per RMC to address the effects of the worsening problems in Manila Bay (Manila Bay Site Coordinating Management Office Annual Report, 2017). Each RMCs crafted their Area-Based Management Plans (ABMPs) which were later consolidated into Metro Manila River System Management

Plan (MMRSMP). This is grounded on the results of series of consultation workshops and group discussions participated by LGUs, representatives from concerned NGAs, and other stakeholders. The MMRSP consists of identified developmental goals and strategies, and action plans that are aimed at restoring the water quality of the three (3) major river systems.

With regards to Marikina River, based on the complexity of issues across the management and sectoral concerns in this ecosystem, there is a need to implement the following development policy directions:

- Intensify waste and pollution control through adaptive waste management systems primarily for domestic sources.
- Promote security of riverside dwellers from flooding, health risks and minimize economic displacement through effective flood control programs and appropriate resettlement plans;
- Rationalize micro-watershed management policies based on validated threshold and vulnerability to sustain its ecological functions focused on incentive-based regulations and adaptive co-management systems, specifically standardize the approval of development plans and permits in the sub-basin.

Enhance management of waste and continue developing incentives and awards as well as promotion of new technologies.

With the pressing concern on the increasing volume of waste produced, its undeniable impact on climate change, and the relatively sustainable SWM interventions among LGUs are some of the reasons why the government should push for the appropriate waste to energy (WTE) technologies. WTE is a various technologies that convert non-recyclable waste into usable forms of energy with minimal residues brought to SLFs.

Engage LGUs to PPP option, joint venture, and financial schemes for the establishment of large-scale waste treatment technologies.

Continue developing an incentives and awards system for LGUs that have best practices on ecological solid waste management to inspired them in continuing and enhancement their activities.

Effectively enforce RA 9003 particularly in the waste management solutions such as waste segregation and recycling/recovery/ composting and pass an ordinance prohibiting the use of non-biodegradable and single-use plastic bags and containers to lessen the volume of solid waste being disposed.

Strengthen and effectively enforce the existing environmental laws, policies and regulations. Existing environmental laws and policies such as Ecological Solid Waste Management Law, Clean Air Act, Clean Water Act, Renewable Energy Law, among others shall be strictly enforced particularly in the local level to avoid further deterioration of the environmental quality in the region. To ensure that the said laws, policies and regulations are able to address the challenges and issues of the changing times, they shall be reviewed and recommend for updating whenever necessary.

Full implementation and compliance of LGUs to RA 9003 in relation to the establishment of material recovery facilities and treatment facilities, closure and rehabilitation of remaining dumpsites and formulation of local solid waste management plans will help in the reduction of water pollution.

Strengthen the coordination between national and local government units to harmonize policies and programs and avoid conflicting implementation.

To sustain biodiversity and functioning of ecosystem services

To sustain the provision of ecosystem services and meet both the present and future demand, the following key strategies will be implemented: (a) restore and effectively manage the remaining protected areas; (b) enhancement of green spaces; (c) intensify forest protection; (d) safeguard coastal resources; and (e) mainstream ecosystem values into local development planning.

Restore and effectively manage the remaining protected areas. Many of the human activities that modify or destroy natural ecosystems may cause deterioration of ecological services. Conservation mechanisms thru ecosystem restoration should be implemented particularly for degraded habitats like mangroves in LLPCHEA and other parts in Metro Manila. Eco-tourism livelihood has a lot of potential in the river to uplift the living conditions of fisherfolks and village residents if only properly implemented. The plant and animal species found in the protected areas should maintain the integrity of these habitats while encouraging the people to participate in the conservation efforts through tree planting and mangroves rehabilitation programs.

Safeguard and maintain the remaining protected areas in Metro Manila such as LPPCHEA, La Mesa Watershed, Marikina Watershed, among others to sustain the services needed by the community and for the maintenance of all other habitat.

Enhance green spaces and intensify greening programs. In order to meet the required number of green spaces and expand the implementation of greening programs, there should be harmonization of greening efforts among government agencies including national agencies and local government units. There should be an integrated greening plan in Metro Manila incorporating all

spatial targets of each agency for the next five (5) years (until 2022). Specifically, the following strategies should be achieved: (a) proclamation of forest parks in each LGU areas (b) enhance all urban parks in coordination with LGUs, other government agencies and communities; (c) enhance available open spaces in schools, government compound and other public spaces; (d) improve connectivity of all parks and other green spaces; (e) rehabilitate easement areas along rivers and creek with appropriate biological measures; and (f) promote alternative urban greening strategies like vertical and roof gardening through local legislative instruments. Local legislative instruments should be the provision of technical assistance in the formulation of draft municipal green plan which includes the enhancement of parks, tree management program, heritage tree program, inventory/mapping/classification of urban parks for adaption and provision/ distribution of ornamental and agroforestry plants.

Intensify forest protection. Modern technologies have to be adopted to intensify the enforcement operations in the remaining forest cover/areas of the region. Taking advantage of the use of drone, smart/android phones and online systems in the surveillance and monitoring operations should be considered. Facilities and personnel at the major entry and exit points of forest products including wildlife, both in the airport (i.e. Ninoy Aquino International Airport) and seaport (i.e. Manila North Harbor), have to be modernized and enhanced to ensure that no illegal forest product will come in and out of the region. Moreover, cooperation, linkages and networking among the government enforcement agencies have to enhance to improve accuracy of intelligence sources and effectiveness of the conduct of operations.

Safeguard coastal resources. To ensure safety and preservation of the remaining coastal resources of the region, maintenance and protection activities has to be intensified in areas with remaining mangrove stands in Las Piñas, Parañaque, Navotas and Malabon. The participation of the local government including barangays and communities have to be strengthened to ensure effectiveness of monitoring operations against illegal activities like poaching, cutting of mangrove, and among others. Development and adoption of livelihood opportunities which are biodiversity-friendly should be prioritized to lessen impact to Manila Bay and coastal resources. Linkages with the concerned government agencies like DENR, DOST and DTI should be enhanced to ensure that these livelihood programs will turn into a small enterprise. In addition, forest parks with high biodiversity or with presence of endemic trees can be declared as an Urban Biodiversity Conservation Area as potential source of progenies.

Mainstream ecosystem values into local development planning. In the development planning, mainstreaming accounting and valuation of the services and products sourced from natural resources is necessary to ensure the importance and appropriate management that will be given to these finite resources. Recognizing the true value of the resources will facilitate informed decision making of political leaders to provide better alternatives and trade-offs, and generate income and employment in the rural areas which will create wealth for the nation.

The ecosystem valuation and natural resource monitoring system should be institutionalized. The institution will monitor the indicators of the ecosystem services to trace the benefits of ecosystem as well as the impact of threats like climate change in the environment.

Develop a policy to institutionalize payment for ecosystem services that will provide incentives in the management of ENR and also provide an alternative source of income to the local communities.

To increase adaptive capacities and resilience of ecosystems

Strategies will be geared toward strengthening the implementation of CCA and DRR actions across all sectors as well as in the local level.

Intensify institutional response to disasters. Strengthen mechanism to conduct a post or rapid disaster need assessment through continuous capacity-building programs for concerned agencies.

Revisit and review existing policies related to disasters specifically on the procurement and development of land. Housing and resettlement areas should be in appropriate land-use and will not encroach on environmentally-critical areas and conservation sites.

Strengthen the monitoring and evaluation of the effectiveness of CCA and DRRM actions at the local level. Ecosystem service indicators which help in the assessment of the conditions of natural resources will be identified and integrated in existing monitoring systems to measure adaptive capacity and resilience.

Develop a database to measure emission reduction per sector particularly in the transport sector pursuant to Executive Order 174, stated that there is a need to “provide direction and guidance for the accounting and reporting of GHG emissions from identified key sources”. This will contribute in monitoring, reporting and verification of the country’s GHG emissions.

PART VI

FROM PLAN TO ACTION



15

Plan Implementation and Monitoring

Plan Implementation and Monitoring

The implementation of the RDP-NCR requires close coordination between and among the regional line agencies, local government units as well as private sector and civil society organizations.

During the implementation period (2017-2022), the effectiveness and efficiency of the identified strategies in realizing the chapter objectives will be monitored and evaluated for every chapter of the plan.

Each line government agencies with support from the private sector and other development partners will implement programs, activities and projects in accordance with the strategies as enumerated in each of the thematic areas in support of the plan priority targets.

The operationalization of the Plan shall be monitored and evaluated based on the Plan's targets in accordance with the Result Matrices identified in each of the plan chapters.

There will also be a midterm Plan updating sometime 2020 to consider relevant development requirements and emerging trends.

The Office of the Assistant General Manager for Planning of the Metro Manila Development Authority, as the overall RDP-NCR secretariat, will closely work with the different implementing agencies of the RDC in monitoring the progress of implementation of this Plan through:

(1) The **Planning Committee on Economic Development** which shall oversee the implementation of the strategies enumerated in **Chapter 5: Promoting Awareness and Cultural Diversity**, **Chapter 6: Expanding Economic Opportunities in Industry and Services through *Trabaho at Negosyo***, and **Chapter 11: Vigorously Advancing Science Technology and Innovation**, in close coordination with its member agencies:

- a. National Commission for Culture and Arts
- b. National Historical Commission of the Philippines
- c. Department of Trade and Industry - NCR
- d. Department of Tourism - NCR
- e. Bureau of Small and Medium Enterprise Development
- f. Philippine Ports Authority
- g. Manila International Container Terminal
- h. Asian Terminal, Inc.
- i. Overseas Workers Welfare Administration

- j. Philippine Overseas Employment Administration
- k. Commission on Filipino Overseas
- l. Department of Labor and Employment - NCR
- m. Department of Science and Technology - NCR
- n. Philippine Statistics Authority - NCR

(2) The **Planning Committee on Social Development A** which shall oversee the implementation of the strategies enumerated in **Chapter 7: Accelerating Human Capital Development, Chapter 9: Building Safe, Secure and Sustainable Communities, and Chapter 10: Prospecting for a Demographic Dividend**, in close coordination with its member agencies:

- a. Commission on Higher Education
- b. Department of Education
- c. Department of Health
- d. National Nutrition Council
- e. Technical Education and Skills Development Authority - NCR
- f. Housing and Urban Development Coordinating Council
- g. Housing and Land Use Regulatory Board - NCR
- h. National Housing Authority
- i. Home Development Mutual Fund
- j. Home Guaranty Corporation
- k. Social Housing Finance Corporation
- l. Population Commission - NCR

(3) The **Planning Committee on Social Development B** which shall oversee the implementation of the strategies enumerated in **Chapters 8: Reducing Vulnerability of Individuals and Families, Chapter 12: Ensuring Security, Public Order and Safety, and Chapter 14: Ensuring Ecological Integrity, Clean and Healthy Environment**, in close coordination with its member agencies:

- a. Office of the Civil Defense - NCR
- b. Department of Interior and Local Government - NCR
- c. Philippine National Police - NCR
- d. Bureau of Fire Protection
- e. Bureau of Jail Management and Penology
- f. Department of Environment and Natural Resources - NCR
- g. DENR - Environmental Management Bureau
- h. Pasig River Rehabilitation Commission
- i. Laguna Lake Development Authority
- j. Manila Bay Coordinating Office

(4) The **Planning Committee on Infrastructure Development** which shall oversee the implementation of the strategies enumerated in **Chapter 13: Accelerating Infrastructure Development**, in close coordination with its member agencies:

- a. Department of Public Works and Highways (DPWH) - National Capital Region
- b. DWPH - Unified Project Management Office Road Management Cluster

- c. DPWH - Unified Project Management Office Flood Control Management Cluster
- d. DPWH - Unified Project Management Office Bridge Mangement Cluster
- e. DPWH - Public Private Partnership Services
- f. Department of Transportation
- g. Department of Information and Communications
- h. Toll Regulatory Board
- i. Department of Energy
- j. Manila Water Company Inc.
- k. Maynilad Water Services Inc.

The various agencies mentioned were actively involved in the formulation of the RDP-NCR 2017-2022. Furthermore, these agencies participated in the series of stakeholder workshop, where bilateral consultations, write shops and learning opportunities were spearheaded by the secretariat. , thereby providing inputs in the identification of assessment and challenges and in recommending strategies to address them.

Thus, the regional line agencies, as implementing agencies, also have vital roles to play in monitoring their work progress to realize the regional development plan objectives in support of the Philippine Development Plan. Regular reporting of their implementation progress shall be required of them.



List of Acronyms

4Ps	Pantawid Pamilyang Pilipino Program
ABTC	Animal Bite Treatment Center
ABMP	Area-Based Management Plans
ADB	Asian Development Bank
AGLS	Airport Ground Lighting System
AICS	Assistance to Individuals under Crisis Situation
ALS	Alternative Learning System
AMS	ASEAN Member States
ASEAN	Association of Southeast Asian Nations
ATI	Asian Terminals, Inc.
AWTIP	Angat Water Transmission Improvement Project
BALAI	Building Adequate, Liveable, Affordable and Inclusive Filipino Communities
BCPCs	Barangay Council for the Protection of Children
BFP	Bureau of Fire Protection
BIR	Bureau of Internal Revenue
BIST	Business Innovation through Science and Technology
BJMP	Bureau of Jail Management and Penology
BOD	Biological Oxygen Demand
BOT	Build-Operate-Transfer
BPLO	Business Permits and Licensing Office
BPO	Business Process Outsourcing
BPO – IT	Business Process Outsourcing – Information Technology
CaMaNaVa	Caloocan-Malabon-Navotas-Valenzuela
CAR	Children at Risk
CAPEX	Capacity Expansion
CARE-KSA	Cash Relief Program for OFW-Availees from KSA
CATV	Cable Television
CCA	Climate Change Adaptation
CCP	Cultural Center of the Philippines
CCT	Conditional Cash Transfer
CCTV	Closed Circuit Television
CDD	Community-Driven Development
CDP	Comprehensive Development Plan
CED	Chronic Energy Deficiency
CFO	Commission on Filipinos Overseas
CHED	Commission on Higher Education
CIA	Community Initiative Approach
CLUP	Comprehensive Land Use Plan

CMI	City Momentum Index
CMTS	Cellular Mobile Telephone System
CNSP	Children in Need of Special Protection
COA	Commission on Audit
CPDEP	Comprehensive Pre-Departure Education Program
CPF	Common Purpose Facility
CPH	Census of Population and Housing
CRADLE	Collaborative Research and Development to Leverage Philippine Economy
CSO	Civil Society Organization
DED	Detailed Engineering Design
DENR	Department of Environment and Natural Resources
DepEd	Department of Education
DICT	Department of Information and Communications Technology
DILG	Department of Interior Local Government
DOH	Department of Health
DOAS	Differential Optical Absorption Spectroscopy
DOE	Department of Energy
DOLE	Department of Labor and Employment
DOST	Department of Science and Technology
DOT	Department of Tourism
DOTr	Department of Transportation
DPWH	Department of Public Works and Highways
DRR	Disaster Risk Reduction
DSWD	Department of Social Welfare and Development
DTI	Department of Trade and Industry
DTT	Digital Terrestrial Television
DTTB	Digital Terrestrial Television Broadcasting
DWIN	Domestic Wideband Information Network
EBT	Enterprise Based Training
ECMTS	Estimated Cellular Mobile Telephone Subscribers
EDLP	Enterprise Development Loan Program
EDSP	Education for Development Scholarship Program
EGS	Estimated Green Spaces
EGDI	E-Government Development Index
EHV	Extra High Voltage
EICC	Energy Investment Coordinating Council
ELAP	Education and Livelihood Assistance Program
EMB	Environmental Management Bureau
ENR	Environment and Natural Resources
EO	Executive Order

EPAs	Environmental Preservation Areas
EPD	Eastern Police District
EPIRA	Electric Power Industry Reform Act
EWBS	Early Warning Broadcast System
FDI	Foreign Direct Investment
FIMS	Fugitive Inmates Monitoring System
FIT	Feed-In Tariff
FLEMMS	Functional Literacy, Education and Mass Media Survey
FHSIS	Field Health Services Information System
FNRI-DOST	Food and Nutrition Research Institute of the Department of Science and Technology
FSIC	Fire Safety Inspection Certificate
GAA	General Appropriations Act
GCI	Global Competitiveness Index
GDP	Gross Domestic Product
GERD	Gross Expenditures on Research and Development
GHG	Greenhouse Gas
GIA	Grants-in-Aid
GIDAs	Government Intervention in Deprived Areas
GOCC	Government Owned and Controlled Corporation
GSIS	Government Service Insurance System
G2B	Government-to-Business
G2C	Government-to-Citizen
G2G	Government-to-Government
HARP	HIV/AIDS & ART Registry of the Philippines
HDHP	High Density Housing Program
HDMF	Home Development Mutual Fund
HEIs	Higher Education Institutions
HHs	Households
HIV	Human Immunodeficiency Virus
HGC	Home Guaranty Corporation
HLURB	Housing and Land Use Regulatory Board
HR	Human Resource
HRMIS	Human Resource Management Information Systems
HUDCC	Housing and Urban Development Coordinating Council
ICT	Information and Communications Technology
ICTO	Information and Communications Technology Office
ICTSI	International Container Terminals Inc.
IDD	Iron Deficiency Disorder
IDI	ICT Development Index
IDRC	International Development Research Centre

IEC	Information, Education and Communications
iGovPhil	Integrated Government Philippines
IKSPs	Indigenous Knowledge Systems and Practices
IMR	Infant Mortality Rate
INSEAD	“ <i>Institut Européen d’Administration des Affaires</i> ” or European Institute of Business Administration
IP	Indigenous People
IRR	Implementing Rules and Regulations
ISDB-T	Integrated Service Digital Broadcasting-Terrestrial
ISFs	Informal Settler Families
ISPs	Internet Service Providers
IT-BPM	Information Technology-Business Process Management
IVI	Intensified Verification Inspection
IWD	Inmates Welfare & Development
JICA	Japan International Cooperation Agency
JLL	Jones Lang Lasalle
JSBAC	Joint Special Bids and Awards Committee
KSA	Key Shelter Agencies
kV	Kilovolt
LDRRMP	Local Disaster Risk Reduction and Management Plan
LGU	Local Government Unit
LLED	Laguna Lakeshore Expressway Dike
LLDA	Laguna Lake Development Authority
LMWR	La Mesa Watershed Reservation
LPG	Liquefied Petroleum Gas
LPPCHEA	Las Piñas – Parañaque Critical Habitat and Ecotourism Area
LRT	Light Rail Transit
LRV	Light Rail Vehicles
LTFRB	Land Transportation Franchising and Regulatory Board
MANATUTI	Malabon-Navotas-Tullahan-Tinajeros
MAPALLA	Manila Bay-Pasig River-Laguna Lake
MBSCMC	Manila Bay Site Coordinating and Management Committee
MCP	Maternity Care Package
MDG	Millennium Development Goals
Meralco	Manila Electric Company
MICE	Meetings, Incentives, Conferences and Exhibitions
MICT	Manila International Container Terminal
MITHI	Medium-Term Information and Communications Technology Harmonization Initiative
MGB	Mines and Geosciences Bureau
MLD	Million of Liters Per Day

MM	Metro Manila
MMARAS	Metro Manila Accident Recording and Analysis Systems
MMDA	Metropolitan Manila Development Authority
MMDRRMC	Metro Manila Disaster Risk Reduction and Management Council
MMR	Maternal Mortality Rate
MMRRSMP	Metro Manila River System Management Plan
MNCHN	Maternal Neonatal Child Health and Nutrition
MOA	Memorandum of Agreement
MPD	Manila Police District
MRF	Materials Recovery Facility
MRT	Metro Rail Transit
MSME	Micro, Small and Medium Enterprise
MuntiParLasTaPat	Muntinlupa-Parañaque-Las Piñas-Taguig-Pateros
MuntiParLasPiZap	Muntinlupa-Parañaque-Las Piñas-Zapote
MV	Motor Vehicle
MVIS	Motor Vehicles Inspection System
MWCI	Manila Water Company Inc.
MWSI	Maynilad Water Services Inc.
MWSS	Metropolitan Waterworks and Sewerage System
NAIA	Ninoy Aquino International Airport
NEDA	National Economic and Development Authority
NCDA	National Council for Disability Affairs
NCCA	National Commission for Culture and the Arts
NCCAP	National Climate Change Action Plan
NCR	National Capital Region
NDHS	National Demographic and Health Survey
NDRRMC	National Disaster Risk Reduction and Management Council
NFA	National Food Authority
NGAs	National Government Agencies
NGCP	National Grid Corporation of the Philippines
NGDC	National Government Data Center
NGOs	Non-Government Organizations
NGP	National Government Portal
NH	North Harbor
NHA	National Housing Authority
NHCP	National Historical Commission of the Philippines
NHCS	Napindan Hydraulic Control Structure
NHIP	National Health Insurance Program
NHTS	National Household Targeting System
NICER	Niche Centers in the Regions for Research and Development

NIMS	National Inmates Monitoring System
NISUS	National Informal Settlements Upgrading Strategy
NLEX	North Luzon Expressway
NMR	Neonatal Mortality Rate
NOH	National Objectives for Health
NPD	Northern Police District
NPO	National Printing Office
NREP	National Renewable Energy Program
NSLF	Navotas Sanitary Landfill
NTC	National Telecommunications Commission
NTESDP	National Technical Education and Skills Development Plan
NTSI	Nautical Transport Services Inc.
NUDHF	National Urban Development and Housing Framework
NYC	National Youth Commission
OFG	Omnibus Franchising Guidelines
OFWDS	OFW Dependents Scholarship Program
OFWs	Overseas Filipino Workers
Oplan-LIKAS	Oplan-Lumikas para Iwas Kalamidad at Sakit
ORD	Office of the Regional Director
OSCA	Office for Senior Citizens Affairs
OWWA	Overseas Workers Welfare Administration
PAGASA	Philippine Atmospheric, Geophysical and Astronomical Services Administration
PaMaMariSan	Pasig-Mandaluyong-Marikina-San Juan
PCB	Primary Care Benefit
PCT	Patent Cooperation Treaty
PDAO	Persons with Disability Affairs Office
PDEP	Pre-Departure Education Program
PDL	Persons Deprived of Liberty
PDOS	Pre-Departure Orientation Seminar
PEATC	Public Estate Authority Tollway Corporation
PEZA	Philippine Economic Zone Authority
PHA	Philippine Health Agenda
Philhealth	Philippine Health Insurance
PIDS	Philippine Institute for Development Studies
PM10	Particulate Matter measuring 10 micrometers in diameter
PNDHS	Philippine National Demographic Health Survey
PNP	Philippine National Police
PNR	Philippine National Railway
POEA	Philippine Overseas Employment Administration
PPA	Philippine Port Authority

PPAN	Philippine Plan of Action on Nutrition
PPASC	Philippine Plan of Action for Senior Citizens
PPE	Personnel Protective Equipment
PPP	Public-Private Partnerships
PRRC	Pasig River Rehabilitation Commission
PRUMS	Pasig River Unified Monitoring System
PSA	Philippine Statistics Authority
PSFTP	Public Safety Field Training Program
PTIMC	Public Transport Information and Management Center
PAVO	Philippine Veterans Affairs Office
PWD	Persons with Disability
QCPD	Quezon City Police District
QCSLF	Quezon City Sanitary Landfill
RA	Republic Act
R&D	Research and Development
RAP-KSA	Relief Assistance Program for affected OFWs in KSA
RCOA	Retail Competition Open Access
RDC	Regional Development Council
RDIs	Research Development Institutes
RDLead	Research and Development Leadership Program
RET	Rapid Exit Taxiways
RFP	Request for Proposal
RIAC-SC	Regional Inter-Agency Committee for Senior Citizens
RMC	River Management Committees
RPPF	Responsible Parenthood and Family Planning
RPHAU	Regional Personnel Holding Accounting Unit
RPSLF	Rizal Provincial Sanitary Landfill
RRAP	Regional Resettlement Action Plan
RRDI	Regional Research, Development & Innovation Committee
RSE	Researchers, Scientists and Engineers
RTC	Regional Trial Court
RTESDP	Regional Technical Education and Skills Development Plan
RTI	Research Triangle Institute
S&T	Science and Technology
SC	Senior Citizens
SDN	Service Delivery Networks
SESP	Skills-for-Employment Scholarship Program
SETUP	Small Enterprise Technology Upgrading Program
SH	South Harbor
SHFC	Social Housing Finance Corporation
SLF	Sanitary Landfill

SLEX	South Luzon Expressway
SLP	Sustainable Livelihood Program
SP	Social Protection
SPD	Southern Police District
SSS	Social Security System
STEM	Science, Technology, Engineering, and Mathematics
STI	Science, Technology, and Innovation
STI	Sexually Transmitted Infection
SUP	Seafarers Upgrading Program
SWD	Social Welfare and Development
SWM	Solid Waste Management
SWMP	Solid Waste Management Plan
TABS	Terminal Appointment Booking System
TB	Tuberculosis
TB DOTS	Tuberculosis Directly Observed Treatment Short Course
TCMP	Therapeutic Community Modality Program
TESDA	Technical Education and Skills Development Authority
TEUs	Twenty-Foot Equivalent Unit
TFR	Total Fertility Rate
TOPCOP	Tourist-Oriented Police for Community Order and Protection
TSP	Total Suspended Particles
TVET	Technical and Vocational Education and Training
TVI	Technical Vocational Institution
UFMR	Under-Five Mortality Rate
UNESCO	United Nations Educational Scientific and Cultural Organization
UNICEF	United Nations Children's Fund
UNICEF EAPRO	UNICEF East Asia and Pacific Regional Office
UNISDR	United Nations International Strategy for Disaster Reduction
USAID-STRIDE	USAID and RTI International titled Science, Technology, Research and Innovation for Development
VAW	Violence Against Women
WAP	Welfare Assistance Program
WB	World Bank
WEF-GCR	World Economic Forum-Global Competitiveness Report
WESM	Wholesale Electricity Spot Market
WHO-UN	World Health Organization of the United Nations
WTE	Waste to Energy
UN	United Nations
UN WTO	UN World Tourism Organization
WQMA	Water Quality Monitoring Areas
YAFS 4	Young Adult Fertility and Sexuality Study
ZO	Zoning Ordinance

Glossary of Terms

Annual Average Daily Traffic (AADT)	A measure used primarily in transportation planning and transportation engineering. Traditionally, it is the total volume of vehicle traffic of a highway or road for a year divided by 365 days. AADT is a useful and simple measurement of how busy the road is.
Build, Build, Build” Program	President Duterte administration’s medium-term goal to increase infrastructure spending from 5.4 percent of the country’s Gross Domestic Product (GDP) in 2017, to 7.3 percent by the end of President Rodrigo Duterte’s term in 2022
Child	Person below 18 years of age, or a person over 18 years of age but is unable to fully care for himself/herself or protect himself/herself from abuse, neglect, cruelty, exploitation or discrimination because of a physical or mental disability or condition
Cohort Survival Rate	The percentage of enrollees at the beginning grade or year in a given school year who reached the final grade or year of the elementary/secondary level
Contraceptive Prevalence Rate	The percentage of women of reproductive age who are currently using, or whose sexual partner is currently using, at least one contraceptive method, regardless of the method used
Comprehensive Land Use Plan	A planning document prepared by LGUs to rationalize the allocation and proper use of land resources
Crime Incidence	The number of crimes reported as to index or non-index crimes within a given period
Crime Rate	Number of crimes committed per 100,000 population
Crude Death Rate	The number of deaths occurring among the population of a given geographical area during a given year, per 1,000 mid-year total population of the given geographical area during the same year
Cultural Mapping	A valuable tool for identifying a community’s strengths and its resources. This process can help as communities move into the planning and implementation phase by identifying early the resources, efficiencies and links between arts and cultural groups, as well as their common aspirations and values.
Culturally Sensitive and Gendered Approach	Can empower marginalized individuals and communities to participate in cultural and political life. New imaginations of the urban can transform citizens’ sense of place and sense of self

Danger Areas	Esteros, railroad tracks, garbage dumps, riverbanks, shorelines, waterways, and other public places such as sidewalks, roads, parks, and playgrounds (RA 7279).
Dependency Ratio	A measure showing the ratio of the number of dependents aged zero to 14 and over the age of 65 to the total population aged 15 to 64.
Drop-Out Rate	The “event” dropout rate reports the percentage of young people ages 15 through 24 who dropped out of grades 10 through 12 in the past year. The “cohort” dropout rate measures the percentage of an entering ninth-grade class that drops out before the end of the twelfth-grade year.
Educational Facilities	Refers to all the physical properties of a school, consisting of the grounds, buildings, and the various facilities within the school and inside the school buildings as describe by Educational Facilities Manual of the DepEd
Environmental Buffer	is a work done on how environment can be purified or at least stabilized by using entities that naturally make environment go back to its original state more or less. It includes some environmental buffers that can be used to reduce air, water and noise pollution.
Employment Rate	Defined as a measure of the extent to which available labour resources (people available to work) are being used. They are calculated as the ratio of the employed to the working age population.
Enrolment Rate	Expressed as net enrolment rates, which are calculated by dividing the number of students of a particular age group enrolled in all levels of education by the size of the population of that age group
Environmental Buffer	A work done on how environment can be purified or at least stabilized by using entities that naturally make environment go back to its original state more or less. It includes some environmental buffers that can be used to reduce air, water and noise pollution.
E-Government Development Index (EGDI)	A composite index based on the weighted average of the three normalized indices, namely Telecommunications Infrastructure Index (TII), Human Capital Index (HCI) and Online Service Index (OSI). Jointly developed by United Nations Department of Economic and Social Affairs (UNDESA), Division for Public Administration and Development Management (DPADM) and many external experts, EGDI measures the readiness and capacity of national administrations in using ICT as a means to deliver public services.

Fertility Rate	The number of children who would be born per woman (or per 1,000 women) if she/they were to pass through the childbearing years bearing children according to a current schedule of age-specific fertility rates.
Fire Incidence	Refers to the volume of incidents caused by fire and the deaths, injuries and total value of property loss due to this incident
Food Threshold	Minimum income required by an individual to meet the basic food needs and satisfy the nutritional requirements set by the Food and Nutrition Research Institute (FNRI), while remaining economically and socially productive.
Foreign Direct Investment (FDI)	FDI is an investment in the form of a controlling ownership in a business in one country by an entity based in another country. It is thus distinguished from a foreign portfolio investment by a notion of direct control
Global Competitiveness Report (GCR)	Annual report published by the World Economic Forum. Since 2004, the Global Competitiveness Report ranks countries based on the Global Competitiveness Index. The Global Competitiveness Index integrates the macroeconomic and the micro/business aspects of competitiveness into a single index. The report “assesses the ability of countries to provide high levels of prosperity to their citizens. This in turn depends on how productively a country uses available resources. Therefore, the Global Competitiveness Index measures the set of institutions, policies, and factors that set the sustainable current and medium-term levels of economic prosperity.”
Global Cybersecurity Index (GCI)	A composite index combining 25 indicators into one benchmark measure to monitor and compare the level of ITU Member States cybersecurity commitment with regard to the five pillars (Legal, Technical, Organizational, Capacity Building, and Cooperation) identified by the High-Level Experts Group and endorsed by the ITU Global Cybersecurity Agenda (GCA).
Global Innovation Index (GII)	An annual ranking of countries by their capacity for, and success in, innovation. It is published by Cornell University, INSEAD, and the World Intellectual Property Organization, in partnership with other organisations and institutions, and is based on both subjective and objective data derived from several sources, including the International Telecommunication Union, the World Bank and the World Economic Forum.
Gross Domestic Product (GDP)	GDP is a monetary measure of the market value of all the final goods and services produced in a period (quarterly or yearly) of time. Nominal GDP estimates are commonly used to determine the economic performance of a whole country or region, and to make international comparisons.

ICT Development Index (IDI)	A unique benchmark of the level of ICT development in countries across the world developed by International Telecommunication Union (ITU). IDI combines eleven indicators on ICT access, use and skills, capturing key aspects of ICT development in one measure that allows for comparisons to be made between countries over time.
ICT sector	Refers to those engaged in providing goods and services intended to fulfil or enable the function of information processing and communication by electronic means. The ICT sector includes telecommunications and broadcast information operators, ICT equipment manufacturers, multimedia content developers, and providers, ICT solution providers, internet service providers, ICT training institutions, software developers and ICT-Enabled Services (ICT-ES) providers
Index Crimes	Crimes which are sufficiently significant and which occur with sufficient regularity to be meaningful. Crimes which include the following: murder, physical injury, robbery, theft and rape
Infant and Young Child Feeding (IYCF) Strategy	A global strategy for Infant and Young Child Feeding (IYCF) was issued jointly by the World Health Organization (WHO) and the United Nations Children's Fund (UNICEF) in 2002, to reverse the disturbing trends in infant and young child feeding practices.
Infant Mortality Rate	The number of deaths under one year of age occurring among the live births in a given geographical area during a given year, per 1,000 live births occurring among the population of the given geographical area during the same year. Probability of dying between birth and exactly 1 year of age, expressed per 1,000 live births
Informal Sector	The informal sector, informal economy, or grey economy is the part of an economy that is neither taxed nor monitored by any form of government. Unlike the formal economy, activities of the informal economy are not included in a country's gross national product (GNP) or gross domestic product (GDP).
Information and Communications Technology (ICT)	The totality of electronic means to access, create, collect, store, process, receive, transmit, present and disseminate information
Labor Force	refers to the population 15 years old and over who contribute to the production of goods and services in the country. It comprises the employed and unemployed
Laguna Lake	The largest lake in the Philippines and the third-largest freshwater lake in South East Asia located east of Metro Manila between the provinces of Laguna to the south and Rizal to the north. It has a surface area of 911–949 km ² , with an average depth of about 2.8 meters and an elevation of about 1 meter above sea level.

Macronutrient Deficiency	pathological state resulting from lack of energy, carbohydrate, protein and/or fat
Malnutrition	is the state of disease caused by sustained deficiency, excess, or imbalance in the intake of calories, nutrients, or both
Manila Electric Company (MERALCO)	The largest electric power distribution company in the Philippines covering 36 cities and 75 municipalities, including Metro Manila
Manila Water Company Inc. (MWCI)	A company that holds the exclusive right to provide water and wastewater services to the East Zone of Metro Manila, under a Concession Agreement entered into between the company and MWSS.
Maternal Death	Death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes
Maynilad Water Services Inc. (MWSI)	The water and wastewater services provider in the West Zone of the greater Metro Manila area under a Concession Agreement entered into between the company and MWSS
Metro Manila Accident Recording and Analysis System (MMARAS)	Created and managed by the Road Safety Unit (RSU) of the MMDA-Traffic Discipline Office-Traffic Engineering Center (MMDA-TDO-TEC), with the cooperation and assistance of the Traffic Enforcement Unit (TEU) of the Philippine National Police (PNP). The objective is to compile and maintain a database of Fatal, Non Fatal Injury and Damage to Property road crashes, which can indicate areas where safety improvements are needed to be made. The system will also allow the impact of improvement measures that needs to be monitored. This report is intended to provide brief information on “Fatal”, “Non Fatal Injury” and “Damage to Property” road crashes that have been recorded by the MMDA-Road Safety Unit thru the Police Blotter of the PNP each year. The information is presented in tabular form, which provides a readily identifiable pattern of accident locations and causation patterns.
Micronutrient Deficiency	pathological state resulting from lack of a certain nutrient e.g. vitamin A deficiency, iron deficiency, iodine deficiency disorders

Morbidity Rate	The frequency or proportion with which a disease appears in a population. Morbidity rates are used in actuarial professions, such as health insurance, life insurance and long-term care insurance to determine the premiums to charge to customers
Mortality Rate	A measure of the number of deaths in a particular population, scaled to the size of that population, per unit of time
National Demographic and Health Survey (NDHS)	A part of the worldwide Demographic and Health Surveys (DHS) program designed to collect, analyze, and disseminate demographic data on fertility, family planning, and maternal and child health.
Neonatal Death	A neonatal death is defined as a death during the first 28 days of life (0-27 days).
Networked Readiness Index (NRI)	A composite indicator made up of four main categories (subindexes), 10 subcategories (pillars), and 53 individual indicators distributed across the different pillars. NRI is a joint undertaking of World Economic Forum (WEF) and INSEAD.
Non-Index Crimes	All types of crimes not considered as index crimes
Obesity	uses weight-for-height as indicator; classifies those whose weight for their height
Overnutrition	pathological state resulting from the consumption of an excessive quantity of food over, and hence a calorie excess, over an extended period of time e.g. obesity
Overweight	uses weight-for-height as indicator; classifies those whose weight are above the standard for their height
Pacific Ring of Fire	Area in the basin of the Pacific Ocean wherein earthquakes and tsunamis occur frequently
Population Growth Rate	The rate at which the number of individuals in a population increases in a given time period, expressed as a fraction of the initial population. Specifically, population growth rate refers to the change in population over a unit time period, often expressed as a percentage of the number of individuals in the population at the beginning of that period.

Philippine Economic Zone Authority (PEZA)	A government agency in the Philippines attached to the Department of Trade and Industry created to help promote investments in the export-oriented manufacturing industry into the country by assisting investors in registering and facilitating their business operations and providing tax incentives. PEZA also assists investors who locate in service facilities inside selected areas in the country (areas are called PEZA Special Economic Zones) which are usually business process outsourcing and knowledge process outsourcing firms. Other activities also eligible for PEZA registration and incentives include establishment and operation within special economic zones for tourism, medical tourism, logistics and warehousing services, economic zone development and operation and facilities providers.
Poverty Incidence	Proportion of families/individuals with per capita income/ expenditure less than the per capita poverty threshold to the total number of families/individuals
Poverty Threshold or Poverty Line	Minimum level of income required by an individual to meet the basic non-food needs such as clothing, housing, transportation, health, and education expenses, among others
Putatan Treatment Plant	The largest membrane-based water treatment plant in the Philippines using large-scale microfiltration and reverse osmosis.
Resettlement Action Plan (RAP)	The document in which a project sponsor or other responsible entity specifies the procedures that it will follow and the actions that it will take to mitigate adverse effects, compensate losses, and provide development benefits to persons and communities affected by an investment project (World Bank).
Service Delivery Networks (SDN)	Refers to the network of health facilities and providers within the city-wide health system offering core packages of health care services in an integrated and coordinated manner to priority population.
Subsistence Incidence	Proportion of families/individuals with per capita income/ expenditure less than the per capita food threshold to the total number of families/ individuals
Sustainable Development Goals (SDGs)	Also known as the Global Goals, are a universal call to action to end poverty, protect the planet and ensure that all people enjoy peace and prosperity
Stunting or Underweight	uses height-for-age as indicator; classifies those whose height are below the standard for their age

Technical and Vocational Education and Training (TVET)	Education and training which provides knowledge and skills for employment. TVET uses formal, non-formal and informal learning. TVET is recognised to be a crucial vehicle for social equity, inclusion and sustainable development.
Total fertility rate	The number of children who would be born per woman (or per 1,000 women) if she/they were to pass through the childbearing years bearing children according to a current schedule of age-specific fertility rates.
Total Suspended Particles	The fraction sampled with high-volume samplers, approximately particle diameters <50-100 µm.
Twenty-foot Equivalent Unit (TEU)	An inexact unit of cargo capacity often used to describe the capacity of container ships and container terminals
Underemployment	Underemployment is the under-use of a worker due to a job that does not use the worker's skills, or is part time, or leaves the worker idle.
Undernutrition	pathological state resulting from the consumption of an inadequate quantity of food over an extended period of time
Underweight	uses weight-for-age as indicator; classifies those whose weight are below the standard for their age
Under-five Mortality Rate	Probability of dying between birth and exactly 5 years of age, expressed per 1,000 live births. Infant mortality rate:
Unemployment rate	Unemployment rate is a measure of the prevalence of unemployment and it is calculated as a percentage by dividing the number of unemployed individuals by all individuals currently in the labor force. During periods of recession, an economy usually experiences a relatively high unemployment rate.
Usufruct	Usufruct is a limited real right (or in rem right) found in civil-law and mixed jurisdictions that unites the two property interests of usus and fructus: Usus (use) is the right to use or enjoy a thing possessed, directly and without altering it. Fructus (fruit, in a figurative sense) is the right to derive profit from a thing possessed: for instance, by selling crops, leasing immovables or annexed movables, taxing for entry, and so on.

Wasting or thinness	uses weight-for-height as indicator; classifies those whose weight are below the standard for their height
West Valley Fault (WVF)	The west segment, known as the is one of the two major fault segments of the Valley Fault System which runs through Metro Manila to the cities of Marikina, Quezon City, Pasig, Makati, Taguig and Muntinlupa and moves in a dominantly dextral strike-slip motion. The West Valley Fault segment traverses from Doña Remedios Trinidad to Calamba with a length of 129.47 kilometres (80.45 mi). The West Fault is capable of producing large scale earthquakes on its active phases with a magnitude of 7 or higher.
Young Adult Fertility and Sexuality (YAFS) Study	The Young Adult Fertility and Sexuality (YAFS) Study is a series of national surveys on the Filipino youth, conducted since 1982 by the University of the Philippines Population Institute (UPPI) and the Demographic Research and Development Foundation. Gathering data from Filipino youth aged 15 to 24, YAFS is one of the primary sources of information on sexual and non-sexual risk behaviors and its determinants in the Philippines, at the national and regional levels. YAFS covers a wide range of topics that are relevant to this age group such as education trajectories, labor force participation, relationships and roles in society, values and attitudes, in addition to the risk behaviors. Findings from the YAFS series have been widely used in education and health and had provided the evidence base for health programs for young people by government and non-government organizations nationwide.
Zoning Ordinance	A zoning ordinance is a written regulation and law that defines how property in specific geographic zones can be used. Zoning ordinances specify whether zones can be used for residential or commercial purposes, and may also regulate lot size, placement, bulk (or density) and the height of structures

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Asec. Jose V. Campo
Assistant General Manager for Planning

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Ms. Arnie L. Geven	Admin Assistant
Mr. Michael L. Jacinto	Driver III
Mr. Benhur P. Parnaso	Heavy Equipment Operator II
Mr. Antonio P. Gatpo	Driver III

Schedule of RDP Consultations

Bilateral Consultations:

1. MWCI and MWSI	-	June 20, 2017
2. DILG, BJMP, BFP, and NCRPO	-	June 23, 2017
3. DOST-NCR	-	June 27, 2017
4. DSWD-NCR	-	June 28, 2017
5. DOLE-NCR	-	June 29, 2017
6. DOT, NCCA, and NHCP	-	July 4, 2017
7. DOH, NNC, DepEd, and CHED	-	July 5, 2017
8. NEDA RDS	-	July 5, 2017
9. PSA	-	July 6, 2017
10. POPCOM	-	July 6, 2017
11. DTI and BSMED	-	July 7, 2017
12. DENR and EMB	-	July 11, 2017
13. NHA, HUDCC, HDMF, HLURB, HGC, and SHFC	-	July 13, 2017
14. DPWH, UPMO, DOTr, DOE, DICT, and TRB	-	August 8, 2017
15. MICT, POEA, OWWA, and DOLE	-	September 27, 2017
16. OWWA	-	September 29, 2017
17. POEA	-	October 4, 2017
18. PPA South Harbor and ATI	-	October 5, 2017
19. PPA PMO North Harbor	-	October 11, 2017
20. TESDA	-	December 1, 2017
21. NCCA	-	December 14, 2017
22. DICT	-	December 21, 2017
23. Commission on Filipino Workers	-	February 7, 2018
24. Infrastructure Planning Committee	-	February 12, 2018
25. Environment Planning Committee	-	February 13, 2018
26. NEDA DIS and RDS	-	May 4, 2018

Writeshops:

27. Writeshop with DOH	-	September 6, 2017
28. Writeshop with DSWD	-	September 7, 2017
29. Writeshop with DOH, POPCOM, and NNC	-	September 18, 2017
30. Validation Workshop with RLAs and Metro Manila LGUs	-	January 29, 2018
31. NCCA Planning and Workshop/Writeshop	-	March 23, 2018

Learning Events:

32. Learning Events with MWCI	-	September 20, 2017
33. Learning Events with MWSI	-	September 20, 2017

Others:

1. RSET Quarterly Meeting	-	November 2, 2016
2. Climate Change and HFEP Summit	-	May 17, 2017
3. Presentation during the MMC 13 th Regular Meeting	-	June 21, 2017
4. 28 th National Statistics Month	-	October 3, 2017
5. Regional Workshop on Social Protection	-	September 24, 2018
6. Localization of the NCR RDP's Result Matrices	-	October 10, 2018